Statistical Architecture Framework

The Framework, complete document (version 0.9)

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Preface

The Statistical Architecture Framework (SAF) was developed to support the modernisation of official statistics. This framework responds to the need of national statistical institutes and their partners for shared guidelines and standards. Producing statistics must remain credible, secure, and efficient. At the same time, organisations must adapt quickly to new data sources, higher user expectations, and tighter regulations and budgets. The SAF provides a solid foundation for this and help organisations guide these changes in a controlled and structured way.

This new framework replaces the older CSPA and CSDA standards, but it reuses the elements that are still current and valuable. The SAF combines the core ideas from these previous framework into a single, logical, and coherent model. While the framework continues to focus on cooperation, reuse, and standards, the focus has shifted heavily toward a practical application of the framework in an organization. It is specifically designed to be used directly within ongoing programmes and projects. This helps teams make concrete choices when designing and implementing new processes and solutions.

The fundamental principle of the framework is that data and its descriptions are the most important assets of a statistical organisation. Information is not just a tool to run processes, but the core of the organisation. The SAF therefore treats this information as an NSIs most valuable asset that must be professionally managed throughout its entire lifecycle. It must support systems that can work with both traditional and modern, new data sources and information flows. In addition, it offers a practical guide for collecting, combining, and publishing data.

The SAF builds on a modular approach where systems are constructed from loose, flexible components. It promotes the reuse of these digital building blocks, both within an organisation and between different organisations. This prevents projects from independently building the same tools, which reduces duplication and lowers costs. Furthermore, it ensures a clear separation between business methods, data, and the underlying technology. This separation is essential for keeping complex IT landscapes manageable and successfully scaling them up. The framework is designed to be used at multiple levels at the same time. It helps with strategic planning, designing specific IT solutions, and monitoring how different projects align with one another.

The SAF does not replace existing statistical models, but rather works closely alongside them. It forms the connecting link between organisational strategy, business processes, and technology. This connection helps organisations make consistent decisions and makes sure that organization policy is translated into practice. To improve communication between teams, the SAF uses different visual models. By using a standard notation style, organisations can easily compare and reuse each other's designs. Finally, the SAF is a living framework that will continuously evolve based on practical experience, community feedback, and new technological developments.

Acknowledgements

TODO

Introduction

The Statistical Architecture Framework (SAF) provides a common architectural basis for the design, delivery, and evolution of statistical production in National Statistical Institutes (NSIs) and related organisations. It brings together architectural concepts, reference structures, and practical guidance that support consistent decision-making across portfolios and programmes. SAF aims to help statistical organisations modernise with control, while preserving the institutional qualities that make official statistics credible: transparency, quality, impartiality, reproducibility, and public trust.

Statistical organisations are facing a period of accelerated change. Data ecosystems are evolving and increasingly depend on partnerships and shared infrastructures. Artificial intelligence and rapid technological advances are changing how data can be collected, processed, and interpreted, while also introducing new risks and new expectations. At the same time, trust, scientific rigor, and data protection requirements are becoming more prominent and more demanding. These changes occur in an uncertain and fragmented global

environment, where information integrity and societal confidence cannot be taken for granted. All of this is happening under sustained resource constraints, requiring organisations to be more agile and deliberate in how they prioritise, reuse, and industrialise capabilities. SAF is intended to support organisations in quickly responding to these drivers without fragmenting their architecture, governance, and implementation landscape in the process.

SAF replaces earlier reference documentation by consolidating core architectural ideas into a single framework that is more usable in practice. It is not designed as a theoretical reference alone. It is designed to be applied. SAF supports the translation of strategy into capabilities, services, and implementation choices. It supports alignment across business, methodology, data, applications, and technology. It also supports collaboration between organisations by promoting common patterns, shared viewpoints, and comparable architectural descriptions.

Purpose

The purpose of SAF is to provide an authoritative, consistent, and actionable framework for developing and governing statistical architectures. It aims to support coherent modernisation across NSIs by providing shared structures for describing needs, designing solutions, and managing change. SAF helps ensure that investments in platforms, data management, methods, and delivery practices strengthen each other instead of competing. It also helps organisations to demonstrate traceability from policy and strategy to implemented solutions, including the controls required for confidentiality, integrity, accountability, and lawful processing.

SAF is intended to enable better outcomes in three areas. First, interoperability and reuse, both within an organisation and across organisational boundaries.

Second, quality and robustness, by making architectural choices explicit and reviewable, and by promoting consistent handling of data, metadata, and methodological assets. Third, overall statistical production effectiveness, by providing a common language and set of expectations that connects architects, methodologists, engineers, managers, and oversight functions.

Definition of Scope

SAF covers the enterprise architecture concerns that are most relevant to statistical production and dissemination, including the supporting functions that enable them. It addresses the design and evolution of capabilities, information assets, applications and services, and underlying infrastructure and deployment arrangements. It also considers the cross-cutting concerns that are essential for official statistics, such as governance, security, privacy, auditability, reproducibility and the responsible use of data and advanced analytics.

To ensure consistency and practical applicability, SAF is organised according to a TOGAF-aligned structure. It addresses motivation and principles, strategic direction and target outcomes, and the architecture domains that connect intent to implementation. This includes business architecture, data architecture, application architecture, and technology and infrastructure considerations, as well as deployment and operationalisation aspects. SAF also includes viewpoints that are tailored to distinct roles and activities within an NSI, enabling the same underlying architecture to be communicated and assessed from multiple perspectives without losing coherence.

SAF does not prescribe a single operating model, a single technology stack, or a single organisational design. It recognises that NSIs differ in mandate, maturity, and context. Instead, it provides a shared architectural foundation that can be adapted to local realities while remaining comparable across organisations. It also assumes that terminology and detailed definitions are managed through a dedicated glossary, which is maintained alongside the framework.

Target Audience

This documentation is primarily written for enterprise and data architects, IT architects, methodologists, and IT delivery teams. It is also intended for senior decision-makers such as CIOs and CTOs, and for privacy officers, policy makers, and managers who carry responsibility for compliance, traceability, and responsible data use.

These groups and individuals can use this framework as a reference and as a working instrument. It can be applied when shaping strategy and roadmaps, when designing new capabilities and platforms, and when assessing initiatives for alignment and risk. It can be used to structure architectural work in programmes and projects, and to support governance decisions by making assumptions, dependencies, and trade-offs explicit. It can also be used to support communication across disciplines by establishing shared architectural concepts and consistent viewpoints.

Different roles will use SAF in different ways, while relying on the same underlying content. Enterprise and data architects will use it to define target states, standards, patterns, and architecture decisions across the organisation. IT architects and delivery teams will use it to translate those decisions into solution designs, implementation plans, and operational controls. Methodologists will use it to connect methods and process design to the enabling data and technology arrangements, and to support repeatability and scientific rigor. CIOs, CTOs, managers, and policy makers will use it to understand the implications of strategic choices, to prioritise investment, and to oversee risk and compliance. Privacy officers and assurance functions will use it to assess whether designs incorporate adequate safeguards, traceability, and accountability.

SAF is most effective when it is used consistently across the lifecycle of change. It supports early-stage framing by clarifying drivers, objectives, and constraints. It supports design by providing reference structures and viewpoints that guide architecture development. It supports delivery by improving alignment between architecture and implementation decisions. It supports operation and evolution by making dependencies and responsibilities explicit, enabling controlled change and measurable improvement.

Expected value

The expected value of SAF is a stronger and more coherent statistical architecture practice across NSIs. It supports faster and safer modernisation by reducing ambiguity and rework. It supports reuse by encouraging common building blocks and comparable solution descriptions. It supports trust by embedding governance and assurance concerns into architecture, rather than treating them as afterthoughts. It also supports collaboration across organisations, where shared approaches to interoperability, data governance, and service orientation reduce the effort required to work together. In providing a common framework with practical orientation, SAF aims to help statistical organisations remain resilient and responsive. It supports them in delivering high-quality statistics under changing conditions, while maintaining the safeguards and institutional discipline that official statistics require.

Constraints & Assumptions

Use of TOGAF and ArchiMate in SAF

The SAF builds on two open standards of The Open Group, a global, vendor-neutral consortium that develops technology and architecture standards: the TOGAF® Standard and the ArchiMate® Specification. This section explains what these standards are, why the SAF uses them, and how they relate to each other and to the statistical standards of the UNECE community.

What TOGAF is

The TOGAF Standard is a framework for developing and governing enterprise architecture. It provides:

- **A method.** The Architecture Development Method (ADM) is an iterative cycle of phases. It runs from establishing the architecture capability and an architecture vision, through business, information systems, and technology architecture, to migration planning, implementation governance, and change management.

- **A content framework.** A common structure for the artefacts that architecture work produces, such as principles, stakeholder maps, capability models, and architecture building blocks.
- **Guidance for an architecture capability.** Advice on governance, roles, and maturity, so that architecture becomes a lasting practice rather than a one-off project.

The SAF refers to the TOGAF Standard, 10th Edition. More information is available at The Open Group: TOGAF Standard.

Why the SAF uses TOGAF

The SAF uses TOGAF for three reasons.

- **A shared structure:** Many NSIs already use TOGAF, or a framework derived from it, in their own architecture practice. Aligning the SAF with TOGAF lets NSIs map SAF content onto their existing repositories, governance processes, and training without translation.
- **A proven method:** TOGAF describes how to move from motivation to implementation in controlled steps. The SAF follows the same order: motivation, strategy, business architecture, data, application, and technology architecture, and implementation. This gives NSIs a recognisable path from intent to solution.
- **A common language across organisations:** TOGAF terminology is widely understood by architects, IT suppliers, and international partners. Using it reduces ambiguity when NSIs compare architectures or collaborate on shared solutions.

The SAF uses TOGAF as a **structuring reference, not as a prescription**. It tailors the TOGAF structure to official statistics and does not require NSIs to follow every ADM phase or to hold TOGAF certification. The SAF does not reproduce TOGAF content; readers are referred to the standard for its full description.

What ArchiMate is and how it relates to TOGAF

ArchiMate is a modelling language for enterprise architecture. Where TOGAF describes *how* to develop an architecture and *which* content to produce, ArchiMate describes *how to express* that content in consistent, unambiguous models and views. The two standards are maintained by The Open Group and are designed to work together:

Aspect	TOGAF	ArchiMate
Nature	Framework and method	Modelling language and notation
Answers	How is architecture developed and governed?	How is architecture described and visualised?
Main content	ADM phases, content framework, architecture capability	Elements, relationships, viewpoints
Role in the SAF	Structures the chapters and the order of the framework	Notation for the models and diagrams in the SAF

The structures of the two standards correspond. The TOGAF Preliminary Phase, Phase A, and Requirements Management correspond to the ArchiMate motivation and strategy elements. TOGAF Phases B, C, and D correspond to the ArchiMate business, application, and technology layers. TOGAF Phases E to H correspond to the ArchiMate implementation and migration elements. This correspondence is the basis for the position of each SAF chapter (see *Position in the framework* in the Motivation chapter).

More information is available at The Open Group: ArchiMate.

Relationship with statistical standards

TOGAF and ArchiMate are generic standards. The SAF complements them with the standards of the statistical community: GSBPM for statistical business processes, GAMSO for the activities of statistical organisations, GSIM for statistical information objects, and SDMX for the exchange of statistical data and metadata. TOGAF and ArchiMate provide the architectural structure and notation; the statistical standards provide the domain content. The SAF connects the two.

Assumptions and constraints

ID	Type	Statement
SAF-A1	Assumption	The SAF is structured according to the TOGAF Standard, 10th Edition, tailored to official statistics.
SAF-A2	Assumption	Models and diagrams in the SAF use the ArchiMate 3.2 Specification. Alignment with the ArchiMate 4 Specification will be assessed in a later version of the SAF.
SAF-A3	Assumption	NSIs can apply the SAF without TOGAF or ArchiMate certification. Basic familiarity with both standards helps but is not required.
SAF-A4	Assumption	Terminology and definitions are maintained in a separate SAF glossary, which takes precedence over definitions in this document.
SAF-C1	Constraint	The SAF refers to TOGAF and ArchiMate but does not reproduce their content. Use of the standards is subject to the licence terms of The Open Group.
SAF-C2	Constraint	The SAF does not prescribe a technology stack, tool, or organisational design (see *Definition of Scope*).

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Reading guide

This reading guide explains how the SAF is built, in which order to read it, and how an NSI can start using it. It also explains how the SAF fits into an NSI's own architecture, how it combines with other reference models, and how legislation enters the architecture.

1. How the SAF is structured

The SAF follows the layers of TOGAF and ArchiMate (see *Constraints & Assumptions*). Each layer answers one question. Each layer is justified by the layer above it, and each layer gives direction to the layer below it.

PLACEHOLDER: Overview of the SAF layers and chapters

Chapter	Question it answers	Main content	Status [**RG4**]
Motivation	*Why* does the NSI change, and what does it want to achieve?	Stakeholders, drivers, assessments, goals, outcomes, principles, requirements, constraints, meaning, value	Developed
Strategy	*What* must the NSI be able to do, and how does it get there?	Capability model, value streams, resources, courses of action and strategies	Developed
Business Architecture	*How* is the statistical work organised?	Business domains, reference models for the data value chain (Steady-State, Medallion), steady states, value stream detail, roles	In development
Data Architecture	*Which* data and metadata are needed, and how are they structured?	To be developed	Planned
Application Architecture	*Which* applications and services support the work?	To be developed	Planned
Technology Architecture	*On which* platforms and infrastructure does it run?	To be developed	Planned
Implementation and Tooling	*How* is the architecture realised and governed?	To be developed	Planned

Every section that describes an architecture concept follows the same pattern:

1. **The concept.** What the concept means, based on ArchiMate and TOGAF.
2. **How the SAF applies it.** Why the SAF uses the concept, and how it fits into an NSI's own architecture.
3. **Relationships with other concepts.** How the concept connects to the rest of the framework.
4. **SAF elements or examples.** The reference content of the SAF, or illustrative examples where the content is specific to each NSI.
5. **Toolkit.** A reference to templates and instruments in the SAF toolkit.

This pattern makes it possible to read a single section without reading the whole document, and still see how it connects to the rest.

The SAF contains two kinds of content. **Reference content** is shared by most NSIs and can be adopted directly, such as the drivers, goals, principles, capability model and value streams. **Illustrative content** shows the form of elements that each NSI defines for itself, such as assessments, requirements, constraints, resources and courses of action. Illustrative content is never a requirement.

2. Reading order

The SAF is best read top-down, at least once: from Motivation, through Strategy, to the architecture domains. This order follows the logic of the framework. A business architecture choice only makes sense once the goals and capabilities behind it are clear.

Applying the SAF is different from reading it. Application is iterative. An NSI does not complete the Motivation layer before it starts on its business architecture. It works in cycles, like the TOGAF Architecture Development Method: each cycle refines the motivation, strategy and architecture a little further, and each change in a driver or goal can start a new cycle.

Different readers can follow different routes through the document (see *Target Audience* for the roles).

Reader	Start with	Then read	Focus
CIO, CTO, board member	Introduction, Motivation	Strategy: capability model, courses of action	Why change is needed, what it delivers, how to prioritise
Enterprise or data architect	Introduction, Motivation	Strategy, Business Architecture, later domains	The full chain, and how to adopt it in the NSI architecture
IT architect, delivery team	Motivation: principles	Business Architecture, later domains	Design rules and reference models for solutions
Methodologist	Strategy: value streams	Business Architecture: steady states	Where methods are applied, and how quality and lineage are secured
Privacy officer, auditor	Motivation: principles, constraints	Business Architecture: steady states; VS-07	Where protection and compliance are built into the chain
Policy maker, manager	Introduction, Motivation: goals, value	Strategy: courses of action	What the framework means for the organisation

3. Positioning the SAF in your own architecture

TOGAF describes a range of architectures, from generic to specific, called the *Enterprise Continuum* [**RG2**]. It runs from foundation architectures, which apply to any organisation, through common systems and industry architectures, to the architecture of one specific organisation.

Level in the Enterprise Continuum	Example	Who owns it
Foundation architecture	TOGAF, ArchiMate	The Open Group
Common systems architecture	Generic reference architectures for, for example, data platforms or identity management	Industry and standards bodies
Industry architecture	**The SAF**, for official statistics	UNECE statistical community
Organisation-specific architecture	The enterprise architecture of an NSI	The NSI

The SAF is an industry architecture. It is not the architecture of any single NSI. Each NSI builds its own organisation-specific architecture, using the SAF as its reference. In practice, this works as follows:

1. **Adopt, adapt or extend.** For each SAF element, the NSI decides whether to adopt it as it is, adapt it to the national context, or add NSI-specific elements (see *How NSIs use this chapter* in the Motivation chapter).
2. **Keep the SAF identifiers.** When an NSI adapts an SAF element, it keeps a reference to the SAF identifier, for example SAF-P07 or capability 2.1.2. This keeps the NSI architecture comparable with the SAF and with other NSIs, and makes it easier to adopt future SAF versions.
3. **Layer the architecture.** The NSI enterprise architecture sets the direction for solution architectures in programmes and projects. Solution architectures do not refer to the SAF directly, but to the NSI architecture that applies it.

PLACEHOLDER: From SAF reference architecture to NSI enterprise architecture to solution architecture

4. Combining the SAF with other reference models

The SAF rarely stands alone. NSIs also work with statistical standards, with reference models of their own government, and with generic frameworks. These models complement each other, but they can also overlap or conflict.

Family	Examples	Role next to the SAF
Statistical standards and models	GSBPM, GAMSO, GSIM, SDMX; the ESS Enterprise Architecture Reference Framework (ESS EARF)	Provide the statistical domain content. The SAF connects them in one architecture
Government reference models	The European Interoperability Framework (EIF) and Reference Architecture (EIRA); national reference models such as those listed below	Often mandatory or leading for public organisations. The SAF fits within them
Generic frameworks and bodies of knowledge	TOGAF, ArchiMate, DAMA-DMBOK	Provide structure, notation and professional practice

Examples of national government reference models [**RG3**]:

Country	Reference model	Owner
Netherlands	NORA, Netherlands Government Reference Architecture	Dutch government (ICTU)
United Kingdom	GovS 005 Digital, Government Functional Standard	Cabinet Office, Government Digital and Data
Canada	Government of Canada Enterprise Architecture Framework	Treasury Board of Canada Secretariat
Australia	Australian Government Architecture (AGA)	Digital Transformation Agency
New Zealand	Government Enterprise Architecture for New Zealand (GEA-NZ)	Government Chief Digital Officer
Norway	National architecture principles for digitalisation of the public sector	Norwegian Digitalisation Agency (Digdir)
United States	Federal Enterprise Architecture Framework (FEAF)	Office of Management and Budget

The SAF works best when the NSI combines models at the level of elements, not documents:

- **Principles.** Map the SAF principles to the principles of the national reference model. Many will be equivalent. Where both exist, the NSI uses one set and records the mapping.
- **Capabilities and functions.** Map the SAF capability model to the capability or function model of the national framework, where one exists. This makes the NSI comparable with other government organisations as well as with other NSIs.
- **Standards.** Check which standards the national model prescribes, for example for exchange formats, identity or security. These standards become requirements or constraints in the NSI architecture.

When models conflict, a simple order of precedence helps:

1. Legislation comes first.
2. Mandatory government standards and reference models come next.
3. Within the statistical domain, the SAF and the statistical standards apply.
4. Generic frameworks and practices fill the remaining gaps.

A deviation from a higher level is a decision for the NSI's architecture governance, and is documented.

PLACEHOLDER: Mapping templates for SAF and national reference models to be added to the toolkit

5. Legislation and regulation

Official statistics operate under strict legal frameworks. Legislation shapes what an NSI may collect, how it must protect data, and how it must publish. The SAF does not describe legislation, because it differs per country. It does describe where legislation enters the architecture.

Legislation enters the architecture through the motivation elements:

A legal provision that…	Is recorded as…	Example
Creates pressure to change	Driver	A new regulation that requires access to privately held data for statistics
Sets a limit on the design	Constraint	Statistical microdata may only be processed within the NSI's controlled environment
Imposes an obligation on a solution	Requirement	Personal data in a data acquisition process is minimised to what is needed
Expresses a lasting rule for all decisions	Principle	Privacy by design (SAF-P11)

This approach makes legislation traceable. From a legal provision, the NSI can follow the path to the requirements and constraints it creates, to the solutions that meet them, and to the evidence that shows compliance. In the other direction, every design decision that is driven by law can be traced back to its legal basis.

Three levels of legislation are relevant for most NSIs:

Level	Examples
International statistical frameworks	UN Fundamental Principles of Official Statistics; European Statistics Code of Practice; Regulation (EC) No 223/2009 on European statistics
Horizontal EU legislation	GDPR; Data Governance Act; Data Act; AI Act; NIS2 Directive; Interoperable Europe Act (Regulation (EU) 2024/903)
National legislation	National statistics act; archives legislation; national rules on government information security and open government

For NSIs outside the EU, the horizontal level consists of the corresponding national or regional legislation.

Compliance is not a separate layer. It is built into the architecture: legal requirements are part of the quality gates of the steady states, of the principles, and of the governance in value stream VS-07 *Governance and Compliance*. A legal register helps to keep the overview: a list of applicable legal provisions, with the drivers, constraints, requirements and principles they lead to.

PLACEHOLDER: Legal register template to be added to the toolkit

6. Conventions used in this document

Convention	Meaning
SAF-ST, SAF-D, SAF-G, SAF-O, SAF-P, SAF-M, SAF-V	Identifiers of reference elements in the Motivation layer
1.2.1, 2.1.2, …	Capability numbers in the capability model
VS-01 to VS-10, VS-01.1, …	Value streams and their stages
#0 to #6	Steady states of data and their boundary positions
SAF-A, SAF-C	Assumptions and constraints of the SAF itself (see *Constraints & Assumptions*)
Illustrative	Examples that show the form of an element. They are not requirements
PLACEHOLDER	Content that will be added, such as an ArchiMate view or a toolkit reference
Toolkit	The SAF toolkit, a separate set of templates and instruments for applying the framework

Terms are defined in the SAF glossary, which takes precedence over definitions in this document (SAF-A4). The SAF uses British English spelling.

Motivation

Purpose of this chapter

The Motivation chapter explains why the SAF exists. It also explains why architecture decisions in an NSI are taken the way they are. It records the forces that act on statistical organisations, the results they want to achieve, the rules that guide their decisions, and the parties whose concerns must be addressed.

Without an explicit motivation, architecture is a set of models without a reason. With it, every capability, process, data state, application or platform choice can be traced back to a need and a result. That traceability makes architecture defensible, whether in a governance board, a budget discussion or an international review.

Position in the framework

Motivation is the first layer of the SAF and the foundation for the other layers. The Strategy layer translates motivation into capabilities and value streams. The architecture domains (business, data, application and technology) then translate strategy into design.

The SAF is structured according to TOGAF and modelled in ArchiMate. The figure below shows how the phases of the TOGAF Architecture Development Method (ADM) relate to the ArchiMate layers. The figure has three areas:

- **Strategy & Motivation:** the Preliminary Phase, Phase A (Architecture Vision), and Requirements Management.
- **Core layers:** Phases B, C, and D, which produce the Business, Application (Information Systems), and Technology architectures.
- **Implementation & Migration:** Phases E to H, which plan, govern, and change the realisation of the architecture.

PLACEHOLDER: Logical model of the different layers to be added

The Motivation layer sits in the Strategy & Motivation area, at the start of the ADM cycle. It holds the content that TOGAF produces before any architecture domain is designed. That content keeps steering the architecture in all later phases.

ADM phase	Role of motivation in this phase	SAF sections
Preliminary	Establish the architecture capability: who the architecture serves and which rules guide decisions	Stakeholders, Architecture Principles
A. Architecture Vision	Explain why change is needed, what must be achieved and what value it brings	Drivers, Goals, Outcomes, Meaning, Value
Requirements Management	Turn goals into requirements and constraints, and keep them traceable through every phase	Requirements, Constraints
B, C, D. Core architectures	Justify design choices in business, application and technology architecture	Traceability from each design decision back to a goal and a driver
E to H. Implementation & Migration	Prioritise solutions and work packages; reassess when drivers change	Drivers and goals as criteria for prioritisation and change requests

Three consequences follow from this position.

- **Motivation comes first.** In TOGAF, no architecture domain is designed until stakeholders, drivers, goals and principles are agreed. The SAF follows the same order. Every later element must be traceable to the motivation.
- **Motivation stays active throughout the cycle.** Requirements Management sits at the centre of the ADM. Goals are refined into requirements there, and those requirements are checked in every phase. Motivation is the reference against which each architecture decision is tested, not a one-off introduction.
- **Motivation closes the loop.** In Phase H (Architecture Change Management), a new or changed driver triggers a change request. Examples are a new regulation or a new data source. The cycle then returns to the Preliminary Phase or Phase A, and the motivation elements become the starting point of the next iteration.

The Strategy layer shares the Strategy & Motivation area with the Motivation layer. Motivation explains *why*. Strategy describes *what* the NSI must be able to do (capabilities) and *how* it creates value (value streams). Strategy is the bridge to Phase B, Business Architecture.

How NSIs use this chapter

The motivation elements in this chapter are a reference set. They describe internal and external forces and intentions that most NSIs share. Modelling the motivation layer is not mandatory, but the SAF strongly recommends it.

Most NSIs have a motivation, but it is implicit. It lives in strategy documents, board decisions and the experience of staff. It is rarely written down in a way that connects it to architecture decisions. Making it explicit, in SAF terms, helps an NSI in four ways.

First, it justifies investment. If every programme can be traced to a driver and a goal, an investment proposal rests on shared reasoning, not on individual preference. Boards and ministries can see what a proposal responds to, and what is at stake if it is not funded. The same links help to set priorities. Work that addresses several drivers, or contributes to a key outcome, comes before work that addresses none.

Second, it makes decisions consistent. Principles are derived from the motivation and turn intentions into decision rules. Similar questions then get similar answers, whichever team or architect is involved. When circumstances change, for example through new legislation, the documented links show which goals, principles and capabilities are affected. The NSI can review those specific decisions instead of starting over.

Third, it preserves knowledge. Staff move on and projects end. A documented motivation keeps the reasoning behind past decisions available. New statisticians and IT specialists can see why the architecture looks the way it does.

Fourth, it supports collaboration between NSIs. When NSIs use the same concepts and identifiers, they can quickly see where their interests overlap. That makes it easier to start joint projects, share components and compare approaches. A documented motivation is also useful evidence in external assessments, such as peer reviews.

An NSI can use each SAF motivation element in one of three ways: 1. **Adopt.** Use an SAF element as is, when it matches the national situation. 2. **Adapt.** Refine an SAF element with national specifics, such as a national statistics act or a government-wide digital strategy. 3. **Extend.** Add elements that are specific to the NSI, while keeping a reference to the closest SAF element where one exists.

PLACEHOLDER: Maturity Assessement to be added to the toolkit

Motivation concepts in the SAF

SAF section	ArchiMate element	Question it answers	Identifier
Drivers	Driver	Which forces make change necessary?	SAF-D
Goals	Goal	What does the NSI intend to achieve?	SAF-G
Meaning	Meaning	What does the framework signify to its users?	SAF-M
Outcomes	Outcome	Which concrete results show that a goal is achieved?	SAF-O
Architecture Principles	Principle	Which rules guide architecture decisions?	SAF-P
Value	Value	What is the framework worth, and to whom?	SAF-V
Assessment	Assessment	What does a driver mean for this NSI?	*TBD by NSI*
Requirements	Requirement	What must a solution do or provide to realise a goal?	*TBD by NSI*
Constraints	Constraint	Which limits apply to the design or its realisation?	*TBD by NSI*
Stakeholders	Stakeholder	Whose concerns must the architecture address?	SAF-ST

The concepts form a chain. Stakeholders have concerns about drivers. An assessment records what a driver means for one NSI. Drivers motivate goals. Outcomes make goals measurable. Principles guide how goals are pursued, and requirements and constraints make them concrete for a solution. Meaning and value express why the result matters to stakeholders.

Drivers

The concept

A **driver** is an external or internal condition that motivates an organisation to set goals and to change. A driver is a force, not a problem and not a solution. "Rapid technological change" is a driver. "Our methods cannot use administrative data" is an assessment of that driver for one NSI. "Integrate new data sources into production" is a goal.

Three characteristics set drivers apart from other motivation elements:

- **They exist independently of the NSI.** An NSI can respond to a driver, but cannot decide whether it exists.
- **They are external or internal.** External drivers come from the environment: technology, legislation, society, politics or the economy. Internal drivers come from within: legacy systems, skills or organisational structure.
- **They are relatively stable.** A driver usually stays relevant for several years. Short-lived events, such as an incident or a budget round, are better recorded as assessments.

How the SAF applies drivers

Why the SAF records drivers

Drivers make the SAF a response to real pressures instead of an abstract ideal. With them, an NSI can explain to its board, its ministry or a peer review why change is needed, and what happens without it. Drivers also help to set priorities. An investment that responds to several drivers deserves more attention than one that responds to none.

How drivers fit into an NSI's own architecture

The SAF drivers are common to the UNECE statistical community. An NSI applies them in four steps:

1. **Recognise.** Decide which SAF drivers apply to the national situation, and add national drivers where needed.
2. **Assess.** Analyse what each driver means for the NSI: strengths, weaknesses, risks and opportunities.
3. **Respond.** Derive or confirm the goals that answer the assessed drivers.
4. **Trace.** Link capabilities, programmes and architecture decisions to the drivers they address.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Stakeholders (SAF-ST)	Stakeholders have concerns about drivers and often raise them	CIOs and CTOs (SAF-ST01) raise resource constraints (SAF-D5)
Assessment	An assessment records what a driver means for one NSI	"Two thirds of our processing runs on platforms out of vendor support" assesses SAF-D5
Goals (SAF-G)	Drivers motivate goals	Resource constraints (SAF-D5) motivate cost efficiency (SAF-G1)
Principles (SAF-P)	Drivers explain why a principle exists	Trust and data protection (SAF-D3) explain privacy by design (SAF-P11)
Capabilities (Strategy)	Capabilities are developed in response to drivers, through goals	AI and technological acceleration (SAF-D2) drives investment in methodology (capability 2.2.3)
Value streams (Strategy)	Value streams show where a driver affects the production chain	Evolving data ecosystems (SAF-D1) affect data acquisition (VS-01)

SAF drivers

ID	Driver	Type	Description	Architectural response in the SAF
SAF-D1	Evolving data ecosystems and partnerships	External	Statistics no longer rely on surveys alone. They draw on administrative, commercial, platform and community data. This creates opportunities, but also dependencies on actors outside the NSI's control.	Secure data sharing, standard interfaces, open-source collaboration and transparent governance, so that NSIs can take part in national data platforms and European data spaces.
SAF-D2	Artificial intelligence and technological acceleration	External	AI, automation and fast technological change alter how data is created, processed and used. NSIs must adopt these technologies and make their own outputs AI-ready.	A risk-based approach: tight control of methodologically critical processes, more automation where risk is lower. Modular, standards-based design to adopt new technology at a controlled pace.
SAF-D3	Trust, scientific rigour and data protection	External	Misinformation, privacy concerns and unregulated data providers put public confidence under pressure. Trust needs more than compliance. It needs rigour, transparency and accountability.	Protection built into the design: confidentiality controls, methodological integrity, traceability, and oversight in proportion to sensitivity.
SAF-D4	Uncertain and polarised global environment	External	Global tensions and political polarisation affect how official statistics are demanded and perceived. Cross-border data flows require openness and interoperability.	Interoperable architectures based on shared standards. Design principles that protect independence, comparability and credibility.

ID	Driver	Type	Description	Architectural response in the SAF
SAF-D5	Resource constraints and organisational agility	Internal and external	Budgets are tight, systems are ageing and specialist skills are scarce. Demand for timely and detailed output keeps growing.	Modular design, shared services, reuse and step-by-step modernisation, so that capacity can be shifted without large reinvestment.

Driver details

SAF-D1 Evolving data ecosystems and partnerships

Official statistics are moving from isolated data collection to a wider data ecosystem. Surveys used to be the main source. Now they sit next to administrative, commercial, platform and community data. This opens up new forms of collaboration. It also makes NSIs dependent on actors, technologies and infrastructures they do not control.

Open source is part of this ecosystem. It gives NSIs a shared base for innovation, transparency and lower costs. NSIs can develop tools and methods together with peers, academia and private partners.

The SAF responds with architectures for secure data sharing, standard interfaces and transparent governance. These let NSIs take part in national data platforms and European data spaces, where production may span several environments and jurisdictions. Openness is combined with strong governance and clear accountability, so that collaboration does not compromise the integrity of official statistics.

SAF-D2 Artificial intelligence and technological acceleration

AI, automation and rapid technological progress are changing how data is created, processed and used. New tools can automate routine tasks, improve analysis and strengthen quality assurance. They also raise the question of what relevant and trusted statistics mean when users rely on intelligent systems.

NSIs must both use AI and make their outputs usable by AI. Internally, that requires architectures and skills for automation, machine learning and advanced analytics, within secure and transparent limits. Externally, statistics must be machine-readable, machine-understandable and interoperable with the platforms that policy makers, researchers and citizens use.

The SAF takes a risk-based approach. Processes that demand methodological rigour and control stay tightly governed. Less sensitive areas can be automated further. Modular, standards-based design lets an NSI adopt new technology at its own pace, while it stays fully accountable for quality, data protection and independence.

SAF-D3 Trust, scientific rigour and data protection

Trust is the basis of official statistics. Misinformation, privacy concerns and unregulated data providers put it under pressure. Compliance alone does not maintain public confidence. Scientific rigour, transparency and clear accountability for the handling of data are also needed.

The SAF builds trust and protection into the architecture. It promotes governance and technical controls that protect confidentiality, keep methods sound and make processes traceable. Oversight is strict where sensitivity and quality demands are high. Where risks are lower, more automation and shared solutions are possible.

SAF-D4 Uncertain and polarised global environment

Global tensions, political polarisation and shifting power relations affect how official statistics are demanded and perceived. At the same time, cross-border data flows require statistical systems to stay open and interoperable, without giving up national and institutional integrity.

The SAF offers a stable base for cooperation. Interoperable architectures based on shared standards let NSIs exchange data and methods securely. Design principles that protect independence and transparency help NSIs stay credible and comparable, even when facts themselves are contested.

SAF-D5 Resource constraints and organisational agility

Many NSIs must do more with limited budgets, ageing systems and a shortage of specialist staff. Demand for timely, detailed and cross-domain output keeps growing. Meeting that demand requires efficiency and the ability to adapt quickly.

The SAF makes flexibility and reuse core principles. It supports modular design, shared services and scalable solutions that reduce duplication. Clear governance of priorities lets NSIs shift capacity, modernise step by step and keep delivering quality output under financial and staffing pressure.

PLACEHOLDER: Drivers Template and Application of Drivers to be added to the toolkit

Goals

The concept

A goal is a statement of intent. It is the direction or end state an organisation wants to achieve. A goal says *what* the organisation wants to achieve. It does not say *how*, or *how much*.

Three characteristics set goals apart:

- **They are qualitative and directional.** "Modernise statistical systems" sets a direction. A measurable result, such as "retire all platforms out of vendor support by 2029", is an outcome.
- **They are long-term.** A goal usually spans several planning cycles and outlives individual programmes and projects.
- **They are owned by stakeholders.** A goal only has meaning if a stakeholder recognises it as their intention.

The SAF goals cover operational and strategic needs. Some focus on efficiency, modernisation and compliance, which keep an NSI sustainable. Others support innovation and collaboration, such as the use of new data sources or advanced analytics. Together these goals make sure that modernisation does not weaken the core values of official statistics: neutrality, accuracy and reliability.

How the SAF applies goals

Why the SAF records goals

Goals define what success looks like, for the SAF and for the NSIs that apply it. They give direction to the Strategy layer: capabilities and value streams are developed because they contribute to a goal. Without explicit goals, modernisation can turn into a series of technical upgrades that nobody can explain.

How goals fit into an NSI's own architecture

The SAF goals are shared by the UNECE statistical community. An NSI relates them to its own strategic plan. It then translates each relevant goal into measurable outcomes and, where needed, into objectives for programmes and projects. Goals that matter nationally but have no SAF counterpart are added as national goals. Each NSI can follow this path at its own pace, while working towards the same shared direction.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Drivers (SAF-D)	Drivers motivate goals	Resource constraints (SAF-D5) motivate cost efficiency (SAF-G1)
Outcomes (SAF-O)	Outcomes make goals measurable	Increased efficiency (SAF-O5) shows progress on cost efficiency (SAF-G1)

Related concept	Relationship	Example in the SAF
Principles (SAF-P)	Principles guide how goals are pursued	Open-source participation (SAF-P01) guides how cost efficiency is achieved
Stakeholders (SAF-ST)	Stakeholders own goals	CIOs and CTOs (SAF-ST01) own modernisation (SAF-G3)
Capabilities (Strategy)	Capabilities are developed to realise goals	Methodology (capability 2.2.3) supports data innovation (SAF-G4)
Value streams (Strategy)	Value streams show where a goal changes the production chain	Sustainable compliance (SAF-G5) is put into practice in VS-07

SAF goals

ID	Goal	Description	Architectural direction in the SAF
SAF-G1	Cost efficiency and resource sharing	Work as efficiently as possible with limited budgets and growing demand, by sharing services, reusing solutions and pooling resources.	Shared services, reusable and open-source components, modular architectures and step-by-step modernisation.
SAF-G2	Harmonised reference framework	Provide shared standards, patterns and practices that NSIs adapt locally while staying interoperable.	Common reference models, consistent design principles, and harmonisation as a condition for shared infrastructures and data spaces.
SAF-G3	Modernisation of statistical systems	Move from fragmented legacy environments to modular, service-based architectures that are secure, scalable and maintainable.	Service-based design, risk-based automation, and principles for access control, data protection and accountability where several parties work together.
SAF-G4	Support for data innovation	Bring new data sources, methods and products into production without loss of quality or consistency.	Controlled integration (versioning, validation, automated testing), shared platforms that link innovation and production, and safe reuse of open components.
SAF-G5	Sustainable compliance	Align modernisation with legal, security, quality and ethical frameworks, inside and outside the statistical domain.	Statistical standards (GSBPM, GSIM, SDMX) combined with standards for data management, AI and cloud. Compliance is designed in from the start.

Goal details

SAF-G1 Cost efficiency and resource sharing

Budgets are limited and demand keeps growing. NSIs must therefore work as efficiently as possible. The SAF promotes shared services, reuse of solutions and better interoperability. By pooling resources and using common components, NSIs gain capabilities they could not build alone, and they avoid duplicate work.

The SAF also encourages open-source components and community platforms where they improve transparency, collaboration and cost. Modular architectures and clear governance let NSIs adapt to new needs, shift capacity and modernise step by step, without large reinvestments. Efficiency then supports quality and long-term sustainability instead of competing with them.

SAF-G2 Harmonised reference framework

Without a shared architectural reference, NSIs build isolated solutions. These reduce comparability and raise long-term costs. The SAF offers a common base of standards, patterns and practices. NSIs adapt it to their own context and stay interoperable across borders and domains.

More and more collaboration takes place on shared infrastructures, such as national data platforms and international data spaces. Harmonisation is a condition for taking part. Consistent design principles help NSIs protect the integrity of official statistics, and reuse shared and open solutions safely.

SAF-G3 Modernisation of statistical systems

Many NSIs still depend on fragmented legacy IT that slows down innovation and response. The SAF guides the move to modular, service-based architectures that are secure, scalable and easier to maintain. Modernisation reduces technical debt, improves reliability and speeds up the delivery of new and better statistics.

Modernisation more often happens in settings with several parties, from national infrastructures to cross-border data spaces. The SAF sets principles for access control, data protection and accountability in those settings. It also takes a risk-based approach to automation and new technology. Critical processes stay tightly governed to protect methodological rigour. Less sensitive steps can be automated or shared.

SAF-G4 Support for data innovation

New data and methods are central to the future of official statistics. Yet many NSIs struggle to introduce a new data source or product. Their production systems are built for predictability and stability. Those qualities are necessary, but they slow down change.

The SAF supports architectures that combine robustness with flexibility. New components, data and workflows can then enter production faster, while quality and compliance stay under control. Versioning, validation and automated testing keep production reliable as it changes. Shared platforms and processes help prototypes move into operational use.

Open source matters here too. NSIs can build on each other's work, which increases transparency and the collective capacity of the community. The SAF supports this with patterns for the safe reuse of open components.

SAF-G5 Sustainable compliance

Compliance with laws, standards and ethical principles underpins the trust in and legitimacy of official statistics. Within the statistical domain, GSBPM, GSIM and SDMX remain the basis for consistent processes, metadata management and data exchange.

Statistical production is now part of a wider data and technology ecosystem. Compliance must therefore extend beyond statistical standards. The SAF encourages the use of relevant standards from data management, artificial intelligence and cloud computing. NSIs then stay interoperable with modern platforms and benefit from progress made elsewhere.

With consistent governance and open, verifiable technology, compliance becomes an enabler instead of a brake. Innovation, data sharing and collaboration take place in controlled and auditable environments. Transparency, scientific rigour and privacy by design are part of every process.

PLACEHOLDER: Goals Template and Application of Goals to be added to the toolkit

Meaning

The concept

Meaning is the knowledge or interpretation that a concept has in a given context. Meaning answers the question *what does this signify?* Value answers a different question: *what is it worth, and to whom?*

The SAF uses meaning at the level of the framework itself. The meaning elements describe how the statistical community should understand the SAF. Two interpretations matter most. First, the SAF connects strategy and technology. Every technology decision, from a new platform to a redesigned process, can be traced to a strategic objective. Next, the SAF is a common language for the UNECE statistical community. Shared models, practices and principles reduce duplicate work and make collaboration easier. This makes the SAF relevant beyond IT, to statisticians, policy makers and the citizens who depend on reliable statistics and information.

How the SAF applies meaning

Why the SAF records meaning

Architecture frameworks are easily misread. IT staff may see a technical blueprint. Managers may see administrative overhead. Explicit meaning elements set the intended interpretation, so that all stakeholders read the framework in the same way.

How meaning fits into an NSI's own architecture

An NSI uses the meaning elements when it introduces the SAF internally, for example in an architecture charter, an onboarding programme or a management briefing. An NSI may also record what its own core concepts mean, such as "statistical product" or "steady state". That use belongs in the glossary.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Value (SAF-V)	Meaning explains how value is interpreted	The SAF as common language (SAF-M4) explains why cross-border comparability (SAF-V1) is achievable
Stakeholders (SAF-ST)	Meaning is shaped for and by stakeholders	IT and data professionals (SAF-ST04) read the SAF through its notation (SAF-M4)
Goals (SAF-G)	Meaning frames how goals are understood	The SAF as a foundation for transformation (SAF-M3) frames modernisation (SAF-G3)
Glossary	The meaning of individual terms is recorded in the glossary	The definition of "steady state"

SAF meaning elements

ID	Meaning	Description
SAF-M1	Bridge between strategy and technology	The SAF links strategic objectives, such as relevance and efficiency, to technology choices. IT investments are justified by what they contribute to the mission.
SAF-M2	Catalyst for collaboration	The SAF is a common framework for the statistical community. It makes joint platforms, interoperable systems and harmonised standards possible, and strengthens what UNECE members can do together.
SAF-M3	Foundation for digital transformation	The SAF is more than a set of technical models. It is the basis for digital transformation in official statistics, without loss of methodological rigour or institutional trust.
SAF-M4	Common language and notation	Through TOGAF and ArchiMate, stakeholders share one language and notation. This reduces misunderstanding and raises architectural maturity across the community.

PLACEHOLDER: Meaning Template and Application of Meaning to be added to the toolkit

Outcomes

The concept

An outcome is an end result that has been, or is intended to be, achieved. Outcomes are more concrete than goals. A goal sets a direction. An outcome describes what will be different once that direction has been followed.

Three characteristics set outcomes apart:

- **They are results, not activities.** "Introduce a metadata catalogue" is an activity. "Every registered dataset has complete metadata and lineage" is an outcome.
- **They are measurable, or at least observable.** An outcome without an indicator cannot be used to steer.
- **They are realised by the strategy and architecture layers.** Capabilities, value streams and solutions exist to produce outcomes.

SAF outcomes are technical as well as societal. On the technical side, NSIs gain simpler processes, better metadata management and faster system development. On the societal side, the SAF supports trust, transparency and comparability, which informed policy and public accountability depend on. Outcomes also change over time. As an NSI matures, it sets new targets and adjusts its course.

How the SAF applies outcomes

Why the SAF records outcomes

Outcomes show stakeholders what adopting the framework returns. They connect long-term goals to concrete changes in capabilities, processes and systems. They also make progress measurable, for one NSI and across the community.

How outcomes fit into an NSI's own architecture

An NSI selects the SAF outcomes that match its goals. It defines indicators and target values for its own context, and monitors them in its planning and control cycle. The indicative measures below are examples. The SAF toolkit will offer them as a starting point in SAF-TK-M05.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Goals (SAF-G)	Outcomes make goals measurable	Improved interoperability (SAF-O3) shows progress on the harmonised reference framework (SAF-G2)
Value (SAF-V)	Outcomes deliver value to stakeholders	Enhanced data quality (SAF-O2) delivers trust and legitimacy (SAF-V6)
Capabilities (Strategy)	Capabilities realise outcomes	Metadata management (capability 2.1.2) realises enhanced data quality (SAF-O2)
Value streams (Strategy)	Value streams produce outcomes in the production chain	The steady-state handovers in VS-01 to VS-06 contribute to improved interoperability (SAF-O3)
Steady states (Business Architecture)	Quality gates make outcomes observable per state	Lineage and metadata requirements per state provide evidence for SAF-O2

SAF outcomes

ID	Outcome	Description	Indicative measures (illustrative)
SAF-O1	Accelerated modernisation	NSIs modernise faster and with less rework, because they start from shared reference models, patterns and building blocks instead of designing from scratch.	Lead time from decision to production for a new statistical product; share of solution designs that reuse SAF building blocks
SAF-O2	Enhanced data quality	Governance, metadata and lineage are part of the architecture. Statistical output therefore meets high standards of quality and reliability. Quality is designed in, not checked afterwards.	Share of datasets with complete metadata and lineage; revision rate of published figures

ID	Outcome	Description	Indicative measures (illustrative)
SAF-O3	Improved interoperability	NSIs reach higher interoperability across systems, domains and borders. This makes international collaboration easier, improves comparability and simplifies the use of external data sources.	Share of data exchanges using standard interfaces (e.g. SDMX); number of components shared between NSIs
SAF-O4	Improved trust and transparency	Openness, security and transparency in the architecture help to build and keep public trust. This strengthens the reputation of NSIs and the legitimacy of their output.	Results of public trust surveys; number of confidentiality or security incidents
SAF-O5	Increased efficiency	Reuse, harmonisation and standardisation reduce duplicate work. The result is lower IT costs, shorter development cycles and more efficient processing.	Number of redundant systems retired; processing time per production cycle; IT cost per statistical product
SAF-O6	Shared knowledge base	A community of practice across UNECE members exchanges lessons learned, reusable components and knowledge. This leads to continuous improvement at lower cost.	Number of reusable components and patterns published; number of NSIs contributing

PLACEHOLDER: Outcome Template and the application of Outcomes to be added to the toolkit

Architecture Principles

The concept

An **architecture principle** is a general rule that guides how an organisation designs, builds and changes its information provision. Principles are durable and rarely change. They apply to all architecture decisions within their scope, not to one project only.

Principles matter in official statistics because production runs in a closely connected environment. A choice in one place affects many others. Principles keep such choices consistent. They also prevent short-term convenience from undermining long-term sustainability, a real risk for NSIs with legacy systems and tight budgets. Finally, principles create accountability. When stakeholders know the principles behind a decision, investments and trade-offs are easier to explain.

TOGAF recommends describing each principle in four parts:

- **Name**: a short, memorable label.
- **Statement**: the rule itself, in one or two sentences.

- **Rationale**: why the rule exists and what it brings.
- **Implications**: what following the rule requires in cost, resources and activities.

The SAF follows this template. A principle is only useful if it is applied. A principle that never changes a decision is a slogan.

How the SAF applies principles

Why the SAF records principles

Principles turn goals into decision rules. When an NSI chooses between technologies, solutions or designs, principles make the choice consistent and explainable. They also give an architecture board a shared basis for reviewing projects.

How principles fit into an NSI's own architecture

An NSI takes the SAF principles into its own principle set, adapts the wording to national policy where needed, and adds national principles. Principles only take effect when they are part of governance. Architecture reviews test compliance, and deviations follow a documented dispensation process. The business layer of the SAF shows how each value stream applies the principles.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Goals (SAF-G)	Principles guide how goals are pursued	Loose coupling (SAF-P07) guides modernisation (SAF-G3)
Drivers (SAF-D)	Drivers explain why a principle exists	Trust and data protection (SAF-D3) explain privacy by design (SAF-P11)
Requirements (SAF-REQ) [**R5**]	Principles are refined into concrete requirements	SAF-P09 becomes the minimum metadata set of each steady state
Steady states (Business Architecture)	Principles shape the design rules of the steady-state model	SAF-P07, P08, P09 and P11 underpin the six design rules
Value streams (Business Architecture)	Each value stream applies a subset of the principles	VS-07 enforces SAF-P09, P10 and P11 across the chain

SAF architecture principles

ID	Principle	Statement
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	Statistical organisations build statistical capabilities on shared open-source software, contribute to its communities, and release software they develop for statistical production as open source by default. Non-statistical processes use standard applications.
SAF-P02	Active support	Systems have an active user community or vendor support. As a rule of thumb, the latest or second-to-last major version is used.
SAF-P03	Active life cycle management	Systems and components are kept up to date according to an agreed life cycle management policy.
SAF-P04	Maximise the benefits for the organisation	IT decisions maximise the benefit for the organisation as a whole.

ID	Principle	Statement
SAF-P05	IT and technology is everyone's business	All parts of the organisation take part in the IT decisions needed to reach organisational goals.
SAF-P06	Business continuity	Work continues despite system or staff outages.
SAF-P07	Implement loosely coupled processes and systems	Processes and systems are built as independently as possible and share functionality through standard interfaces.
SAF-P08	Data is shared property	Statistical data that adds value for internal or external users is shared.
SAF-P09	No data without metadata and classification	All shared data has metadata describing its content and meaning.
SAF-P10	Security by design	Security is an integral part of solutions across all layers of the architecture.
SAF-P11	Privacy by design	Privacy is an integral part of system architecture and business processes.

Principle details

SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes

- *Statement.* Statistical organisations build statistical capabilities on shared open-source software, contribute to its communities, and release software they develop for statistical production as open source by default. Non-statistical processes use standard applications.
- *Rationale.* Critical statistical methods stay under the collective control of the official-statistics community rather than vendors, and open methods are transparent and reproducible. Sharing costs and contributing, not only consuming, keeps common components alive.
- *Implications.* Contribution is planned and funded, improvements go back to the community project, and choices follow the order reuse, open source, buy, build, with building justified. Own components are released under an open licence unless security or confidentiality prevents it, and purchased software must support open standards, APIs and data export.

SAF-P02 Active support

- *Statement.* Hardware and software have an active user community or vendor support. As a rule of thumb, the latest or second-to-last major version is used.
- *Rationale.* Business continuity requires fast and competent support when something breaks. Security requires that known vulnerabilities are fixed in time.
- *Implications.* Support status is a selection criterion for every component. Components that approach end of support are flagged in the life cycle plan.

SAF-P03 Active life cycle management

- *Statement.* Systems and components are kept up to date according to an agreed life cycle management (LCM) policy.
- *Rationale.* Without LCM, systems become outdated and vulnerable, and technical debt builds up. Paying off that debt later takes disproportionate effort and creates continuity risks.
- *Implications.* Every component has an owner and a life cycle plan. LCM has a structural budget, not only when problems arise.

SAF-P04 Maximise the benefits for the organisation

- *Statement.* IT decisions maximise the benefit for the organisation as a whole.
- *Rationale.* Decisions taken from an organisation-wide view have more long-term value than decisions taken from a departmental view. The principle stands for "service above self".
- *Implications.* Local optimisations need justification against organisation-wide priorities. Shared solutions are preferred over departmental ones, even when they fit one department less well.

SAF-P05 IT and technology is everyone's business

- *Statement.* All parts of the organisation take part in the IT decisions needed to reach organisational goals.
- *Rationale.* The users of IT are its main stakeholders. IT only fits the business when business experts and technical staff set goals and objectives together.
- *Implications.* Business ownership of each information system is assigned explicitly. Architecture and change decisions are taken in joint business and IT forums.

SAF-P06 Business continuity

- *Statement.* Work continues despite system or staff outages.
- *Rationale.* The more an NSI depends on its systems, the more their reliability matters. Hardware failures, disasters and data corruption must not stop statistical production or release.
- *Implications.* Continuity requirements are part of every design. Critical processes have fallback arrangements, and recovery is tested regularly.

SAF-P07 Implement loosely coupled processes and systems

- *Statement.* Processes and systems are built as independently as possible. They share functionality and logic through standard interfaces.
- *Rationale.* Loose coupling makes it easier to change or replace a single process or system. It also improves scalability and robustness, and it makes different security and privacy requirements easier to enforce.
- *Implications.* Domains exchange data through defined interfaces, such as the steady-state handovers in the business architecture. Direct dependencies on another domain's internal data or logic are avoided.

SAF-P08 Data is shared property

- *Statement.* Statistical data that adds value for internal or external users is shared.
- *Rationale.* Sharing enables reuse. In data collection, reuse reduces response burden. In processing, it avoids duplicate work on cleaning and analysing data.
- *Implications.* Data is registered and findable in a catalogue. Before new data is collected, reuse of existing data is checked. Access is governed, not withheld by default.

SAF-P09 No data without metadata and classification

- *Statement.* All shared data has metadata that describes its content and meaning as well as possible, and an AIC classification (availability, integrity, confidentiality).
- *Rationale.* Metadata makes data FAIR. An AIC classification sets the technical and organisational measures that laws, regulations and security standards require.
- *Implications.* Metadata and classification are recorded when data is registered, not afterwards. Data without the required metadata is not handed over.

SAF-P10 Security by design

- *Statement.* Security is an integral part of every solution. Security mechanisms are consistent across all layers of the architecture and scale from small to large objects.
- *Rationale.* Security added afterwards costs more, works less well and leaves gaps between layers.
- *Implications.* Security requirements are set in the design phase of every solution and tested before production.

SAF-P11 Privacy by design

- *Statement.* Privacy is an integral part of system architecture and business processes.
- *Rationale.* Privacy added afterwards conflicts with functionality and leaves risks. Built in from the start, it becomes part of the core functionality without limiting it.

- *Implications.* Privacy impact is assessed in the design phase. Measures such as pseudonymisation, data minimisation and disclosure control are placed at defined points in the value chain.

PLACEHOLDER: Principles Template and the application of Principles to be added to the toolkit

Value

The concept

Value is the worth, usefulness or importance of something to a stakeholder. Value is always value *for someone*. The same outcome can be worth a lot to one stakeholder and little to another.

Value differs from the other motivation elements:

- An outcome is a result. Value is what that result is worth to a stakeholder.
- A goal is an intention of the organisation. Value is the reason a stakeholder cares whether the goal is reached.
- Meaning is how something is interpreted. Value is how much it is worth.

The SAF creates value at several levels. For NSIs, it means lower risk, more agility and better alignment between IT and business. For international organisations, it means harmonisation and comparability. For citizens and policy makers, it means statistics they can rely on. Value also has a time dimension. Without a framework, modernisation tends to be fragmented, costly and short-lived. The SAF gives a stable base that reduces technical debt and makes investments last.

How the SAF applies value

Why the SAF records value

Value makes the case for architecture in terms that stakeholders recognise. Architecture competes with other priorities for money and attention. It only gets support when its long-term worth is clear.

How value fits into an NSI's own architecture

An NSI uses the SAF value elements to build the business case for architecture work: in investment proposals, annual reports and briefings for its ministry or board. It selects the values that matter to its key stakeholders and links them to the outcomes it measures.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Stakeholders (SAF-ST)	Stakeholders perceive value	Policy makers (SAF-ST03) value trust and legitimacy (SAF-V6)
Outcomes (SAF-O)	Outcomes deliver value	Improved interoperability (SAF-O3) delivers cross-border comparability (SAF-V1)
Meaning (SAF-M)	Meaning explains how value is interpreted	The SAF as bridge between strategy and technology (SAF-M1) underpins strategic alignment (SAF-V4)
Value streams (Strategy)	Value streams create value for stakeholders, stage by stage	VS-06 delivers released statistics to users and society

SAF value elements

ID	Value	Description
SAF-V1	Cross-border comparability	Statistics are worth more when they can be compared across borders. Common standards, models and processes make that comparison meaningful.
SAF-V2	Innovation enablement	The SAF lowers the barriers for AI, machine learning, real-time analytics and big data. NSIs can experiment safely and keep their methods sound.
SAF-V3	Risk reduction	Compliance, security and governance are built into the architecture. This reduces the risk of system failures, data breaches and regulatory non-compliance, and protects the trust of citizens and stakeholders.
SAF-V4	Strategic alignment	IT initiatives contribute directly to the mission of producing official statistics. This improves governance, prioritisation and the impact of IT investments.
SAF-V5	Sustainability	Efficient, modern architectures with less technical debt give NSIs a clear path forward, without repeated and costly overhauls.
SAF-V6	Trust and legitimacy	Transparency, ethical practice and robust security strengthen the position of NSIs as authoritative providers of public data.

PLACEHOLDER: Value Template and the application of Values to be added to the toolkit

Assessment

The concept

An assessment is the result of analysing a driver for one specific organisation. It records what the driver means for that organisation: a strength, a weakness, an opportunity or a threat. A driver describes a force. An assessment describes how that force affects this NSI, today.

Three characteristics set assessments apart:

- **They are specific to one organisation.** "Resource constraints" is a driver for many NSIs. "Two thirds of our processing runs on platforms out of vendor support" is an assessment for one NSI.
- **They are based on evidence.** An assessment rests on facts, such as an application portfolio scan, an audit finding, a maturity assessment or a staff survey. An opinion without evidence is not an assessment.
- **They change over time.** A driver often stays the same for years. An assessment changes as the NSI acts on it. It is therefore reviewed regularly, for example once a year.

The SAF does not define assessments. A generic assessment would contradict the concept itself. This section explains how assessments are used, and the toolkit provides a template to record them.

How the SAF applies assessments

Why the SAF uses assessments

Assessments are the link between the generic SAF drivers and the goals of an NSI. Without them, an NSI jumps from "the world is changing" straight to "we must modernise", and nobody can see why this particular goal has priority. An assessment makes that step explicit. It also gives a baseline: if the next assessment of the same driver is better, the NSI can show progress.

How assessments fit into an NSI's own architecture

An NSI writes one or more assessments for each driver it recognises. The second step of applying drivers, *Assess*, produces them (see *Drivers*). A SWOT analysis is a common format: each finding is recorded as a strength, weakness, opportunity or threat. Assessments then feed the third step, *Respond*, in which the NSI derives or confirms its goals. In TOGAF terms, assessments are part of Phase A, Architecture Vision, and of the baseline description of each architecture domain.

Relationships with other concepts

Related concept	Relationship	Example
Drivers (SAF-D)	An assessment analyses a driver for one NSI	An assessment of SAF-D5 records the share of platforms out of vendor support
Stakeholders (SAF-ST)	Stakeholders contribute evidence and recognise the findings	CIOs and CTOs (SAF-ST01) own the assessment of the application portfolio
Goals (SAF-G)	Assessments influence which goals an NSI sets and how it prioritises them	A weakness on SAF-D5 makes modernisation (SAF-G3) a priority
Outcomes (SAF-O)	A new assessment shows whether outcomes have been reached	The share of unsupported platforms falls after a modernisation programme (SAF-O1)
Capability Model (Strategy)	A capability maturity assessment is a structured form of assessment	A low score on metadata management (capability 2.1.2)

Examples (illustrative)

The examples show the form of an assessment and its links. They are not SAF assessments and do not apply to every NSI.

Driver	Assessment	Type	Evidence	Leads to
SAF-D5 Resource constraints and organisational agility	Two thirds of our processing runs on platforms that are out of vendor support	Weakness	Application portfolio scan	Priority for SAF-G3 Modernisation of statistical systems
SAF-D1 Evolving data ecosystems and partnerships	A national data space is being set up, and our NSI is invited to join as a participant	Opportunity	Government programme plan	Priority for SAF-G2 Harmonised reference framework
SAF-D3 Trust, scientific rigour and data protection	We cannot show, for each published figure, which source data version it is based on	Weakness	Internal audit finding	Priority for SAF-O2 Enhanced data quality (lineage)

PLACEHOLDER: Assessment Template and the application of Assessments to be added to the toolkit

Requirements

The concept

A requirement is a statement of need that a solution must meet. A goal says what the NSI wants to achieve. A requirement says what a specific solution, process or platform must do or provide to contribute to that goal.

Three characteristics set requirements apart:

- **They apply to a specific solution.** "Modernise statistical systems" is a goal. "The new processing platform records the lineage of every registered dataset" is a requirement.
- **They are testable.** It must be possible to check whether a solution meets the requirement, for example in an architecture review or an acceptance test.
- **They are derived, not invented.** A good requirement can be traced to a goal, an outcome or a principle. A requirement without that trace needs a justification.

The SAF does not define requirements for NSIs. Requirements depend on the solution, the context and the maturity of each NSI. This section explains how requirements relate to the other motivation elements. The toolkit provides a template to derive and record them.

How the SAF applies requirements

Why the SAF uses requirements

Principles and goals are deliberately general. They only change a design when they are made concrete. Requirements are that concrete form. They connect the motivation layer to the architecture domains, and they let an architecture board test whether a solution really contributes to the goals it claims to serve.

How requirements fit into an NSI's own architecture

In TOGAF, requirements are managed in Requirements Management, at the centre of the ADM. They are identified in every phase and checked against each architecture decision. An NSI derives its requirements from three sources:

1. **Goals and outcomes.** What must the solution provide to contribute to a goal, and how will the outcome be measured?
2. **Principles.** What does each relevant SAF principle mean for this solution?
3. **SAF reference content.** Which SAF building blocks apply, and what do they require?

The SAF already contains content that works as a set of requirements. The steady states in the business architecture each have quality criteria and a minimum metadata set. An NSI that adopts the steady-state model can use these directly as requirements for the processes and platforms that produce and store each state. The same applies to the principle tables in each value stream. They show which principles apply at which stage of the production chain.

Relationships with other concepts

Related concept	Relationship	Example
Goals (SAF-G)	Requirements make goals concrete for a solution	SAF-G4 Support for data innovation leads to a requirement for automated testing in the production pipeline
Outcomes (SAF-O)	Requirements contribute to outcomes and make them measurable in a solution	Lineage recording contributes to SAF-O2 Enhanced data quality

Related concept	Relationship	Example
Principles (SAF-P)	Requirements apply principles to a specific solution	SAF-P09 becomes the requirement that no dataset is handed over without its minimum metadata
Constraints	Requirements describe what a solution must do; constraints limit how it may be done	A requirement for scalable processing within the constraint that data stays on premises
Steady states (Business Architecture)	Quality criteria and minimum metadata per state work as reusable requirements	The criteria for registering data in State #1 Raw Data

Examples (illustrative)

The examples show how a requirement is derived and traced. They are not SAF requirements.

Source	Requirement	Applies to	How to verify
SAF-P09 No data without metadata and classification	No dataset is registered in a steady state without its minimum metadata set and an AIC classification	Data catalogue and registration process	Registration is refused when mandatory metadata is missing
SAF-G4 Support for data innovation	A new method can move from the innovation environment to production without manual re-implementation	Analytical platform and deployment pipeline	Time and effort for one pilot method to reach production
SAF-P07 Implement loosely coupled processes and systems	Business domains exchange data only through registered steady states, not through each other's internal data	All production processes	Architecture review of data flows between domains

PLACEHOLDER: Requirements Template and the application of Requirements to be added to the toolkit

Constraints

The concept

A constraint is a factor that limits how goals can be realised. A requirement describes what a solution must do. A constraint describes the boundaries within which it must be done. Constraints are usually not chosen by the architect. They are given by law, policy, budget, existing contracts or the current landscape.

Three characteristics set constraints apart:

- **They limit the solution space.** A constraint rules out options. "Personal data may not be processed outside the EU" excludes certain cloud services, whatever their other merits.
- **They come from outside the design.** Typical sources are legislation, government policy, budget, staffing, existing contracts and technical dependencies.
- **They differ in how fixed they are.** A legal constraint cannot be negotiated. A budget or policy constraint may change. Recording the source of a constraint shows which ones can be challenged.

This section is about constraints on an NSI's own architecture. It is different from the *Constraints & Assumptions* in the Introduction, which describe the boundaries of the SAF itself (for example SAF-C1 and SAF-C2).

The SAF does not define constraints for NSIs, because legal and organisational contexts differ. Some constraints, however, apply to almost every NSI, such as data protection law and the statistical confidentiality required by national statistical law. They are used below as examples.

How the SAF applies constraints

Why the SAF uses constraints

Constraints that stay implicit cause late surprises. A design that looks right fails in a procurement, a privacy assessment or a security review, because a limit was not known in time. Recording constraints early, and linking them to the goals they affect, keeps the architecture realistic. It also makes trade-offs visible: when a constraint blocks a goal, the board can decide whether to accept that or to challenge the constraint.

How constraints fit into an NSI's own architecture

In TOGAF, constraints are identified in Phase A, Architecture Vision, and managed together with requirements in Requirements Management. An NSI records its constraints with their source and whether they are fixed or negotiable. Each constraint is linked to the goals, requirements or solutions it affects. Constraints are reviewed when their source changes, for example after new legislation or a new government policy on cloud use.

Several parts of the SAF help to identify constraints. *Definition of Scope* states that the SAF does not prescribe a technology stack or organisational design, so those choices are left to the NSI and its constraints. The principles on security and privacy by design (SAF-P10 and SAF-P11) point to the legal and security frameworks that usually impose constraints.

Relationships with other concepts

Related concept	Relationship	Example
Drivers (SAF-D)	Some drivers are the origin of constraints	SAF-D3 Trust, scientific rigour and data protection is the origin of data protection constraints
Goals (SAF-G)	Constraints limit how goals can be realised	A ban on public cloud for statistical microdata limits how SAF-G3 Modernisation is realised
Requirements	Constraints bound the solutions that can meet a requirement	A scalability requirement must be met within on-premises infrastructure
Principles (SAF-P)	Principles are applied within the limits that constraints set	SAF-P01's reuse–open source–buy–build order applies within public procurement rules
Stakeholders (SAF-ST)	Stakeholders often own or impose constraints	A privacy officer applies data protection law; a ministry sets the budget

Examples (illustrative)

The examples show the form of a constraint. They are not SAF constraints, and their content differs per country.

Constraint	Source	Fixed or negotiable	Affects
Statistical microdata is processed only within the NSI's own controlled environment	National statistical law and confidentiality policy	Fixed	Platform and hosting choices for SAF-G3; data sharing under SAF-G2
Personal data is processed in line with the GDPR, including data minimisation and purpose limitation	EU legislation	Fixed	Design of data acquisition and standardisation; SAF-P11 Privacy by design
Public cloud services are used only for business support functions, not for statistical production	Government or NSI cloud policy	Negotiable	Technology choices for SAF-G3 and SAF-G4

PLACEHOLDER: Constraints Template and the application of Constraints to be added to the toolkit

Stakeholders

The concept

A **stakeholder** is an individual, group or organisation with an interest in the effects of the architecture [**R8**]. Stakeholders have *concerns*: questions or interests they want the architecture to address, such as cost, risk, quality or compliance.

A stakeholder is not the same as a business actor or role. An actor or role performs work in the architecture, for example a data steward who registers datasets. A stakeholder has an interest in the architecture, whether or not it performs work in it. The same person or organisation can be both.

SAF stakeholders range from CIOs and technical experts within NSIs to international organisations, policy makers, researchers and society as a whole. Their interests differ, and the framework has to balance them. IT professionals turn SAF principles into systems. Policy makers care about the results those systems make possible. UNECE keeps the SAF harmonised and relevant across countries.

How the SAF applies stakeholders

Why the SAF records stakeholders

Architecture is never developed in isolation. Knowing the stakeholders makes clear whose concerns the architecture must address, who should take part in governance and decisions, and which views each audience needs. Involvement also builds support. People who take part in architecture decisions are more likely to back them.

How stakeholders fit into an NSI's own architecture

An NSI maps the SAF stakeholders onto its own stakeholder landscape. It identifies who represents each stakeholder and records their concerns. The concerns determine which views the NSI produces. A CIO needs a capability and investment view. A privacy officer needs a view of data flows and protection measures. The stakeholder map is reviewed at the start of each architecture cycle.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Drivers (SAF-D)	Stakeholders have concerns about drivers and raise them	CIOs and CTOs (SAF-ST01) raise resource constraints (SAF-D5)
Goals (SAF-G)	Stakeholders own goals	NSIs (SAF-ST05) own modernisation (SAF-G3)
Value (SAF-V)	Stakeholders perceive value	Researchers and academia (SAF-ST06) value innovation enablement (SAF-V2)
Viewpoints	Stakeholder concerns determine which views are produced	Privacy officers need data-flow and protection views
Business actors and roles (Business Architecture)	A stakeholder may also act in the architecture	Data providers are stakeholders and actors in VS-01

SAF stakeholders

ID	Stakeholder	Category	Key concerns	What the SAF offers
SAF-ST01	CIOs and CTOs of NSIs	Within the NSI	Strategic IT investment, alignment of IT with business goals, compliance	A framework to compare options, align IT with business goals and show compliance with international standards
SAF-ST02	Citizens and society	Society	Trustworthy, transparent and modern statistics	Statistics that support democratic accountability and informed public debate
SAF-ST03	Government policy makers	Society	Timely, accurate and comparable data in accessible formats	Better statistical production and dissemination (indirect benefit)
SAF-ST04	IT and data professionals	Within the NSI	Common patterns, tools and practices; collaboration across institutions	Shared patterns and practices that save effort and shorten the learning curve
SAF-ST05	National Statistical Institutes (NSIs)	Statistical community	Structured guidance for enterprise architecture; efficiency; quality	Guidance tailored to official statistics and aligned with international practice
SAF-ST06	Researchers and academia	Society	Access to high-quality data and metadata for research	Systems designed for openness, data sharing and rich metadata
SAF-ST07	UNECE and other international bodies	Statistical community	Relevance, coherence and adoption of the framework	A custodial role for UNECE; alignment with the modernisation work of Eurostat and the UN Statistics Division

Strategy

Purpose of this chapter

The Strategy chapter translates motivation into what an NSI must be able to do and how it creates value. Motivation explains *why* change is needed. Strategy describes *what* the organisation needs: its capabilities, the resources behind them, the value streams they support and the courses of action that develop them.

These elements are deliberately independent of organisation structure, processes and systems. Departments, processes and applications change often. Capabilities and value streams change slowly. Anchoring architecture work in this layer lets an NSI compare itself with peers, plan investments over several years, and reorganise or replace technology without losing sight of what it must deliver.

Position in the framework

The Strategy layer shares the Strategy & Motivation area of the TOGAF ADM with the Motivation layer (see *Position in the framework* in the Motivation chapter). It receives direction from goals and outcomes. It gives direction to the business architecture: value stream stages are realised by business processes, and capabilities are realised by people, processes, information and technology in the lower layers.

PLACEHOLDER: Position of the Strategy layer between Motivation and the architecture domains

In TOGAF terms, the Strategy chapter supports capability-based planning. Capabilities and value streams are part of the Architecture Vision (Phase A) and of the baseline and target Business Architecture (Phase B). Courses of action and resources return in Phases E and F, where the NSI decides how and in which order capabilities are developed.

How NSIs use this chapter

The capability model and the value streams in this chapter are a reference. Most NSIs perform the same core activities, so these elements are largely shared across the community. An NSI uses them in the same three ways as the motivation elements: adopt, adapt or extend.

The capability model is the most practical starting point. An NSI can assess the maturity of each capability, mark where its goals require improvement, and use the result to prioritise investment. Value streams show where in the production chain a capability creates value, and which stakeholders notice a weak capability first.

Resources and courses of action are specific to each NSI. The SAF explains these concepts and gives examples. The toolkit provides templates to record them.

PLACEHOLDER: Capability maturity assessment and capability heatmap to be added to the toolkit

Strategy concepts in the SAF

SAF section	ArchiMate element	Question it answers	Identifier
Capability Model	Capability	What must the NSI be able to do?	Capability number (e.g. 1.2.1)
Value Streams	Value Stream	Which sequence of stages creates value for a stakeholder?	VS-01 to VS-10

SAF section	ArchiMate element	Question it answers	Identifier
Resources	Resource	Which assets does the NSI have or need to perform its capabilities?	Defined by the NSI
Courses of Action	Course of Action	Which approach will the NSI take to develop its capabilities and reach its goals?	Defined by the NSI

The concepts form a chain. Capabilities are what the NSI can do. Resources are what it uses to do it: people, data, knowledge, technology and money. Value streams show where capabilities create value for a stakeholder. Courses of action describe how the NSI will strengthen its capabilities and resources to reach its goals.

Capability Model

The concept

Within the Statistical Architecture Framework (SAF), business capabilities define the fundamental abilities of a statistical organization to perform its mission , independent of processes, organizational structures, or technologies. A business capability represents what the organization must be able to do to deliver value and achieve its objectives, providing a stable and long-term view of the enterprise's functional landscape.

Capabilities serve as the enabling building blocks of the statistical organization. They do not change frequently, even when the underlying processes, applications, or organizational units evolve. For example, the capability "Manage Statistical Data Quality" remains essential regardless of whether data validation is manual, automated, or AI-driven. This stability makes capabilities the preferred unit of analysis for strategic planning, transformation roadmaps, and architecture alignment within the SAF.

Three characteristics set capabilities apart:

- **They are stable.** Statistical quality management (2.1.3) remains essential whether quality checks are manual, automated or AI-supported. Processes and applications change; the capability stays.
- **They are independent of organisation.** A capability can be spread over several departments, or one department can hold several capabilities. The capability model does not follow the organisation chart.
- **They are hierarchical.** High-level capabilities are decomposed into more detailed ones. In the SAF, *Data management* (2.1) is decomposed into, among others, *Metadata management* (2.1.2).

Each capability describes a specific business purpose and outcome, which is realized through one or more business functions and operationalized by business processes. Capabilities therefore serve as the conceptual bridge between strategy and execution. They translate strategic goals into operational competence. Within the SAF, capabilities are often organized hierarchically, showing decomposition from high-level domains (such as Data Governance or Statistical Production) into more granular sub-capabilities (such as Metadata Management or Sampling Design).

How the SAF applies capabilities

Why the SAF uses a capability model

Capabilities are the preferred unit for strategic planning in the SAF. Because they are stable, they make a reliable basis for multi-annual roadmaps. Because they are independent of organisation and technology, they let NSIs compare themselves with each other. A capability view also makes gaps visible: where a goal requires a capability that is weak or missing, investment is needed.

How capabilities fit into an NSI's own architecture

An NSI maps the SAF capability model onto its own organisation. It checks which capabilities it has, which it lacks and which are specific to its mandate. It then assesses the maturity of each relevant capability and

links capabilities to its goals and outcomes. Capabilities with a low maturity and a high contribution to goals become candidates for investment. Each capability can then be traced to the processes, applications and platforms that realise it.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Goals and outcomes (Motivation)	Capabilities realise goals and outcomes	Metadata management (2.1.2) realises enhanced data quality (SAF-O2)
Assessment (Motivation)	A capability maturity assessment is a structured form of assessment	A low maturity score for Statistical quality management (2.1.3)
Value Streams	Capabilities enable value stream stages	Primary data collection (1.2.1) enables stage *Collect or Receive Data* of VS-01
Resources	Capabilities are performed with resources	Methodology (2.2.3) depends on staff with methodological expertise
Courses of Action	Courses of action develop capabilities	A shared platform with partner NSIs develops IT infrastructure (3.2.5)
Business functions and processes (Business Architecture)	Business functions and processes realise capabilities	Business function *Classify Data* realises Data standardisation (1.3.1)
Steady states (Business Architecture)	Capabilities produce or govern steady states	Data standardisation (1.3.1) produces State #2 Standardized Data

Capability map

The SAF capability model has four main levels.

PLACEHOLDER: SAF capability map (ArchiMate)

ID	Main level	Description	Reference
1	Primary statistical capabilities	The ability to perform the statutory statistical tasks of an NSI.	GAMSO
2	Supporting data capabilities	The ability to manage statistical data and to carry out statistical research.	
3	Supporting business capabilities	The ability to make sure that the right people, information and resources are available to support the statistical and governance capabilities.	

ID	Main level	Description	Reference
4	Governance capabilities	The ability to give direction to the organisation, by setting its strategy and the framework within which change takes place.	GAMSO 1.2, Strategy and Leadership

Capability catalogue

1. Primary statistical capabilities

ID	Capability	Description	Reference
1.1	**Statistical design and build**	The ability to identify information needs and set up statistical processes to meet them.	GSBPM 5.2, Design
1.1.1	Needs inventory	The ability to identify the demand for statistical information and translate it into the societal questions the NSI will respond to. Needs are often identified through strategic relationships with users and stakeholders. The result is usually laid down in the NSI's multi-annual programme and the process that prepares it.	GSBPM

ID	Capability	Description	Reference
1.1.2	Design statistics product	The ability to translate information needs into concrete designs of datasets, process and workflow models, and rule sets and methodology, for the datasets that are released. This covers both the statistical product and the process that produces it. The process consists of process steps, rule sets for each step, the methodology, the process flow, the designs of the required datasets and the designs of shared connection points for reuse. If a required dataset is not available, a data collection assignment can follow. For primary data collection, designs then follow for sampling, questionnaires and similar instruments.	GSBPM, Design phase
1.1.3	Build statistics product	The ability to build, test and manage a statistical product until the end of its life cycle. The outputs of the design are assembled and configured into a complete operational environment in which the process can run.	GSBPM
1.2	**Data collection**	The ability to collect data for official statistics.	GSBPM 5.2, Collect

ID	Capability	Description	Reference
1.2.1	Primary data collection	The ability to set up and manage surveys up to completion, and to transfer the collected data and metadata for processing and analysis. Primary data collection uses one or more collection modes. The target population and sample size are set together with the client, based on the information need. Person surveys and business surveys are distinguished. A survey consists of, among other things, a sample, a questionnaire and an approach strategy.	ESS capability model
1.2.2	Secondary data collection	The ability to collect data and metadata from third parties for the production of statistics. Data that third parties have already collected is used as much as possible.	ESS capability model
1.3	**Processing**	The ability to process raw data into statistical microdata and statistics.	GSBPM, Process phase
1.3.1	Data standardisation	The ability to standardise data without loss of content. Standardisation covers data formats, category codes, linkage keys and units of measurement, such as weight, distance and time. It takes place directly after the raw data is received. Standardising as early as possible makes the rest of the process more efficient.	

ID	Capability	Description	Reference
1.3.2	Deriving variables and units	The ability to create new units, and values of new variables, from existing units and values of existing variables. New variables are usually derived to describe reality better. The data source was often not designed for that purpose.	
1.3.3	Check content	The ability to check datasets for irregularities, such as missing data, possible errors and outliers. Detected errors can lead to adjustment (see 1.3.4). They can also lead to a quality indicator that is used in quality reporting. Detected outliers lead to an outlier indicator, which matters for a good estimation method (see 1.3.5).	GSBPM
1.3.4	Adjusting and imputing	The ability to adjust incorrect values and to fill in missing values. Together with *Check content* (1.3.3), this capability is often called editing. Checks use editing rules, and adjustment is often based on imputation. Imputed values do not yet have to meet the editing rules. A final adjustment therefore often follows, with the editing rules as constraints.	GSBPM, ESS capability model

ID	Capability	Description	Reference
1.3.5	Estimating population parameters	The ability to estimate population parameters from microdata. Estimated population parameters often appear as observations, the cells, in a dimensional dataset such as a table. There are several estimation methods, including the traditional "weight and aggregate". Weights can be derived with several techniques. Sometimes weights are adjusted to give detected outliers less influence.	
1.3.6	Integrate statistics	The ability to integrate separate statistics into one consistent whole. Examples are integration systems such as the national accounts, sector accounts and labour accounts.	
1.3.7	Statistical disclosure	The ability to check datasets for privacy sensitivity and to adjust them where confidentiality would be violated.	GSBPM 6.4
1.4	**Analysis**	The ability to examine and understand data before it is disseminated. In NSIs, statistical analysis is also called statistical research. It draws connections or conclusions from relationships in the data, such as correlations.	GSBPM 5.2, Analyse
1.4.1	Statistical (output) analysis	The ability to validate, interpret and explain statistical output, and to finalise it for dissemination using shared methods and processes.	ESS capability model
1.5	**Dissemination**	The ability to manage the release of statistical products to internal and external users.	GSBPM 5.2, Disseminate

ID	Capability	Description	Reference
1.5.1	Publication management	The ability to manage the release of statistical output and related content according to release calendars, so that users have predictable and equal access.	ESS capability model
1.5.2	Promotion to use	The ability to promote statistical output to potential users, and to inform the press and other interested parties about it.	ESS capability model
1.5.3	Facilitate data access	The ability to make statistical data and metadata available to people and machines through multiple channels.	ESS capability model
1.5.4	Statistical information management	The ability to prepare and distribute publications about statistical output, such as press releases, interpretations and reports.	ESS capability model

2. Supporting data capabilities

ID	Capability	Description	Reference
2.1	**Data management**	The ability to manage data and metadata.	
2.1.1	Data lifecycle management	The ability to manage data throughout its life cycle.	
2.1.2	Metadata management	The ability to manage metadata throughout its life cycle.	ESS capability model
2.1.3	Statistical quality management	The ability to assess quality, put control mechanisms in place and start quality improvements across the statistical value chain.	
2.1.4	Statistical register management	The ability to set up, maintain and provide register services for collecting and integrating data. This includes the management of master data.	ESS capability model

ID	Capability	Description	Reference
2.2	**Research**	The ability to build and record knowledge about statistical methods, products, sources and techniques.	
2.2.1	Data source exploration	The ability to explore new data sources.	ESS capability model
2.2.2	Product innovation	The ability to create new statistical products from existing data sources, and to explore new data sources that are useful and important to users. Product innovation can precede the design of a statistical product (see 1.1.2).	ESS capability model
2.2.3	Methodology	The ability to investigate new methods for the statistical process.	
2.2.4	Data services	The ability to provide data for statistical or scientific research. This applies to both internal and external datasets. These services are provided through, among other things, microdata services.	
2.2.5	User research	The ability to collect and assess user needs, and translate them into statistical output.	

3. Supporting business capabilities

ID	Capability	Description	Reference
3.1	**Business operations**	The ability to manage personnel, purchasing and contracts, finance and housing, including the related systems and communication.	
3.1.1	Personnel management (HR)	The ability to make sure that enough qualified staff are available to carry out all business processes.	
3.1.2	Purchasing and contract management	The ability to acquire goods and services, and to monitor the agreements made with suppliers.	

ID	Capability	Description	Reference
3.1.3	Financial management	The ability to plan, organise, design, record and control the financial activities of the organisation.	
3.1.4	Communication management	The ability to communicate correctly with internal and external parties.	
3.1.5	Administrative support	The ability to support administrative tasks.	
3.1.6	Facility management	The ability to integrate people, place and process within the built environment. The description is based on ISO 41011. Facility management aims to improve the quality of life of people and the productivity of the core business.	ISO 41011
3.1.7	Legal support	The ability to handle requests and applications, assess agreements and advise on relevant laws and regulations.	
3.1.8	Information management and archive management	The ability to keep the organisation's information sustainably accessible (FAIR). This concerns information created in supporting and management processes, not statistical information from the primary statistical process.	
3.2	**Computerisation and automation**	The ability to translate the organisation's information needs into functionality of the information provision, and to keep information systems available to support business processes.	
3.2.1	Application development	The ability to develop applications in-house. This capability is needed to fill gaps in functionality that the statistical process requires.	

ID	Capability	Description	Reference
3.2.2	Functional management	The ability to manage applications so that they support business processes as well as possible. An example is collecting user requirements and translating them into new application functionality.	
3.2.3	Technical management	The ability to manage system software and applications. This covers installing, configuring and managing hardware and software so that it keeps working correctly, including all measures to resolve disruptions in time.	
3.2.4	Digital workplace management	The ability to manage the resources and services that staff use in their digital workplace. Examples are laptops and desktops, mobile devices, virtual desktops, video and audio conferencing, printing, VPN and proxy services.	
3.2.5	IT infrastructure	The ability to develop, maintain and manage IT infrastructure.	
3.3	**Security and privacy**	The ability to carry out all security and privacy activities.	
3.3.1	Privacy management [**S7**]	The ability to develop, manage, apply and test the NSI's privacy policy, in order to comply with privacy laws and regulations.	
3.3.2	Security planning	The ability to identify, analyse and plan measures to secure systems, networks, the organisation and its facilities.	
3.3.3	Security incident management	The ability to identify, analyse, respond to and recover from security incidents.	

ID	Capability	Description	Reference
3.3.4	Logical identity and access control	The ability to make sure that a person, organisation or IT facility can only use the automated functions for which access rights have been granted.	
3.3.5	Physical security and access control	The ability to provide controlled and safe access to buildings and spaces, and to keep the people and goods in them safe.	
3.3.6	IT security services	The ability to provide IT services that support information security. Examples are endpoint protection, PKI and network security management.	
3.4	**Customer and chain interaction**	The ability to interact with customers and chain partners: providing them with the right data and information, and receiving notifications, requests and data.	
3.4.1	Services management	The ability to design, develop, implement and manage the services that the organisation provides to customers and chain partners.	
3.4.2	Contact management	The ability to maintain relationships with customers and chain partners.	GAMSO 1.2, Manage Consumers **[S9]**
3.4.3	Service agreement	The ability to draw up, manage and enforce service agreements with clients and chain partners. Service agreements apply to clients that receive services from the NSI, such as access for external researchers or additional statistical services. They also apply to parties that supply data for primary or secondary data collection. If a party does not comply, the NSI can take enforcement action.	

ID	Capability	Description	Reference
3.4.4	Self-service	The ability to let customers and chain partners answer questions about the services they use, or make justified changes to them, without help from staff.	

4. Governance capabilities

ID	Capability	Description	Reference
4.1	**Strategy and leadership**	The ability to set the direction of the organisation and to manage change towards it.	GAMSO 1.2, Strategy and Leadership
4.1.1	Strategy and governance	The ability to develop a vision, and to set up and monitor the organisation and its management.	
4.1.2	Policy and planning	The ability to translate the strategy into concrete objectives, principles and plans.	
4.1.3	Change management	The ability to define and manage major changes, so that they contribute as much as possible to the objectives.	
4.1.4	Improvement management	The ability to manage daily operations, identify areas for improvement and implement improvements based on daily operations.	
4.1.5	Accountability	The ability to report to stakeholders inside and outside the organisation on how far obligations and agreements are met.	

ID	Capability	Description	Reference
4.1.6	Collaboration	The ability to maintain and strengthen strategic relationships with external stakeholders. Collaboration covers coordination with other statistical organisations and other external stakeholders. Within a statistical system, this coordination can follow a geographical hierarchy (local, regional, national, multinational) or a division of responsibilities between organisations. It includes identifying new opportunities for data exchange and integration. It also gives the statistical community opportunities to exchange knowledge, improve statistical infrastructure and practice, and influence statistical standards.	GAMSO 1.2, Manage Strategic Collaboration and Cooperation
4.1.7	Architecture Management	The ability to develop, maintain and govern the enterprise architecture of the NSI, and to use it in decisions on strategy, investment and change.	

PLACEHOLDER: Capability Model Template and the application of Capabilities to be added to the toolkit

Value Streams [T1]

The concept

A value stream is a sequence of stages that together create an overall result for a stakeholder. Each stage adds a specific piece of value. The value stream ends when the stakeholder receives the result.

Three characteristics set value streams apart:

- **They are seen from the stakeholder.** A value stream starts with a need and ends with value for someone. It does not follow the organisation chart or the order of systems.
- **They describe value, not work.** A stage says *what value* is added. *How* the work is done is described by business processes in the Business Architecture.
- **They are enabled by capabilities.** Each stage needs one or more capabilities. A weak capability shows up as a weak stage.

Value streams and capabilities are two views on the same organisation. Capabilities show *what* the NSI can do. Value streams show *where* that ability turns into value.

How the SAF applies value streams

Why the SAF uses value streams

Value streams connect the capability model to the stakeholders. They make clear which capabilities matter for which result, and which stakeholders notice a weak capability first. They also give the Business Architecture its structure: each stage is realised by business processes, and the primary value streams follow the steady states of data.

How value streams fit into an NSI's own architecture

Most NSIs produce statistics in a similar sequence, so the SAF value streams fit most of them. An NSI checks whether the stages match its own production chain, adapts names where needed, and links its own capabilities to each stage. It can then use the value streams to assess where in the chain improvements create the most value for users.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Stakeholders (Motivation)	A value stream creates value for a stakeholder	VS-06 creates value for users of statistics (SAF-ST02, SAF-ST03, SAF-ST06)
Value (Motivation)	A value stream delivers value elements	VS-06 contributes to trust and legitimacy (SAF-V6)
Capability Model	Capabilities enable value stream stages	Primary data collection (1.2.1) enables VS-01.3 *Collect or Receive Data*
Courses of Action	Courses of action improve value streams by developing capabilities	A web-first collection strategy improves VS-01
Steady states (Business Architecture)	Each primary value stream produces one steady state	VS-02 produces State #2 Standardized Data
Business processes (Business Architecture)	Business processes realise value stream stages	The execution processes of the Standardization domain realise the stages of VS-02

The processes that realise each stage, and the steady states between them, are described in the Business Architecture chapter.

Value stream overview [T2] [T3]

The SAF has ten value streams. Six primary value streams form the statistical value chain. Four supporting value streams run across the chain.

ID	Value stream	Type	Receives value	Value created	Steady state in → out
VS-01	Data Acquisition and Ingestion	Primary	Statistical production; data providers	Trust and readiness: data acquired correctly and ready for use	#0 → #1
VS-02	Data Standardization and Integration	Primary	Statistical production; reusers of microdata	Coherence and comparability	#1 → #2

ID	Value stream	Type	Receives value	Value created	Steady state in → out
VS-03	Data Editing and Enrichment	Primary	Statistical production; methodologists	Reliability and analytical readiness	#2 → #3
VS-04	Estimation, Analysis and Statistical Modeling	Primary	Statistical product owners	Analytical insight and statistical credibility	#3 → #4
VS-05	Statistical Product Composition	Primary	Statistical product owners; users	Coherence and interpretability	#4 → #4 (approved)
VS-06	Dissemination and Service Delivery	Primary	Users: policy makers, researchers, media, businesses, the public	Impact and accessibility	#4 → #5
VS-07	Governance and Compliance	Supporting	Management, supervisory bodies, society	Integrity and accountability	Across all states
VS-08	Innovation and Methodological Development	Supporting	Statistical production	Continuous improvement	Across all states
VS-09	Infrastructure and Platform Enablement	Supporting	All value streams	Operational capability and continuity	Across all states
VS-10	Customer and Stakeholder Engagement	Supporting	Users, data providers, partners	Relevance and trust	#5 → (#6), feedback to VS-01

Capabilities per value stream [T4]

PLACEHOLDER: ArchiMate view of capabilities serving value stream stages

Primary Value Streams

VS-01 Data Acquisition and Ingestion

The Data Acquisition & Ingestion Value Stream is the first step in the statistical value chain. It focuses on bringing raw data into the statistical system in a secure, traceable, and compliant way. The purpose of this value stream is to make sure that data, no matter where the data comes from, enters the organization in a state that is authentic, well-documented, and ready for use. This value stream sets the foundation for everything that follows. It makes sure that the right data is collected, that legal requirements and ethical guidelines are observed, and that the data is safely stored and traceable from its source.

PLACEHOLDER: Add ArchiMate view of VS-01

Stage	Name	Value added
VS-01.1	Identify Data Needs and Sources	The information need is clear, and suitable sources are known
VS-01.2	Establish Data Agreements and Access Rights	Data can be obtained lawfully, under agreed conditions

Stage	Name	Value added
VS-01.3	Collect or Receive Data	Data and its metadata are in NSI custody
VS-01.4	Validate and Log Data Intake	The delivery is complete, intact and traceable
VS-01.5	Store and Register Raw Data Artefacts	Raw data is registered and available for reuse (State #1)

The value created here lies in trust and readiness , trust that the data has been acquired correctly, and readiness for the data to move forward into later stages of the statistical process.

Once data has been safely acquired and registered, it is ready to move into the next phase - the Data Standardization & Integration Value Stream. Here, the focus shifts from collecting data to structuring it, ensuring that all incoming data can be compared, combined, and analyzed.

VS-02 Data Standardization and Integration

The Data Standardization & Integration Value Stream is the second step in the statistical value chain. After raw data has been collected and securely stored, it now needs to be standardized, harmonized, and prepared for further processing. In this value stream, the organization ensures that data from different sources - surveys, administrative registers, or external datasets - can work together seamlessly. This is achieved through consistent structures, common formats, and shared definitions. The main goal is interoperability: data that looks different at the source must be aligned so it can be understood, compared, and reused across systems, domains, and time periods.

PLACEHOLDER: Add ArchiMate view of VS-02

Stage	Name	Value added
VS-02.1	Profile and Assess Data Structure	Differences from the target model are known
VS-02.2	Apply Standard Classifications and Formats	Data uses common codes, formats and identifiers
VS-02.3	Align Metadata and Semantics	Definitions and descriptions are consistent
VS-02.4	Validate Structural and Semantic Conformance	Conformance to standards is confirmed
VS-02.5	Register Standardized Artefacts	Standardized data is registered and available for reuse (State #2)

The value created here is coherence and comparability, transforming raw data into a structured and standardized asset that can flow smoothly into editing, analysis, and modeling.

Once data is standardized and validated, it can safely move to the Data Editing & Enrichment Value Stream. In that stage, attention shifts from structure and standards to quality and completeness. Here the organization begins improving data accuracy, filling gaps, and enriching datasets with derived or corrected information.

VS-03 Data Editing and Enrichment

The Data Editing & Enrichment Value Stream is the third step in the statistical value chain. After data has been standardized and aligned, the next task is to improve its quality, completeness, and usability. At this stage, the organization focuses on cleaning, correcting, and enriching data so that it becomes trustworthy and analytically sound. Errors are detected, inconsistencies are corrected, missing values are filled, and new variables are derived to add more meaning and analytical value. This is where raw, standardized data begins to transform into high-quality statistical input , ready for estimation, modeling, and analysis.

PLACEHOLDER: Add ArchiMate view of VS-03

Stage	Name	Value added
VS-03.1	Conduct Error Detection and Quality Diagnostics	Errors, duplicates, gaps and outliers are known
VS-03.2	Perform Data Editing and Correction	Incorrect values are corrected
VS-03.3	Impute Missing or Non-Response Data	Missing values are filled with documented methods
VS-03.4	Derive Additional Variables and Indicators	Data describes reality better than the source alone
VS-03.5	Perform Quality Evaluation and Versioning	Quality is documented, and processed microdata is registered (State #3)

The main value created here is data reliability and analytical readiness. By systematically improving data quality, the organization ensures that every subsequent result , every estimate, model, and publication, rests on a solid foundation.

VS-04 Estimation, Analysis and Statistical Modeling

The Estimation & Analysis Value Stream is the fourth stage of the statistical value chain. Here, the focus shifts from improving data quality to extracting insight and meaning. At this stage, the organization applies statistical methods, estimation techniques, and analytical models to transform edited and enriched data into representative results that describe real-world phenomena. This is where data becomes statistical information, summarized, interpreted, and validated to reflect populations, trends, and behaviors. Every decision taken in this stage must balance methodological detail with transparency and reproducibility.

PLACEHOLDER: Add ArchiMate view of VS-04

Stage	Name	Value added
VS-04.1	Confirm Estimation and Analytical Approach	The method fits the design and the purpose
VS-04.2	Apply Weighting and Calibration	Bias from sampling, non-response and coverage is reduced
VS-04.3	Estimate and Aggregate Data	Statistics and indicators are available
VS-04.4	Perform Model-Based Analysis	Model-based results complement direct estimates
VS-04.5	Validate and Document Results	Results are sound and documented (State #4)

The main value created here is analytical insight and statistical credibility. Through careful estimation and analysis, data is turned into knowledge that policymakers, researchers, and society can trust.

After estimation and analysis, the organization moves to the Statistical Product Composition Value Stream. Here, the focus shifts from individual analyses to building integrated and composite products. In that next stage, data from multiple sources and analyses are combined into coherent statistical publications, dashboards, and reports, turning analytical results into accessible products that provide a complete picture.

VS-05 Statistical Product Composition

The Statistical Product Composition Value Stream is the fifth stage in the statistical value chain. After estimation and analysis, the focus now shifts to bringing multiple analytical results together to create

coherent, meaningful, and high-value statistical products. At this stage, data from different sources, domains, and analyses are combined, refined, and formatted into composite datasets, time series, and indicators that can be used for reporting, policymaking, and research. This value stream transforms intermediate analytical outputs into final statistical products , ready for validation, interpretation, and dissemination.

PLACEHOLDER: Add ArchiMate view of VS-05

Stage	Name	Value added
VS-05.1	Select and Integrate Source Datasets	Relevant results are brought together
VS-05.2	Construct Composite Indicators or Indices	Indices, ratios and aggregates are available
VS-05.3	Perform Consistency and Coherence Checks	The product is internally consistent
VS-05.4	Generate Analytical and Interpretive Outputs	Results are explained and visualised
VS-05.5	Approve and Register Statistical Products	The product is approved for release (State #4, approved)

The key value created here is coherence and interpretability. Through careful composition, integration, and quality control, statistical outputs become consistent, accessible, and ready to serve their intended purpose.

After products are approved and registered, they move into the Dissemination & Service Delivery Value Stream. Here, the focus shifts from producing statistics to sharing them, making them available through publications, APIs, dashboards, and data services. That next stage ensures that the work done throughout the entire value chain ultimately reaches users in a secure, accessible, and user-friendly way.

VS-06 Dissemination and Service Delivery

The Dissemination & Service Delivery Value Stream is the final stage in the statistical value chain. At this point, the statistical products created in the previous stages are published, distributed, and made accessible to users through trusted and secure channels. The main goal of this value stream is to ensure that information reaches users in a way that is accurate, timely, and easy to understand. This includes policymakers, researchers, journalists, businesses, and the general public. Dissemination is not only about releasing data, it is about providing services that help users access, interpret, and apply statistical information responsibly. This step transforms statistical outputs into societal value by enabling informed decision-making, evidence-based policy, and transparency.

PLACEHOLDER: Add ArchiMate view of VS-06

Stage	Name	Value added
VS-06.1	Plan Dissemination and Access Strategy	Target users, channels and formats are set
VS-06.2	Prepare Data for Dissemination	Output is disclosure-safe and in the right format
VS-06.3	Publish and Provide Access Services	Users can find and access the output (State #5)
VS-06.4	Support and Engage Users	Users understand and use the output correctly
VS-06.5	Monitor Usage and Feedback	The NSI knows how its output is used

The key value created here is impact and accessibility, ensuring that statistics and data services are usable, trusted, and beneficial to all. The Dissemination & Service Delivery Value Stream relies heavily on several cross-cutting value streams:

- Governance and Compliance ensures that all dissemination complies with data protection and confidentiality laws.
- Infrastructure and Platform Enablement provides the technical platforms for portals, Customer and Stakeholder Engagement ensures that dissemination services meet real user needs and expectations.

Together, these supporting streams ensure that dissemination is not just a technical process , but a strategic, user-centered activity that delivers societal value through trustworthy data.

Supporting value streams

VS-07 Governance and Compliance

The Governance and Compliance Value Stream runs horizontally across all stages of the statistical value chain. Its purpose is to make sure that every process, dataset, and product within the organization operates under clear rules, standards, and accountability mechanisms. This value stream builds trust , both inside and outside the organization , by ensuring that all data activities are ethical, transparent, and compliant with legal, methodological, and quality requirements. It provides the framework of control and assurance that keeps the statistical system reliable, consistent, and aligned with national and international standards such as the European Statistics Code of Practice, GDPR, GSBPM, GAMSO and ISO data quality principles.

PLACEHOLDER: Add ArchiMate view of VS-07

Stage	Name	Value added
VS-07.1	Define Governance Frameworks and Policies	Rules for data, metadata, ethics, confidentiality and quality are set
VS-07.2	Establish Legal and Regulatory Compliance Mechanisms	Applicable rules are known and translated into procedures
VS-07.3	Implement Data and Metadata Governance	Ownership, stewardship and lineage are assigned
VS-07.4	Monitor Quality and Compliance Indicators	Quality and compliance status is visible
VS-07.5	Audit, Certify and Report Compliance	Compliance is demonstrated
VS-07.6	Update and Improve Governance Mechanisms	Governance keeps up with new rules and lessons learned

The value created here is integrity and accountability making sure that the NSI acts responsibly, protects privacy, and delivers products users can trust.

The Governance and Compliance Value Stream supports all other value streams by acting as the foundation of trust and assurance. It ensures that Data Acquisition & Ingestion follows lawful and ethical standards. It guarantees that Editing, Estimation, and Dissemination meet quality benchmarks. It provides the governance structure for Infrastructure, Innovation, and Stakeholder Engagement. In short, it supports the entire SAF value chain, ensuring that every activity is compliant, transparent, and aligned with the mission of producing trustworthy statistics.

VS-08 Innovation and Methodological Development

The Innovation and Methodological Development Value Stream makes sure the statistical organization does not stand still. Its purpose is to discover, test, and introduce new ways of working , new data sources, new methods, new tools, and new technologies. This value stream helps the organization move from “we have always done it this way” to “we can do this better, faster, and smarter.” It connects research, methodology,

IT, and business users. It also makes sure that innovation is controlled , so that only methods that are sound, documented, and compliant end up in production.

PLACEHOLDER: Add ArchiMate view of VS-08

Stage	Name	Value added
VS-08.1	Identify Innovation Needs and Opportunities	Promising developments are known
VS-08.2	Conduct Research and Proof-of-Concepts	Ideas are tested in a limited scope
VS-08.3	Evaluate Methodological Soundness and Feasibility	Only sound and feasible innovations proceed
VS-08.4	Develop and Document New Methodologies or Tools	Innovations are ready for reuse
VS-08.5	Integrate Innovation into Production Environments	Innovations run in production under governance
VS-08.6	Promote Knowledge Sharing and Capability Building	Staff can apply the new methods

The value created here is continuous improvement. Thanks to this stream, the SAF remains future-proof, able to work with new data (IoT, admin, data spaces), new techniques (ML, small-area estimation), and new operating models (secure processing environments, data sharing). How This Supports the Other Value Streams. - It gives Data Acquisition new ways to connect to data spaces and new sources. - It gives Standardization & Integration new mapping or classification techniques. - It gives Editing & Enrichment smarter imputation and outlier detection. - It gives Estimation & Analysis new models and ML-based methods. - It gives Dissemination new channels (APIs, linked data, interactive views).

So, this value stream keeps the entire SAF alive and evolving.

VS-09 Infrastructure and Platform Enablement

The Infrastructure and Platform Enablement Value Stream provides the technical and digital foundation that supports every part of the statistical organization. Where the other value streams focus on data, methods, or governance, this one ensures that all those activities can actually run safely, efficiently, and at scale. It delivers the IT environments, data platforms, tools, and integration services that make statistical production possible, from data collection to dissemination. It ensures that the systems are secure, resilient, and compliant, while also flexible enough to handle new technologies and data sources as the organization modernizes. This value stream enables the organization to work faster, smarter, and with confidence. It transforms infrastructure from a background function into an active enabler of the statistical mission, ensuring that every user, process, and dataset has the tools and environment it needs to perform reliably.

PLACEHOLDER: Add ArchiMate view of VS-09

Stage	Name	Value added
VS-09.1	Define Digital and Technical Requirements	Infrastructure needs follow from business needs
VS-09.2	Design Platform and Infrastructure Architecture	Platforms are designed for storage, processing, integration and secure access
VS-09.3	Provision and Configure Platforms	Environments are available with the required controls
VS-09.4	Enable Integration and Orchestration Services	Processes and services are connected

Stage	Name	Value added
VS-09.5	Ensure Performance, Security and Continuity	Platforms are reliable and protected
VS-09.6	Evolve and Modernise the Technical Ecosystem	The technical landscape stays current

The value created here is operational capability and continuity, a stable, modern, and adaptive foundation for the entire SAF ecosystem.

Connecting to Other Value Streams The Infrastructure and Platform Enablement Value Stream supports every other part of the SAF. It underpins the business and data processes by providing the digital environment in which they operate. - It gives Data Acquisition the secure pipelines and storage needed for data collection. - It gives Standardization, Editing, and Analysis the computing and orchestration needed to process large datasets efficiently. - It gives Dissemination the platforms, portals, and APIs through which users access results. - It ensures Governance and Compliance have the tools to monitor security, quality, and data lineage, and - It enables Innovation by offering scalable, modern platforms for experimentation and research.

VS-10 Customer and Stakeholder Engagement

The Customer and Stakeholder Engagement Value Stream ensures that the statistical organization remains connected, relevant, and responsive to the needs of those it serves. It focuses on building and maintaining mutual understanding and trust between the organization and its users, data providers, partners, and policy stakeholders. In the SAF, this value stream is essential because official statistics exist to create public value. To achieve that, the organization must understand what stakeholders need, communicate transparently, and ensure that products, data services, and policies remain aligned with real-world demands.

This value stream is not a one-way process. It is a continuous dialogue, listening to users, understanding their challenges, and translating those insights into better products, improved services, and stronger relationships.

PLACEHOLDER: Add ArchiMate view of VS-10

Stage	Name	Value added
VS-10.1	Identify and Segment Stakeholders	The NSI knows its users, providers and partners
VS-10.2	Collect and Analyse User Needs	Needs are translated into requirements
VS-10.3	Define and Manage Service Portfolios	Services and products match the needs
VS-10.4	Deliver Services and Manage Relationships	Users and partners are served
VS-10.5	Engage and Communicate with Stakeholders	Stakeholders understand the output
VS-10.6	Monitor Impact and Satisfaction	The NSI knows its impact and acts on it

The value created here is relevance and trust, ensuring that the organization's work is understood, used, and appreciated by its partners, users, and society at large.

Connecting to Other Value Streams The Customer and Stakeholder Engagement Value Stream is the voice and mirror of the statistical organization. - It connects external expectations with internal capabilities, closing the feedback loop across the SAF. - It informs Data Acquisition by revealing what data sources or topics matter most to users. - It influences Product Composition and Dissemination by identifying which outputs and formats users prefer. - It strengthens Governance and Compliance through transparent communication about data ethics, confidentiality, and quality. It supports Innovation and Methodology by gathering ideas and feedback from real users and partners.

Through this value stream, the organization ensures that it doesn't just produce data, it produces public value. It turns statistics into impact by aligning what the organization creates with what stakeholders need.

PLACEHOLDER: Value Stream Template and the application of Value Streams to be added to the toolkit

Resources [T6]

The concept

A **resource** is an asset that an organisation owns or controls. Resources are what the organisation uses to perform its capabilities. A capability without the right resources exists on paper only.

The SAF distinguishes six types of resources:

Type	Examples in an NSI
People and skills	Methodologists, data engineers, subject-matter statisticians, architects
Data and information	Registers, survey data, administrative data deliveries, classifications, metadata
Knowledge	Statistical methods, process documentation, quality frameworks
Technology	Data platforms, processing tools, collection systems, secure processing environments
Financial	Structural budget, project funding, international grants
Relationships and agreements	Data sharing agreements, partnerships with other NSIs, supplier contracts

Three characteristics set resources apart:

- **They are owned or controlled.** A resource is something the NSI has, or has secured through an agreement. Access to administrative data under a provision agreement is a resource.
- **They are limited.** People, money and data are scarce. Resources therefore force choices between capabilities.
- **They are shared.** The same resource often serves several capabilities. A data platform supports collection, processing and dissemination at the same time.

For an NSI, data is a resource in its own right. Registers, long time series and trusted relationships with data providers are often harder to replace than any system.

How the SAF applies resources

Why the SAF uses resources

Resources make the capability model realistic. A maturity assessment that only looks at processes misses the most common bottleneck: not enough people with the right skills, or no access to a needed data source. Linking resources to capabilities shows where a capability depends on a single expert, an outdated platform or an agreement that expires.

How resources fit into an NSI's own architecture

Resources are specific to each NSI. The SAF does not list them. An NSI records the resources behind its key capabilities, notes where resources are missing or at risk, and uses this in its courses of action. Resources also link strategy to the lower layers: technology resources are described in more detail in the application and technology architecture, and data resources in the data architecture.

Relationships with other concepts

Related concept	Relationship	Example
Capability Model	Resources are assigned to capabilities	Staff with methodological expertise are assigned to Methodology (2.2.3)
Courses of Action	Courses of action configure or develop resources	A workforce strategy trains data engineers
Assessment (Motivation)	An assessment can record a resource gap	"Two of our three methodologists for small area estimation retire within two years"
Constraints (Motivation)	Constraints limit which resources can be used	Public cloud may not be used for statistical microdata
Data and technology architecture	Data and technology resources are detailed there	The data platform as a technology resource

Examples (illustrative)

Resource	Type	Supports capabilities	Note
Access to the national population register	Data and information; relationships and agreements	1.2.2 Secondary data collection; 2.1.4 Statistical register management	Depends on a provision agreement that must be renewed
Team of methodologists for estimation	People and skills	1.3.5 Estimating population parameters; 2.2.3 Methodology	Small team; knowledge concentrated in few people
Shared data processing platform	Technology	1.3 Processing; 2.1.1 Data lifecycle management	Shared by several statistical domains

PLACEHOLDER: Resource Template and the application of Resources to be added to the toolkit

Courses of Action [T6] [T7]

The concept

A **course of action** is an approach or plan for configuring capabilities and resources to reach a goal. It answers the question *how will we get there?*

Courses of action range from broad to specific. ArchiMate calls the broad, long-term ones **strategies** and the narrower, shorter-term ones **tactics**. There is no hard line between the two. A strategy is usually broken down into several tactics, and tactics are realised by developing capabilities and resources.

Three characteristics set courses of action apart:

- **They are choices.** A course of action chooses one approach over others. "Collect data web-first" rules out paper-first. A plan without a choice is not a course of action.
- **They serve goals.** Each course of action contributes to one or more goals and outcomes. A course of action that serves no goal needs a justification.
- **They change capabilities and resources.** A course of action builds, strengthens, shares or retires capabilities and resources.

Strategy, architecture and roadmap are related but different. A strategy sets the direction and the main choices. The architecture describes the structure that results. A roadmap sets the order and timing of the changes.

How the SAF applies courses of action

Why the SAF uses courses of action

Goals say where the NSI wants to go. Capabilities say what it must be able to do. Courses of action connect the two: they describe the choices the NSI makes to close the gap. Recording them explicitly prevents a common problem: programmes that each make sense on their own, but pull the organisation in different directions.

How courses of action fit into an NSI's own architecture

Courses of action are specific to each NSI. The SAF does not define them. Most NSIs work with three levels:

1. **Organisation strategy.** The overall direction of the NSI, usually in a multi-annual plan approved by its board or ministry.
2. **Domain strategies.** Strategies for a specific area, such as data, metadata, IT, cloud, security or privacy. Each one refines the organisation strategy for its domain.
3. **Tactics and roadmaps.** Concrete approaches and their order in time, for example in a programme plan.

Each NSI names its own strategies. A small NSI may have one IT strategy that covers applications, infrastructure and cloud. A large NSI may have separate strategies for each. What matters is that the strategies are consistent with each other and traceable to the motivation.

The SAF elements help to make strategies consistent. A domain strategy can be described with the same building blocks used in this framework: the drivers and assessments that make it necessary, the goals and outcomes it serves, the principles it applies, the courses of action it chooses, the capabilities and resources it develops, and the roadmap that sets the order. When all strategies use these building blocks, overlaps and conflicts between them become visible. The toolkit offers templates for this.

Choosing a reference model for the data value chain, such as the Steady-State or the Medallion architecture (see Business Architecture), is itself a course of action.

Relationships with other concepts

Related concept	Relationship	Example
Goals and outcomes (Motivation)	Courses of action serve goals and contribute to outcomes	A shared-platform strategy serves cost efficiency (SAF-G1)
Drivers and assessments (Motivation)	Assessments show why a course of action is needed	An assessment of SAF-D5 leads to a decision to retire legacy platforms
Principles (Motivation)	Courses of action apply principles	An application strategy applies SAF-P01 *Actively participate in open-source communities to maintain collective control over statistical processes*
Capability Model	Courses of action are realised by developing capabilities	A metadata strategy develops Metadata management (2.1.2)
Resources	Courses of action configure resources	A workforce strategy develops data engineering skills
Value Streams	Courses of action improve value streams	A web-first collection strategy improves VS-01
Capability 4.1.2 *Policy and planning*	This capability produces the strategies	Translating the organisation strategy into domain strategies

Examples of strategies (illustrative)

The examples show which domain strategies NSIs often develop. They are not a required set. Each NSI decides which strategies it needs and how to combine them.

Strategy	Typical question it answers	Mainly develops
Data strategy	How do we manage and use data as an asset?	2.1 Data management; VS-01 to VS-06
Metadata strategy	How do we make data findable, understandable and reusable?	2.1.2 Metadata management
IT strategy	What do we reuse, buy or build, and on which platforms?	3.2 Computerisation and automation; VS-09
Cloud strategy	Which workloads may run where, and under which conditions?	3.2.5 IT infrastructure; 3.3 Security and privacy
Information security strategy	How do we protect information and systems?	3.3 Security and privacy; VS-07
Privacy and confidentiality strategy	How do we protect respondents and keep statistical confidentiality?	3.3.1 Privacy management; 1.3.7 Statistical disclosure

Other strategies NSIs develop include strategies for data acquisition, dissemination and open data, artificial intelligence, innovation, workforce and skills, and sourcing and partnerships.

Reference strategies

The following published strategies and guidance documents can serve as reference when an NSI develops its own strategies.

Document	Organisation	Type	Relevant for
Statistics Canada Data Strategy	Statistics Canada	Data strategy of an NSI	Data strategy; shows links to IT, cloud, privacy and metadata strategies
Data Strategy	Office for National Statistics (UK)	Data strategy of an NSI	Data strategy
ABS Data Strategy 2025–2030	Australian Bureau of Statistics	Data strategy of an NSI	Data strategy
Data Strategy of the Secretary-General	United Nations	Data strategy of an international organisation	Data strategy
Cloud for Official Statistics (2024)	UNECE	Guidance	Cloud strategy
Privacy-Enhancing Technologies Guide	UN Committee of Experts on Big Data and Data Science	Guidance	Privacy and confidentiality strategy

PLACEHOLDER: Strategy Template and the application of Courses of Action to be added to the toolkit

Business Architecture

Purpose of this chapter

The business architecture describes how a statistical organisation is organised to turn data into official statistics. It shows who is involved, which processes and functions they perform, which services they offer, what triggers their work, and which information they create and exchange.

The Strategy layer describes *what* the NSI must be able to do and *where* it creates value. The business architecture describes *how* that work is organised. It is the layer where capabilities and value streams become concrete: in roles that people take on, processes they carry out, and information they hand over to each other.

For official statistics, one question deserves special attention: where does responsibility for data pass from one part of the organisation to another? A statistical value chain is long. It involves many teams and often confidential data. An untraceable change or uncontrolled reuse at any point can undermine every statistic built on that data. The SAF business architecture therefore pays explicit attention to the points where data is stored, quality-assured and handed over. These handover points are the backbone of the reference models at the end of this chapter.

The business architecture is independent of technology. It does not describe applications or platforms. It does describe the business needs that applications and platforms must support, and it is the starting point for the data, application and technology architecture.

Position in the framework

The business architecture is the first of the core layers in TOGAF and ArchiMate (see *Position in the framework* in the Motivation chapter). In the TOGAF Architecture Development Method, it is developed in Phase B, Business Architecture. It receives direction from the Strategy layer and gives direction to the layers below it.

PLACEHOLDER: Position of the Business Architecture between the Strategy layer and the data, application and technology layers

Direction	Relationship	Example in the SAF
From Strategy	Business processes realise value stream stages	The execution processes of the Standardization domain realise the stages of VS-02
From Strategy	Business functions and processes realise capabilities	The function *Classify Data* realises Data standardisation (1.3.1)
From Motivation	Principles and requirements shape the business design	SAF-P07 leads to loose coupling of domains through steady states
To Data Architecture	Business objects are realised by data objects	A registered *Raw Data* version is realised by a dataset in a data store
To Application Architecture	Business processes are served by application services	Data validation at intake is served by a validation service
To Technology Architecture	Business requirements drive platform requirements	Retention and versioning per steady state drive storage requirements

In TOGAF, Phase B produces a baseline and a target business architecture, and the gap between them. The SAF provides a reference for the target. An NSI describes its own baseline and uses the SAF to define where it wants to go.

How NSIs use this chapter

The SAF business architecture contains two kinds of content.

Reference content. Most NSIs perform the same statistical work and follow a comparable production process. The SAF therefore offers reference content for business processes, business functions, business services and business roles. NSIs use this content in the same three ways as the other SAF elements: adopt, adapt or extend.

Concepts with examples. Business events, business objects, contracts, representations and products are specific to each NSI. They depend on its products, its legal basis and its users. For these elements, the SAF explains the concept, how it is used, and gives examples. The toolkit provides templates.

The data value chain needs a separate choice. NSIs organise their chain in different ways, depending on their size, legal context and platforms. The SAF therefore offers more than one reference model for the chain: the Steady-State Architecture and the Medallion Architecture. An NSI chooses the model that fits, or combines them. This choice is a course of action in the sense of the Strategy chapter.

The business architecture is also the place where the NSI's own organisation meets the framework. An NSI maps its departments, teams and job profiles onto the SAF actors, roles and functions. The mapping shows which parts of the organisation carry which responsibilities, where they overlap, and where they are missing.

PLACEHOLDER: Business architecture templates and mapping instruments to be added to the toolkit

Relationship with the statistical standards

The SAF business architecture builds on two models that the statistical community already uses.

GSBPM, the Generic Statistical Business Process Model, describes statistical production in phases and sub-processes, from *Specify Needs* to *Evaluate*. The SAF process model is based on GSBPM.

GAMSO, the Generic Activity Model for Statistical Organisations, describes the activities of a statistical organisation as a whole: strategy and leadership, capability development, production, and the activities that support them. GAMSO places GSBPM in a wider context: GSBPM describes the production activities of GAMSO in detail. The SAF uses GAMSO as the frame of reference for its business functions. The functions are described in SAF terms, so that NSIs can apply them together with their own national function models. An NSI that already works with GAMSO will recognise the main areas.

GSIM, the Generic Statistical Information Model, describes the information objects used in statistical production. It is the reference for the SAF business objects.

Business architecture concepts in the SAF

The SAF uses the business layer elements of ArchiMate, and adds concepts that are specific to the statistical data value chain.

ArchiMate business elements used in the SAF

Concept	ArchiMate element	Question it answers	Example	SAF content
Business actor	Business Actor	Who is involved: a person, team or organisation?	Data provider, NSI, research institute	Reference
Business role	Business Role	Which responsibility does an actor take on?	Data owner, data steward, methodologist	Reference

Concept	ArchiMate element	Question it answers	Example	SAF content
Business collaboration	Business Collaboration	Which actors or roles work together?	A data sharing partnership between an NSI and a register holder	Reference
Business process	Business Process	Which sequence of activities produces a defined result?	Collect, Process, Disseminate	Reference
Business function	Business Function	What does the organisation do continuously, grouped by required knowledge and skills?	Manage Quality	Reference
Business service	Business Service	Which behaviour does the organisation offer to others?	Microdata access for researchers	Reference
Business interface	Business Interface	Through which channel is a service offered?	A web portal, a secure data transfer channel, a helpdesk	Reference
Business event	Business Event	What triggers a process, or marks its result?	*Raw Data Registered*; start of a reference period	Concept and examples
Business object	Business Object	Which information does the business use and produce?	Raw Data, Statistical Product, Classification	Concept and examples
Contract	Contract	Which formal agreement governs a service or exchange?	A data provision agreement with a register holder	Concept and examples
Representation	Representation	In which perceptible form is information made available?	A press release, a statistical table, an API response	Concept and examples
Product	Product	Which coherent set of services and information is offered to users?	A statistical product with its publication and data services	Concept and examples

SAF-specific concepts for the data value chain

Concept	Definition
Business domain	A coherent part of the statistical process that adds one specific kind of statistical value to data and is governed as a unit. In ArchiMate, a business domain is modelled as a grouping.
Data state	A defined condition of data, characterised by the processing it has undergone and the quality criteria it meets.
Steady state	A data state that is stored, versioned, quality-assured and registered, so that it can be handed over and reused.

Concept	Definition
Quality gate	The set of criteria that data must meet before it may be registered in a state.
Quality version	An immutable, identified version of a dataset that has passed a quality gate.
Handover interface	The agreement between a producing and a consuming domain: the artefact, its required metadata, the triggering event and the transfer of responsibility.
Reference model	A generic arrangement of domains, states and handover points that an NSI can adopt and adapt, such as the Steady-State or Medallion Architecture.

How the business architecture is built up

ArchiMate describes each layer with three aspects: *who* acts (active structure), *what* happens (behaviour) and *what* is acted upon (passive structure). The SAF business architecture follows these aspects, and then brings them together in two integrating views.

PLACEHOLDER: Structure of the SAF business architecture: three ArchiMate aspects and two integrating views

Part	Elements	Question
Active structure	Actors, roles, collaborations, interfaces	*Who* does the work, and through which channels?
Behaviour	Processes, functions, services, events	*What* happens, in which order, and what triggers it?
Passive structure	Business objects, contracts, representations, products	*Which* information is used, agreed and delivered?
Integrating view 1	Process and function architecture and value chain	*How* do processes, functions and value streams fit together?
Integrating view 2	Steady-State and Medallion Architectures	*Where* is data handed over, and with which quality?

Functions and processes

Functions and processes are easily confused, but they are different.

A **business function** groups behaviour by the knowledge, skills and resources it needs. Functions are stable. They have no start or end, and they do not depend on the order of work. *Manage Quality* is a function: it exists as long as the NSI exists.

A **business process** is a sequence of activities that leads to a defined result. A process has a trigger, a start and an end. *Collect* is a process: it starts when a collection period opens and ends when the data is stored and checked.

The two cross each other. One process uses several functions, and one function contributes to several processes. In ArchiMate, functions and processes are connected through realisation and assignment relationships. Statistical production is the clearest example: the production function is realised by the production process, which the SAF describes in detail.

Processes and the data value chain

The process view and the data value chain look at the same production from two angles. The process follows the life cycle of one statistical product, from need to evaluation. The data value chain follows the data, from source to release, across products.

In the Steady-State Architecture, each business domain uses four process types. They connect the two angles:

Process type in a business domain	Provides	Corresponding GSBPM phases
Design	Metadata: designs, rules, specifications	2 Design
Implementation	The execution process: instruments, components, configured workflows	3 Build
Execution	Statistical data, registered in a steady state	4 Collect, 5 Process, 6 Analyse, 7 Disseminate
Management	Quality reports and improvements	8 Evaluate, and the overarching quality and metadata management

The GSBPM phases 4 to 7 run across several business domains. *Process* (phase 5), for example, spans standardization, editing and part of estimation. The steady states mark where, within that flow, data is handed over with a defined quality.

Generic and model-specific content

Actors, roles, processes, functions, services, events and objects are generic. They apply to every NSI, whatever reference model it chooses for its data value chain.

The data value chain is model-specific. The business domains, steady states and quality gates differ between the Steady-State and the Medallion Architecture. That is why the SAF describes them per reference model, at the end of this chapter.

Structure of this chapter

Section	Content	SAF content
Business Actors, Business Roles and Business Collaborations	Who is involved in statistical production, and with which responsibilities	Reference
Business Processes	The SAF process model, based on GSBPM	Reference
Business Functions	The SAF function model	Reference
Business Services and Business Interfaces	The services an NSI offers, and the channels through which it offers them	Reference
Business Events	Triggering events: what they are, how they are used, examples	Concept and examples
Business Objects, Contracts, Representations and Products	Information objects, agreements, forms of output and products: what they are, how they are used, examples	Concept and examples
Process and Function Architecture and Value Chain	How processes, functions and value streams fit together, per value stream	Reference

Section	Content	SAF content
Steady-State and Medallion Architectures	Reference models for the data value chain: comparison, selection criteria, combination, and each model in detail	Reference

Readers who want to understand the statistical data value chain can go straight to the last section. Readers who want to map their own organisation onto the SAF start with actors and roles, and with the function model.

Business Actors, Business Roles and Business Collaborations

The concepts

A **business actor** is a person, team or organisation that can perform behaviour. An actor is a real entity. It exists whether or not it takes part in a specific process. Examples are a ministry that supplies a register, a research institute, or the NSI itself. In the SAF, actors are described as types, not as named organisations.

A **business role** is a responsibility for performing specific behaviour. A role says what someone is accountable for, not who that someone is. One actor can take on several roles, and one role can be filled by several actors. A methodologist in a small NSI may also act as statistical designer. A large NSI may have a whole team in the role of data steward.

A **business collaboration** is a group of actors or roles that work together for a shared purpose. A collaboration performs behaviour that none of its members can perform alone. It can be permanent, such as a data governance board, or temporary, such as a project team. Collaborations can also cross the boundaries of the NSI. A data sharing partnership between an NSI and a register holder is an example.

Actors, roles and collaborations are different from stakeholders (see *Stakeholders* in the Motivation chapter). A stakeholder has an interest in the architecture. An actor or role performs work in it. The same organisation can be both. A data provider is a stakeholder with concerns about burden, and an actor that delivers data in VS-01.

How the SAF applies actors, roles and collaborations

Why the SAF describes actors, roles and collaborations

Roles make responsibilities explicit. A steady state needs a data owner. A quality gate needs someone who decides. A release needs someone who approves. Without defined roles, these decisions are taken by whoever happens to be involved. That makes accountability unclear and audits difficult. Defined roles also make NSIs comparable. When two NSIs both use the role *data steward*, they can compare practices and share solutions.

How actors, roles and collaborations fit into an NSI's own architecture

The SAF offers a reference set of actor types, roles and collaborations. An NSI maps its own organisation onto this set. It links its departments and teams to actor types, and its job profiles to roles. One job profile often covers several SAF roles, especially in small NSIs. The mapping shows which roles are filled, which are combined, and which are missing. An NSI can then assign roles to processes, for example with a RACI matrix (responsible, accountable, consulted, informed).

The SAF roles are deliberately generic. The value streams use more specific role names, such as *Estimation Methodologist* or *Imputation Specialist*. These are specialisations of the reference roles. The mapping table at the end of this section shows how they relate.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Stakeholders (Motivation)	A stakeholder can also be an actor	A data provider is a stakeholder and an actor in VS-01
Business processes	Roles are assigned to processes	The data steward is assigned to *Register Raw Data*
Business functions	Roles are assigned to functions	The quality manager is assigned to *Manage Quality*
Business services and interfaces	Actors and roles offer or use services through interfaces	A researcher uses the microdata access service through a secure portal
Steady states	Each steady state has an owner, a steward and a custodian	The data owner of State #3 decides on access to processed microdata
Capabilities (Strategy)	Roles are part of the resources that realise capabilities	Methodologists are a key resource for Methodology (2.2.3)

PLACEHOLDER: ArchiMate view of SAF actors, roles and collaborations

SAF business actors

The SAF distinguishes nine actor types. They are grouped by their position relative to the NSI.

ID	Actor	Position	Description
ACT-01	National Statistical Institute	Internal	The NSI is the organisation that produces official statistics under a legal mandate. It is the central actor in the SAF. The NSI owns the statistical processes and is accountable for the quality and confidentiality of its output. It enters into agreements with data providers, users and partners. In the architecture, the NSI as a whole is the actor that offers business services to society.

ID	Actor	Position	Description
ACT-02	Organisational unit	Internal	An organisational unit is a department, division or team within the NSI. Units are the actors that actually carry out the work. Each unit fills one or more roles, such as data collection, methodology or dissemination. The SAF does not prescribe an organisation structure. An NSI maps its own units onto the SAF roles, so that it can see which unit is responsible for which part of the chain.
ACT-03	Data provider	External, supplies data	A data provider is an organisation that supplies data it holds to the NSI. Examples are government agencies with administrative registers, businesses with transaction data, and platform operators. The data is collected for another purpose and reused for statistics. Delivery usually takes place under a legal obligation or a data provision agreement. The NSI cannot control the quality at the source, but it can influence it through agreements and feedback.
ACT-04	Respondent	External, supplies data	A respondent is a person, household or business that provides data directly to the NSI, usually through a survey. Respondents answer questionnaires on the web, by telephone, face to face or on paper. Their burden is a constant concern for the NSI. Respondents are protected by statistical confidentiality: the NSI may not disclose their individual data. The NSI also has a duty to explain why it needs the data and how it uses it.

ID	Actor	Position	Description
ACT-05	Data user	External, uses output	A data user is a person or organisation that uses the output of the NSI. Users include policy makers, researchers, journalists, businesses and members of the public. Their needs differ widely. A minister needs a clear figure, a researcher needs microdata under strict conditions, and a developer needs a reliable API. The NSI groups users into segments, so that it can design services and channels for each segment.
ACT-06	Other producer of official statistics	Statistical system	Many countries have more than one producer of official statistics. Examples are central banks, ministries with statistical units and regional statistical offices. The NSI often coordinates this national statistical system. Producers exchange data, align methods and share infrastructure. In the architecture, they are partners in collaborations and sometimes providers or users of each other's data.

ID	Actor	Position	Description
ACT-07	International statistical organisation	Statistical system	International statistical organisations set standards, coordinate statistics across countries and collect national data. Examples are UNECE, Eurostat, the OECD, the IMF and the UN Statistics Division. They require NSIs to deliver data in agreed formats and on agreed dates. They also develop the shared models that the SAF builds on, such as GSBPM and GSIM. For the NSI, they are users, standard setters and partners in collaboration.
ACT-08	Supervisory and legal authority	External, oversees	A supervisory or legal authority oversees whether the NSI acts within the law. Examples are the data protection authority, a national statistics council, the national audit office and the courts. Some authorities grant the legal basis for data collection. Others assess compliance after the fact. The NSI must be able to show these authorities how it handles data, which is one reason why traceability is a core concern of the SAF.

ID	Actor	Position	Description
ACT-09	Supplier and service provider	External, supports	A supplier delivers goods or services that the NSI needs but does not produce itself. Examples are software vendors, cloud and hosting providers, field work agencies and trusted third parties for pseudonymisation. Suppliers work under contracts that set requirements for security, confidentiality and continuity. When a supplier processes statistical data, the NSI remains accountable for it. The choice of suppliers is guided by SAF-P01 and by the NSI's sourcing strategy.

SAF business roles

The SAF reference set contains 24 roles in seven groups. The column *Process type* shows in which of the four process types of a business domain (design, implementation, execution, management) the role mainly works.

Governance and accountability

ID	Role	Process type	Description
ROL-01	Process owner	Management	The process owner is accountable for the end-to-end performance of a business domain or value stream. This role sets the objectives of the process and decides on changes to it. The process owner makes sure the process has the people, methods and systems it needs. The role receives the quality reports of the management process and acts on them. In the steady-state model, each business domain has one process owner.

ID	Role	Process type	Description
ROL-02	Data owner	Management	The data owner is accountable for a dataset or a steady state. This role decides who may access the data, for which purpose and under which conditions. The data owner also approves the retention period and the classification of the data. The role is usually held by a manager in the domain that produces the data. Data ownership is a business role, not an IT role: the data owner answers for the content, not for the storage.
ROL-03	Data steward	Execution	The data steward takes care of data on behalf of the data owner. This role registers datasets in the catalogue and maintains their metadata. The data steward checks that data meets the quality gate before it is registered. The role also handles questions from users and resolves exceptions. Data stewards are the daily point of contact for anyone who wants to find or use a dataset.
ROL-04	Data custodian	Execution	The data custodian is responsible for the safe storage and technical management of data. This role runs storage, backup and access control according to the decisions of the data owner. The custodian carries out retention and disposal. The role does not decide on the use of the data, but makes sure that decisions are applied correctly. In many NSIs, the custodian role is held by an IT or platform team.

ID	Role	Process type	Description
ROL-05	Chief data officer	Management	The chief data officer is accountable for data governance across the whole NSI. This role sets the data policy and the governance framework. It assigns the data owner, steward and custodian roles to people and units. The chief data officer chairs or supports the data governance board. The role is the escalation point for conflicts about data that cross domain boundaries.

Design

ID	Role	Process type	Description
ROL-06	Statistical designer	Design	The statistical designer translates information needs into the design of a statistical product. This role defines the outputs, the variables, the population and the required data. The designer specifies the processes that produce the product, in close cooperation with the methodologist. The role is the link between users and production. In some NSIs this role is called product manager.

ID	Role	Process type	Description
ROL-07	Methodologist	Design	The methodologist designs and assesses the statistical methods. This role covers sampling, collection methods, editing and imputation, estimation and quality measurement. The methodologist makes sure methods are sound, documented and reproducible. The role also evaluates new methods before they enter production. Specialisations exist for each step of the chain, such as estimation or editing.
ROL-08	Metadata and data architect	Design	The metadata and data architect designs the structure of data and metadata. This role defines data models, metadata schemas and the minimum metadata set of each steady state. The architect makes sure that data from different sources can be combined and compared. The role works with standards such as GSIM and SDMX. It is the bridge between the business architecture and the data architecture.

ID	Role	Process type	Description
ROL-09	Classification steward	Design	The classification steward manages the classifications and code lists that the NSI uses. This role maintains national versions of international classifications, such as NACE or ISCO. The steward manages correspondence tables between versions. The role makes sure that all domains use the same version at the same time. Changes to a classification are announced and documented before they take effect.

Implementation

ID	Role	Process type	Description
ROL-10	Data engineer	Implementation	The data engineer builds the technical processes that move and transform data. This role develops ingestion pipelines, collection channels and processing workflows. The data engineer implements the designs of the statistical designer and the methodologist. The role also builds the automated checks that support the quality gates. The data engineer works closely with the platform engineer, who provides the environment.

ID	Role	Process type	Description
ROL-11	Platform engineer	Implementation	The platform engineer provides and runs the environments in which the processes operate. This role configures platforms, storage, connectors and access controls. The platform engineer keeps environments secure, available and up to date. The role supports all value streams but is anchored in VS-09. In many NSIs, the role is also called system configurator or DevOps engineer.
ROL-12	Test manager	Implementation	The test manager plans and coordinates the testing of new or changed processes. This role covers technical tests, integration tests and field tests. The test manager checks that a process produces the expected result before it goes into production. The role reports test results to the process owner. Testing is required before any change to a process that produces a steady state.

Execution

ID	Role	Process type	Description
ROL-13	Data collection manager	Execution	The data collection manager runs data collection campaigns. This role plans the collection period, coordinates field staff and monitors response. The manager follows up on non-response and late deliveries. The role also handles contacts with respondents and data providers during collection. The data collection manager reports on collection progress and quality to the process owner.
ROL-14	Data quality analyst	Execution	The data quality analyst checks data during processing. This role runs validation checks, assesses results against the quality gate and logs exceptions. The analyst investigates errors and outliers and proposes corrections. The role documents quality indicators for each version of a dataset. Data quality analysts are active at every handover point in the chain.
ROL-15	Subject-matter statistician	Execution	The subject-matter statistician has in-depth knowledge of a statistical domain, such as labour market or prices. This role interprets results, checks plausibility and explains developments. The statistician carries out or supervises editing, estimation and analysis in the domain. The role writes or reviews the texts that accompany a release. Subject-matter statisticians are the experts that users and journalists turn to.

ID	Role	Process type	Description
ROL-16	Disclosure control specialist	Design, Execution	The disclosure control specialist protects the confidentiality of respondents in the output. This role designs and applies disclosure control methods, such as suppression, rounding and noise. The specialist assesses the disclosure risk of tables, microdata and services. The role works at State #4 and at the gate to State #5. No output is released without a disclosure check that this role has approved.
ROL-17	Release manager	Execution	The release manager controls the publication of statistical output. This role makes sure that releases follow the release calendar and the embargo rules. The release manager checks that the output has been approved and has passed disclosure control. The role coordinates the release across channels, such as the website, the API and press releases. Equal access for all users at the moment of release is a key responsibility.

Assurance

ID	Role	Process type	Description
ROL-18	Quality manager	Management	The quality manager is responsible for the quality framework of the NSI. This role defines quality indicators and monitors them across the chain. The quality manager produces quality reports and leads improvement actions. The role supports process owners in the management process of their domains. It also prepares the NSI for quality reviews, such as peer reviews against the European Statistics Code of Practice.
ROL-19	Compliance officer	Management	The compliance officer makes sure that the NSI complies with laws and regulations. This role covers statistical law, data protection law and other applicable rules. The officer checks new processes and methods against compliance criteria before they go into production. The role keeps the legal register up to date. In many NSIs the data protection officer is a specific form of this role.
ROL-20	Security officer	Management	The security officer is responsible for information security and continuity. This role defines security measures and monitors their application. The officer manages security incidents and leads recovery when something goes wrong. The role assesses the security of new solutions and suppliers. It works closely with the platform engineer and the compliance officer.

ID	Role	Process type	Description
ROL-21	Auditor	Management	The auditor checks independently whether processes and controls work as intended. This role can be internal, such as an internal audit department, or external, such as the national audit office or a certification body. The auditor reviews evidence, such as logs, quality reports and lineage records. The role reports findings to management and to supervisory authorities. Its work depends on the traceability that the SAF builds into the chain.

Users and partners

ID	Role	Process type	Description
ROL-22	Dissemination and service manager	Design, Execution	The dissemination and service manager designs and manages the services through which users receive output. This role covers publications, open data, APIs and microdata access. The manager defines channels and formats for each user segment. The role manages the portfolio of services and their service levels. It works with the release manager on the timing and with the relationship manager on user needs.

ID	Role	Process type	Description
ROL-23	Relationship manager	Execution	The relationship manager maintains the relationships with key external actors. These include data providers, major users, partners and international organisations. The manager collects needs, feedback and complaints, and passes them to the right roles. The role negotiates and maintains agreements, such as data provision agreements. Relationship managers make sure that the NSI speaks with one voice to each partner.

Support

ID	Role	Process type	Description
ROL-24	Enterprise architect	Design, Management	The enterprise architect develops and maintains the enterprise architecture of the NSI. This role translates strategy into architecture and uses the SAF as its reference. The architect advises on investments and reviews projects against the architecture principles. The role supports the architecture board and keeps the architecture repository up to date. It is the role that realises the capability *Architecture management*.

SAF business collaborations

ID	Collaboration	Members	Type	Description
COL-01	Data provision partnership	NSI, data provider, relationship manager, data owner	External, permanent	A data provision partnership is the long-term cooperation between the NSI and a data provider. It goes beyond a single delivery. The partners agree on content, format, timing and quality of the data. They also agree on how to handle changes at the source, such as a new system or a new definition. A good partnership reduces the burden for the provider and improves the quality for the NSI.
COL-02	National statistical system	NSI, other producers of official statistics	External, permanent	The national statistical system is the cooperation between all producers of official statistics in one country. The NSI usually coordinates it. The members align methods, classifications and release practices, so that national statistics are consistent. They exchange data and sometimes share infrastructure. The cooperation is often based on the national statistics law.

ID	Collaboration	Members	Type	Description
COL-03	International statistical cooperation	NSI, international statistical organisations, other NSIs	External, permanent	International statistical cooperation is the collaboration between NSIs and international organisations. It covers the development of standards, the exchange of data and the joint development of tools. The SAF itself is a product of this cooperation. Members take part in working groups, such as those of UNECE and the European Statistical System. The cooperation helps NSIs avoid building the same solution many times.
COL-04	Research access partnership	NSI, research institutes, data owner, disclosure control specialist	External, permanent	A research access partnership gives researchers access to microdata under strict conditions. The NSI, research institutes and sometimes funding bodies take part. The partners agree on who may access which data, for which purpose and in which environment. Output is checked for disclosure risk before it leaves the secure environment. The partnership supports science while protecting the confidentiality of respondents.

ID	Collaboration	Members	Type	Description
COL-05	Statistical product team	Statistical designer, methodologist, data engineer, subject-matter statistician	Internal, permanent or temporary	A statistical product team brings together the roles needed to design, build and run a statistical product. It typically includes a statistical designer, a methodologist, a data engineer and a subject-matter statistician. The team can be permanent for a product or temporary for a redesign. Working as one team shortens the path from design to production. It also makes sure that methodological and technical choices are made together.
COL-06	Data governance board	Chief data officer, data owners, data stewards, compliance officer	Internal, permanent	The data governance board steers data management across the NSI. The chief data officer chairs it, and data owners, data stewards and the compliance officer take part. The board sets data policies, resolves conflicts between domains and monitors compliance. It decides on access requests that cross domain boundaries. The board makes sure that the steady-state model is applied consistently.

ID	Collaboration	Members	Type	Description
COL-07	Architecture board	Enterprise architect, process owners, CIO or CTO	Internal, permanent	The architecture board steers the enterprise architecture of the NSI. The enterprise architect, process owners and the CIO or CTO take part. The board reviews projects against the architecture principles and the target architecture. It decides on deviations through a documented dispensation process. The board keeps the architecture aligned with the strategy and with the SAF.

Role names in the value streams

The value streams use specific role names that fit their context. The table shows how they relate to the reference roles. A specialised role inherits the responsibilities of its reference role and adds those specific to its value stream.

Role name in the value streams	Reference role
Process Owner; Process Owners (VS-01…VS-10)	ROL-01 Process owner
Data Owner / Steward / Custodian	ROL-02 Data owner; ROL-03 Data steward; ROL-04 Data custodian
Chief Data / Governance Officer	ROL-05 Chief data officer
Statistical Designer; Product Manager; Model Designer; Derivation Designer; Coherence Designer	ROL-06 Statistical designer
Methodologist; Estimation Methodologist; Editing Methodologist; Imputation Specialist; Indicator/Index Methodologist; Head of Methodology / Innovation	ROL-07 Methodologist
Metadata Architect; Data Architect	ROL-08 Metadata and data architect
Classification Steward; Coding Specialist	ROL-09 Classification steward
Data Engineer; Integration Specialist; Integration Engineer; Tool / Platform Engineer	ROL-10 Data engineer
System Configurator; DevOps / Platform Engineer; Platform / Cloud Architect; Service Desk / Operations	ROL-11 Platform engineer
Test Manager	ROL-12 Test manager
Data Collection Manager	ROL-13 Data collection manager
Data Quality Analyst	ROL-14 Data quality analyst

Role name in the value streams	Reference role
Subject-Matter Analyst; Subject-Matter Reviewer; Subject-Matter Author; Subject-Matter Expert; Survey Statistician; Analyst / Modeler; Researcher / Data Scientist	ROL-15 Subject-matter statistician
Disclosure-Control Designer	ROL-16 Disclosure control specialist
Release Manager; Approval Authority	ROL-17 Release manager
Quality Manager	ROL-18 Quality manager
Compliance Officer; Legal Counsel	ROL-19 Compliance officer
Security & Continuity Officer	ROL-20 Security officer
Internal / External Auditor	ROL-21 Auditor
Dissemination Manager; Channel/Service Designer; Open-Data / API Product Owner; Service / Product Portfolio Manager; User-Support Lead; Service Desk / User Support	ROL-22 Dissemination and service manager
Account / Relationship Manager; Engagement / Communications Officer; Communications / Outreach Officer; Head of Communication / Engagement; User Researcher; Impact Analyst	ROL-23 Relationship manager
Chief Technology / Infrastructure Officer	Actor: management of the NSI (ACT-01); member of COL-07
Trainer / Knowledge Manager	Not included in the reference set
Data Provider; Survey Respondent; Data User; Legal Authority; Standards Body; Methodology Authority	Actors ACT-03, ACT-04, ACT-05, ACT-08, ACT-07

PLACEHOLDER: Role mapping and RACI templates to be added to the toolkit

Business Processes

The concept

A **business process** is a sequence of activities that produces a defined result for a customer, a stakeholder or another process. A process has a trigger, a start and an end. It has inputs and outputs, and it uses the knowledge and resources of one or more business functions. Processes describe *how* work flows, step by step. They are the part of the business architecture that people recognise most directly in their daily work.

Processes can be decomposed. A high-level process consists of sub-processes, which can in turn be broken down further. The SAF describes processes at two levels: process phases or process groups, and the sub-processes within them. More detailed levels, such as work instructions, are specific to each NSI and fall outside the SAF.

A process is not the same as a value stream stage. A value stream stage describes the value that is added. A process describes the work that adds it. One stage is usually realised by several processes, and one process can contribute to several stages.

How the SAF applies business processes

Why the SAF describes business processes

Processes are where architecture meets daily operations. A shared process model lets NSIs compare how they work, reuse each other's solutions and discuss improvements in the same terms. It also links the strategic layer to the systems: value streams are realised by processes, and processes are supported by application services. Without a process model, it is hard to show which system supports which part of the work, or where a change in method affects other steps.

How business processes fit into an NSI's own architecture

The core of the SAF process model is GSBPM, which most NSIs already know. The SAF adds three process groups that GSBPM does not describe in detail: data management and lifecycle, microdata services and data spaces. An NSI adopts the process model as a reference, maps its own processes onto it and adds its own lower levels. A national process for, say, the monthly price index can then be traced to the SAF sub-processes it implements. The SAF does not prescribe the order of sub-processes. GSBPM is not a strict sequence: sub-processes can be skipped, repeated or run in parallel.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Value streams (Strategy)	Processes realise value stream stages	PRD-4.3 *Run collection* realises VS-01.3 *Collect or Receive Data*
Capabilities (Strategy)	Processes realise capabilities	PRD-5.4 *Edit and impute* realises Adjusting and imputing (1.3.4)
Business functions	Processes realise functions; functions are assigned to processes	The production process realises the production function
Business roles	Roles are assigned to processes	ROL-14 *Data quality analyst* is assigned to PRD-5.3 *Review and validate*
Business events	Events trigger processes and mark their results	*Collection period opened* triggers PRD-4.3
Business objects	Processes access business objects	PRD-5.8 *Finalise data files* writes the processed microdata
Steady states	Processes produce and consume steady states	DLM-3 *Register and catalogue data* registers a dataset in a steady state
Application services	Application services serve processes	A validation service serves PRD-5.3

Process groups

Group	IDs	Basis	Scope
A. Statistical production	PRD-1 to PRD-8	GSBPM phases and sub-processes	The production of a statistical product, from need to evaluation
B. Overarching processes	OVA-1 to OVA-6	GSBPM overarching processes	Activities that run across all production phases
C. Data management and lifecycle	DLM-1 to DLM-11	SAF	The management of data assets and steady states across products
D. Microdata services	MDS-1 to MDS-9	SAF	Controlled access to microdata for research
E. Data spaces	DSP-1 to DSP-6	SAF	Participation in data spaces, as data provider and data user

PLACEHOLDER: ArchiMate view of the SAF process model

A. Statistical production processes

The statistical production processes follow the eight phases of GSBPM. Each phase is described first, followed by its sub-processes. The descriptions are paraphrased from GSBPM, a model of UNECE published under the Creative Commons Attribution 4.0 licence (CC BY 4.0).

PRD-1 Specify Needs

ID	Process	Description
PRD-1	**Specify Needs**	This phase starts when a new statistical need is identified, or when an existing product needs review. It establishes what users need and whether the NSI can meet that need. The phase explores options, including the use of data that already exists. It ends with a decision on whether to proceed, based on a business case. For existing products that run regularly, this phase is usually carried out only when a review is due.
PRD-1.1	Identify needs	This sub-process investigates which statistics are needed and what they are needed for. The trigger can be a new policy question, a legal requirement or a user request. It also looks at changes in the needs for existing statistics. Input comes from users, stakeholders, legislation and earlier evaluations. The result is a first description of the need.
PRD-1.2	Consult and confirm needs	This sub-process consults users and stakeholders to understand the need in detail. It checks what exactly they want to know, at which level of detail and how often. It also explores which constraints apply, such as timing or budget. The NSI and the users confirm the need together. The result is a confirmed and documented set of needs.
PRD-1.3	Establish output objectives	This sub-process translates the confirmed needs into output objectives. It defines which statistics will be produced and which quality they must meet. Legal and confidentiality constraints are taken into account. The objectives are realistic for the data and resources that are available. The result is the basis for the design of the outputs.

ID	Process	Description
PRD-1.4	Identify concepts	This sub-process clarifies the concepts to be measured, from the point of view of the users. A concept such as “household income” or “unemployment” needs a clear meaning. The sub-process checks whether existing statistical concepts can be reused. Concepts may not yet match the definitions that data sources use. That mismatch is resolved later, in the design phase.
PRD-1.5	Check data availability and suitability	This sub-process checks whether existing data sources can meet the need. It looks at internal data, administrative data, data spaces and other external sources. For each source, it assesses quality, coverage, timeliness and legal conditions. Reusing existing data comes before new collection, in line with SAF-P08. The result shows which gaps remain that new data must fill.
PRD-1.6	Prepare and submit business case	This sub-process documents the findings of the phase in a business case. The business case describes the need, the proposed solution, the costs and the benefits. It also lists risks and alternatives. The business case is submitted to management for a decision. Approval starts the design phase.

PRD-2 Design

ID	Process	Description
PRD-2	**Design**	This phase designs the statistical product and the process that produces it. It covers outputs, variables, collection, sampling, processing, analysis and dissemination. It also defines the metadata and the quality criteria. Designs reuse existing standards, methods and components wherever possible. The results are the specifications for the build phase.

ID	Process	Description
PRD-2.1	Design outputs and dissemination	This sub-process designs the statistical outputs in detail. It defines tables, datasets, indicators and their metadata. It also designs how and through which channels the outputs will be disseminated. Disclosure control requirements are taken into account from the start. The design follows the output objectives from PRD-1.3.
PRD-2.2	Design variables	This sub-process defines the variables to be collected and to be derived. It specifies their definitions, the types of statistical units and the classifications that apply. Existing variables and classifications are reused where possible. Each variable gets metadata that describes its meaning. The result is a variable list that all later phases use.
PRD-2.3	Design collection	This sub-process determines how the data will be collected. It chooses the sources, the collection modes and the instruments, such as questionnaires or data interfaces. For administrative data and data spaces, it designs the agreements and delivery formats. Respondent burden is kept as low as possible. The design also covers the metadata and paradata to be captured.
PRD-2.4	Design frame and sample	This sub-process defines the target population and the frame from which units are selected. It designs the sampling method and the sample size, where a sample is used. For register-based statistics, it defines how the population is derived from the registers. Coverage problems in the frame are identified and addressed. The result is the method used later in PRD-4.1.

ID	Process	Description
PRD-2.5	Design processing and analysis	This sub-process designs the methods for processing and analysis. It covers integration, coding, validation, editing, imputation, derivation, weighting and estimation. It also designs the methods for disclosure control. Methods are documented so that results can be reproduced. The designs are reviewed by methodologists before they are built.
PRD-2.6	Design production systems and workflows	This sub-process designs the end-to-end workflow of the statistical process. It defines which systems and components are needed for each step. It avoids gaps and duplication between steps. Reuse of existing components and shared platforms comes first, in line with SAF-P01. The result is the blueprint for the build phase.

PRD-3 Build

ID	Process	Description
PRD-3	**Build**	This phase builds and tests the production solution. It assembles the components, configures the workflows and tests the whole process. Components are reused where possible and built only where needed. The phase ends when the process is ready for production. For regular products, this phase is only needed when something changes.
PRD-3.1	Reuse or build collection instruments	This sub-process builds or reuses the instruments for data collection. Examples are electronic questionnaires, data delivery interfaces and connectors to data spaces. It follows the design from PRD-2.3. Instruments are tested for usability and correctness. The result is a set of instruments ready for collection.

ID	Process	Description
PRD-3.2	Reuse or build processing and analysis components	This sub-process builds or reuses the components for processing and analysis. Examples are validation rules, imputation routines and estimation programs. The components implement the methods designed in PRD-2.5. Shared components that other products already use are preferred. Each component is documented and version-controlled.
PRD-3.3	Reuse or build dissemination components	This sub-process builds or reuses the components for dissemination. Examples are table templates, publication workflows and API endpoints. It follows the design from PRD-2.1. Shared dissemination platforms are used where possible. The components are tested together with the output systems.
PRD-3.4	Configure workflows	This sub-process configures the workflows that connect the components. It sets up the order of steps, the data flows and the transformations. It also configures the registration of data in the steady states. Configuration is preferred over custom code. The result is an executable workflow.
PRD-3.5	Test production systems	This sub-process tests the assembled systems and their interactions. It checks that components work together and handle errors correctly. It also tests performance and security. Test results are documented and reported to the process owner. Defects are fixed before the process goes live.
PRD-3.6	Test statistical business process	This sub-process tests the whole statistical process in practice. It can take the form of a pilot or a field test with real data. It checks that the process produces results of the expected quality. Problems found in the test lead to changes in design or build. The process owner decides whether the process is ready.

ID	Process	Description
PRD-3.7	Finalise production systems	This sub-process puts the process and its systems into production. It completes the documentation and trains the staff involved. It transfers the systems to the teams that will run and maintain them. The metadata of the process is registered. From this moment, the process can run in the collect to disseminate phases.

PRD-4 Collect

ID	Process	Description
PRD-4	**Collect**	This phase collects or acquires the data, metadata and paradata needed for the product. It covers surveys, administrative data, data from data spaces and other sources. Data is received securely and checked for completeness. The phase ends when the data is stored and registered as raw data. In the steady-state model, this phase produces State #1.
PRD-4.1	Create frame and select sample	This sub-process creates the frame for this iteration of the process. It applies the method designed in PRD-2.4 to the current population. Where a sample is used, it draws the sample. The quality of the frame and the sample is checked. The result is the list of units from which data will be collected.
PRD-4.2	Set up collection	This sub-process makes sure that people, processes and systems are ready to collect. It prepares the instruments, trains interviewers and informs respondents or data providers. It checks that agreements and access rights are in place. Collection channels are tested before collection starts. The collection period is planned and communicated.
PRD-4.3	Run collection	This sub-process carries out the collection. Respondents complete questionnaires, and data providers deliver their data. Progress and response are monitored continuously. Non-response is followed up within the agreed strategy. Paradata about the collection is recorded.

ID	Process	Description
PRD-4.4	Finalise collection	This sub-process loads the collected data and metadata into a secure environment. It checks integrity, completeness and structure at intake. It registers the data as a new version of raw data. Delivery problems are reported to the data provider. The collection is closed when all expected data has been received or written off.

PRD-5 Process

ID	Process	Description
PRD-5	**Process**	This phase turns the collected data into microdata that is ready for analysis. It integrates, classifies, checks, corrects and derives data. It also calculates weights and aggregates. The phase runs across several business domains of the steady-state model. It produces States #2 and #3, and the first aggregates of State #4.
PRD-5.1	Integrate data	This sub-process combines data from one or more sources into a consistent dataset. It links records that describe the same unit, using agreed keys. It resolves overlaps and conflicts between sources. Linkage uses pseudonymised keys where personal data is involved. The result is an integrated dataset with documented linkage quality.
PRD-5.2	Classify and code	This sub-process assigns codes from standard classifications to the data. Examples are codes for economic activity, occupation or region. Coding can be automatic, manual or a mix of both. The classification versions used are recorded in the metadata. Coding quality is checked and documented.

ID	Process	Description
PRD-5.3	Review and validate	This sub-process checks the data for errors, outliers and inconsistencies. It applies the validation rules designed in PRD-2.5. It flags records that need attention. Some checks run automatically, others need expert review. The result is a dataset with quality flags and a list of issues.
PRD-5.4	Edit and impute	This sub-process corrects incorrect values and fills in missing values. Corrections follow editing rules, and missing values are imputed with documented methods. Every change is recorded, so that the original value can always be traced. The impact of editing and imputation on the results is measured. The result is a complete and consistent dataset.
PRD-5.5	Derive new variables and units	This sub-process creates variables and units that were not collected directly. Examples are age groups, household composition or enterprise groups. Derivations follow the specifications from PRD-2.2. Derived variables get their own metadata. The derivation rules are documented and reusable.
PRD-5.6	Calculate weights	This sub-process calculates weights for the records in the dataset. Weights correct for the sample design, non-response and coverage. They follow the method designed in PRD-2.5. The quality of the weights is checked. The weights are stored with the microdata.
PRD-5.7	Calculate aggregates	This sub-process calculates totals and other aggregates from the microdata. It applies the weights and the estimation methods. It can also combine lower-level aggregates into higher levels. Accuracy measures are calculated alongside the estimates. The result is the first version of the statistics.

ID	Process	Description
PRD-5.8	Finalise data files	This sub-process combines the results of the phase into the files used for analysis. It checks that the files are complete and consistent. It registers the processed microdata as a new version in the steady state. The metadata of the files is completed. The files are now ready for the analysis phase.

PRD-6 Analyse

ID	Process	Description
PRD-6	**Analyse**	This phase turns the processed data into statistical outputs and checks that they are fit for release. It prepares draft outputs, validates them and explains them. It also applies disclosure control. The phase ends with the approval of the content for release. In the steady-state model, it completes State #4.
PRD-6.1	Prepare draft outputs	This sub-process transforms the data files into draft statistical outputs. Examples are tables, indicators, time series and charts. It also prepares the methodological notes that go with them. Drafts follow the output design from PRD-2.1. They are the input for validation and interpretation.
PRD-6.2	Validate outputs	This sub-process checks the draft outputs for quality and plausibility. It compares results with earlier periods, other sources and expectations. It checks them against the quality framework. Unexpected results are investigated before they are accepted. The result is a set of validated outputs with documented quality.
PRD-6.3	Interpret and explain outputs	This sub-process interprets the outputs and explains what they mean. Subject-matter experts describe developments and their causes. Deeper analyses may be carried out to support the explanation. The interpretation must be neutral and based on the data. The result is the content for publications and press releases.

ID	Process	Description
PRD-6.4	Apply disclosure control	This sub-process makes sure that outputs do not reveal information about individual respondents. It applies the disclosure control methods designed in PRD-2.5. Examples are suppression of cells, rounding and adding noise. The disclosure risk is assessed and documented. No output is released without passing this check.
PRD-6.5	Finalise outputs	This sub-process makes sure that the statistics are fit for use. It completes the outputs and their metadata. It checks that all quality and confidentiality requirements are met. The content is approved for release by the responsible role. The approved outputs are registered as statistics ready for dissemination.

PRD-7 Disseminate

ID	Process	Description
PRD-7	**Disseminate**	This phase publishes the statistical products and supports users. It updates the output systems, produces the products and releases them. It also promotes the products and answers user questions. Equal access for all users at the moment of release is a core rule. In the steady-state model, this phase produces State #5.
PRD-7.1	Update output systems	This sub-process loads the approved data and metadata into the systems used for dissemination. Examples are databases behind the website, open data portals and APIs. It checks that the data is loaded correctly and remains embargoed until release. Format and structure follow the dissemination design. The result is output that is ready for release.

ID	Process	Description
PRD-7.2	Produce dissemination products	This sub-process produces the products that users receive. Examples are press releases, publications, tables, visualisations and datasets. It follows the design from PRD-2.1. Products are checked for correctness and consistency before release. The result is a set of products ready to publish.
PRD-7.3	Release dissemination products	This sub-process releases the products to users. It follows the release calendar and the embargo rules. All channels are updated at the same moment, so that no user gets an advantage. Errors found after release are corrected openly. The release is logged for accountability.
PRD-7.4	Promote dissemination products	This sub-process brings the products to the attention of users. It uses channels such as social media, newsletters and events. It targets the user segments that have an interest in the product. Promotion helps users find and understand the statistics. Its effect is monitored.
PRD-7.5	Provide user support	This sub-process answers questions and requests from users. Questions are registered and answered within agreed times. Frequent questions lead to better documentation or products. Requests for tailor-made data are handled according to confidentiality rules. Feedback from users is passed on to the product owners.

PRD-8 Evaluate

ID	Process	Description
PRD-8	**Evaluate**	This phase evaluates a specific run of the statistical process. It collects evidence, assesses quality and performance, and agrees on improvements. Evaluation can take place after every run or at set intervals. Its results feed back into the next run or into a redesign. The phase is part of the management process type of each business domain.

ID	Process	Description
PRD-8.1	Gather evaluation inputs	This sub-process collects the material for the evaluation. Examples are quality indicators, paradata, process logs, user feedback and staff comments. The inputs come from all earlier phases. They are brought together in one place. The result is a complete evidence base for the evaluation.
PRD-8.2	Conduct evaluation	This sub-process analyses the inputs and draws conclusions. It assesses the quality of the outputs and the efficiency of the process. It compares results with targets and with earlier runs. Strengths and weaknesses are identified. The result is an evaluation report or dashboard.
PRD-8.3	Agree an action plan	This sub-process agrees on actions to improve the process. It sets priorities, owners and deadlines. Actions can lead to small changes or to a full redesign. Progress on the actions is monitored. The action plan is the input for the next cycle.

B. Overarching processes

Overarching processes run across all phases of statistical production. They are not steps in the sequence. They apply to every phase, and to every product. They are based on the overarching processes of GSBPM.

ID	Process	Description
OVA-1	Quality Management	This process assesses and controls the quality of products and processes. It applies the quality framework of the NSI in every phase. It defines quality indicators, measures them and reports on them. Quality problems lead to corrective actions. The process supports the NSI-wide quality management function.

ID	Process	Description
OVA-2	Metadata Management	This process creates, updates, uses and archives statistical metadata in all phases. Metadata describes concepts, variables, classifications, methods and processes. It is captured when it is created, not afterwards. Metadata is shared through a catalogue, so that others can find and understand the data. The process applies SAF-P09 *No data without metadata and classification*.
OVA-3	Data Management	This process manages data throughout its life cycle in production. It covers storage, security, versioning, retention and disposal. It makes sure that data is available to the right people at the right time. The SAF details this process in group C, *Data management and lifecycle processes*. The process applies the design rules of the steady-state model.
OVA-4	Process Data Management	This process records and uses data about the execution of the statistical process. Examples are paradata from collection, logs of processing steps and timings. Process data shows how the process actually ran. It is used for monitoring, evaluation and audit. It is also the basis for lineage: the trace from output back to source.
OVA-5	Knowledge Management	This process maintains the knowledge needed to run the production process. It keeps documentation, methods and procedures up to date. It makes sure that recurring processes can be repeated in the same way. It also captures lessons learned from evaluations. Knowledge management reduces the dependence on individual staff members.

ID	Process	Description
OVA-6	Data Supplier Management	This process manages the technical and operational relationship with data suppliers during production. It covers delivery schedules, formats, quality feedback and problem resolution. It keeps track of the burden placed on suppliers. It works within the agreements made at corporate level. Good supplier management improves data quality and reduces delays.

C. Data management and lifecycle processes

These processes manage data as an asset, across products and over its whole life. They detail the overarching process OVA-3 *Data Management*. They follow the life cycle of a steady state: promotion, gate, registration, use, revision and retirement. They apply to every reference model for the data value chain.

ID	Process	Description
DLM-1	Plan and design data assets	This process decides which data assets the NSI needs and how they are structured. It defines datasets, their owners, their classification and their place in the steady states. It is triggered by a new product, a new source or a change in the data architecture. Planning prevents duplicate datasets and unclear ownership. The result is a documented design for each data asset.
DLM-2	Promote data through a quality gate	This process submits a dataset to the quality gate of a steady state. It checks the dataset against the gate criteria, such as completeness, structure and required metadata. The outcome is acceptance, conditional acceptance with quality flags, or rejection. Rejected data returns to the producing domain or the data provider. The process makes sure that only data of known quality enters a steady state.

ID	Process	Description
DLM-3	Register and catalogue data	This process registers an accepted dataset as a new, immutable version. It assigns a persistent identifier and records the metadata in the catalogue. It publishes an event that tells consuming domains a new version is available. Registration makes the data findable for authorised users. Without registration, data cannot be handed over.
DLM-4	Classify and protect data	This process assigns a security classification to each dataset, for availability, integrity and confidentiality. It applies the protection measures that follow from the classification. It also applies pseudonymisation or other privacy measures where required. Classification is reviewed when data moves to the next steady state. The process applies SAF-P09, SAF-P10 and SAF-P11.
DLM-5	Store and version data	This process stores each registered version of a dataset safely. Earlier versions are kept, not overwritten. Backups and recovery are arranged according to the availability requirements. Storage follows the retention policy of each steady state. The process makes it possible to reproduce any statistic from the data it was built on.
DLM-6	Provide data for reuse	This process gives authorised internal users access to registered data. A user requests access, and the data owner decides on the request. Access is granted for a stated purpose and a set period. Use is logged. The process applies SAF-P08 *Data is shared property*, within the limits of confidentiality.
DLM-7	Revise data	This process handles corrections and redeliveries of data that is already registered. A correction creates a new version that passes the same quality gate. Consumers of the earlier version are notified. They decide whether to reprocess their own data. The reason for each revision is recorded.

ID	Process	Description
DLM-8	Monitor data quality, lineage and use	This process monitors registered data over time. It tracks quality indicators, the lineage from source to output, and how the data is used. It detects problems, such as declining quality or unauthorised use. It reports to data owners and the data governance board. The results feed into improvements and audits.
DLM-9	Manage reference and master data	This process manages the data that other data depends on. Examples are classifications, code lists, statistical registers and population frames. It controls versions and publishes changes in advance. It makes sure that all domains use the same version at the same time. Errors in reference data affect many products, so changes are tested carefully.
DLM-10	Archive and preserve data	This process transfers data that must be kept long-term to an archive. It selects what must be preserved, based on legal obligations and reproducibility needs. It makes sure that archived data remains readable and understandable, with its metadata. Access to archived data is controlled. The process follows national archives legislation.
DLM-11	Dispose of data	This process deletes data at the end of its retention period. It checks that no statistic or obligation still depends on the data. Deletion covers all copies, including backups and working copies. Keys used for pseudonymisation are destroyed when they are no longer needed. Each disposal is recorded, so that it can be shown in an audit.

D. Microdata services processes

Microdata services give researchers controlled access to microdata for statistical and scientific research. Access takes place under strict conditions, usually in a secure processing environment. The NSI remains responsible for confidentiality throughout. These processes realise part of VS-06 *Dissemination and Service Delivery* and work within the collaboration COL-04 *Research access partnership*.

ID	Process	Description
MDS-1	Authorise institutions and researchers	This process decides which institutions and researchers may use microdata services. It checks the legal status of the institution and the qualifications of the researchers. Authorised institutions sign an agreement that sets the conditions of use. Researchers confirm that they will respect confidentiality. Authorisations are reviewed periodically and can be withdrawn.
MDS-2	Assess access applications	This process handles an application to use microdata for a specific research project. The applicant describes the purpose, the data needed and the expected outputs. The NSI checks the legal basis, the public interest and whether the data requested is necessary. It also checks for special categories of data that need extra approval. The result is an approval with conditions, or a reasoned refusal.
MDS-3	Prepare research datasets	This process prepares the datasets for an approved project. It selects only the data that the project needs. It replaces identifiers with project-specific pseudonyms, so that data cannot be linked across projects. Where the project brings its own data, the NSI links it under the same rules. The prepared dataset is registered with its metadata and lineage.
MDS-4	Provide a secure processing environment	This process sets up the environment in which researchers work with the data. The environment prevents data from leaving without control. Access is limited to the authorised researchers of the project. Tools and computing power are provided as agreed. All activity in the environment is logged.

ID	Process	Description
MDS-5	Facilitate privacy-preserving analysis	This process supports analysis without moving the data into one place. Examples are federated analysis, where the algorithm goes to the data, and secure multi-party computation. These techniques are used when partners cannot or may not share their data. The NSI defines the trust model, the linkage method and the output rules for each setup. The process extends microdata services to data that the NSI does not hold itself.
MDS-6	Support researchers	This process helps researchers use the data and the environment correctly. It answers questions about content, metadata and methods. It explains the rules for confidentiality and output. It handles technical problems in the environment. Good support reduces errors and speeds up research.
MDS-7	Check and release output	This process checks every output before it leaves the secure environment. It applies disclosure control rules to tables, models and other results. It also checks for risks across outputs, such as differencing between similar results. Outputs that pass are released to the researcher. Outputs that fail are returned with an explanation.
MDS-8	Monitor use and compliance	This process monitors whether institutions and researchers follow the conditions of use. It reviews logs, outputs and incidents. It also tracks the use of query or disclosure budgets where these apply. Breaches lead to measures, up to withdrawal of the authorisation. The results are reported to the data owner and to supervisory authorities where required.

ID	Process	Description
MDS-9	Close projects	This process ends a research project in a controlled way. It removes the project's data and workspace. It destroys the project-specific pseudonymisation keys. It archives what must be kept for reproducibility, such as the code and the approved outputs. The closure is recorded, so that it can be shown in an audit.

E. Data space processes

A data space is a shared environment in which organisations exchange data under common rules, while each keeps control over its own data. Examples are the European data spaces for health, mobility or public administration. An NSI can take part as a data provider, as a data user, or both. Some NSIs also act as an access body that grants access to data on behalf of other data holders. That role is not covered by the SAF reference processes. The data space processes realise parts of VS-01 *Data Acquisition and Ingestion* and VS-06 *Dissemination and Service Delivery*.

ID	Process	Description
DSP-1	Join a data space	This process prepares the NSI to take part in a data space. It assesses the rules of the data space against the NSI's legal mandate and policies. It signs the participation agreement and obtains the required identity and trust credentials. It sets up the technical connection, for example through a connector. The NSI decides in advance whether it joins as provider, as user, or as both.
DSP-2	Publish data offerings	This process makes NSI data visible in the data space. It describes datasets in the data space catalogue, using agreed metadata standards. It states the conditions under which the data may be used. Only data that the NSI may share under its legal mandate is offered. The offering is kept up to date when the data or the conditions change.

ID	Process	Description
DSP-3	Agree data use conditions	This process agrees on the terms of a specific data exchange. Provider and user agree on purpose, duration, permitted use and security measures. The terms are recorded in a contract or usage policy that machines can check where possible. The NSI checks that the terms fit its confidentiality rules. Without agreed terms, no data is exchanged.
DSP-4	Exchange data under usage control	This process carries out the agreed data exchange. Data is transferred securely, or analysed where it resides without transfer. Usage control enforces the agreed conditions where the technology allows it. Every exchange is logged. Incoming data enters the NSI through the data acquisition process and is registered as raw data.
DSP-5	Monitor compliance and report	This process monitors whether the NSI's own exchanges follow the rules of the data space. It checks agreements, logs, incidents and the use of data. It reports security incidents within the legal deadlines. It also checks whether partners respect the conditions agreed for NSI data. Breaches lead to measures, including suspension of the exchange.
DSP-6	Participate in data space governance	This process covers the NSI's role in steering the data space. The NSI takes part in governance bodies and contributes to the rules and standards. It brings in statistical expertise on metadata, classifications, quality and confidentiality. It also evaluates whether participation still serves its mandate. When the NSI leaves a data space, this process arranges a controlled exit.

PLACEHOLDER: Process templates and process mapping instruments to be added to the toolkit

Business Functions

The concept

A **business function** is a grouping of behaviour based on the knowledge, skills and resources it needs. A function describes *what* the organisation does on a continuous basis. It has no start or end, and it does not depend on the order in which work is done. *Manage metadata* is a function: it exists as long as the NSI exists, whatever process happens to use it. Functions are stable, because the competences an organisation needs change much more slowly than its procedures.

Functions differ from processes. A process is a sequence of activities with a trigger and a result. A function is a lasting area of competence that processes draw on. One process uses several functions, and one function serves several processes. The mapping at the end of this section shows how they relate.

Functions also differ from capabilities. A capability, in the Strategy layer, describes an *ability* the NSI has or needs. A function, in the business layer, describes the *behaviour* that delivers that ability. Functions realise capabilities. In practice, the two often have similar names, but they serve different purposes: capabilities support planning and investment, functions support the design of the organisation and its processes.

How the SAF applies business functions

Why the SAF describes business functions

Functions give a stable picture of the organisation that does not change with every reorganisation or new system. They are the natural basis for assigning responsibilities: each function has an owner who maintains the competence. They also help to compare NSIs. Two NSIs may have very different processes and departments, but they will recognise the same functions. Finally, functions make gaps visible. A function that no one owns is a competence that the organisation risks losing.

How business functions fit into an NSI's own architecture

An NSI maps its departments and teams onto the SAF functions. The mapping shows which unit carries which function, and where functions are spread over several units. An NSI can also map the SAF functions onto a national function model, such as a government-wide model for business support. Functions that are specific to the NSI's mandate are added as extensions. The SAF function areas for statistical production, data management, microdata services and data spaces are specific to official statistics. The business support area is generic and can be replaced by a national model where one exists.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Capabilities (Strategy)	Functions realise capabilities	FUN-7.2 *Manage metadata* realises Metadata management (2.1.2)
Business processes	Processes realise functions; functions are assigned to processes	PRD-5.4 *Edit and impute* uses FUN-3.3 *Process statistical data*
Business roles	Roles are assigned to functions	ROL-03 *Data steward* is assigned to FUN-7.3 *Manage data lifecycle and storage*
Business actors	Organisational units carry functions	A methodology department carries FUN-4.1
Business services	Functions realise business services	FUN-8.1 realises the microdata access service
Application services	Application services serve functions	A metadata catalogue serves FUN-7.2

Function areas

Area	Name	Scope
FUN-1	Strategy and Leadership	Direction, governance and strategic relations of the NSI
FUN-2	Capability Development	Planning, developing and maintaining the abilities of the NSI, including architecture
FUN-3	Statistical Production	Designing, producing and disseminating statistics
FUN-4	Methodology and Research	Statistical methods, research and innovation
FUN-5	Quality Management	The quality framework and quality assurance
FUN-6	Data Supplier Management	Relations, agreements and burden for data providers and respondents
FUN-7	Data Management	Data and metadata as managed assets across their life cycle
FUN-8	Microdata Services	Controlled access to microdata for research
FUN-9	Data Space Participation	Taking part in data spaces as provider and user
FUN-10	Business Support	The generic support functions of the organisation

PLACEHOLDER: ArchiMate view of the SAF function model

FUN-1 Strategy and Leadership

ID	Function	Description	Capabilities
FUN-1	**Strategy and Leadership**	This area gives direction to the NSI. It sets the vision, the strategy and the priorities. It governs the organisation and accounts for its results. It maintains relationships with the government, the statistical community and society. All other functions work within the direction this area sets.	4.1

ID	Function	Description	Capabilities
FUN-1.1	Define vision and strategy	This function defines the mission, vision, goals and values of the NSI. It analyses the national and international environment, including the drivers in the Motivation layer. It translates the vision into a strategy and domain strategies. It sets priorities for the product and service portfolio. The results guide investment and change for several years.	4.1.1, 4.1.2
FUN-1.2	Govern and steer the organisation	This function makes sure the NSI is run according to its strategy and its legal mandate. It sets up governance bodies, decision rights and the planning and control cycle. It allocates resources to programmes and departments. It monitors performance and intervenes when needed. It also accounts to the ministry, supervisory bodies and the public.	4.1.1, 4.1.5
FUN-1.3	Manage strategic collaboration	This function maintains the strategic relationships of the NSI with other organisations. It covers the national statistical system, international statistical organisations and other governments. It identifies opportunities for joint work, data exchange and shared solutions. It represents the NSI in working groups and governance bodies. It makes sure the NSI speaks with one voice.	4.1.6

ID	Function	Description	Capabilities
FUN-1.4	Build public confidence	This function builds and protects public trust in official statistics. It explains the role and independence of the NSI to the public. It promotes the value of statistics and of providing data to the NSI. It responds to misuse or misinterpretation of statistics. Trust is the basis for both response and use, so this function supports the whole chain.	1.5.2, 4.1.5

FUN-2 Capability Development

ID	Function	Description	Capabilities
FUN-2	**Capability Development**	This area develops and improves the abilities of the NSI. It plans improvements, carries them out and checks whether they deliver. It covers methods, processes, data, systems and skills. It also maintains the enterprise architecture that keeps improvements coherent. This area turns the strategy into concrete change.	4.1.3, 4.1.4
FUN-2.1	Plan capability improvements	This function identifies which capabilities need to improve and in which order. It uses capability assessments, evaluations and strategic priorities. It proposes programmes and projects with a business case. It coordinates the plans across the organisation to avoid overlap. The result is a prioritised portfolio of improvements.	4.1.3

ID	Function	Description	Capabilities
FUN-2.2	Develop capability improvements	This function carries out approved improvements. It specifies, designs, builds and tests the change. It follows the architecture principles and reuses existing solutions where possible. It involves the people who will use the result. The function ends when the improvement is ready to be taken into use.	4.1.3, 3.2.1
FUN-2.3	Monitor capabilities	This function checks whether capabilities perform as intended. It measures maturity and the benefits of past investments. It identifies capabilities that are declining or no longer needed. It reports to management and feeds the next planning round. Without monitoring, investments cannot be shown to pay off.	4.1.4
FUN-2.4	Transfer capability improvements to operations	This function hands over a finished improvement to the teams that will run and support it. It arranges documentation, training and support agreements. It checks that the operating teams are ready before the handover. It closes the project or programme. A clean transfer prevents improvements from falling back into project mode.	3.2.2

ID	Function	Description	Capabilities
FUN-2.5	Manage enterprise architecture	This function develops and maintains the enterprise architecture of the NSI. It uses the SAF as its reference. It advises on strategy, investments and projects, and reviews them against the architecture principles. It supports the architecture board and keeps the architecture repository up to date. It keeps the NSI architecture aligned with the SAF as both evolve.	

FUN-3 Statistical Production

ID	Function	Description	Capabilities
FUN-3	**Statistical Production**	This area produces official statistics, from the first design to the release. It is the core of the NSI's mandate. The statistical production processes (PRD) realise this area. Its functions follow the main kinds of statistical work: design, collection, processing, estimation, confidentiality and dissemination. Each function brings together the specialist knowledge for that kind of work.	1
FUN-3.1	Design and build statistical products	This function designs statistical products and the processes that produce them. It translates information needs into outputs, variables, methods and workflows. It also builds and tests the production solution. It works closely with methodology and with the users of the product. The function owns the design of each product over its life cycle.	1.1

ID	Function	Description	Capabilities
FUN-3.2	Collect and acquire data	This function obtains the data that statistical products need. It runs surveys, receives administrative data and acquires data from other sources, including data spaces. It makes sure that data arrives complete, on time and under the right agreements. It keeps respondent burden as low as possible. The function delivers data into the first steady state.	1.2
FUN-3.3	Process statistical data	This function turns raw data into reliable microdata. It integrates, standardises, codes, validates, edits and imputes data. It also derives new variables and units. It applies the methods designed for each product. The function produces standardized and processed microdata.	1.3.1, 1.3.2, 1.3.3, 1.3.4
FUN-3.4	Estimate and analyse	This function produces statistics from microdata and checks that they are sound. It calculates weights, estimates and aggregates. It integrates statistics into consistent wholes, such as the national accounts. It validates and interprets the results. The function produces the statistics that are ready for release.	1.3.5, 1.3.6, 1.4

ID	Function	Description	Capabilities
FUN-3.5	Protect statistical confidentiality	This function makes sure that no output reveals information about individual respondents. It sets the confidentiality rules and the methods for disclosure control. It applies them to tables, microdata files and services. It assesses disclosure risks in new types of output. The function works across all products and channels.	1.3.7
FUN-3.6	Disseminate statistics	This function brings statistics to users. It manages the release calendar, the publication channels and the products. It makes sure that all users get access at the same moment. It also supports users in finding and understanding the statistics. The function produces released data.	1.5

FUN-4 Methodology and Research

ID	Function	Description	Capabilities
FUN-4	**Methodology and Research**	This area provides the scientific basis for statistical production. It develops and maintains statistical methods. It explores new data sources, new techniques and new products. It makes sure that innovations are sound before they enter production. This area protects the methodological quality that makes official statistics credible.	2.2

ID	Function	Description	Capabilities
FUN-4.1	Manage statistical and analytical methodology	This function develops, documents and maintains the methods used in production. It covers sampling, collection, editing, imputation, estimation, disclosure control and analysis. It includes methods based on machine learning and AI. It advises production teams and reviews their methods. Methods are standardised and shared across products where possible.	2.2.3
FUN-4.2	Research and innovate	This function explores new possibilities for statistics. It investigates new data sources and new ways to use existing data. It develops prototypes of new products and methods. It evaluates whether innovations are feasible and sound. Successful innovations are handed over to production through capability development.	2.2.1, 2.2.2

FUN-5 Quality Management

ID	Function	Description	Capabilities
FUN-5	**Quality Management**	This area makes sure that the NSI delivers statistics of the required quality. It maintains the quality framework and the tools to apply it. It assures quality in products and processes. It prepares the NSI for reviews and audits. This area is the corporate counterpart of the quality management process in production.	2.1.3

ID	Function	Description	Capabilities
FUN-5.1	Manage the quality framework	This function sets the quality standards of the NSI. It defines quality dimensions, indicators and reporting requirements. It aligns the framework with international standards, such as the European Statistics Code of Practice. It provides tools and guidance to production teams. The framework is reviewed and improved regularly.	2.1.3
FUN-5.2	Assure and review quality	This function checks whether products and processes meet the quality standards. It monitors quality indicators and reviews quality reports. It carries out quality reviews and self-assessments. It follows up on findings with improvement actions. It also supports external reviews, such as peer reviews.	2.1.3, 4.1.5

FUN-6 Data Supplier Management

ID	Function	Description	Capabilities
FUN-6	**Data Supplier Management**	This area manages the relationships with everyone who supplies data to the NSI. It covers respondents, administrative data holders and other data providers. It sets up and maintains agreements. It monitors the burden that the NSI places on suppliers. Good supplier management improves both the quality of data and the willingness to provide it.	3.4

ID	Function	Description	Capabilities
FUN-6.1	Manage data supplier relationships	This function maintains contact with data providers and respondent groups. It explains why the NSI needs the data and how it is used. It handles questions, complaints and feedback on data quality. It coordinates all contacts, so that a supplier deals with one NSI and not with many departments. The function builds long-term cooperation.	3.4.2
FUN-6.2	Manage data provision agreements	This function draws up, maintains and enforces agreements with data providers. Agreements cover content, format, timing, quality, security and legal basis. The function keeps a register of all agreements. It renews or adjusts agreements when sources or needs change. It also handles enforcement when obligations are not met.	3.4.3
FUN-6.3	Manage response burden	This function measures and limits the burden of statistical data collection. It tracks how much time and effort respondents and data providers spend. It looks for ways to reduce burden, such as reusing existing data. It coordinates surveys so that the same units are not approached too often. It reports on burden to management and government.	1.2, 2.2.5

FUN-7 Data Management

ID	Function	Description	Capabilities
FUN-7	**Data Management**	This area manages data and metadata as assets of the NSI. It covers governance, metadata, the data life cycle, reference data and data protection. It works across all statistical products and domains. It makes sure that data can be found, understood, trusted and reused. The data lifecycle management processes (DLM) realise this area.	2.1
FUN-7.1	Govern data	This function sets the rules for data management in the NSI. It defines data policies, standards and responsibilities. It assigns data owners, data stewards and data custodians. It resolves issues that cross domain boundaries. It monitors compliance with the data policies.	2.1.1, 4.1.2
FUN-7.2	Manage metadata	This function captures, maintains and publishes metadata. Metadata describes concepts, variables, classifications, datasets, methods and processes. The function maintains the catalogue in which data can be found. It makes sure that metadata is captured when data is created. It applies international standards, such as GSIM and SDMX.	2.1.2

ID	Function	Description	Capabilities
FUN-7.3	Manage data lifecycle and storage	This function manages data from creation to disposal. It covers quality gates, registration, versioning, storage, retention and disposal. It keeps track of lineage, from output back to source. It makes sure that earlier versions remain available when they are needed. The function applies the design rules of the steady-state model.	2.1.1
FUN-7.4	Manage reference and master data	This function manages the data that other data depends on. Examples are classifications, code lists, statistical registers and population frames. It controls versions and announces changes in advance. It makes sure that all domains use the same version. The function is the single source of truth for reference data.	2.1.4
FUN-7.5	Protect data	This function protects data according to its classification. It applies measures for availability, integrity and confidentiality. It manages pseudonymisation and the keys used for it. It keeps keys separate from the data they protect. The function works closely with information security and privacy management.	3.3.1, 3.3.4

FUN-8 Microdata Services

ID	Function	Description	Capabilities
FUN-8	**Microdata Services**	This area gives researchers controlled access to microdata. It covers authorisation, data preparation, secure environments and output control. It also supports analysis without moving data, using privacy-preserving techniques. The NSI remains responsible for confidentiality throughout. The microdata services processes (MDS) realise this area.	2.2.4
FUN-8.1	Manage research access	This function decides who may use microdata, for which purpose and under which conditions. It authorises institutions and researchers. It assesses applications for research projects. It monitors whether users keep to the conditions. It withdraws access when rules are broken.	2.2.4, 3.4.3
FUN-8.2	Prepare research data	This function prepares the datasets for each approved project. It selects only the data that the project needs. It applies project-specific pseudonyms, so that data cannot be linked across projects. It links data that researchers bring in, under the same rules. It documents the prepared data with metadata and lineage.	2.2.4, 3.3.1
FUN-8.3	Operate secure research environments	This function provides and runs the environments in which researchers work. It controls who may enter and what may leave. It provides tools and computing power. It logs all activity. It removes data and workspaces when projects end.	2.2.4, 3.2.5

ID	Function	Description	Capabilities
FUN-8.4	Control research output	This function checks all results before they leave the secure environment. It applies disclosure control rules to tables, models and other output. It checks for risks across outputs, such as differencing between similar results. It keeps a record of every release. The function protects respondents while allowing research to proceed.	1.3.7
FUN-8.5	Enable privacy-preserving analysis	This function makes analysis possible without bringing all data together. It uses techniques such as federated analysis and secure multi-party computation. It defines the trust model, the linkage method and the output rules for each setup. It works with partners who cannot or may not share their data. The function extends microdata services beyond the data the NSI holds itself.	2.2.4, 2.2.3

FUN-9 Data Space Participation

ID	Function	Description	Capabilities
FUN-9	**Data Space Participation**	This area enables the NSI to take part in data spaces. The NSI can act as a data provider, as a data user, or both. The area covers agreements, technical connection and governance. It makes sure that participation fits the NSI's legal mandate and confidentiality rules. The data space processes (DSP) realise this area.	4.1.6, 3.4

ID	Function	Description	Capabilities
FUN-9.1	Manage data space participation	This function decides in which data spaces the NSI takes part, and in which role. It assesses the rules of each data space against the NSI's mandate. It manages participation agreements, data offerings and usage conditions. It monitors whether exchanges follow the agreed rules. It reviews participation regularly.	4.1.6, 3.4.3
FUN-9.2	Operate data space connectivity	This function provides the technical connection to data spaces. It runs the connectors, identity credentials and catalogue entries. It carries out data exchanges under the agreed usage conditions. It logs every exchange. Incoming data is passed on to data collection and enters the first steady state.	3.4.1, 3.2.5
FUN-9.3	Contribute to data space governance	This function represents the NSI in the governance of data spaces. It contributes to rules, standards and rulebooks. It brings in statistical expertise on metadata, classifications, quality and confidentiality. It aligns positions with other NSIs where possible. It also arranges a controlled exit when participation ends.	4.1.6

FUN-10 Business Support

ID	Function	Description	Capabilities
FUN-10	**Business Support**	This area provides the people, money, information, technology and facilities that the other functions need. Its functions are similar in every public organisation. The SAF describes them only at a high level. An NSI can replace this area with the function model of its own government. The functions are still part of the SAF, because statistical work depends on them.	3
FUN-10.1	Manage human resources	This function makes sure the NSI has enough qualified staff. It covers recruitment, development, performance and the departure of staff. It plans the skills the NSI will need in the future. It supports managers in leading their teams. Skills such as data science and methodology are scarce, which makes this function critical.	3.1.1
FUN-10.2	Manage finance and procurement	This function manages the budget, the accounts and the purchase of goods and services. It plans and controls spending. It runs procurement procedures and manages supplier contracts. It reports on the financial position to management and government. It makes sure that spending follows the rules for public money.	3.1.2, 3.1.3

ID	Function	Description	Capabilities
FUN-10.3	Manage information technology	This function provides and manages the IT the NSI needs. It covers applications, platforms, infrastructure and the digital workplace. It develops, configures and maintains systems. It runs IT services and resolves disruptions. It plans the evolution of the technical landscape.	3.2
FUN-10.4	Manage information and records	This function manages the information created in management and support processes. It keeps records according to archives legislation. It makes information findable and accessible over time. It also manages the knowledge base of the organisation. Statistical data itself is managed by FUN-7 *Data Management*.	3.1.8
FUN-10.5	Manage legal affairs	This function provides legal advice and support. It interprets the statistics law, data protection law and other rules. It assesses agreements and contracts. It handles legal procedures and objections. It keeps the overview of legal obligations, for example in a legal register.	3.1.7
FUN-10.6	Manage communication	This function manages internal and external communication. It covers press relations, the website, social media and internal information. It makes sure the NSI communicates clearly and consistently. It supports the release of statistics with press releases and explanations. It also handles questions from the media.	3.1.4

ID	Function	Description	Capabilities
FUN-10.7	Manage facilities	This function provides the buildings, workplaces and physical services the NSI needs. It covers housing, maintenance, catering and physical access. It makes sure that buildings and spaces are safe. It supports field staff with the means they need. It plans facilities in line with ways of working, such as hybrid work.	3.1.6, 3.3.5
FUN-10.8	Manage information security and privacy	This function protects the NSI's information and the privacy of the people it concerns. It sets security and privacy policies and monitors compliance. It manages incidents and leads recovery. It assesses risks in new solutions and suppliers. It includes the role of the data protection officer.	3.3
FUN-10.9	Manage users and services	This function manages the relationship with the users of statistics. It identifies user groups and their needs. It manages the portfolio of products and services and their service levels. It handles questions, requests and complaints. It measures user satisfaction and passes feedback on to production.	3.4.1, 3.4.2, 2.2.5

Process and function mapping

The table shows which functions each process uses. The *main function* carries the knowledge the process mainly depends on. *Supporting functions* contribute specific expertise or resources. In ArchiMate, the process realises part of the main function, and the supporting functions are related through assignment or serving relationships.

PLACEHOLDER: ArchiMate view of the process and function mapping

A. Statistical production processes

Process	Name	Main function	Supporting functions
PRD-1	**Specify Needs**	FUN-3.1	FUN-10.9
PRD-1.1	Identify needs	FUN-3.1	FUN-10.9
PRD-1.2	Consult and confirm needs	FUN-3.1	FUN-10.9
PRD-1.3	Establish output objectives	FUN-3.1	FUN-5.1
PRD-1.4	Identify concepts	FUN-3.1	FUN-4.1
PRD-1.5	Check data availability and suitability	FUN-3.1	FUN-6.1, FUN-7.2
PRD-1.6	Prepare and submit business case	FUN-3.1	FUN-1.2, FUN-2.1
PRD-2	**Design**	FUN-3.1	FUN-4.1
PRD-2.1	Design outputs and dissemination	FUN-3.1	FUN-3.5, FUN-3.6
PRD-2.2	Design variables	FUN-3.1	FUN-7.2, FUN-7.4
PRD-2.3	Design collection	FUN-3.1	FUN-3.2, FUN-6.1
PRD-2.4	Design frame and sample	FUN-3.1	FUN-4.1, FUN-7.4
PRD-2.5	Design processing and analysis	FUN-4.1	FUN-3.1, FUN-3.5
PRD-2.6	Design production systems and workflows	FUN-3.1	FUN-2.5, FUN-10.3
PRD-3	**Build**	FUN-3.1	FUN-10.3
PRD-3.1	Reuse or build collection instruments	FUN-3.1	FUN-3.2
PRD-3.2	Reuse or build processing and analysis components	FUN-3.1	FUN-4.1, FUN-10.3
PRD-3.3	Reuse or build dissemination components	FUN-3.1	FUN-3.6, FUN-10.3
PRD-3.4	Configure workflows	FUN-3.1	FUN-7.3, FUN-10.3
PRD-3.5	Test production systems	FUN-3.1	FUN-10.3
PRD-3.6	Test statistical business process	FUN-3.1	FUN-5.2
PRD-3.7	Finalise production systems	FUN-3.1	FUN-2.4, FUN-7.2
PRD-4	**Collect**	FUN-3.2	FUN-6.1
PRD-4.1	Create frame and select sample	FUN-3.2	FUN-4.1, FUN-7.4
PRD-4.2	Set up collection	FUN-3.2	FUN-6.2
PRD-4.3	Run collection	FUN-3.2	FUN-6.1, FUN-6.3
PRD-4.4	Finalise collection	FUN-3.2	FUN-7.3
PRD-5	**Process**	FUN-3.3	FUN-3.4
PRD-5.1	Integrate data	FUN-3.3	FUN-7.4, FUN-7.5
PRD-5.2	Classify and code	FUN-3.3	FUN-7.4
PRD-5.3	Review and validate	FUN-3.3	FUN-5.2
PRD-5.4	Edit and impute	FUN-3.3	FUN-4.1
PRD-5.5	Derive new variables and units	FUN-3.3	FUN-7.2
PRD-5.6	Calculate weights	FUN-3.4	FUN-4.1
PRD-5.7	Calculate aggregates	FUN-3.4	FUN-4.1
PRD-5.8	Finalise data files	FUN-3.3	FUN-7.3
PRD-6	**Analyse**	FUN-3.4	FUN-3.5
PRD-6.1	Prepare draft outputs	FUN-3.4	FUN-3.6
PRD-6.2	Validate outputs	FUN-3.4	FUN-5.2

Process	Name	Main function	Supporting functions
PRD-6.3	Interpret and explain outputs	FUN-3.4	FUN-10.6
PRD-6.4	Apply disclosure control	FUN-3.5	FUN-4.1
PRD-6.5	Finalise outputs	FUN-3.4	FUN-5.2
PRD-7	**Disseminate**	FUN-3.6	FUN-10.9
PRD-7.1	Update output systems	FUN-3.6	FUN-10.3
PRD-7.2	Produce dissemination products	FUN-3.6	FUN-10.6
PRD-7.3	Release dissemination products	FUN-3.6	FUN-1.4
PRD-7.4	Promote dissemination products	FUN-3.6	FUN-10.6, FUN-1.4
PRD-7.5	Provide user support	FUN-10.9	FUN-3.6
PRD-8	**Evaluate**	FUN-5.2	FUN-2.3
PRD-8.1	Gather evaluation inputs	FUN-5.2	FUN-7.3
PRD-8.2	Conduct evaluation	FUN-5.2	FUN-2.3
PRD-8.3	Agree an action plan	FUN-2.1	FUN-5.2

B. Overarching processes

Process	Name	Main function	Supporting functions
OVA-1	Quality Management	FUN-5.2	FUN-5.1
OVA-2	Metadata Management	FUN-7.2	FUN-7.4
OVA-3	Data Management	FUN-7.3	FUN-7.1
OVA-4	Process Data Management	FUN-7.3	FUN-5.2
OVA-5	Knowledge Management	FUN-10.4	FUN-4.1
OVA-6	Data Supplier Management	FUN-6.1	FUN-6.3

C. Data lifecycle management processes

Process	Name	Main function	Supporting functions
DLM-1	Plan and design data assets	FUN-7.1	FUN-2.5
DLM-2	Promote data through a quality gate	FUN-7.3	FUN-5.2
DLM-3	Register and catalogue data	FUN-7.2	FUN-7.3
DLM-4	Classify and protect data	FUN-7.5	FUN-10.8
DLM-5	Store and version data	FUN-7.3	FUN-10.3
DLM-6	Provide data for reuse	FUN-7.1	FUN-7.3
DLM-7	Revise data	FUN-7.3	FUN-7.2
DLM-8	Monitor data quality, lineage and use	FUN-7.1	FUN-5.2
DLM-9	Manage reference and master data	FUN-7.4	FUN-7.2
DLM-10	Archive and preserve data	FUN-10.4	FUN-7.3
DLM-11	Dispose of data	FUN-7.3	FUN-7.5

D. Microdata services processes

Process	Name	Main function	Supporting functions
MDS-1	Authorise institutions and researchers	FUN-8.1	FUN-10.5
MDS-2	Assess access applications	FUN-8.1	FUN-10.5, FUN-7.1

Process	Name	Main function	Supporting functions
MDS-3	Prepare research datasets	FUN-8.2	FUN-7.5
MDS-4	Provide a secure processing environment	FUN-8.3	FUN-10.3, FUN-10.8
MDS-5	Facilitate privacy-preserving analysis	FUN-8.5	FUN-4.1, FUN-7.5
MDS-6	Support researchers	FUN-8.3	FUN-10.9
MDS-7	Check and release output	FUN-8.4	FUN-3.5
MDS-8	Monitor use and compliance	FUN-8.1	FUN-10.8
MDS-9	Close projects	FUN-8.3	FUN-7.3, FUN-7.5

E. Data space processes

Process	Name	Main function	Supporting functions
DSP-1	Join a data space	FUN-9.1	FUN-10.5, FUN-10.8
DSP-2	Publish data offerings	FUN-9.1	FUN-7.2
DSP-3	Agree data use conditions	FUN-9.1	FUN-10.5
DSP-4	Exchange data under usage control	FUN-9.2	FUN-3.2, FUN-7.3
DSP-5	Monitor compliance and report	FUN-9.1	FUN-10.8
DSP-6	Participate in data space governance	FUN-9.3	FUN-1.3

Functions without a reference process

Six functions are not used by any SAF reference process. They are carried out through management, change or support processes that are specific to each NSI. Many other functions appear only as supporting functions, which is normal for strategic and support functions.

Function	Carried out through
FUN-1.1 Define vision and strategy	The strategic planning cycle of the NSI
FUN-2.2 Develop capability improvements	Change programmes and projects
FUN-4.2 Research and innovate	Innovation projects (see VS-08)
FUN-10.1 Manage human resources	Generic business support processes
FUN-10.2 Manage finance and procurement	Generic business support processes
FUN-10.7 Manage facilities	Generic business support processes

Mapping to value stream stages

The mapping connects the Strategy layer to the business layer. A value stream stage describes the value that is added. The processes describe the work that adds it. The functions describe the competences that the work depends on.

PLACEHOLDER: ArchiMate view of value stream stages, processes and functions

Primary value streams

Stage	Name	Realised by processes	Main functions
VS-01.1	Identify Data Needs and Sources	PRD-1.1, PRD-1.5, PRD-2.3	FUN-3.1, FUN-6.1
VS-01.2	Establish Data Agreements and Access Rights	OVA-6, DSP-1, DSP-3	FUN-6.2, FUN-9.1, FUN-10.5
VS-01.3	Collect or Receive Data	PRD-4.1, PRD-4.2, PRD-4.3, DSP-4	FUN-3.2, FUN-9.2
VS-01.4	Validate and Log Data Intake	PRD-4.4, DLM-2	FUN-3.2, FUN-7.3
VS-01.5	Store and Register Raw Data Artefacts	DLM-3, DLM-4, DLM-5	FUN-7.2, FUN-7.3, FUN-7.5
VS-02.1	Profile and Assess Data Structure	PRD-5.3	FUN-3.3
VS-02.2	Apply Standard Classifications and Formats	PRD-5.2, DLM-9	FUN-3.3, FUN-7.4
VS-02.3	Align Metadata and Semantics	OVA-2	FUN-7.2
VS-02.4	Validate Structural and Semantic Conformance	PRD-5.3, DLM-2	FUN-3.3, FUN-7.3
VS-02.5	Register Standardized Artefacts	DLM-3, DLM-4, DLM-5	FUN-7.2, FUN-7.3, FUN-7.5
VS-03.1	Conduct Error Detection and Quality Diagnostics	PRD-5.3	FUN-3.3
VS-03.2	Perform Data Editing and Correction	PRD-5.2, PRD-5.4	FUN-3.3
VS-03.3	Impute Missing or Non-Response Data	PRD-5.4	FUN-3.3, FUN-4.1
VS-03.4	Derive Additional Variables and Indicators	PRD-5.1, PRD-5.5	FUN-3.3
VS-03.5	Perform Quality Evaluation and Versioning	PRD-5.8, DLM-2, DLM-3	FUN-3.3, FUN-5.2, FUN-7.3
VS-04.1	Confirm Estimation and Analytical Approach	PRD-2.4, PRD-2.5	FUN-3.1, FUN-4.1
VS-04.2	Apply Weighting and Calibration	PRD-5.6	FUN-3.4
VS-04.3	Estimate and Aggregate Data	PRD-5.7	FUN-3.4
VS-04.4	Perform Model-Based Analysis	PRD-6.1, PRD-6.3	FUN-3.4
VS-04.5	Validate and Document Results	PRD-6.2, PRD-6.4, OVA-2	FUN-3.4, FUN-3.5, FUN-7.2
VS-05.1	Select and Integrate Source Datasets	PRD-5.1, PRD-6.1	FUN-3.4
VS-05.2	Construct Composite Indicators or Indices	PRD-5.5, PRD-5.7	FUN-3.3, FUN-3.4
VS-05.3	Perform Consistency and Coherence Checks	PRD-6.2	FUN-3.4, FUN-5.2
VS-05.4	Generate Analytical and Interpretive Outputs	PRD-6.3, PRD-7.2	FUN-3.4, FUN-3.6
VS-05.5	Approve and Register Statistical Products	PRD-6.4, PRD-6.5, DLM-3	FUN-3.4, FUN-3.5, FUN-7.2

Stage	Name	Realised by processes	Main functions
VS-06.1	Plan Dissemination and Access Strategy	PRD-2.1	FUN-3.1, FUN-3.6
VS-06.2	Prepare Data for Dissemination	PRD-6.4, PRD-7.1	FUN-3.5, FUN-3.6
VS-06.3	Publish and Provide Access Services	PRD-7.2, PRD-7.3, MDS-1 to MDS-5, MDS-7, DSP-2	FUN-3.6, FUN-8.1 to FUN-8.5, FUN-9.1
VS-06.4	Support and Engage Users	PRD-7.4, PRD-7.5, MDS-6	FUN-3.6, FUN-8.3, FUN-10.9
VS-06.5	Monitor Usage and Feedback	PRD-8.1, MDS-8, MDS-9	FUN-8.1, FUN-8.3, FUN-10.9

Supporting value streams

Stage	Name	Realised by processes	Main functions
VS-07.1	Define Governance Frameworks and Policies	DLM-1	FUN-1.2, FUN-5.1, FUN-7.1
VS-07.2	Establish Legal and Regulatory Compliance Mechanisms	DLM-4	FUN-10.5, FUN-10.8
VS-07.3	Implement Data and Metadata Governance	OVA-2, OVA-3, DLM-1, DLM-6	FUN-7.1, FUN-7.2
VS-07.4	Monitor Quality and Compliance Indicators	OVA-1, DLM-8, MDS-8, DSP-5	FUN-5.2, FUN-7.1
VS-07.5	Audit, Certify and Report Compliance	OVA-1, PRD-8.2	FUN-1.2, FUN-5.2
VS-07.6	Update and Improve Governance Mechanisms	PRD-8.3, DLM-10, DLM-11	FUN-2.1, FUN-7.1
VS-08.1	Identify Innovation Needs and Opportunities	PRD-1.5	FUN-4.2
VS-08.2	Conduct Research and Proof-of-Concepts	No reference process (NSI-specific)	FUN-4.2
VS-08.3	Evaluate Methodological Soundness and Feasibility	PRD-3.6	FUN-4.1, FUN-5.2
VS-08.4	Develop and Document New Methodologies or Tools	PRD-3.2, OVA-5	FUN-4.1, FUN-10.4
VS-08.5	Integrate Innovation into Production Environments	PRD-3.7	FUN-2.4, FUN-3.1
VS-08.6	Promote Knowledge Sharing and Capability Building	OVA-5	FUN-10.1, FUN-10.4
VS-09.1	Define Digital and Technical Requirements	PRD-2.6	FUN-2.5, FUN-3.1
VS-09.2	Design Platform and Infrastructure Architecture	PRD-2.6	FUN-2.5, FUN-10.3
VS-09.3	Provision and Configure Platforms	PRD-3.4, MDS-4	FUN-8.3, FUN-10.3
VS-09.4	Enable Integration and Orchestration Services	PRD-3.4, DSP-4	FUN-9.2, FUN-10.3

Stage	Name	Realised by processes	Main functions
VS-09.5	Ensure Performance, Security and Continuity	PRD-3.5, DLM-5	FUN-10.3, FUN-10.8
VS-09.6	Evolve and Modernise the Technical Ecosystem	No reference process (NSI-specific)	FUN-2.2, FUN-10.3
VS-10.1	Identify and Segment Stakeholders	PRD-1.2, DSP-6	FUN-1.3, FUN-10.9
VS-10.2	Collect and Analyse User Needs	PRD-1.1, PRD-1.2	FUN-3.1, FUN-10.9
VS-10.3	Define and Manage Service Portfolios	PRD-2.1	FUN-10.9
VS-10.4	Deliver Services and Manage Relationships	PRD-7.5, MDS-6, OVA-6	FUN-6.1, FUN-10.9
VS-10.5	Engage and Communicate with Stakeholders	PRD-7.4	FUN-1.4, FUN-10.6
VS-10.6	Monitor Impact and Satisfaction	PRD-8.1, PRD-8.2	FUN-5.2, FUN-10.9

PLACEHOLDER: Function templates and organisation-to-function mapping instruments to be added to the toolkit

Business Services and Business Interfaces

Business Events

Business Objects, Contracts, Representations and Products

Reference Models for the Statistical Data Value Chain

Why the SAF offers more than one model NSIs differ in size, mandate, legal context, data volumes, and technology platform. A single prescribed arrangement would fit some offices well and others poorly. The SAF therefore describes reference models: alternative, internally consistent ways of organising the value chain. Each model answers the same four questions:

1. At which points is data persisted and made available for reuse?
2. Which quality criteria must data meet at each point?
3. Who is accountable for the data at each point?
4. How are confidentiality and traceability guaranteed along the chain?

The SAF currently describes two reference models. Both follow the same underlying idea, progressive improvement of data through defined layers, but they differ in granularity and in where they place governance.

Model A: Steady-State Architecture

The Steady-State Architecture organises the value chain into business domains that are loosely coupled through five numbered steady states (#1 Raw, #2 Standardized, #3 Processed, #4 Statistics, #5 Released). Each state represents a specific kind of statistical value and has its own quality gate, ownership, and handover interface. The model is derived from production practice in NSIs with many statistical domains and extensive microdata reuse. It is described in full in the remainder of this chapter.

Model B: Medallion Architecture

The Medallion Architecture organises data into three layers of increasing quality, usually called Bronze (raw), Silver (cleansed and conformed), and Gold (curated for consumption). The pattern originates from

lakehouse platforms and is a logical design pattern rather than a product. Its strength lays in its simplicity. There are three layers, a clear direction of flow, and the ability to rebuild any layer from the one below it.

As a data-engineering pattern, Medallion does not by itself define statistical semantics, accountability, or confidentiality rules per layer.

Comparison

Characteristic	Steady-State Architecture	Medallion Architecture
Number of governed handover points	Five (#1 to #5)	Three (Bronze, Silver, Gold)
Origin	Statistical production practice	Lakehouse data engineering
Granularity of quality gates	Fine: one gate per kind of statistical value	Coarse: several kinds of value combined per layer
Separation of standardization and editing	Explicit (#2 vs. #3)	Combined in Silver
Separation of internal statistics and released data	Explicit (#4 vs. #5)	Combined in Gold unless subdivided
Governance overhead	Higher: more states to own, register, and audit	Lower
Fit with platform tooling	Platform-neutral; mapping required	Native in lakehouse platforms
Typical adopters	Larger NSIs, many domains, high microdata reuse	Smaller NSIs, platform-led modernisation

Selection criteria

The criteria below help an NSI decide which model fits best. They are indicative. No single criterion is decisive, and the SAF toolkit will provide a structured assessment.

Criterion	Points towards Steady-State	Points towards Medallion
Organisational size and number of statistical domains	Many domains, many teams handing over data	Few domains, small central data team
Reuse of intermediate microdata across domains	High; standardized and processed microdata reused by several products	Limited; most data flows within one product chain
Legal requirements on pseudonymisation timing	Law or policy requires a separate, auditable pseudonymisation point	Pseudonymisation can be applied at the Silver boundary
Archival and reproducibility obligations	Specific states (e.g., processed microdata) must be retained long term	Retention can be defined per layer
Separation of internal statistics from released data	Required (embargo, different confidentiality thresholds per use)	Achievable by subdividing Gold
Metadata and governance maturity	Mature data ownership and catalogue practices in place	Governance capacity is limited and must stay lean
Technology platform	Heterogeneous landscape, on-premises and private cloud	Single lakehouse platform
Participation in international data exchange	Many handover points to external partners	Exchange concentrated on final outputs

Practice within the SAF community indicates that smaller NSIs often adopt a Medallion arrangement, while larger NSIs with many domains tend towards finer-grained states. This observation is to be substantiated through the SAF toolkit assessments.

Combining the models

The models are not mutually exclusive. The steady states can be mapped onto Medallion layers, which allows an NSI to use Medallion as its platform convention while governing data through steady states, or to start with Medallion and introduce finer states as it grows.

Steady state	Medallion layer	Note
#1 Raw Data	Bronze	Direct correspondence
#2 Standardized Data	Silver (lower sub-layer)	In Medallion often merged with #3
#3 Processed Data	Silver	Main reusable microdata layer
#4 Statistics	Gold (internal sub-layer)	Not yet released
#5 Released Data	Gold (published sub-layer)	Released after final disclosure control
#6 Data Products	Gold (consumption product sub-layer)	Packaged for consumption

Steady-State Architecture

Overview

Business domains

The statistical process is divided into three main business domains (Observation, Processing/Analysis, and Dissemination). Processing/Analysis is divided into four subdomains, which results in six business domains. Each domain adds one specific kind of statistical value and produces one steady state.

#	Business domain	Value added	Business functions (ArchiMate view)	Value stream	GSBPM	State in → out
1	Observation/Collection	Data in NSI custody, authentic and traceable	Receive Data, Extract Data, Collect Survey	VS-01	4	#0 → #1
2	Standardization without loss of content	Common structure, format, and classifications	Transform Data, Anonymize Data, Classify Data	VS-02	5.1–5.2	#1 → #2
3	Editing, derivation, and non-response correction	Reliable, complete, enriched microdata	Imputate Data, Integrate Data	VS-03	5.3–5.5, 5.8	#2 → #3
4	Estimation and balancing	Representative statistics with accuracy	Estimate Data, Aggregate Data	VS-04	5.6–5.7, 6.1–6.4	#3 → #4

#	Business domain	Value added	Business functions (ArchiMate view)	Value stream	GSBPM	State in → out
5	Statistically composed and composite products	Coherent, approved products	Compose Data	VS-05	6.5, 7.1–7.2	#4 → #4 (approved)
6	Dissemination	Disclosure-safe, accessible released data	Data Delivery, Data Publication	VS-06	6.4, 7.3–7.5	#4 → #5
—	Consumption (external)	Use of statistics by society	Data Consumption, Consumption Optimization	VS-10 (feedback)	8	#5 → (#6)

The execution processes between the business domains are loosely coupled. Each has its own design and implementation. For the mutual exchange of data, they are loosely linked via steady state interfaces. To this end, five steady state phases are recognized as handover points at the business-domain boundaries described above. This means that the business domain "Observation" produces "Raw Data", business domain "Standardization without loss of content" produces "standardized data" etc. 1. raw data, 2. standardized data, 3. processed data, 4. statistics, 5. released data, and 6. Consumable data products.

A steady state phase is therefore an environment where high-quality versions of data are brought together for exchange, but it also represents a value in the value chain of this data.

The four process types

Each business domain uses the same four process types:

1. **Design** of the execution processes, which provides metadata.
2. **Implementation** of the execution processes, which provides the execution process.
3. **Execution** of the execution processes, which provides the statistical data.
4. **Management** of the execution processes, which provides quality reports.

How an execution process adds statistical value, and therefore how it is designed and implemented, falls outside the scope of the SAF business architecture. That design follows from, among other things, the NSI's methodology.

Design rules of the steady-state model

The model rests on six design rules. They apply the SAF architecture principles to the data value chain.

1. **Loose coupling through states (SAF-P07).** Domains exchange data only through registered steady states, never through each other's internal working data. A domain can change its methods or systems without affecting its neighbours, as long as the state it produces meets its quality gate.
2. **One state, one kind of value.** Each state represents exactly one kind of added statistical value. This keeps quality gates testable and accountability unambiguous.
3. **Immutable quality versions.** Data registered in a state is never overwritten. Corrections and redeliveries create a new version, so every statistic can be traced to the exact data it was built on.
4. **No state without metadata (SAF-P09).** Registration requires the minimum metadata set of the state. Data without it does not enter the state.

5. **Progressive confidentiality (SAF-P11).** Protection increases along the chain: direct identifiers are removed or pseudonymised at #2, disclosure risk is assessed at #3, initial disclosure control is applied at #4, and final disclosure control gates #5.
6. **Reuse before reprocessing (SAF-P08).** An authorised consumer that needs data at a given level of value reuses the registered state rather than repeating the upstream processing.

Steady states at a glance

Position	Name	Produced by	Kind of value	Mandatory?
(#0)	Source Data	External provider	Outside NSI custody, boundary position	n/a
#1	Raw Data	VS-01	Authentic, traceable custody	When the product collects its own data
#2	Standardized Data	VS-02	Comparability, values unchanged	Optional
#3	Processed Data	VS-03	Reliability and completeness of microdata	Always
#4	Statistics	VS-04, VS-05	Representative, approved statistics	Always
#5	Released Data	VS-06	Disclosure-safe, accessible output	When output is released
(#6)	Consumed Data	External users	Outside NSI control, boundary position	n/a

State lifecycle

Every steady state follows the same lifecycle:

1. **Promotion.** The producing domain submits a dataset to the quality gate of the target state.
2. **Gate outcome.** The dataset is accepted, conditionally accepted with explicit quality flags, or rejected. Rejected data does not enter the state; it returns to the producing domain or, for #1, to the provider.
3. **Registration.** An accepted dataset receives a persistent identifier, its metadata is registered in the catalogue, and a registration event notifies consuming domains.
4. **Consumption.** Authorised consuming domains read the registered version. Responsibility for further processing transfers on receipt of the event; custody of the registered version stays with the producer.
5. **Revision.** A correction produces a new version that passes the same gate. Consumers are notified and decide whether to reprocess. Earlier versions remain available for as long as statistics depend on them.
6. **Retirement.** A version is disposed of according to the retention policy of the state, after its lineage obligations have been met.

How each state is described

Each state below is described with the same ten attributes. NSIs can use this template to document their own implementation; the SAF toolkit will provide it as a fill-in form.

#	Attribute	Question it answers
1	Definition and value added	What is this state and why does it exist?

#	Attribute	Question it answers
2	Producing and consuming domains	Who creates it, and who may use it?
3	Quality gate	What must data satisfy to be registered?
4	Data and metadata characteristics	What does the data look like, and which metadata must accompany it?
5	Security, privacy, and access	How is the data protected, and who may see it?
6	Ownership and governance	Who is accountable?
7	Versioning, retention, and lineage	How are versions managed, how long are they kept, and how is the trace preserved?
8	Reuse and handover	How is the state made available to consumers?
9	Traceability to the framework	Which capabilities, value-stream stages, and GSBPM sub-processes relate to it?
10	Examples and anti-patterns	What does good practice look like, and what should be avoided?

Boundary Position #0: Source Data

Description: Source data is data as it exists at the data holder, before it enters the NSI: a register at a government agency, a company's accounting system, a respondent's answers before submission, or a web page. The NSI has no custody of source data and cannot guarantee its quality. It can influence it only through agreements, questionnaire design, and cooperation with the data holder.

Why it is described: The SAF describes #0 so that the boundary of the NSIs responsibility is explicit. Everything the NSI knows about source data is captured as metadata when the data crosses into #1: who provided it, under which agreement, through which channel, and in what condition.

State #1: Raw Data

1. Definition and value added

Raw data is data received from an external data holder or respondent, placed under NSI custody in its original form, and verified for technical integrity and authenticity. It is the first point of possession. At this moment the NSI does not have access to the data at the source, so #1 is the earliest point at which the NSI can guarantee what it holds.

The value of #1 is **authentic custody**. It proves that the NSI holds exactly what the provider delivered, when it was delivered, under which legal basis, and through which channel. That proof is the foundation for all downstream traceability. If a statistic is ever questioned, the chain of evidence ends here.

2. Producing and consuming domains

Role	Domain	Note
Producer	Observation/Collection (VS-01)	Functions: Receive Data, Extract Data, Collect Survey
Primary consumer	Standardization (VS-02)	Triggered by the *Raw Data Registered* event
Secondary consumer	Editing (VS-03)	Only where #2 is skipped for this product

Role	Domain	Note
Occasional consumer	Innovation and Methodology (VS-08)	Source exploration, under specific authorisation
Assurance	Governance and Compliance (VS-07)	Audit of provenance and legal basis

3. Quality gate

The gate for #1 checks technical and legal conditions only. It does not assess content; content quality is the concern of #3.

Criterion	Check	Typical outcome on failure
Legal basis	A valid agreement or statutory basis covers this delivery	Reject; escalate to legal
Sender authenticity	Sender verified (certificate, credentials, signed delivery)	Reject
Integrity	Checksum or digital signature matches	Reject; request redelivery
Security	Malware and security scan passed	Quarantine; reject
Structural conformance	Delivery matches the agreed technical specification (file format, schema, encoding)	Reject or conditionally accept with flag
Completeness against agreement	Record counts and delivery scope within agreed tolerance	Conditionally accept with flag, or reject
Timeliness	Delivered within the agreed window	Accept with flag; report to relationship management
Metadata	Minimum metadata set captured (see attribute 4)	Hold until complete

Conditional acceptance is a deliberate option. A late or slightly incomplete delivery may still be the best available data. The flag ensures every downstream user knows this.

4. Data and metadata characteristics

Data. Raw data is stored in its original format, or in a lossless technical equivalent. Permitted technical operations are those that change representation without changing content and that can be reversed: decryption, decompression, character-encoding normalisation, and container conversion. Operations that change values, codes, or record selection are not permitted at #1. The original byte stream, or a verifiable hash of it, is retained.

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this delivery version
Provider	Organisation or respondent group that supplied the data
Agreement reference and legal basis	Agreement identifier and the applicable legal ground
Channel and method	Transfer channel (file transfer, API, data space connector, survey mode) and instrument version
Receipt timestamp	Moment of receipt into NSI custody
Integrity proof	Checksum or signature value and algorithm
Volume	Record count and size
Reference period	Period the data describes, as stated by the provider
Structural description	Schema or technical specification the delivery conforms to

Metadata element	Description
Gate outcome	Accepted or conditionally accepted, with quality flags
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, #1 typically instantiates a *Data Set* received from an *Information Provider* through an *Exchange Channel* under a *Provision Agreement*.

5. Security, privacy, and access

Raw data usually contains direct identifiers and therefore carries the highest confidentiality rating in the chain. Access is limited to the roles that operate the intake and the standardization process. Analysts and subject-matter staff do not work on #1.

Pseudonymisation is normally applied at the transition to #2. Some NSIs pseudonymise at the intake boundary, or receive data already pseudonymised by a trusted third party. Both variants are compatible with the model, provided the choice is documented and the key management is separated from the data. Data minimisation starts before #1: the agreement and the collection design determine that only necessary data is received.

6. Ownership and governance

Role	Responsibility for #1
Data owner	Process owner of Observation/Collection; accountable for lawful acquisition and fitness of the intake
Data steward	Maintains metadata and catalogue entries; resolves gate exceptions
Data custodian	Operates storage, access control, and retention
Relationship manager	Handles provider follow-up on rejected or flagged deliveries (with VS-10)

7. Versioning, retention, and lineage

Every delivery, including a redelivery or correction by the provider, is a new immutable version. A redelivery never overwrites an earlier version, because statistics may already be built on it.

Retention of #1 is governed by two needs that pull in opposite directions. Reproducibility and audit favour keeping raw data; data minimisation and storage cost favour disposal. A common arrangement is to keep #1 until the dependent #3 version is verified and the audit period has passed, while #3 is the state subject to long-term archival requirements. The exact period is set by national law and NSI policy.

Lineage at #1 is the provenance record: the link from the registered version to the provider, the agreement, and the delivery event. Every downstream state must be able to trace back to a #1 version or, where a product starts from another domain's data, to the state it consumed.

8. Reuse and handover

Registration of #1 publishes the *Raw Data Registered* event and a catalogue entry. Consumers discover the delivery through the catalogue and access it through the handover interface, never through direct access to the intake system.

One delivery can serve several statistical products. The register-based census and the labour market statistics may both draw on the same population register delivery. Under SAF-P08, products reuse the

registered delivery rather than requesting the same data again from the provider. This also reduces response burden.

Handover element	Content
Artefact	Registered Raw Data version
Required metadata	Minimum metadata set (attribute 4)
Trigger event	*Raw Data Registered*
Responsibility transfer	VS-01 keeps custody; the consumer takes responsibility for further processing on receipt of the event

9. Traceability to the framework

Framework element	References
Capabilities	1.2.1 Primary data collection, 1.2.2 Secondary data collection, 2.1.1 Data lifecycle management, 2.1.2 Metadata management, 3.3.4 Logical identity and access control, 3.4.3 Service agreement
Value-stream stages	VS-01.3 Collect, Receive, or Extract Data; VS-01.4 Validate and Log Data Intake; VS-01.5 Store and Register Raw Data Artefacts
GSBPM	4.3 Run collection, 4.4 Finalise collection
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11

10. Examples and anti-patterns

Examples

- *Survey.* Responses from a web questionnaire are received with paradata (mode, completion time, instrument version). The delivery is registered as one #1 version per collection batch.
- *Administrative register.* A quarterly tax file arrives by file transfer, with a manifest. Checksum, schema, and record count are checked against the agreement; 14 unknown business identifiers are not a #1 issue and pass through for handling downstream.
- *Web data.* Daily scraped price data is registered with the scraper version and page fingerprint per retailer, so a change in a website's structure can be traced to the affected deliveries.

Anti-patterns

- Correcting obvious errors "while the file is open" at intake. Any value change belongs in #3, where it is documented.
- Overwriting a delivery with the provider's corrected version. Statistics built on the first version lose their trace.
- Giving analysts direct access to raw data for convenience. This bypasses the pseudonymisation at #2.
- Keeping only a converted copy and discarding the original. The proof of authentic custody is lost.
- Registering a delivery without its agreement reference. The legal basis can no longer be demonstrated in an audit.

State #2: Standardized Data

1. Definition and value added

Standardized data is raw data that has been brought into the common structure, format, and classifications of the NSI. Its values are unchanged. Only the representation changes: fields are renamed and restructured, types are cast, source codes are mapped one-to-one to standard codes, and descriptive metadata is attached.

Direct identifiers are removed or replaced with pseudonymous keys. #2 is therefore the first state in which data from different sources can be compared and combined, and the first state without direct identifiers.

The value of #2 is **comparability without loss of content**. Data that looked different at the source now uses the same structure, the same codes, and documented definitions, units, and reference periods. Because no value has changed, the trace back to the authentic delivery in #1 stays simple and deterministic. Every interpretation, correction, or conversion is left to #3, where it is documented as an edit.

#2 is an optional state. A product can go directly from #1 to #3, for example when a source already arrives in the NSI's standard structure and coding. In that case the NSI records in the catalog that #2 was skipped, and why (see *Applying the steady states pragmatically*). **[SS1]**

2. Producing and consuming domains

Role	Domain	Note
Producer	Standardization without loss of content (VS-02)	Functions: Transform Data, Anonymize Data, Classify Data
Primary consumer	Editing (VS-03)	Triggered by the *Standardized Data Registered* event
Secondary consumer	Other statistical products	Reuse of the same standardized source under SAF-P08, with authorization
Occasional consumer	Innovation and Methodology (VS-08)	Exploration of sources and methods, under specific authorization
Assurance	Governance and Compliance (VS-07)	Audit of mapping lineage and pseudonymization

3. Quality gate

The gate for #2 checks conformance and preservation. It does not assess whether values are correct; that is the concern of #3. A value that cannot be mapped is not an error of standardization. It is a fact about the source, so it is flagged and handed to VS-03.

Criterion	Check	Typical outcome on failure
Structural conformance	Fields, types, formats, and cardinalities match the target data model	Reject; re-map
Classification conformance	Every mapped code exists in the stated classification version; no orphan codes	Reject; correct the correspondence table
Mapping coverage	Share of source codes mapped one-to-one is above the agreed threshold	Conditionally accept with flag; unmapped values listed for VS-03
Semantic metadata	Unit, currency, reference-period basis, and definition recorded for every variable	Hold until complete
No loss of content	Record counts and key measures identical to the #1 version; no value changed	Reject
Identifier protection	Direct identifiers removed or pseudonymized according to the privacy design	Reject; escalate to the privacy officer
Referential integrity	Linkage keys exist in the reference register (e.g., the statistical business register)	Conditionally accept with flag

Criterion	Check	Typical outcome on failure
Metadata and lineage	Minimum metadata set captured (see attribute 4); field-level lineage complete	Hold until complete

Conditional acceptance is the normal outcome for many sources. Unmapped codes and unmatched keys are often caused by known lags, such as newly registered businesses. The flag ensures every downstream user knows exactly which values still need work.

4. Data and metadata characteristics

Data. Standardized data is stored in the canonical target data model. Standard codes are stored next to the original source values, so the original remains visible. Permitted operations are renaming, restructuring, type casting, reformatting (for example from CSV to a standard exchange format), key alignment, one-to-one code mapping through maintained correspondence tables, removal or pseudonymization of direct identifiers, and dropping variables that are not needed for the statistical purpose. Operations that change values are not permitted at #2: unit or currency conversion, reference-period alignment, reconciliation of differing definitions, derivation, interpretive coding of free text, editing, and imputation. These belong to #3. [**SS2**]

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this standardized version
Source reference	The #1 version or versions this dataset was produced from
Target model version	Version of the canonical data model applied
Mapping ruleset and engine version	Versions of the mapping rules and the tool that applied them
Classifications applied	Scheme and version per coded variable (e.g., NACE Rev. 2, ISCED 2011)
Mapping coverage and unmapped values	Coverage per variable and the list of values routed to VS-03
Units, currency, and reference-period basis	Recorded per variable as received, not converted
Concept links and definition differences	Link to the canonical concept; recorded differences with other sources
Pseudonymization record	Identifiers removed or pseudonymized, method, key reference (never the key itself)
Conformance assessment	Structural, classification, and semantic results; no-loss-of-content confirmation
Gate outcome	Accepted or conditionally accepted, with quality flags
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, #2 typically instantiates a *Unit Data Set* that conforms to a *Data Structure*, whose *Represented Variables* take their values from *Statistical Classifications* and *Code Lists*.

5. Security, privacy, and access

#2 is the point where direct identifiers leave the working data. **Removal** deletes an identifier and ends the possibility to link on it. **Pseudonymization** replaces it with a key that still allows records to be linked. The privacy design decides per identifier which of the two applies, and the choice is recorded in the pseudonymization record.

Pseudonymization keys are managed by the data custodian, separately from the data and under separate access rights. Staff who work on #2 and later states never have access to the keys. Some NSIs pseudonymize

at intake, or receive data already pseudonymized by a trusted third party. In that case #2 confirms and records the protection that is already in place.

Standardized data is still detailed microdata. Indirect identifiers, such as date of birth, place of residence, or occupation, remain, and combinations of them can identify a person or business. The confidentiality rating of #2 therefore stays high. Access is limited to the roles that standardize and edit the data, and to authorized reusers with a stated purpose. The formal assessment of disclosure risk starts at #3. **[SS3]**

6. Ownership and governance

Role	Responsibility for #2
Data owner	Process owner of Standardization; accountable for conformance and for applying the privacy design
Data steward	Registers versions, maintains catalog entries, and resolves gate exceptions
Data custodian	Operates storage, access control, and retention; manages pseudonymization keys separately from the data
Classification steward	Maintains the correspondence tables and the classification versions used in mapping
Metadata and data architect	Owns the target data model and the links to canonical concepts

7. Versioning, retention, and lineage

Every standardization run produces a new immutable version. A new version also arises when the input stays the same but the rules change: a new correspondence table, a new classification version, or a revised target model. The earlier version is never overwritten, because #3 versions may already be built on it. Each #2 version records the exact #1 version and rule versions it was produced from.

Standardization is deterministic, so a #2 version can be regenerated from its #1 version and the recorded rule versions. Many NSIs therefore keep #2 only as long as the dependent #3 versions are produced and reviewed. This only works while #1 and the rule versions are themselves retained. If #1 is disposed of earlier, #2 must be kept for as long as reproducibility requires. The exact periods are set by national law and NSI policy.

Lineage at #2 is field-level. Each standardized value can be traced to the raw value it came from and to the mapping rule that was applied. The pseudonymization record shows which identifiers were replaced, without revealing the keys.

8. Reuse and handover

Registration of #2 publishes the *Standardized Data Registered* event and a catalog entry. Consumers discover the version through the catalog and access it through the handover interface, never through the standardization working environment.

One standardized source can serve many statistical products. A population register delivery is standardized once and is then used by the census, labour market statistics, and income statistics. Under SAF-P08, products reuse the registered version instead of standardizing the same source again. Repeating the work per product wastes effort and leads to inconsistent coding of the same source.

The list of unmapped values and the recorded units and periods travel with the handover. VS-03 then knows exactly which interpretive coding and conversions remain to be done.

Handover element	Content
Artifact	Registered Standardized Data version

Handover element	Content
Required metadata	Minimum metadata set (attribute 4), including unmapped values and recorded units and periods
Trigger event	*Standardized Data Registered*
Responsibility transfer	VS-02 keeps custody; the consumer takes responsibility for editing, interpretive coding, and conversion on receipt of the event

9. Traceability to the framework

Framework element	References
Capabilities	1.3.1 Data standardization, 2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.4 Statistical register management, 3.3.1 Privacy management, 3.3.4 Logical identity and access control
Value-stream stages	VS-02.1 Profile and Assess Data Structure; VS-02.2 Apply Standard Classifications and Formats; VS-02.3 Align Metadata and Semantics; VS-02.4 Validate Structural and Semantic Conformance; VS-02.5 Register Standardized Artifacts
SAF processes [**SS4**]	PRD-5.2 Classify and code (deterministic mapping only), PRD-5.3 Review and validate (structural checks only), DLM-2 Promote data through a quality gate, DLM-3 Register and catalog data, DLM-4 Classify and protect data, DLM-9 Manage reference and master data
GSBPM	5.1 Integrate data (key alignment only; record linkage takes place in #3), 5.2 Classify and code
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11
Medallion layer [**SS4**]	Silver, lower sub-layer; in a Medallion arrangement often merged with #3

10. Examples and anti-patterns

Examples

- *Survey.* Labour Force Survey returns are restructured into the canonical person and household model. Source education codes are mapped one-to-one to ISCED 2011, and region codes to NUTS-3. The 42,000 free-text occupation answers are not coded here; they are listed as unmapped and handed to VS-03. The personal identifier is replaced with a pseudonymous key.
- *Administrative register.* The tax authority's internal activity codes are mapped to NACE Rev. 2 through a maintained correspondence table; 0.6% of codes have no correspondence and are flagged. Turnover stays in thousands of euro on a fiscal-year basis, and this basis is recorded as metadata. Conversion to units on a calendar-year basis takes place in #3.
- *Web data.* Scraped price records are reshaped into the canonical price record. Currency, unit, and scrape timestamp are recorded as metadata, and prices are kept exactly as received. Most products only have a free-text name and are passed to VS-03 for classification to COICOP.

Anti-patterns

- Converting currencies or units at #2 "because the data is open anyway". The value changes without an edit record.
- Coding free text with a classifier at #2. Interpretive coding needs review and logging, which happens in #3.
- Silently dropping records with unknown codes. This breaks the no-loss-of-content guarantee.

- Overwriting a #2 version after a correspondence table update. Statistics built on the earlier version lose their trace.
- Storing pseudonymization keys together with the data. The protection of #2 is then only on paper.
- Skipping #2 without recording the reason. Auditors can no longer see why the state is missing.

State #3: Processed Data

1. Definition and value added

Processed data is microdata whose content has been checked, corrected, completed, and enriched. Errors, duplicates, outliers, and inconsistencies have been detected and resolved. Missing values have been imputed, free text has been coded, and units, currencies, and reference periods have been converted to a common basis. Records from different sources have been linked, and new variables have been derived. Unlike in #2, values do change here, but every change is documented, attributable, and reversible.

The value of #3 is **reliability and analytical readiness**. Every estimate, model, and publication downstream rests on this microdata. The guarantee is not that values stay the same, but that no value changes without a record of the original value, the rule or decision applied, the reason, the actor, and the time. That record makes it possible to explain, reproduce, or reverse any change.

#3 is the only steady state that is mandatory for all statistical production. It is the microdata on which everything downstream is verifiably built. It is also the state that national archival requirements usually apply to: without it, the statistics built on it cannot be re-verified. **[SS5]**

Processing is primarily driven by the individual statistical domains. Depending on the type of data, it may also be performed by a central unit for efficiency and consistency. Either way, the result is registered as #3 under the same quality gate.

2. Producing and consuming domains

Role	Domain	Note
Producer	Editing, derivation, and non-response correction (VS-03)	Functions: Impute Data, Integrate Data
Primary consumer	Estimation and Analysis (VS-04)	Triggered by the *Processed Data Registered* event
Secondary consumer	Other statistical products	Reuse of processed microdata under SAF-P08, with authorization
Secondary consumer	Microdata services (VS-06)	Research datasets are prepared from #3 and released as #5 **[SS6]**
Occasional consumer	Innovation and Methodology (VS-08)	Testing of new editing, imputation, or coding methods, under specific authorization
Assurance	Governance and Compliance (VS-07)	Audit of edit lineage, methodology compliance, and disclosure risk

3. Quality gate

The gate for #3 checks content quality and traceability. It is the first gate that judges whether values are plausible and complete. A dataset may pass with known remaining quality issues, as long as they are measured and documented.

Criterion	Check	Typical outcome on failure
Rule conformance	No record violates the validation and editing rules after correction	Reject; re-edit
Edit lineage	Every changed value has its original value, rule or decision, reason, actor, and time recorded	Hold until complete
Imputation flags	Every imputed value carries a flag and a method code; no observed value is overwritten	Reject
Coding	Free-text values are coded or explicitly left uncoded with a reason; low-confidence codes are reviewed	Conditionally accept with flag
Conversion and reconciliation	Every unit, currency, or reference-period conversion and every reconciliation has its rule and original value recorded	Hold until complete
Linkage quality	Linkage rate and match quality are measured and within agreed limits	Conditionally accept with flag, or reject
Derived variables	Every derived variable has a formula, inputs, unit, and classification	Hold until complete
Quality improvement	Completeness and accuracy measured against the #2 baseline (or #1 where #2 was skipped)	Conditionally accept with flag
Disclosure risk	Re-identification risk of the enriched and linked data assessed against the privacy design	Reject; reduce detail or restrict access
Reproducibility	The version can be regenerated from its inputs plus the recorded rules and models	Reject

Conditional acceptance is common for imputation and linkage. A linkage rate slightly below target may still give the best available data. The flag makes sure estimation in VS-04 takes the issue into account, for example in its variance estimation.

4. Data and metadata characteristics

Data. Processed data contains the processed value of every variable. The original value stays recoverable through the edit lineage, and imputed values carry a flag. Permitted operations are error detection, editing and correction, imputation, interpretive coding of free text, conversion of units, currencies, and reference periods, reconciliation of values between sources, record linkage, and derivation of new variables and units. These are the value-changing operations that #2 deliberately left open. Operations that turn microdata into statistics are not permitted at #3: weighting, calibration, estimation, and aggregation belong to #4. **[SS5]**

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this quality version
Source reference	The #2 version or versions (or #1, where #2 was skipped) this dataset was produced from
Rule and model versions	Versions of the editing rules, imputation models, coding service, and derivation specifications applied
Edit lineage	Per changed value: original value, new value, rule or decision, reason code, actor, and time

Metadata element	Description
Imputation record	Per imputed value: flag, method code, and the predictors or donors used
Coding record	Per coded value: source text, assigned code, method, and confidence
Conversion and reconciliation record	Rule, factor or rate, and original value for every conversion or reconciliation
Linkage record	Keys used, matching method, linkage rate, and match quality
Derived variables	Formula, inputs, unit, and classification per derived variable
Quality assessment	Edit rate, imputation rate, coding completion, and improvement against the baseline
Disclosure-risk assessment	Outcome of the re-identification risk assessment and the resulting access conditions
Gate outcome	Accepted or conditionally accepted, with quality flags
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, #3 typically instantiates a *Unit Data Set*, extended with derived *Represented Variables*. The edits are described as *Rules* applied in *Process Steps*, which gives the edit lineage its formal structure.

5. Security, privacy, and access

Processed data is pseudonymized microdata, but it is richer than #2. Linking sources and deriving variables adds detail, and more detail means a higher risk that a person or business can be recognized. #3 is therefore the point where the disclosure risk of microdata is formally assessed. The outcome sets the conditions for access and reuse.

Linkage uses the pseudonymous keys created at #2. The keys themselves remain with the data custodian, separately from the data. Access to #3 is limited to the roles that edit, estimate, and analyze the data, and to authorized reusers with a stated purpose. External researchers never access #3 directly. Their access runs through the microdata services in VS-06, which prepare a project-specific dataset with its own pseudonyms and release it as #5.

The edit audit trail is itself sensitive and must be tamper-evident. It shows who changed which value and why, which is essential evidence in an audit.

6. Ownership and governance

Role	Responsibility for #3
Data owner	Process owner of Editing; accountable for the quality version and for the correct application of approved methods
Data steward	Registers quality versions, maintains catalog entries, and resolves gate exceptions
Data custodian	Operates storage, access control, retention, and archival; keeps pseudonymization keys separate from the data
Methodologist	Owns the editing rules, imputation methodology, and derivation specifications
Subject-matter statistician	Adjudicates flagged records and signs off domain-specific corrections

7. Versioning, retention, and lineage

Every completed editing cycle produces a new immutable quality version. A new version also arises when a source is redelivered, when a correction comes in, or when a rule or model changes. The earlier version is never overwritten, because estimates may already be built on it. Many NSIs distinguish preliminary and final versions; both are registered as separate quality versions of #3.

#3 is usually the state with the longest retention. It is the microdata that must be preserved to re-verify published statistics, and archival legislation often applies to it. When #3 is archived, its edit lineage, imputation flags, and derived-variable definitions are archived with it. Without that metadata, the archived values cannot be interpreted.

Lineage at #3 is value-level. Each processed value can be traced back to its original value in #2 or #1, the rule or decision that changed it, and the inputs of any imputation, conversion, or derivation. Linked records keep a reference to the source records they were built from.

8. Reuse and handover

Registration of #3 publishes the *Processed Data Registered* event and a catalog entry. Consumers discover the version through the catalog and access it through the handover interface, never through the editing working environment.

Processed microdata is the most valuable reusable asset in the chain. An integrated person-level dataset, edited and enriched once, can serve many statistics on labor, income, education, and health. Under SAF-P08, products reuse the registered quality version rather than editing the same data again. Editing the same data twice leads to different corrections of the same record, and therefore to statistics that do not agree.

The imputation flags travel with the handover. VS-04 needs them to reflect the extra uncertainty of imputed values in its accuracy measures.

Handover element	Content
Artifact	Registered Processed Data quality version
Required metadata	Minimum metadata set (attribute 4), including edit lineage, imputation flags, and derived-variable definitions
Trigger event	*Processed Data Registered*
Responsibility transfer	VS-03 keeps custody; the consumer takes responsibility for weighting, estimation, and analysis on receipt of the event

9. Traceability to the framework

Framework element	References
Capabilities	1.3.2 Deriving variables and units, 1.3.3 Check content, 1.3.4 Adjusting and imputing, 2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management
Value-stream stages	VS-03.1 Conduct Error Detection and Quality Diagnostics; VS-03.2 Perform Data Editing and Correction; VS-03.3 Impute Missing or Non-Response Data; VS-03.4 Derive Additional Variables and Indicators; VS-03.5 Perform Quality Evaluation and Versioning

Framework element	References
SAF processes	PRD-5.1 Integrate data, PRD-5.2 Classify and code (interpretive coding), PRD-5.3 Review and validate, PRD-5.4 Edit and impute, PRD-5.5 Derive new variables and units, PRD-5.8 Finalize data files, DLM-2, DLM-3, DLM-7 Revise data, DLM-10 Archive and preserve data
GSBPM	5.1 Integrate data, 5.2 Classify and code, 5.3 Review and validate, 5.4 Edit and impute, 5.5 Derive new variables and units, 5.8 Finalize data files
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11
Medallion layer	Silver; the main reusable microdata layer

10. Examples and anti-patterns

Examples

- *Survey.* The 42,000 free-text occupation answers handed over from #2 are coded to ISCO-08: 88% automatically, the rest computer-assisted or by experts. Item non-response on income (9%) is imputed by regression and flagged. Each coded and imputed value carries its method, so it can be reviewed or revised later.
- *Administrative register.* Turnover, received in thousands of euro on a fiscal-year basis, is converted to units on a calendar-year basis. The conversion factor and the original value are stored in the lineage. The tax data is then linked to the business survey on the statistical business register key, and a reconciliation rule resolves values where the two sources disagree.
- *Web data.* Products with only a free-text name are classified to COICOP with a machine-learning classifier; low-confidence cases are reviewed. Prices are converted to euro and standard units. Missing daily prices are filled by carry-forward and flagged with a staleness indicator.

Anti-patterns

- Overwriting the original value with the corrected one. The edit can no longer be explained or reversed.
- Imputing without a flag. Estimation then treats imputed values as observed and understates uncertainty.
- Correcting every outlier by hand. Over-editing costs effort, changes little in the results, and can introduce bias.
- Calculating weights or aggregates at #3. That work belongs to #4, where its method and accuracy are documented.
- Fixing values in a spreadsheet outside the editing pipeline. The change has no lineage and cannot be reproduced.
- Linking an extra source without reassessing disclosure risk. The richer dataset may identify people that #2 did not.
- Disposing of #3 once figures are published. The published statistics can then no longer be verified.

State #4: Statistics

1. Definition and value added

Statistics are the results produced from processed microdata: estimates, aggregates, indices, time series, and model outputs. Each result carries the method that produced it and a measure of its accuracy, such as a standard error or confidence interval. State #4 can also hold output microdata with weights, prepared for a specific internal purpose. Initial disclosure control has been applied, so the results can potentially be released. They are not yet released: #4 may still contain detail that is confidential or not suitable for publication.

The value of #4 is **statistical credibility and coherence**. Data has become information about a population, and that information is accurate, transparent, and consistent with related statistics. Any figure

in #4 can be regenerated from #3 and defended, because its method, parameters, weights, and accuracy are recorded.

#4 is produced by two domains and therefore has **two gate moments**. [**SS7**]

Gate moment	Produced by	Status of the version	What it confirms
Analytical gate	Estimation and closed-loop (VS-04)	*Analytical*	Results are methodologically sound, accurate, coherent, and initially disclosure-controlled
Approval gate	Statistically composed and composite products (VS-05)	*Approved*	Results are composed into a coherent product, explained, checked against dissemination and confidentiality standards, and signed off

Both are versions of the same state. Approval does not change an analytical version; it creates a new, immutable version with the status *approved* that refers to the analytical versions it was composed from. #4 is mandatory for all statistical production.

2. Producing and consuming domains

Role	Domain	Note
Producer (analytical)	Estimation and closed-loop (VS-04)	Functions: Estimate Data, Aggregate Data
Producer (approved)	Statistically composed and composite products (VS-05)	Function: Compose Data
Primary consumer (analytical)	Product composition (VS-05)	Triggered by the *Output Data Registered* event
Primary consumer (approved)	Dissemination (VS-06)	Triggered by the *Statistical Product Registered* event
Secondary consumer	Other statistical processes	Internal input for macro statistics and integration systems, such as the national accounts
Assurance	Governance and Compliance (VS-07)	Audit of methodology, reproducibility, and approval

3. Quality gate

The two gate moments check different things. The analytical gate tests the statistics; the approval gate tests the product.

Analytical gate (VS-04)

Criterion	Check	Typical outcome on failure
Approved method	Estimation, weighting, and model methods are approved and documented for this output	Reject; return to design
Weights	Calibration converged; weights within the agreed bounds	Reject; re-weight

Criterion	Check	Typical outcome on failure
Accuracy	Every estimate has an accuracy measure; imputation uncertainty from #3 is reflected	Hold until complete
Model diagnostics	Diagnostics of model-based results are within the agreed limits	Reject, or conditionally accept with flag
Soundness and coherence	Results are plausible and coherent with prior periods and related statistics; large movements are explained	Conditionally accept with flag, or reject
Initial disclosure control	Disclosure risk assessed and initial protection applied	Reject
Reproducibility	Results regenerate from #3 plus the recorded methodology and parameters	Reject
Metadata	Minimum metadata set captured (see attribute 4)	Hold until complete

Approval gate (VS-05)

Criterion	Check	Typical outcome on failure
Component selection	The exact analytical versions used are pinned and recorded	Hold until complete
Composition	Composite indicators, indices, and time series follow their documented specifications	Reject; re-compose
Consistency and coherence	Totals reconcile to components; the product is coherent with related and prior products	Return to VS-04, or conditionally accept with explained flag
Interpretive content	Commentary and visualizations are reviewed and consistent with the data and its accuracy	Hold until complete
Dissemination and confidentiality standards	The product meets the standards, including the disclosure risk of combining outputs	Reject
Sign-off	The designated approval authority has approved the product	Hold until approved

Coherence problems at the approval gate are never solved by adjusting a component estimate in VS-05. If a component is wrong, it goes back to VS-04 and returns as a new analytical version.

4. Data and metadata characteristics

Data. Statistics in #4 are mostly aggregated data: tables, data cubes, indicators, indices, and time series. They can also include weighted output microdata, prepared for a specific internal use. Permitted operations are weighting and calibration, estimation and aggregation, model-based analysis, seasonal adjustment, initial disclosure control, and, at the approval stage, composition of indicators, indices, and time series from analytical versions. Changing microdata values is not permitted at #4: corrections go back to #3. Final disclosure control for release is not part of #4 either; it belongs to the transition to #5. **[SS8]**

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this version
Status	*Analytical* or *approved*
Source reference	The #3 versions (for analytical versions) or the analytical #4 versions (for approved versions) used
Methodology	Estimation, weighting, and model methods, with their versions and parameters
Weights and calibration totals	Calibration method, auxiliary variables, and the source and version of the totals
Accuracy	Standard error, coefficient of variation, or confidence interval per estimate; whether imputation uncertainty is included
Model diagnostics	Fit and prediction intervals for model-based results
Disclosure control	Method applied and outcome of the risk assessment
Coherence results	Outcome of the coherence checks against related and prior statistics
Composition record	For approved versions: composition method and the component versions used
Interpretive content	For approved versions: reference to commentary, key messages, and visualizations
Approval record	For approved versions: approval authority, date, and confidentiality clearance
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, aggregated results in #4 typically instantiate a *Dimensional Data Set*, and weighted output microdata a *Unit Data Set*. Both conform to a *Data Structure* that SDMX can use for exchange.

5. Security, privacy, and access

Initial disclosure control is applied at the analytical gate, so the results could in principle be released. That does not make #4 public. It may still hold detailed cells, internal breakdowns, or output microdata that are only suitable for internal use. Different confidentiality thresholds can apply depending on the intended use, and the version records which threshold was applied.

Composition creates a specific risk. Two tables that are each safe on their own can together reveal a value through differencing. The approval gate therefore checks the disclosure risk of the product as a whole, not only of its components.

Approved products are usually market-sensitive or politically sensitive before their release. Access to approved versions is restricted to the people who need it to prepare the release. Every pre-release access is logged, so equal access at the moment of release can be demonstrated.

6. Ownership and governance

Role	Responsibility for #4
Data owner (analytical)	Process owner of Estimation; accountable for methodological soundness and accuracy
Data owner (approved)	Process owner of Product Composition; accountable for coherence and interpretability of the product
Approval authority	Signs off approved versions against dissemination and confidentiality standards
Data steward	Registers versions, maintains catalog entries, and resolves gate exceptions
Data custodian	Operates storage, access control, retention, and the pre-release access log
Methodologist	Owns estimation, weighting, and model methods

Role	Responsibility for #4
Disclosure control specialist	Owns the disclosure-control rules for internal and potential external use

7. Versioning, retention, and lineage

Every estimation run produces a new immutable analytical version. Every approval produces a new immutable approved version that refers to the analytical versions it was built from. Revisions follow the revision policy: a preliminary figure, a revised figure, and a final figure are three separate versions. Nothing is overwritten, because published figures and other statistics may depend on any of them.

Approved versions that were released are kept for as long as the released figures exist, because they are the direct basis of #5. Analytical versions that never led to a release can be kept for a shorter period, as long as revisions and reproducibility are covered. The exact periods are set by national law and NSI policy.

Lineage at #4 has two steps. An estimate traces back to the #3 version, the method, the parameters, and the weights that produced it. An approved product traces back to the analytical versions it was composed from, and through them to #3.

8. Reuse and handover

#4 has two handovers, one for each gate moment. The analytical version goes to VS-05 for composition; the approved version goes to VS-06 for release. Consumers access versions through the catalog and the handover interface, never through the estimation or composition working environment.

Statistics in #4 are also reused internally. Integration systems, such as the national accounts or labor accounts, use statistics from many domains as input. Under SAF-P08 they reuse the registered version instead of estimating the same figure again. This keeps internal use and later publication consistent.

Handover element	Analytical version (VS-04 → VS-05)	Approved version (VS-05 → VS-06)
Artifact	Registered analytical #4 version	Registered approved #4 version
Required metadata	Methodology, weights, accuracy, model diagnostics, disclosure control, coherence results	Composition record, propagated accuracy, coherence results, interpretive content, approval record
Trigger event	*Output Data Registered*	*Statistical Product Registered*
Responsibility transfer	VS-04 keeps custody; VS-05 takes responsibility for composition and approval	VS-05 keeps custody; VS-06 takes responsibility for final disclosure control, release, and user support

9. Traceability to the framework

Framework element	References
Capabilities	1.3.5 Estimating population parameters, 1.3.6 Integrate statistics, 1.3.7 Statistical disclosure, 1.4.1 Statistical (output) analysis, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Value-stream stages	VS-04.1 to VS-04.5 (analytical gate); VS-05.1 to VS-05.5 (approval gate)

Framework element	References
SAF processes	PRD-5.6 Calculate weights, PRD-5.7 Calculate aggregates, PRD-6.1 Prepare draft outputs, PRD-6.2 Validate outputs, PRD-6.3 Interpret and explain outputs, PRD-6.4 Apply disclosure control (initial), PRD-6.5 Finalize outputs, PRD-7.2 Produce dissemination products (content), DLM-2, DLM-3, DLM-7 Revise data
GSBPM	5.6 Calculate weights, 5.7 Calculate aggregates, 6.1 to 6.5 Analyze
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11
Medallion layer	Gold, internal sub-layer; not yet released

10. Examples and anti-patterns

Examples

- *Survey.* The Labor Force Survey estimates an employment rate of 71.4%, with a 95% confidence interval of ±0.5 percentage points at national level. The variance includes the uncertainty of imputed income from #3. This analytical version is then composed with unemployment and vacancy figures into the quarterly labor market product, which is approved and registered.
- *Administrative register.* Turnover is aggregated by industry and region. For small domains where the direct aggregate is unstable, a small-area model is used, and its results are flagged as model-based. In the approval gate, one domain fails the coherence check; it goes back to VS-04 instead of being adjusted in the product.
- *Web data.* A multilateral price index is calculated from the item prices in #3. The headline index and its main divisions are composed with the period weights, checked for coherence with the prior month and year, and approved.

Anti-patterns

- Approving a product by overwriting the analytical version. The difference between the tested figure and the approved figure disappears.
- Adjusting a component estimate in VS-05 to make the coherence check pass. The estimate no longer matches its documented method.
- Publishing an estimate without an accuracy measure. Users cannot judge whether a movement is meaningful.
- Treating initial disclosure control as final. Release requires the final check of the transition to #5.
- Sharing approved figures before release without logging the access. Equal access can no longer be demonstrated.
- Re-estimating a figure for internal use instead of reusing the registered version. Internal and published figures start to differ.

State #5: Released Data

1. Definition and value added

Released data is statistical output that the NSI has made available outside its own organization. It is based on an approved #4 version that has passed final disclosure control, has been documented for correct interpretation, and has been released through a trusted channel. Released data includes published tables and datasets, open data, data available through an API, deliveries to international organizations, data offered in a data space, and de-identified microdata made available to accredited researchers. #5 is the last point at which the NSI controls the data. Once users have it, it becomes consumed data, outside the NSI's control.

The value of #5 is **accessibility and trust**. Users get accurate, documented statistics through channels they can rely on, at a moment that is the same for everyone. The final disclosure check ensures that no

respondent can be identified. The documentation ensures that users can interpret the figures correctly, including their accuracy and limitations.

#5 is mandatory whenever output is released. If nothing changes between the approved #4 version and the release, the same artifact can be marked as released instead of being registered again. The release steps still take place and are recorded: release authorization, the final disclosure check, and the release itself (see *Applying the steady states pragmatically*).

2. Producing and consuming domains

Role	Domain	Note
Producer	Dissemination (VS-06)	Functions: Data Delivery, Data Publication
Primary consumer	External users	Policy makers, researchers, media, businesses, and the public; use turns #5 into #6
Primary consumer	International organizations	Deliveries under agreed transmission programs, such as to Eurostat or the OECD
Primary consumer	Accredited researchers	Access to de-identified research datasets in a secure processing environment **[SS9]**
Secondary consumer	Customer and Stakeholder Engagement (VS-10)	Usage and feedback data for needs analysis and impact measurement
Assurance	Governance and Compliance (VS-07)	Audit of disclosure control, equal access, and release authorization

3. Quality gate

The gate for #5 is the release gate. Confidentiality is not negotiable at this boundary. Unlike the earlier gates, the disclosure criteria allow no conditional acceptance: a release that fails them is returned for further protection. **[SS10]**

Criterion	Check	Typical outcome on failure
Approved source	The release is based on an approved #4 version	Reject
Final disclosure control	Final protection applied and disclosure-risk criteria met for every table, dataset, and dashboard	Reject; apply further protection
De-identification	Released microdata meets the de-identification standard for its channel	Reject
Documentation	Methodology notes, metadata, and user guidance are complete	Hold until complete
Format and accessibility	Each channel format is valid, machine-readable where required, and meets the accessibility standard	Hold until corrected
Release authorization	The release has been signed off	Hold until authorized
Timing and equal access	Release time follows the release calendar; embargo arrangements are in place and logged	Hold until the release slot
Metadata	Minimum metadata set captured (see attribute 4)	Hold until complete

4. Data and metadata characteristics

Data. One released version can have several representations, one per channel: a published table, an SDMX dataset behind an API, a dashboard, or a research dataset. All representations come from the same released version, so a user always gets the same figure through every channel. Permitted operations are final disclosure control, de-identification of microdata, formatting per channel, packaging, and adding documentation. Changing the statistical content is not permitted at #5. A correction to a released figure goes back to #4 as a new approved version and is released again with an erratum. [**SS11**]

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this release version
Source reference	The approved #4 version this release is based on
Final disclosure control	Method applied, risk assessment outcome, and sign-off
De-identification	For microdata: the standard applied and the project or channel it applies to
Channels and formats	Every channel used, with format and access conditions
Release timestamp	Moment of release and reference to the release calendar
Embargo record	Who had pre-release access, when, and under which agreement
Documentation	References to methodology notes, reference metadata, and user guidance
Accessibility	Accessibility standard met per channel
Terms of use	Licence or terms under which the data may be used
Correction history	Errata issued and the versions they replace
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, #5 is the disseminated form of a *Data Set* together with its *Referential Metadata*. SDMX defines the structure in which it is exchanged with users and international organizations.

5. Security, privacy, and access

After final disclosure control, most released data is public and has a low confidentiality rating. Its integrity rating is high. Users must be able to trust that a figure on the website or in the API is exactly the figure the NSI released. Channels therefore protect released data against tampering, and the NSI can prove which version was released when.

Before the release moment, the same data is highly restricted. Embargoed copies go only to parties with a formal agreement, such as accredited media, and every pre-release access is logged. No user gets the data before the release time outside that policy.

Research microdata follows a different path. It is released only to accredited researchers and only inside a secure processing environment. Each project gets its own dataset with project-specific pseudonyms, so data cannot be linked across projects. Results leave the environment only after an output check.

6. Ownership and governance

Role	Responsibility for #5
Data owner	Process owner of Dissemination; accountable for confidentiality, equal access, and the quality of the release

Role	Responsibility for #5
Release manager	Controls release timing, embargo, and equal access across channels
Disclosure control specialist	Applies and signs off the final disclosure control
Dissemination and service manager	Manages channels, formats, service levels, and the research access service
Data steward	Registers release versions, maintains documentation, and handles errata
Data custodian	Operates channels, access control, integrity protection, and the embargo log

7. Versioning, retention, and lineage

Every release is an immutable version. A correction never silently replaces a released figure. It produces a new approved #4 version, a new release version, and a public erratum that explains what changed and why. Earlier releases remain identifiable, so users can see which figure was available at which moment.

Released statistics are usually kept permanently, as part of the public record. National archival legislation often prescribes this. Research datasets are the exception: they are disposed of when the research project closes, together with their project-specific pseudonymization keys. The approved code and outputs of the project are kept for reproducibility.

Lineage at #5 closes the chain. Each released figure traces back to its approved #4 version, and from there through the analytical versions, #3, and #2 to the authentic delivery in #1. This makes it possible to answer any question about a published figure, from source to release.

8. Reuse and handover

Release publishes the *Statistics Released* event and makes the data available through the agreed channels. #5 has no downstream production stream. Its handover is to users outside the NSI, where the data becomes consumed data (#6). Usage and feedback flow back to VS-06 and VS-10, and through them into the design of the whole chain.

Released data is the most widely reused form of NSI output. Under SAF-P08 it is offered as openly as confidentiality allows, in standard formats and with clear terms of use. Open, documented release reduces the number of individual data requests and lets users combine NSI data with other sources.

Handover element	Content
Artifact	Released data version, in its channel-specific representations
Required metadata	Minimum metadata set (attribute 4), including documentation, terms of use, and correction history
Trigger event	*Statistics Released*
Responsibility transfer	The NSI remains responsible for the correctness and documentation of the release and for issuing errata; users are responsible for how they use the data

9. Traceability to the framework

Framework element	References
Capabilities	1.3.7 Statistical disclosure, 1.5.1 Publication management, 1.5.2 Promotion to use, 1.5.3 Facilitate data access, 1.5.4 Statistical information management, 2.2.4 Data services, 3.4.1 Services management

Framework element	References
Value-stream stages	VS-06.1 Plan Dissemination and Access Strategy; VS-06.2 Prepare Data for Dissemination; VS-06.3 Publish and Provide Access Services; VS-06.4 Support and Engage Users; VS-06.5 Monitor Usage and Feedback
SAF processes	PRD-6.4 Apply disclosure control (final), PRD-7.1 to PRD-7.5 Disseminate, MDS-3 Prepare research datasets, MDS-4 Provide a secure processing environment, MDS-7 Check and release output, MDS-9 Close projects, DSP-2 Publish data offerings, DSP-4 Exchange data under usage control
GSBPM	6.4 Apply disclosure control, 7.1 to 7.5 Disseminate
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11
Medallion layer	Gold, published sub-layer

10. Examples and anti-patterns

Examples

- *Survey.* The quarterly labor market figures are released at 09:00 on the calendar date, at the same moment on the website, the API, and the dashboard. Two small regional cells are suppressed in the final disclosure check. Accredited media received an embargoed copy and publish at the same moment.
- *Administrative register.* Aggregated business statistics are published as open data. De-identified business microdata is released to accredited researchers inside a secure processing environment, with project-specific pseudonyms and output checking.
- *Web data.* The monthly price index is market-sensitive. It is released at a fixed minute with strict equal access, and the documentation explains the main contributions to the monthly change.

Anti-patterns

- Publishing directly from #4 without a recorded final disclosure check. There is no evidence that the release was safe.
- Correcting a published figure silently. Users who already used the old figure cannot see that it changed.
- Giving a ministry or partner early access outside the embargo policy. The NSI's independence and equal access are compromised.
- Releasing figures without methodology notes or accuracy. Users draw conclusions the data cannot support.
- Sending research microdata by email or file transfer to a researcher's own environment. Control over the data is lost.
- Releasing different figures through different channels. Users lose trust when the website and the API disagree.

Boundary Position #6: Consumed Data

Description: Consumed data is released data that users have taken into their own environment and use for their own purposes. Examples are a government departments using an unemployment rate in a policy note, a researcher combining released tables with other sources, a journalist quoting an inflation figure, or an application that pulls a series from the API every day. The NSI has no custody of consumed data and cannot control how it is used, combined, or presented. It can influence correct use only through the quality of its documentation, its terms of use, its user support, and its errata.

Why it is described: The SAF describes #6 so that the end of the NSI's responsibility is explicit, just as #0 marks where it begins. Everything the NSI can do for correct use has to be done at #5: documentation, accuracy measures, terms of use, and a clear correction history. What the NSI learns about consumed data

comes back as usage and feedback, through VS-06.5 and VS-10. That feedback closes the loop: it shapes new information needs and, through them, the next cycle of the value chain from #0 onward.

Statistical Data Value Chain

Value Chains Introduction

Purpose of this chapter

This chapter describes how the statistical value chain works in practice. The Strategy chapter introduced the ten value streams, their stages, and the value they create. The Steady-State Architecture describes the results the chain hands over: the states, their quality gates, and their metadata. This chapter describes the work between those results: the stages, the processes that realize them, the roles that carry them out, and the events that connect them. Every value stream is described with the same structure, so that NSIs can compare them and map their own organization onto them. **[VC1]**

Where to find what

Question	Answered in
Which value streams exist, and what value do they create?	Strategy, *Value Streams*
What must data satisfy at each handover point?	Steady-State Architecture, per state
How is the work in each value stream organized?	This chapter
Which actors, roles, processes, and functions exist?	Business Architecture, reference content
How mature is an NSI, and how does it apply a value stream?	SAF Toolkit

The division of work between states and value streams is deliberate. A state describes the **result**: what the data looks like, which gate it passed, and how it is protected. A value stream describes the **work**: who does what, in which order, and with which controls. Gate criteria and minimum metadata sets are therefore not repeated in this chapter. Each value stream refers to the state it produces.

Key concepts — if you are new to architecture documents This documentation uses a handful of architecture terms. Here is what they mean in plain language:

- **Value stream** — a sequence of steps that together create a statistical output. Think of it as a production line: VS-01 to VS-06 each add a specific kind of value to the data.
- **Value stream stage vs. business process** — the stage (`VS-0x.y`) says *what value* is added; the business process describes *how* your office actually does it. The numbering maps one-to-one.[^archimate]
- **Steady state** — a fixed handover point: a versioned, quality-assured dataset that one stream produces and the next one consumes. The framework defines five numbered steady states, #1 to #5; data still at the external provider (#0) and data in users' hands (#6) sit outside the NSI's control and are boundary positions, not numbered states.
- **The four process types** — every domain is described through the same four groups: **Design** (decide how it should work), **Implementation** (build and test it), **Execution** (run it each production cycle), **Management** (check quality and feed improvements back).
- **Business service / object / event** — what a domain offers to others / the data artefacts it works on / the signals streams send each other (e.g. *"Raw Data Registered"* tells the next stream new data is ready).
- **NSI** — National Statistical Office.
- **GSBPM** — the *Generic Statistical Business Process Model*: the standard, internationally agreed description of the phases of statistical production (specify needs, design, build, collect, process, analyse, disseminate, evaluate).

- **GAMSO** — the *Generic Activity Model for Statistical Organizations*: the companion standard describing management and support activities (strategy, capability development, corporate support).
- **SAF-P01 ... SAF-P11** — the eleven SAF architecture **p**rinciples, defined in the SAF Framework document and referenced throughout.

The chain at a glance

Six primary value streams form the production chain. Each one consumes a registered state version and produces the next. They are loosely coupled (SAF-P07): a value stream needs no knowledge of the internal work of its neighbors, only of the state it receives and the state it delivers. The handover is triggered by a registration event. Four supporting value streams run continuously across the whole chain. They set the rules, develop new methods, provide the platforms, and keep the chain connected to its users and data providers.

VS	Value stream	Type	State in → out	Event out
VS-01	Data Acquisition and Ingestion	Primary	#0 → #1	*Raw Data Registered*
VS-02	Data Standardization and Integration	Primary	#1 → #2	*Standardized Data Registered*
VS-03	Data Editing and Enrichment	Primary	#2 → #3	*Processed Data Registered*
VS-04	Estimation, Analysis and Statistical Modeling	Primary	#3 → #4 (analytical)	*Output Data Registered*
VS-05	Statistical Product Composition	Primary	#4 (analytical) → #4 (approved)	*Statistical Product Registered*
VS-06	Dissemination and Service Delivery	Primary	#4 (approved) → #5	*Statistics Released*
VS-07	Governance and Compliance	Supporting	Across the chain	n/a
VS-08	Innovation and Methodological Development	Supporting	Across the chain	n/a
VS-09	Infrastructure and Platform Enablement	Supporting	Across the chain	n/a
VS-10	Customer and Stakeholder Engagement	Supporting	Across the chain; feedback from #5 and #6	n/a

Not every statistical product passes through every primary value stream. A macro statistic, for example, often starts from statistics another domain has already registered. Such a product enters the chain downstream and its lineage points to the state it consumed (see *Applying the steady states pragmatically*). Data that comes from outside the NSI always enters through VS-01.

The four process types

Every business domain, and therefore every primary value stream, runs the same four process types. **Design** decides how the work is done and delivers metadata: designs, rules, and specifications. **Implementation** builds and tests the solution and delivers the execution process. **Execution** runs the work in each production cycle and delivers the statistical data; the stages of a value stream describe this execution. **Management** monitors the work, assesses quality, and feeds improvements back to design. Design and implementation only run when something changes, execution runs every cycle, and management runs continuously.

The supporting value streams do not follow a production cycle. They run the same pattern as a continuous cycle of plan, build, operate, and improve.

Process type	Provides	GSBPM phases	Equivalent in supporting streams
Design	Metadata: designs, rules, specifications	1 Specify Needs, 2 Design	Plan
Implementation	The execution process: instruments, components, configured workflows	3 Build	Build
Execution	Statistical data, registered in a state	4 Collect, 5 Process, 6 Analyze, 7 Disseminate	Operate
Management	Quality reports and improvements	8 Evaluate; overarching quality and metadata management	Improve

How each value stream is described

Each primary value stream is described with the same ten attributes. The supporting value streams use the same attributes, with two differences: attribute 2 describes the relation to the chain instead of a state transition, and attribute 4 follows the cycle of plan, build, operate, and improve.

#	Attribute	Question it answers
1	Purpose and value created	Why does this value stream exist, and what value does it add?
2	Position and state transition	Which state does it consume, and which does it produce?
3	Stages	Which stages create the value, and what happens in each?
4	The four process types	How is the value stream designed, built, and managed?
5	Actors and roles	Who is involved, and with which responsibility?
6	Capabilities, processes, and functions	Which SAF capabilities, processes, and functions realize the stages?
7	Principles	Which architecture principles weigh most, and how are they applied?
8	Interfaces, services, and events	What does the value stream offer and exchange, and what triggers it?
9	Key performance indicators	How is the performance of the value stream measured?
10	Examples	What does the value stream look like in practice?

Each value stream has three ArchiMate views: (a) the value stream with its stages and the capabilities that serve them, (b) the stages and the processes and functions that realize them, and (c) the state handover and its events. Maturity checklists and detailed end-to-end scenarios for each value stream are part of the SAF Toolkit. **[VC2]**

Identifiers used in this chapter

Identifier	Refers to	Defined in
VS-01 to VS-10, VS-01.1, …	Value streams and their stages	Strategy, *Value Streams*
1.2.1, 2.1.2, …	Capabilities	Strategy, *Capability Model*
ACT, ROL, COL	Business actors, roles, and collaborations	Business Architecture
PRD, OVA, DLM, MDS, DSP	Business processes	Business Architecture, *Business Processes*
FUN	Business functions	Business Architecture, *Business Functions*
#0 to #6	Steady states and boundary positions	Steady-State Architecture
SAF-P01 to SAF-P11	Architecture principles	Motivation, *Architecture Principles*

Steady States of Data

The streams work independently of each other (in architecture terms: they are **loosely coupled**). They exchange data only at defined **steady states** — points where a high-quality, versioned dataset is handed over. In the usual case each primary stream consumes one steady state and produces the next, so a stream needs no knowledge of its neighbors' internals (SAF-P07). The handover works through events: when a stream registers a finished dataset, a notification tells the next stream that new data is ready. Not every process has to pass through every state; see Applying the steady states pragmatically.

Of the five, **processed data (#3) is the only steady state mandatory for all production** — it is the microdata on which everything downstream is verifiably built. **Raw Data (#1)** is also mandatory when a product collects its own data, but not for a macro statistic that starts entirely from another domain's data. **Standardized data (#2)** is always optional, and **Statistics (#4)** may share its artefact with **Released data (#5)** when the two are identical.

Applying the steady states pragmatically

A steady state is an agreement about data: the quality it has reached, who owns it, how it is identified and what metadata comes with it. It does not require a separate physical copy at every state, and it does not require every process to pass through all six. An office may depart from the usual sequence in the three cases below, provided it records the choice (which state was shared or skipped, and why) in the catalogue entry. In other words, the rule is "comply or explain".

1. **Two states can share one artefact.** If nothing happens between State #4 and State #5 (no transformation and no further disclosure control), the #4 artefact does not have to be registered again as a new dataset; it is marked as released instead. The release steps still take place: release authorisation is obtained and the final disclosure check is carried out, and both are recorded even when the check changes nothing.
2. **A process can start downstream.** Not every production process begins at VS-01. Macro statistics, for example, are often built on Processed Data (#3) or Statistics (#4) that another team has already registered, and State #4 above can already be "shared internally for use as input data in other statistical processes". Such a process starts from the registered artefact and its lineage points to it; there is no need to create placeholder #1 or #2 artefacts. Data that comes from outside the office still enters through VS-01 and State #1.
3. **Lineage is about traceability, not duplication.** SAF-P09 requires that every value can be traced back to where it came from. When a stream uses an artefact without changing any values, a logged process event that refers to the existing artefact version meets this requirement, and no identical copy needs to be registered. As soon as a value changes, a new version is registered.

One limit applies to all three cases. States must not be merged where this would hide a change of values (for example, standardisation at #2 and editing at #3 carried out as a single undocumented step) or a disclosure-control step (for example, #4 and #5 when additional disclosure control is applied before release).

Principles to Apply

The SAF defines eleven architecture principles, labeled SAF-P01 to SAF-P11. The six **primary** streams actively apply six of them (marked below); the four **supporting** streams foreground the other five (marked †). The table shows each principle with practical guidance on how to apply it; VS-07 in particular **enforces** the data principles (P09/P10/P11) across the whole chain.

Table 1: SAF Principles

Principle	Statement	How to apply across the value chain
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	Build statistical capabilities on shared, open tooling; contribute back, so NSIs keep collective control of critical methods rather than depending on vendors.
SAF-P07	Implement loosely coupled processes and systems	Exchange data only at steady-state interfaces; each stream consumes one state and produces the next without knowing the others' internals.
SAF-P08	Data is shared property	Once an artefact is registered at a steady state, make it available to all authorized consumers; avoid duplicating collection, processing, or analysis.
SAF-P09	No data without metadata and classification	Every artefact carries provenance (origin), methodology, accuracy, classification, and lineage (the trace back to its inputs) — captured at the point each value is created or changed.
SAF-P10	Security By Design	Embed access control, integrity verification, and tamper-evident audit trails in every pipeline from the start.
SAF-P11	Privacy By Design	Remove/pseudonymize identifiers during standardization; assess disclosure risk on enrichment and apply statistical disclosure control before release.
SAF-P02 †	Active support	Operate and support capabilities as living services, not one-off builds (VS-09 platforms, VS-10 services).
SAF-P03 †	Active Life Cycle management	Version and govern rules, methods, models, and products across their life cycle; retire what is obsolete (VS-07, VS-08).
SAF-P04 †	Maximize the benefits for the organization	Prefer solutions that serve the whole organization and real user value over local optimizations (VS-10, VS-07).
SAF-P05 †	IT and technology is everyone's business	Treat methodology, data, and technology as a shared responsibility across business and IT (VS-08, VS-09).
SAF-P06 †	Business Continuity	Keep the chain running reliably and able to recover from disruption (VS-09).

 = actively applied by the six primary value streams. † = foregrounded by the four supporting value streams (VS-07–VS-10).

Metadata everywhere (SAF-P09): value is only added once its context is captured — what was done, by which rule/method, with what accuracy, and traceable to its inputs.

Confidentiality progressively applied (SAF-P11): identifiers are removed/pseudonymized at standardization (#2) at the latest, or earlier where that suits the process; disclosure risk is assessed on enrichment (#3), disclosure control is applied to outputs (#4), and a final disclosure check gates release (#5).

Primary value streams

VS-01 Data Acquisition and Ingestion

1. Purpose and value created

VS-01 is the entry point of the statistical value chain. It brings data from outside the NSI under NSI custody: survey responses, administrative registers, exchanges in data spaces, and new sources such as web or sensor data. It does not change the content of the data. It makes sure the data is obtained lawfully, received securely, checked for integrity, and registered with its provenance. The value created is **trust and readiness**: trust that the data was acquired correctly, and readiness for use in the rest of the chain. Everything downstream depends on this. If the acquisition of a source cannot be traced, no statistic built on it can be fully defended.

VS-01 corresponds to the business domain **Observation/Collection** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

- **Data collection manager or operations staff:** the stages in attribute 3, which describe what happens in each production cycle.
- **Methodologist or statistical designer:** the Design processes in attribute 4.
- **System builder or IT staff:** the Implementation processes in attribute 4.
- **Quality or compliance manager:** the Management processes in attribute 4.
- **Manager who wants the big picture:** attributes 1, 2, and 10. End-to-end scenarios are part of the SAF Toolkit.

PLACEHOLDER: ArchiMate view VS-01 (a) value stream and capabilities

2. Position and state transition

VS-01 is the first primary value stream. All data from outside the NSI enters the statistical system through it, before any downstream processing takes place.

VS-01 takes data from boundary position #0 to State #1. Before VS-01, the data is held by a provider or respondent, outside the NSI's control. After VS-01, it is a registered, immutable version of raw data, in NSI custody, in its original form, with provenance metadata recorded and basic quality verification completed. The quality gate, the minimum metadata set, and the protection of State #1 are described in *State #1: Raw Data* and are not repeated here. **[VC10]** VS-01 is the only entrance for data from outside the NSI: there is no side door into the chain. A product that starts from data already registered by another domain does not pass VS-01; it enters the chain downstream.

From	To	Gate	Event out
#0 Source Data	#1 Raw Data	Gate of State #1 (legal basis, authenticity, integrity, security, structure, completeness, timeliness, metadata)	*Raw Data Registered*

PLACEHOLDER: ArchiMate view VS-01 (c) state handover and events: Source Data (#0) outside the NSI boundary, the ingestion service at the boundary (metadata addition, quality and schema check), Raw Data (#1) inside the NSI

3. Stages

The stages describe the value that VS-01 adds in every production cycle. Together they form the execution process type of VS-01. In a mature steady state, most of the work has been designed and built in advance (see attribute 4). Execution then focuses on running the cycle, capturing data and metadata, and making sure the quality gate is passed.

Stage	Name	Value added
VS-01.1	Identify Data Needs and Sources	The information need is clear, and suitable sources are known
VS-01.2	Establish Data Agreements and Access Rights	Data can be obtained lawfully, under agreed conditions
VS-01.3	Collect or Receive Data	Data and its metadata are in NSI custody
VS-01.4	Validate and Log Data Intake	The delivery is complete, intact, and traceable
VS-01.5	Store and Register Raw Data Artifacts	Raw data is registered and available for reuse (State #1)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies. **[VC9]**

VS-01.1 Identify Data Needs and Sources

Value created: *Clear understanding of what data is needed and where it can be obtained.*

This stage starts from the information need set in the design of the statistical product. In a mature steady state, the Design process *Design Data Requirements and Source Strategy* has already defined the data requirements and the source strategy. Execution in each cycle therefore focuses on confirmation rather than re-creation. The team confirms or updates the data requirements for the current production cycle. It verifies that the identified sources remain available and accessible, and checks that source metadata is current: contact persons, data formats, and delivery schedules. Before any new collection, it checks whether the data already exists as a registered state version, in line with SAF-P08. It also identifies new sources that may have become available since the last cycle. A significant change in requirements or source availability goes back to design instead of being solved during the cycle.

Activity	Key outputs	SAF process	Other reference
Confirm or update data requirements for this production cycle	Confirmed requirements for the cycle	PRD-1.1 Identify needs	
Check whether the data already exists as a registered state version	Reuse decision	PRD-1.5 Check data availability and suitability	SAF-P08
Verify data source availability and accessibility	Source availability confirmation	PRD-1.5 Check data availability and suitability	
Check that source metadata is current (contacts, formats, schedules)	Updated source metadata	PRD-1.5 Check data availability and suitability; OVA-6 Data Supplier Management	
Identify new sources available since the last cycle	New source assessment (if any)	PRD-1.5 Check data availability and suitability	

VS-01.2 Establish Data Agreements and Access Rights

Value created: *Legal, ethical, and transparent access to data under controlled conditions.*

Before data flows, legal and operational access must be confirmed for the current production cycle. For each source, the legal basis is established: a statutory obligation, a data provision agreement, or a usage policy in a data space. The team activates or renews data sharing agreements. It verifies compliance with data protection law for the current data scope, for example by checking that the records of processing activities (GDPR Article 30) are up to date. It confirms that access credentials and security certificates are valid and have not expired. For sources that are new in this cycle, the full agreement process is carried out. The active agreement status is recorded and linked to the upcoming deliveries. This creates a traceable chain from legal authorization to received data, so the legal basis of each version can be shown in an audit.

Activity	Key outputs	SAF process	Other reference
Activate or renew data sharing agreements for the current cycle	Active agreement record (per source)	OVA-6 Data Supplier Management; DSP-3 Agree data use conditions	3.4.3 Service agreement
Verify data protection compliance: records of processing up to date	Compliance confirmation log	DLM-4 Classify and protect data	SAF-P11
Confirm that access credentials and security certificates are valid	Verified access credentials	PRD-4.2 Set up collection	SAF-P10
For new sources: carry out the full agreement process	New source agreement	OVA-6 Data Supplier Management; DSP-1 Join a data space; DSP-3 Agree data use conditions	SAF-P11
Record the active agreement status, linked to upcoming deliveries	Agreement-delivery link	PRD-4.2 Set up collection	

VS-01.3 Collect or Receive Data

Value created: *Secure and verifiable transfer of authentic data into the organization.*

This stage is the operational heart of the value stream, where data physically enters NSI custody. Primary data collection runs surveys through the designed and built instruments. Web questionnaires (CAWI) open, invitation links are sent, and response rates are monitored. Field interviewers (CAPI) are deployed with managed assignments, and telephone interview schedules (CATI) are carried out with callback management. Non-response is followed up within the agreed strategy. Secondary data collection receives administrative data through agreed channels. Secure file transfer (SFTP) accepts scheduled deliveries, APIs pull data on agreed schedules, and data space connectors support federated exchange. Emerging sources, such as web scraping, sensor data, and commercial feeds, are extracted according to their specification.

Equally critical is the metadata captured at the moment of receipt. For every dataset received, the ingestion process records the fields in the table below. This obligation follows from SAF-P09. It makes sure that no data enters the NSI without the context needed to understand its provenance and quality. At the end of this stage, the data is held by the NSI, but it has not yet been accepted.

Activity	Key outputs	SAF process	Other reference
Create the frame and select the sample for this cycle (surveys)	Frame and sample	PRD-4.1 Create frame and select sample	1.2.1 Primary data collection
Prepare instruments, field staff, and channels; plan the collection period	Collection ready to start	PRD-4.2 Set up collection	

Activity	Key outputs	SAF process	Other reference
Run survey collection and follow up non-response	Survey responses, paradata	PRD-4.3 Run collection	1.2.1 Primary data collection
Receive administrative data and data space deliveries through agreed channels	Received deliveries	PRD-4.3 Run collection; DSP-4 Exchange data under usage control	1.2.2 Secondary data collection
Extract data from emerging sources according to specification	Extracted data	PRD-4.3 Run collection	
Capture ingestion metadata and paradata at the point of receipt	Ingestion metadata record	PRD-4.3 Run collection; OVA-4 Process Data Management	SAF-P09

Ingestion metadata field	Description
Source identifier and organization	Who provided the data
Timestamp of receipt	When it was received
Transfer method and channel	How it was delivered (SFTP, API, CAWI, and so on)
File hash (SHA-256) and format	Integrity verification and format record
Volume (record count, file size)	Scale of the delivery
Agreement reference	Legal basis for the data
Collection mode and instrument version	Survey-specific provenance

PLACEHOLDER: VS-01 collection channel architecture: survey, administrative, and emerging-source channels arriving at one ingestion point, each adding its own metadata (source, timestamp, transfer method, file hash, volume, agreement reference), followed by VS-01.4

VS-01.4 Validate and Log Data Intake

Value created: *Confidence that received data is complete, structured, and fully traceable.*

Once data has been received, it must pass the quality gate of State #1 before it can be accepted. The validation checks completeness: expected record counts are present, required fields are populated, and the delivery matches the service level agreement. It checks structure and format against the raw data schema from *Design Data Model and Metadata Schema*. It checks security and integrity: virus scan, checksum verification (SHA-256), and digital signature validation, in line with SAF-P10. A quality assessment is carried out against the criteria from *Design Validation Rules and Quality Criteria*, producing a completeness rate, a timeliness score, and a conformance rating.

Every validation result, exception, and decision is logged in an audit trail. The trail captures data metadata (what was validated, what passed, what failed) and process metadata (who validated, when, which rules were applied, which tool version). The outcome is one of two business events. *Data Intake Validated* starts storage and registration. *Data Intake Rejected* starts exception handling according to the designed rules: notify the provider through the relationship manager, queue the delivery for manual review, or accept it conditionally with a quality flag.

Activity	Key outputs	SAF process	Other reference
Verify completeness: record counts, required fields, conformance with the service level agreement	Completeness assessment	PRD-4.4 Finalize collection	
Validate structure and format against the raw data schema	Structural validation result	PRD-4.4 Finalize collection; DLM-2 Promote data through a quality gate	
Perform security and integrity checks: virus scan, checksum, signature	Security verification result	PRD-4.4 Finalize collection	SAF-P10
Carry out the quality assessment against the designed criteria	Quality assessment scores	DLM-2 Promote data through a quality gate	
Create the audit trail: log all validation results, exceptions, and decisions	Audit trail record	OVA-4 Process Data Management	GAMSO Quality Management

VS-01.5 Store and Register Raw Data Artifacts

Value created: *Secure, organized, and traceable storage of raw data.*

This stage persists validated data and makes it findable for the rest of the statistical system. Each dataset version receives a unique persistent identifier (URI or UUID). It is stored in the raw data repository in its original format, in line with the loose coupling of SAF-P07. Its metadata is registered in the data catalog, in line with SAF-P09. Each dataset is linked to its provenance record (source, method, timing), its quality assessment from VS-01.4, the governing data sharing agreement, and the classification and coding schemes it uses. Lineage from Source Data (#0) to Raw Data (#1) is recorded. Access permissions are applied according to the access control and privacy design and the decisions of the data owner. A redelivery or correction by the provider always creates a new version; it never overwrites an earlier one. The stage ends by publishing the *Raw Data Registered* event, which tells VS-02 that new raw data is available for standardization.

Output: Raw Data (State #1), the official handover artifact to the rest of the statistical value chain. Its structure and minimum metadata set are described in *State #1: Raw Data.* **[VC10]**

Activity	Key outputs	SAF process	Other reference
Assign a unique persistent identifier (URI or UUID) to the dataset version	Dataset ID	DLM-3 Register and catalog data	
Store the data in the raw data repository in its original format	Stored raw data artifact	DLM-5 Store and version data	SAF-P07
Register metadata in the data catalog: provenance, quality, agreement, classification, lineage	Catalog registration	DLM-3 Register and catalog data; OVA-2 Metadata Management	SAF-P09
Apply access permissions according to the designed framework	Access-controlled dataset	DLM-4 Classify and protect data	SAF-P10, SAF-P11

Activity	Key outputs	SAF process	Other reference
Publish the *Raw Data Registered* event to notify VS-02	Event notification	DLM-3 Register and catalog data	SAF-P07

4. The four process types

VS-01 runs through four process types. They do not form a linear sequence but a governance cycle: each process type produces its own output, and that output feeds the others. Design produces metadata, Implementation produces the execution process, Execution produces the statistical data, and Management produces quality reports that feed back into Design. The generic pattern is described in the Overview of this chapter.

The sub-processes below are specific to VS-01. Their names describe the work in the context of data acquisition. Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. This keeps every VS-01 sub-process traceable to the reference process model in *Business Processes*.

Design: provides metadata

The design process type produces the **metadata artifacts** that govern all other process types. Without a well-designed acquisition process, execution becomes ad hoc, implementation becomes inconsistent, and management has nothing to measure against. Design runs when a new product or source is introduced, or when Management feeds back an improvement.

Design sub-process	Refines	Main outputs
Design Data Requirements and Source Strategy	PRD-1.1 Identify needs; PRD-1.2 Consult and confirm needs; PRD-1.3 Establish output objectives; PRD-1.5 Check data availability and suitability; PRD-1.6 Prepare and submit business case; PRD-2.3 Design collection	Data requirements specification, source assessment, collection strategy
Design Collection Methodology and Instruments	PRD-2.2 Design variables; PRD-2.3 Design collection; PRD-2.4 Design frame and sample	Collection methodology, questionnaire design, sample design, communication plan
Design Data Model and Metadata Schema	PRD-2.2 Design variables; DLM-1 Plan and design data assets; OVA-2 Metadata Management	Raw data schema, ingestion metadata model, catalog template
Design Validation Rules and Quality Criteria	PRD-2.5 Design processing and analysis; DLM-2 Promote data through a quality gate; OVA-1 Quality Management	Validation rule set, quality criteria, exception handling policy
Design Access Control and Privacy Framework	DLM-1 Plan and design data assets; DLM-4 Classify and protect data	Data classification, access control design, privacy impact assessment

Design Data Requirements and Source Strategy

Refines: PRD-1.1 Identify needs; PRD-1.2 Consult and confirm needs; PRD-1.3 Establish output objectives; PRD-1.5 Check data availability and suitability; PRD-1.6 Prepare and submit business case; PRD-2.3 Design collection.

The starting point of any data acquisition effort is understanding what data is needed and where it can be obtained. This sub-process translates statistical objectives into a concrete source strategy: target

populations, required variables, reference periods, and quality expectations. The team inventories and evaluates potential sources across the full spectrum: surveys, administrative registers, APIs, web sources, sensors, data spaces, and commercial data. Each source is assessed for relevance, quality, accessibility, cost, timeliness, and legal availability. Reuse of existing data comes before new collection, in line with SAF-P08. The outcome is a decision on the collection approach: primary collection (run a survey), secondary collection (reuse administrative data), or a combination of sources. The trade-offs behind that decision are documented.

Activity	Key outputs	SAF process	Other reference
Define statistical needs: target population, variables, reference period, quality expectations	Data requirements specification	PRD-1.1 Identify needs; PRD-1.2 Consult and confirm needs; PRD-1.3 Establish output objectives	1.1.1 Needs inventory
Inventory and evaluate potential data sources of all types	Source assessment report	PRD-1.5 Check data availability and suitability	SAF-P08
Assess each source for relevance, quality, accessibility, cost, timeliness, legal availability	Source assessment report	PRD-1.5 Check data availability and suitability	
Decide the collection strategy: primary, secondary, or multi-source	Collection strategy decision record	PRD-2.3 Design collection	
Document the trade-offs between source options	Collection strategy decision record; business case	PRD-1.6 Prepare and submit business case	

Design Collection Methodology and Instruments

Refines: PRD-2.2 Design variables; PRD-2.3 Design collection; PRD-2.4 Design frame and sample.

Once the source strategy is set, the methodology team designs how data will actually be collected. This involves selecting collection modes and defining how each mode works within the overall design. Examples are web, face-to-face, or telephone interviewing (CAWI, CAPI, CATI), secure file transfer (SFTP), automated machine-to-machine interfaces (API), data space connectors, and web scraping. For survey-based collection, the design includes the questionnaire: variable definitions, question wording, skip logic, and code lists. The sampling frame and sample selection method are designed to make the sample statistically representative. Where several modes are used, the team specifies which respondents receive which mode and the fallback sequence for non-response. A respondent communication strategy completes the design: invitation letters, reminders, and help desk scripts. Respondent burden is kept as low as possible.

Activity	Key outputs	SAF process	Other reference
Design collection modes (CAWI, CAPI, CATI, SFTP, API, data space connector, web scraping)	Collection methodology document	PRD-2.3 Design collection	1.2.1 Primary data collection; 1.2.2 Secondary data collection
Design questionnaire content: variable definitions, question wording, skip logic, code lists	Questionnaire design (per mode)	PRD-2.2 Design variables; PRD-2.3 Design collection	
Design the sampling frame and sample selection method	Sample design and selection criteria	PRD-2.4 Design frame and sample	

Activity	Key outputs	SAF process	Other reference
Define the multi-mode strategy and fallback sequences	Collection methodology document	PRD-2.3 Design collection	
Design the respondent communication strategy	Respondent communication plan	PRD-2.3 Design collection	3.4.2 Contact management

Design Data Model and Metadata Schema

Refines: PRD-2.2 Design variables; DLM-1 Plan and design data assets; DLM-9 Manage reference and master data; OVA-2 Metadata Management.

This sub-process defines the target data structure of Raw Data (State #1) and the metadata that must accompany every dataset at the point of ingestion. The data model specifies fields, types, formats, and constraints for the storage area where incoming data first lands, the landing zone. Equally important is the ingestion metadata model. It prescribes what must be captured at each point of receipt: provenance (source organization, contact, agreement reference), method (collection mode, instrument version, sampling parameters), timing (collection period, delivery and processing timestamps), and quality indicators (completeness rate, delivery timeliness, format conformance). Classification and coding schemes are defined, consistent with national and international standards. Finally, the structure of the metadata catalog is designed so that datasets are findable, linked, and traceable. This makes active metadata a foundation for automation downstream, as emphasized by the Metadata Management capability of CSDA (Common Statistical Data Architecture).

Activity	Key outputs	SAF process	Other reference
Define the target data structure of Raw Data (State #1): fields, types, formats, constraints	Raw data schema definition	PRD-2.2 Design variables; DLM-1 Plan and design data assets	
Define the ingestion metadata model: provenance, method, timing, quality indicators	Ingestion metadata model	OVA-2 Metadata Management	SAF-P09
Define classification and coding schemes	Classification mapping rules	DLM-9 Manage reference and master data	SAF-P09
Design the metadata catalog structure for registration and discovery	Catalog registration template	DLM-1 Plan and design data assets; OVA-2 Metadata Management	CSDA Metadata Management

Design Validation Rules and Quality Criteria

Refines: PRD-2.5 Design processing and analysis; DLM-2 Promote data through a quality gate; OVA-1 Quality Management; OVA-6 Data Supplier Management.

Before any data enters the NSI, the acceptance criteria and the rules for handling exceptions must be clear. This sub-process sets completeness thresholds, such as minimum record counts and required field coverage. It sets the parameters of service level agreements: delivery windows, response times, and format specifications. It defines structural validation rules: field types, value ranges, referential integrity, and format conformance. Security and integrity checks are specified in line with SAF-P10: virus scan requirements, checksum algorithms, and encryption standards. The quality dimensions to track are agreed: completeness, timeliness, conformance, and integrity. Exception handling rules distinguish between rejection, conditional acceptance with quality flags, and routing to manual review. Together, these rules make up the VS-01 part of the quality gate of State #1. The Quality Management capability of GAMSO provides the overall control framework.

Activity	Key outputs	SAF process	Other reference
Define completeness thresholds and service level parameters	Quality criteria document, service level template	PRD-2.5 Design processing and analysis; OVA-6 Data Supplier Management	3.4.3 Service agreement
Define structural validation rules: field types, value ranges, referential integrity	Validation rule set	PRD-2.5 Design processing and analysis	
Define security and integrity check specifications	Validation rule set	PRD-2.5 Design processing and analysis	SAF-P10
Define the quality dimensions to track	Quality criteria document	OVA-1 Quality Management	GAMSO Quality Management
Design exception handling rules: rejection, conditional acceptance, manual review	Exception handling policy	DLM-2 Promote data through a quality gate	

Design Access Control and Privacy Framework

Refines: DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-4 Process Data Management.

This last design sub-process makes sure that security and privacy are built in from the start, not added afterwards. Each data source receives a classification for availability, integrity, and confidentiality, in line with SAF-P09. Role-based access controls for the raw data repository are designed in line with SAF-P10: who can read, write, and administer. Privacy protection follows SAF-P11. It covers the moment of pseudonymization (when direct identifiers are removed or replaced), purpose limitation (which uses each agreement allows), the legal basis for processing (for example, which ground of GDPR Article 6 applies to each source), and data minimization (only necessary data is collected). Audit logging requirements, which define the access events that must be recorded, complete the design.

Activity	Key outputs	SAF process	Other reference
Define the data classification (availability, integrity, confidentiality) per source	Data classification policy	DLM-4 Classify and protect data	SAF-P09
Design role-based access control for the raw data repository	Access control design	DLM-1 Plan and design data assets; DLM-4 Classify and protect data	SAF-P10; 3.3.4 Logical identity and access control
Design privacy protection: moment of pseudonymization, purpose limitation, legal basis, data minimization	Privacy impact assessment	DLM-4 Classify and protect data	SAF-P11; 3.3.1 Privacy management
Design audit logging requirements	Audit logging specification	OVA-4 Process Data Management	SAF-P10

PLACEHOLDER: VS-01 design process decomposition: inputs (statistical needs, data sources, framework standards, legal and privacy requirements), the five design sub-processes in order, and the design document each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

The implementation process type takes the metadata artifacts from Design and **builds, configures, and tests** the acquisition process until it is ready for production. Nothing runs operationally until this process type is complete. Components are reused where possible and built only where needed, in line with SAF-P01.

Implementation sub-process	Refines	Main outputs
Build Collection Instruments and Channels	PRD-3.1 Reuse or build collection instruments; DSP-1 Join a data space	Production-ready instruments and channels
Configure Data Ingestion Workflows	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Automated validation, metadata capture, exception routing, events
Build Metadata Registration and Cataloging	PRD-3.4 Configure workflows; DLM-3 Register and catalog data; OVA-2 Metadata Management	Catalog, automated metadata propagation, discovery
Test and Validate the Pipeline	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems	Test reports, production-ready process, operational procedures

Build Collection Instruments and Channels

Refines: PRD-3.1 Reuse or build collection instruments; DSP-1 Join a data space.

This sub-process turns the collection methodology into operational tools and channels. For survey-based collection, it builds or configures the instruments. A CAWI portal gets the designed question logic, support for several languages, and a layout that works on any device. A CAPI application gets offline capability and the designed questionnaire. CATI scripts get the designed routing logic. Paper forms are laid out for automated scanning (OCR) or manual data entry. For secondary and emerging sources, the receiving channels are configured. SFTP endpoints get the designed encryption and authentication, API connections get rate limiting and error handling, and data space connectors are set up according to their specifications. Secure transfer channels with end-to-end encryption are set up in line with SAF-P10. A respondent management system handles invitations, reminders, and tracking.

Activity	Key outputs	SAF process	Other reference
Build or configure survey instruments per mode (CAWI, CAPI, CATI, paper)	Production-ready instruments	PRD-3.1 Reuse or build collection instruments	1.1.3 Build statistics product
Configure data receipt channels (SFTP, API, data space connectors)	Configured channel endpoints	PRD-3.1 Reuse or build collection instruments; DSP-1 Join a data space	
Set up secure transfer channels with end-to-end encryption	Secure channel configuration	PRD-3.1 Reuse or build collection instruments	SAF-P10
Configure the respondent management system (invitations, reminders, tracking)	Respondent management system	PRD-3.1 Reuse or build collection instruments	3.4.2 Contact management

Configure Data Ingestion Workflows

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-2 Promote data through a quality gate.

This sub-process translates the designed validation rules and metadata capture requirements into automated workflows. Structural validation is implemented first: schema conformance, field types, and required fields. Alongside it come integrity checks (checksum verification, virus scanning, digital signature validation), completeness checks (record counts, field coverage thresholds), and service level checks (delivery timeliness, format conformance). Metadata capture at each ingestion point is automated, so that provenance, format, timestamps, and file hashes are extracted without manual work. The landing zone for raw data is set up with the designed access controls and storage policies. Event notifications are configured, so that VS-02 is alerted when raw data is registered. Exception handling workflows route rejected data to the right queue: provider notification, manual review, or conditional acceptance.

Activity	Key outputs	SAF process	Other reference
Implement validation rules as automated checks (structural, integrity, completeness, service level)	Automated validation pipeline	PRD-3.2 Reuse or build processing and analysis components	
Configure metadata capture at each ingestion point	Automated metadata extraction	PRD-3.4 Configure workflows	SAF-P09
Set up the raw data landing zone with access controls and storage policies	Configured repository	PRD-3.4 Configure workflows; DLM-4 Classify and protect data	
Configure event notifications for downstream consumers	Publication of the *Raw Data Registered* event	PRD-3.4 Configure workflows	SAF-P07
Configure exception handling workflows	Rejection routing and manual review queues	PRD-3.4 Configure workflows; DLM-2 Promote data through a quality gate	

Build Metadata Registration and Cataloging

Refines: PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-6 Provide data for reuse; OVA-2 Metadata Management.

This sub-process implements the metadata catalog and the automated registration that make every dataset findable and traceable. Catalog entries are built from the metadata model of *Design Data Model and Metadata Schema*. Metadata is propagated automatically into the catalog from three sources: the collection instruments (variables, code lists, question wording), the transfer channels (source, method, timing), and the validation workflow (validation results, completeness scores). Metadata quality validation makes sure that every registered dataset has all required metadata fields. Search and discovery functions let downstream consumers find raw data and its metadata. As CSDA emphasizes, the ultimate goal is active metadata: metadata that can generate collection instruments, validation rules, and processing specifications automatically.

Activity	Key outputs	SAF process	Other reference
Implement catalog entries based on the designed metadata model	Populated metadata catalog	DLM-3 Register and catalog data; OVA-2 Metadata Management	

Activity	Key outputs	SAF process	Other reference
Configure automatic metadata propagation from instruments, channels, and validation	Automated metadata pipelines	PRD-3.4 Configure workflows	CSDA Metadata Management
Set up metadata quality validation	Metadata completeness checks	OVA-2 Metadata Management	SAF-P09
Implement metadata search and discovery for downstream consumers	Discovery interface	DLM-6 Provide data for reuse	CSDA Metadata Management

Test and Validate the Pipeline

Refines: PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems.

Before the acquisition process goes into production, it must be thoroughly verified. Technical testing covers all individual components: instruments, channels, validators, and catalog. Integration testing checks the end-to-end flow from receipt through validation to catalog registration. A field test or pilot with real or synthetic data runs on a small scale with actual data providers. It verifies that metadata is captured and propagated correctly, that quality criteria are enforced at each checkpoint, and that exception handling works as designed. Performance testing confirms that the process can handle the expected volumes within the agreed delivery windows. Security testing, such as penetration testing of transfer channels and verification of access controls, makes sure the process meets SAF-P10. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	
Test integration: end-to-end flow from receipt to registration	Integration test report	PRD-3.5 Test production systems	
Run a field test or pilot with real or synthetic data	Field test report, validated metadata propagation	PRD-3.6 Test statistical business process	
Test performance against expected volumes	Performance test report	PRD-3.5 Test production systems	
Test security: penetration testing, access control verification	Security test report	PRD-3.5 Test production systems	SAF-P10; 3.3.2 Security planning
Finalize the production system and document operational procedures	Operational procedures manual, production-ready process	PRD-3.7 Finalize production systems	

PLACEHOLDER: VS-01 implementation pipeline: the five design artifacts feeding the three build sub-processes, followed by technical testing, field test, and security and performance testing, resulting in a production-ready acquisition process and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-01 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when a collection period opens or when a provider's delivery arrives. The

stages then run in order, but for continuous sources, such as daily web data, VS-01.3 to VS-01.5 repeat for every delivery. Execution applies the designs and the solution as they are; anything that needs a new rule or a new channel goes back to Design. Every delivery ends in one of three outcomes: registered, registered with quality flags, or returned to the provider.

Aspect	VS-01
Trigger	*Collection period opened*; *Delivery received*
Cadence	Per survey cycle; per delivery for administrative and web data
Performed by	ROL-13 Data collection manager, ROL-14 Data quality analyst, ROL-03 Data steward
Input	Source data (#0), active agreements, production-ready instruments
Output	Registered raw data version (#1) and its intake log
Control point	Gate of State #1 in VS-01.4
Exception path	Rejection to the provider via ROL-23 Relationship manager; structural change at the source back to Design

Management: provides quality reports

The management process type runs alongside and after execution. It monitors the performance of the acquisition process, assesses the quality of the ingested data, makes sure the domain complies with the rules, and feeds improvements back into Design. This closes the governance cycle.

Management sub-process	Refines	Main outputs
Monitor Process Performance	OVA-4 Process Data Management; OVA-6 Data Supplier Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard, service level compliance report
Assess Data Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Data quality report, quality scorecards, trend analysis
Manage Metadata Lifecycle	OVA-2 Metadata Management; DLM-8 Monitor data quality, lineage and use	Metadata quality report, catalog maintenance log, lineage audit
Audit and Compliance Reporting	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Audit trail summary, compliance reports to VS-07
Continuous Improvement	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, improvement tracking log

Monitor Process Performance

Refines: OVA-4 Process Data Management; OVA-6 Data Supplier Management; PRD-8.1 Gather evaluation inputs.

The operational performance of VS-01 must be tracked systematically across production cycles. The process owner and the quality manager collect and aggregate the key performance indicators (see attribute 9). They monitor compliance with the service level agreements of all data providers. They track paradata: who performed each step, when, how long it took, which tools were used, and which errors occurred. This information feeds dashboards that give the process owner a current view of the health of the domain. The approach follows the Quality Management capability of GAMSO for monitoring performance across the production cycle.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per production cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Monitor service level compliance of all data providers	Service level compliance report	OVA-6 Data Supplier Management	3.4.3 Service agreement
Track paradata: operator, timing, tools, errors	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Assess Data Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the ingested data itself is evaluated against the criteria set in *Design Validation Rules and Quality Criteria*. Each dataset is assessed on several quality dimensions. Completeness covers record counts against expectations and field coverage. Timeliness compares the delivery date with the agreed deadline. Structural accuracy covers format conformance and valid value ranges. Consistency covers consistency between fields and with earlier deliveries. Conformance covers alignment with the designed schema and classification schemes. Quality indicators are produced per dataset, per source, and per production cycle. Quality trends are tracked to detect declining sources or modes with consistently lower quality. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Evaluate the quality dimensions per dataset: completeness, timeliness, accuracy, consistency, conformance	Data quality report (per cycle)	OVA-1 Quality Management	CSDA Data Quality
Compare the results with the designed quality criteria	Quality assessment against baseline	OVA-1 Quality Management	2.1.3 Statistical quality management
Produce quality indicators per dataset, per source, per cycle	Quality scorecards (per source, per mode)	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality
Identify quality trends across sources and modes	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management

Manage Metadata Lifecycle

Refines: OVA-2 Metadata Management; DLM-8 Monitor data quality, lineage and use.

The metadata that supports data acquisition must stay complete, accurate, and current throughout the data life cycle. This sub-process verifies metadata completeness: every registered dataset must have all required metadata fields, as a compliance check on SAF-P09. It assesses metadata quality: are provenance records accurate, are classifications applied correctly, and are agreement references current? The metadata catalog is actively maintained. Outdated entries are retired, schemas are updated for new source formats, and metadata models are versioned. Lineage from source through ingestion to raw data is tracked. Reconciliation across systems makes sure that catalog entries match the artifacts actually stored. As CSDA emphasizes, metadata carries all the context needed to understand and reuse data; without it, data is effectively useless.

Activity	Key outputs	SAF process	Other reference
Verify metadata completeness for all registered datasets	Metadata quality report	OVA-2 Metadata Management	SAF-P09
Assess metadata quality: accuracy, currency, correct classification	Metadata quality report	OVA-2 Metadata Management	CSDA Metadata Management
Maintain the metadata catalog: retire entries, update schemas, version models	Catalog maintenance log	OVA-2 Metadata Management	CSDA Metadata Management
Track metadata lineage from source through ingestion to raw data	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Metadata Management
Reconcile metadata across systems	Catalog-artifact reconciliation	OVA-2 Metadata Management	SAF-P09

Audit and Compliance Reporting

Refines: OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use.

This sub-process produces the evidence that VS-01 operates within legal, regulatory, and organizational requirements. Audit trail reports show the full traceability of every dataset, from the external source to the raw data repository. Data protection reporting covers the records of processing activities (GDPR Article 30), reviews of data protection impact assessments for high-risk processing, and the ability to show which data the NSI holds and why. Agreement compliance reporting confirms that all use of data stays within the terms of the governing agreements. Security compliance reporting verifies that all designed security measures have been applied and documents any breaches or incidents. These reports meet the obligation of SAF capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance.*

Activity	Key outputs	SAF process	Other reference
Produce audit trail reports: full traceability from source to raw data	Audit trail summary	OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	4.1.5 Accountability
Report on data protection compliance: records of processing, impact assessments, data subject rights	Records of processing activities	OVA-1 Quality Management	SAF-P11; 3.3.1 Privacy management
Report on agreement compliance	Agreement compliance report	OVA-1 Quality Management	4.1.5 Accountability
Report on security compliance: measures applied, breaches, incidents	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10; 3.3.3 Security incident management

Continuous Improvement

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This last management sub-process closes the governance cycle by analyzing performance and quality data and driving improvements. Quality trends across production cycles are examined: are rejection rates decreasing, is metadata completeness improving, and are processing times getting shorter? Improvement opportunities are identified. Examples are new sources that could replace sources of lower quality, collection modes that reduce burden or increase response, automation of manual ingestion steps, and simplification of validation rules that produce too many false alarms. Recommendations are fed back to the design

sub-processes, prioritized by impact and feasibility, and tracked through the following cycles. Where a new method or source is needed, the recommendation goes to VS-08. This is SAF capability 4.1.4 *Improvement management* in practice. It also connects to the GAMSO idea of transferring mature results of innovation to production. **[VC11]**

Activity	Key outputs	SAF process	Other reference
Analyze quality trends across production cycles	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities: new sources, better modes, automation, simpler rules	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Feed recommendations back to the design sub-processes or to VS-08	Design feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management

PLACEHOLDER: VS-01 management feedback loop: data and paradata from Execution flowing through the five management sub-processes, with dashboards to the process owner, quality reports to the quality manager, compliance reports to VS-07, and improvement recommendations back to Design

Metadata and paradata per process type

Metadata is produced in every process type, not only in Design. The table shows what each process type contributes. Data metadata describes the datasets. Process metadata, also called paradata, describes how the process itself ran: who, when, and with which tools.

Process type	Data metadata produced	Process metadata (paradata) produced
Design	Data models, variable definitions, classification schemes, expected formats, quality thresholds	Log of design decisions, methodology version, review and approval records
Implementation	Schema configurations, parameters of validation rules, catalog templates, mappings from instruments to metadata	Build log, test results, records of configuration changes, deployment approvals
Execution	Provenance, source, timestamp, file hash, volume, quality indicator values, and agreement reference per dataset	Operator identity, timing, volumes, errors, collection mode, instrument version
Management	Quality scores, lineage assessments, completeness reports, life cycle status, trend analyses	Dashboards of key performance indicators, quality trend analyses, improvement decisions, audit findings

PLACEHOLDER: VS-01 business object lifecycle: design artifacts govern implementation assets, which produce the process that turns Source Data (#0) into Raw Data (#1), generating provenance, quality, and audit records that management assesses and feeds back to design

5. Actors and roles

VS-01 involves more external actors than any other value stream, because it is where the NSI meets its data providers. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the system configurator in the original VS-01 description is ROL-11 Platform engineer. The long-term relationship with each provider is organized in the collaboration COL-01 *Data provision partnership*.

Actor or role	ID	Process type	Responsibility in VS-01
Data provider	ACT-03	External	Supplies data under a data sharing agreement: a government agency, a business, or an individual
Respondent	ACT-04	External	Provides data through survey instruments: web (CAWI), face-to-face (CAPI), telephone (CATI), or paper
Supervisory and legal authority	ACT-08	External	Grants or oversees the legal basis for collection, for example under the statistics law or data protection law
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-01
Data owner	ROL-02	Management	Accountable for State #1; decides on access and retention
Data steward	ROL-03	Execution	Registers datasets in the catalog, manages metadata entries, applies access controls, resolves gate exceptions
Data custodian	ROL-04	Execution	Operates storage, access control, and retention
Statistical designer	ROL-06	Design	Translates statistical needs into data requirements and process specifications
Methodologist	ROL-07	Design	Designs collection methodology, sampling strategies, and questionnaire content
Metadata and data architect	ROL-08	Design	Designs data models, metadata schemas, and classification mappings
Data engineer	ROL-10	Implementation	Builds and configures ingestion workflows, collection instruments, and data channels

Actor or role	ID	Process type	Responsibility in VS-01
Platform engineer	ROL-11	Implementation	Configures platforms, SFTP endpoints, API connectors, and access controls
Test manager	ROL-12	Implementation	Plans and carries out testing of the acquisition process: technical tests, field tests, integration tests
Data collection manager	ROL-13	Execution	Oversees collection campaigns, coordinates collection teams, and monitors response
Data quality analyst	ROL-14	Execution	Performs validation checks, assesses data quality at intake, and logs exceptions
Quality manager	ROL-18	Management	Monitors key performance indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures legal and regulatory compliance and produces audit reports
Relationship manager	ROL-23	Execution	Maintains provider relationships and agreements; handles rejected deliveries with the provider

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-01 relies on, and what role they play. NSIs can use it to map the value stream to their own organization.

Capability group	Capabilities	Role in VS-01
1.1 Statistical design and build	1.1.1 Needs inventory, 1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the data requirements and the collection methodology, and build the acquisition process
1.2 Data collection	1.2.1 Primary data collection, 1.2.2 Secondary data collection	Carry out the collection: surveys, intake of administrative data, extraction through APIs
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage data and metadata during ingestion, assess quality, maintain registers
3.1 Business operations	3.1.2 Purchasing and contract management	Manage contracts with data providers that are suppliers

Capability group	Capabilities	Role in VS-01
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain collection instruments and ingestion systems
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.3 Security incident management, 3.3.4 Logical identity and access control	Build privacy and security into all acquisition processes
3.4 Customer and partner relations	3.4.2 Contact management, 3.4.3 Service agreement	Maintain relationships with respondents and providers, and manage data provision agreements
4.1 Governance	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, plan improvements, report on compliance

Stages, processes, and functions

The table shows which SAF business processes realize each stage of VS-01 and which business functions they mainly use.

Stage	Realized by processes	Main functions
VS-01.1 Identify Data Needs and Sources	PRD-1.1 Identify needs; PRD-1.5 Check data availability and suitability; PRD-2.3 Design collection	FUN-3.1 Design and build statistical products; FUN-6.1 Manage data supplier relationships
VS-01.2 Establish Data Agreements and Access Rights	OVA-6 Data Supplier Management; DSP-1 Join a data space; DSP-3 Agree data use conditions	FUN-6.2 Manage data provision agreements; FUN-9.1 Manage data space participation; FUN-10.5 Manage legal affairs
VS-01.3 Collect or Receive Data	PRD-4.1 Create frame and select sample; PRD-4.2 Set up collection; PRD-4.3 Run collection; DSP-4 Exchange data under usage control	FUN-3.2 Collect and acquire data; FUN-9.2 Operate data space connectivity
VS-01.4 Validate and Log Data Intake	PRD-4.4 Finalize collection; DLM-2 Promote data through a quality gate	FUN-3.2 Collect and acquire data; FUN-7.3 Manage data lifecycle and storage
VS-01.5 Store and Register Raw Data Artifacts	DLM-3 Register and catalog data; DLM-4 Classify and protect data; DLM-5 Store and version data	FUN-7.2 Manage metadata; FUN-7.3 Manage data lifecycle and storage; FUN-7.5 Protect data

PLACEHOLDER: ArchiMate view VS-01 (b) stages realized by processes and functions

Framework mappings

This reference table maps each process type of VS-01 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-1.1 to PRD-1.3, PRD-1.5, PRD-1.6, PRD-2.2 to PRD-2.5, DLM-1, DLM-2, DLM-4, DLM-9, OVA-1, OVA-2, OVA-4, OVA-6	Phase 1 Specify Needs (1.1 to 1.3, 1.5, 1.6), Phase 2 Design (2.2 to 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management	1.1.1 Needs inventory, 1.1.2 Design statistics product, 2.1.2 Metadata management
Implementation	PRD-3.1, PRD-3.2, PRD-3.4 to PRD-3.7, DLM-2, DLM-3, DLM-4, DLM-6, OVA-2, DSP-1	Phase 3 Build (3.1, 3.2, 3.4 to 3.7)	Capability Management	Data Ingestion (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-1.1, PRD-1.5, PRD-4.1 to PRD-4.4, DLM-2 to DLM-5, OVA-2, OVA-4, OVA-6, DSP-1, DSP-3, DSP-4	Phase 4 Collect (4.1 to 4.4); 1.1 and 1.5 in their recurrent form	Production (execution)	Data Ingestion (execution)	1.2.1 Primary data collection, 1.2.2 Secondary data collection
Management	OVA-1, OVA-2, OVA-4, OVA-6, DLM-8, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-01 most directly.

Principle	Statement	Application in VS-01
SAF-P01	Reuse before open source, open source before buying, buying before building	Collection instruments, connectors, and ingestion tooling are reused across products first. Where new tooling is needed, open-source solutions, preferably developed with other NSIs, come before purchase or own development. **[VC3]**

Principle	Statement	Application in VS-01
SAF-P07	Implement loosely coupled processes and systems	Data acquisition is independent from downstream processing. Raw Data (State #1) is the decoupling interface: downstream consumers use it without knowing how it was collected.
SAF-P08	Data is shared property	Raw data, once registered, is available to all authorized consumers. Collection is not duplicated across statistical products when the same source data can serve several purposes. This also reduces response burden.
SAF-P09	No data without metadata and classification	At ingestion, every dataset is enriched with provenance (where it came from and how), source, method, timestamp, quality indicators, and a classification for availability, integrity, and confidentiality. This is not a precondition that blocks ingestion, but an obligation the NSI fulfills during acquisition.
SAF-P10	Security by design	Security is built into the acquisition process from the start: encrypted transfer channels, virus scanning, checksum verification, and access controls.
SAF-P11	Privacy by design	Privacy protection is designed into the acquisition process from the start, not added after data has entered: the moment of pseudonymization, purpose limitation, the legal basis for processing, and data minimization.

i Principles that apply across the process types

Several principles apply to all four process types. SAF-P09 applies everywhere: Design defines the target metadata, Execution captures it at ingestion, and Management assesses it and closes remaining gaps. SAF-P10 applies to *Design Access Control and Privacy Framework*, to *Build Collection Instruments and Channels*, and to the stages VS-01.3 to VS-01.5. SAF-P11 starts in *Design Access Control and Privacy Framework* and carries through all process types. SAF-P07 governs the handover of State #1 to VS-02.

8. Interfaces, services, and events

Boundary interfaces

VS-01 is loosely coupled (SAF-P07) to its neighbors through two boundary interfaces. Specifying each boundary explicitly makes the loose-coupling contract concrete and testable: the handover artifact, the metadata it must carry, the event that triggers it, and the point where responsibility transfers. Each interface exchanges a registered, immutable artifact plus its metadata, never internal working data. Neither side therefore needs knowledge of the other's internal processing.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
Provider or respondent → VS-01 (intake)	Delivery or survey response (#0)	Source identifier and organization, transfer method and channel, receipt timestamp, file hash, volume, agreement reference, collection mode and instrument version (surveys)	*Delivery received* or response submitted	The provider is responsible up to receipt; VS-01 takes custody and the obligation to capture metadata (SAF-P09) at the point of receipt
VS-01 → VS-02 (or VS-03) (handover)	Registered raw data version (#1)	Minimum metadata set of State #1: provenance, source, collection method, receipt timestamp, file hash, agreement reference, quality assessment, classification	*Raw Data Registered*	VS-01 keeps custody of the raw artifact; the consumer takes responsibility for further processing on receipt of the event

Business services

Service	Offered to	Realized in	Interfaces
Data collection service	Respondents, data providers	VS-01.3 Collect or Receive Data	Web portals (CAWI), mobile applications (CAPI), telephone systems (CATI), paper, SFTP endpoints, REST APIs, data space connectors
Data agreement service	Data providers, legal authorities	VS-01.2 Establish Data Agreements and Access Rights	Agreement management, relationship manager
Raw data provisioning service	VS-02 Data Standardization and Integration, VS-03 Data Editing and Enrichment, other domains	VS-01.5 Store and Register Raw Data Artifacts	Event notification (*Raw Data Registered*) and a catalog query interface to find and access raw datasets
Ingestion monitoring service	Process owner, quality manager	Management sub-processes *Monitor Process Performance* and *Assess Data Quality*	Dashboards and quality reports
Metadata catalog service	All data consumers in the NSI	Management sub-process *Manage Metadata Lifecycle*	Search and browse interface for metadata, with metadata exchange compliant with SDMX
Design and methodology service	Methodologists, statistical designers, data engineers	All Design sub-processes of VS-01	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artifacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Collection period opened	Start of VS-01.3 for a survey cycle (PRD-4.3 Run collection)
Agreement expired or changed	A review in VS-01.2 before the next delivery
Delivery received	Start of VS-01.4 for that delivery
Data Intake Validated / Data Intake Rejected	Outcome of VS-01.4; rejection starts provider follow-up
Raw Data Registered	Completion of VS-01.5; triggers the consuming value stream
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-01 event choreography between data provider, NSI teams, and process management across Design, Implementation, Execution (agreement, primary or secondary collection, validation passed or failed), and Management (key performance indicators, quality, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-01	Receives from VS-01
VS-07 Governance and Compliance	Legal register, privacy and security policy	Audit trails, compliance evidence
VS-08 Innovation and Methodological Development	New sources and collection methods	Improvement needs
VS-09 Infrastructure and Platform Enablement	Channels, storage, catalog, event interfaces	Capacity and channel requirements
VS-10 Customer and Stakeholder Engagement	Provider relationships and burden insight	Provider feedback and issues

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
Response rate	Share of sampled units that provided data (surveys)	Per survey, per mode, per cycle
Delivery timeliness	Share of deliveries received within the agreed window	Per source, per cycle
Rejection rate	Share of deliveries that failed the gate of State #1	Per source, per cycle
Processing time	Time from receipt to catalog registration	Per delivery, per cycle
Metadata completeness	Share of required metadata fields populated at registration	Per delivery
Exception rate	Frequency of exception handling per check rule	Per rule, per cycle

10. Toolkit

Examples from NSI practice, the maturity checklist, and end-to-end scenarios for VS-01 are part of the SAF Toolkit.

Toolkit: SAF-TK-V01 VS-01 Data Acquisition and Ingestion companion [**VC17**]

VS-02 Data Standardization and Integration

1. Purpose and value created

VS-02 makes data from different sources speak the same language. It takes raw data that is already in NSI custody and brings it into the common structure, format, and classifications of the NSI. It does this without changing any value. Only the representation changes: fields are renamed and restructured, types are cast, source codes are mapped one-to-one to standard codes, and descriptive metadata is attached. Direct identifiers are removed or replaced with pseudonymous keys. The value created is **comparability without loss of content**. Data that looked different at the source now uses the same structure, the same codes, and documented definitions, units, and reference periods. Because no value changes, the trace back to the authentic delivery in State #1 stays simple and deterministic.

VS-02 corresponds to the business domain **Standardization without loss of content** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

- **Data quality analyst or operations staff:** the stages in attribute 3, which describe what happens in each production cycle.
- **Methodologist, classification steward, or data architect:** the Design processes in attribute 4.
- **System builder or IT staff:** the Implementation processes in attribute 4.
- **Quality or compliance manager:** the Management processes in attribute 4.
- **Manager who wants the big picture:** attributes 1, 2, and 8. Examples and end-to-end scenarios are part of the SAF Toolkit.

> **i** Signature discipline: standardization without loss of content
>
> VS-02 never changes a value. It re-expresses the structure and format, adds standard codes through deterministic one-to-one mapping, and attaches metadata. Unit and currency conversion, reference-period alignment, reconciliation of differing definitions, derivation, and interpretive coding of free text are not VS-02 activities. They belong to VS-03, where every change is documented as an edit. The removal or pseudonymization of direct identifiers is not a loss of content, because identifiers are personal data, not statistical content.

PLACEHOLDER: ArchiMate view VS-02 (a) value stream and capabilities

2. Position and state transition

VS-02 is the second primary value stream. It sits between acquisition (VS-01) and editing (VS-03). It receives Raw Data when VS-01 publishes the *Raw Data Registered* event, and it hands over Standardized Data to VS-03.

VS-02 takes data from State #1 to State #2. Before VS-02, the data is a registered raw version in its original form, with direct identifiers still present. After VS-02, it is a registered, immutable version in the canonical target data model, with standard codes stored next to the original source values, with recorded units and reference periods, and without direct identifiers. The quality gate, the minimum metadata set, and the protection of State #2 are described in *State #2: Standardized Data* and are not repeated here.

State #2 is optional. A source that already arrives in the NSI's standard structure and coding can go directly from #1 to #3. In that case the NSI records in the catalog that #2 was skipped, and why (see *Applying the steady states pragmatically*). Where #2 is skipped, the removal or pseudonymization of direct identifiers must still take place before #3, and it must still be recorded. **[VC12]**

From	To	Gate	Event out
#1 Raw Data	#2 Standardized Data	Gate of State #2 (structural conformance, classification conformance, mapping coverage, semantic metadata, no loss of content, identifier protection, referential integrity, metadata and lineage)	*Standardized Data Registered*

PLACEHOLDER: ArchiMate view VS-02 (c) state handover and events: Raw Data (#1) in the upstream domain, the standardization service at the boundary (profiling, one-to-one code mapping, semantic alignment, conformance check), Standardized Data (#2) inside the standardization domain

3. Stages

The stages describe the value that VS-02 adds in every production cycle. Together they form the execution process type of VS-02. In a mature steady state, the target model, the correspondence tables, and the conformance rules have been designed and built in advance (see attribute 4). Execution then focuses on applying them, capturing lineage and conformance metadata, and passing the quality gate without changing any value.

Stage	Name	Value added
VS-02.1	Profile and Assess Data Structure	It is clear how far the raw data is from the target standard
VS-02.2	Apply Standard Classifications and Formats	Source codes are expressed in standard classifications and a common format; no value is changed
VS-02.3	Align Metadata and Semantics	The same concept means the same thing across datasets
VS-02.4	Validate Structural and Semantic Conformance	The data conforms to the standards, and no content was lost
VS-02.5	Register Standardized Artifacts	Standardized data is registered and available for reuse (State #2)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-02.1 Profile and Assess Data Structure

Value created: *A clear view of how far the incoming raw data is from the target standard, so that the right mapping and format coercion can be applied.*

The stage starts when VS-01 publishes the *Raw Data Registered* event. The profiling components analyze the structure of the incoming version. They infer types, measure cardinalities and value distributions, and detect missing fields and structural anomalies. The result is compared with the target data model. This produces a structure-gap assessment: which fields map directly, which need format coercion, which source codes have a standard correspondence, and which values are uncoded and must go to VS-03. A significant

or unexpected change in structure, such as a source that has changed its schema, goes back to Design. It is not solved during the cycle.

Activity	Key outputs	SAF process	Other reference
Profile the incoming raw data: types, cardinalities, distributions, anomalies	Data profiling report	PRD-5.1 Integrate data	1.3.1 Data standardization
Compare the structure with the target data model	Structure-gap assessment	PRD-5.1 Integrate data	
Identify the format coercions and code mappings to apply	Mapping plan for the dataset	PRD-5.1 Integrate data; PRD-5.2 Classify and code	
Flag unexpected structural deviations for design review	Schema-change exception (if any)	PRD-5.1 Integrate data	

VS-02.2 Apply Standard Classifications and Formats

Value created: *Source codes re-expressed in shared, authoritative classifications and in a common format, so that data is comparable across sources, with no value changed.*

This is the core stage of VS-02. The mapping and format-coercion component applies the designed value-preserving operations: renaming, restructuring, type casting, reformatting, and key alignment. The mapping services then translate values that already carry a source code into the standard classification, such as NACE, ISCO, COICOP, NUTS, or SDMX code lists. The translation uses maintained correspondence tables, one-to-one, and the original value is kept next to the standard code. Values without a correspondence, and free text that needs interpretation, are not coded here. They are flagged and listed for VS-03. Every applied mapping and classification version is recorded as lineage metadata, in line with SAF-P09.

Activity	Key outputs	SAF process	Other reference
Apply format coercion: rename, restructure, cast, reformat, align keys	Format-coerced dataset (values unchanged)	PRD-5.1 Integrate data	1.3.1 Data standardization
Map source codes to standard classifications one-to-one through correspondence tables	Standard-coded dataset (original values kept)	PRD-5.2 Classify and code; DLM-9 Manage reference and master data	SAF-P09 No data without metadata and classification
Flag unmapped values and free text, and list them for VS-03 Data Editing and Enrichment	List of unmapped values	PRD-5.2 Classify and code	
Record the applied mappings and classification versions as lineage	Mapping and classification lineage	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

PLACEHOLDER: VS-02 classification mapping: survey, administrative, and emerging-source inputs passing format coercion and one-to-one correspondence tables; values with a source-code correspondence become standard-coded values with the original kept; unmapped values and free text are routed to VS-03 for interpretive coding

VS-02.3 Align Metadata and Semantics

Value created: *Coherent, shared meaning across datasets, so that the same concept means the same thing everywhere.*

Once structure and codes are standardized, the descriptive metadata is harmonized. This happens at the level of metadata only; no value changes. Each source variable is linked to its canonical concept in the concept system. The unit of measure, the currency, and the reference-period basis of each variable are recorded as received. They are not converted. Where two sources define the same concept differently, the difference is recorded as metadata and flagged for VS-03, which decides how to reconcile it. Quality descriptors, such as completeness and confidentiality flags, are standardized and attached. Downstream consumers then receive both the standardized data and the meaning they need to interpret it.

Activity	Key outputs	SAF process	Other reference
Link source variables to canonical concepts	Concept-aligned variables	OVA-2 Metadata Management	CSDA Metadata Management
Record unit, currency, and reference-period basis per variable (no conversion)	Unit and period metadata	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Record differing cross-source definitions and flag them for VS-03 Data Editing and Enrichment	Definition-difference record	OVA-2 Metadata Management	2.1.4 Statistical register management
Standardize and attach quality descriptors	Harmonized quality metadata	OVA-2 Metadata Management; OVA-1 Quality Management	CSDA Metadata Management

VS-02.4 Validate Structural and Semantic Conformance

Value created: *Confidence that the standardized data conforms to the standards and that no content was lost.*

Before standardized data can be accepted, it must pass the quality gate of State #2. The automated checks from *Design Conformance Validation Rules* run on every dataset. Structural validation checks types, formats, ranges, cardinalities, and referential integrity against the target model. Classification conformance checks that every mapped code exists in the stated classification version, that there are no orphan codes, and that mapping coverage is above the threshold. Semantic conformance checks that units, definitions, and reference periods are recorded for every variable. The content-preservation checks confirm that record counts and key measures are identical to the #1 version, and that no value has changed. Identifier protection is checked against the privacy design.

Every result, exception, and decision is logged in an audit trail. The trail captures data metadata (what was checked, what passed, what failed) and process metadata (who checked, when, which rule version and which engine version). The outcome is one of two business events. *Standardization Conformed* starts registration. *Standardization Non-Conformant* starts exception handling according to the designed rules: re-map, correct the correspondence table, or accept conditionally with a conformance flag. Conditional acceptance is a normal outcome for many sources, because unmapped codes and unmatched keys often have a known cause, such as the lag in registering new businesses.

Activity	Key outputs	SAF process	Other reference
Validate the structure against the target model, including referential integrity	Structural validation result	PRD-5.3 Review and validate; DLM-2 Promote data through a quality gate	

Activity	Key outputs	SAF process	Other reference
Validate classification conformance: valid codes, correct version, coverage	Classification conformance result	PRD-5.3 Review and validate; DLM-2 Promote data through a quality gate	
Validate semantic conformance: units, definitions, reference periods	Semantic conformance result	DLM-2 Promote data through a quality gate	
Run the no-loss-of-content checks: record counts and key measures	Content-preservation result	DLM-2 Promote data through a quality gate	GAMSO Quality Management
Check identifier protection against the privacy design	Identifier protection result	DLM-4 Classify and protect data	SAF-P11 Privacy by design
Create the audit trail: log all results, exceptions, and decisions	Audit trail record	OVA-4 Process Data Management	SAF-P10 Security by design

! Boundary with VS-03

No value changes in VS-02. The following activities belong to VS-03 Data Editing and Enrichment: content correction, imputation, enrichment, unit and currency conversion, reference-period alignment, reconciliation of values across sources, derivation of new variables, and interpretive coding of free text.

VS-02.5 Register Standardized Artifacts

Value created: *Secure, organized, and traceable storage of standardized data that downstream consumers can find.*

This stage persists conformant standardized data and makes it findable for the rest of the statistical system. Each dataset version receives a unique persistent identifier (URI or UUID). It is stored in the standardized data repository, in line with the loose coupling of SAF-P07. The removal or pseudonymization of direct identifiers is confirmed and recorded in the pseudonymization record, without ever recording the keys themselves. Metadata is registered in the data catalog, in line with SAF-P09: mapping lineage, classification versions, concept links, recorded units and periods, the list of unmapped values, and the conformance assessment. Field-level lineage from State #1 to State #2 is recorded, so every standardized value can be traced to the raw value and the mapping rule. Access permissions are applied according to the privacy design. A rerun with new rules, such as a new correspondence table, always creates a new version; it never overwrites an earlier one. The stage ends by publishing the *Standardized Data Registered* event, which tells VS-03 that new standardized data is available.

Output: Standardized Data (State #2), the official handover artifact to the rest of the statistical value chain. Its structure and minimum metadata set are described in *State #2: Standardized Data.*

Activity	Key outputs	SAF process	Other reference
Assign a unique persistent identifier (URI or UUID) to the dataset version	Dataset ID	DLM-3 Register and catalog data	
Store the data in the standardized data repository	Stored standardized artifact	DLM-5 Store and version data	SAF-P07 Implement loosely coupled processes and systems

Activity	Key outputs	SAF process	Other reference
Confirm and record identifier removal or pseudonymization	Pseudonymization record	DLM-4 Classify and protect data	SAF-P11 Privacy by design
Register metadata in the data catalog: mapping lineage, classifications, concepts, units and periods, unmapped values, conformance	Catalog registration	DLM-3 Register and catalog data; OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Apply access permissions according to the privacy design	Access-controlled dataset	DLM-4 Classify and protect data	SAF-P10 Security by design, SAF-P11 Privacy by design
Publish the *Standardized Data Registered* event to notify VS-03 Data Editing and Enrichment	Event notification	DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

PLACEHOLDER: VS-02 standardized data artifact structure: the standardized dataset version linked to its mapping lineage, classification record, conformance assessment, and pseudonymization and access record

4. The four process types

VS-02 runs through four process types. They do not form a linear sequence but a governance cycle: each process type produces its own output, and that output feeds the others. Design produces metadata, Implementation produces the execution process, Execution produces the statistical data, and Management produces quality reports that feed back into Design. The generic pattern is described in the Overview of this chapter.

The sub-processes below are specific to VS-02. Their names describe the work in the context of standardization. Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. This keeps every VS-02 sub-process traceable to the reference process model in *Business Processes*.

Design: provides metadata

The design process type produces the **metadata artifacts** that govern all other process types. In VS-02 these are above all the target data model, the correspondence tables, and the conformance rules. Without them, execution produces data that cannot be compared, and management has no baseline to measure against. Design runs when a new product or source is introduced, when a standard classification is revised, or when Management feeds back an improvement.

Design sub-process	Refines	Main outputs
Design Target Data Models and Schemas	PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-1 Plan and design data assets; DLM-9 Manage reference and master data	Target data model, canonical concept catalog, structural conformance rules, integration key design
Design Classification Mapping Strategy	PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	Classification standards register, version policy, correspondence tables, routing rules for unmapped values

Design sub-process	Refines	Main outputs
Design Metadata and Semantic Harmonization Rules	PRD-2.2 Design variables; DLM-1 Plan and design data assets; OVA-2 Metadata Management	Harmonized concept system, unit and period metadata standards, semantic mapping rules, definition-difference policy
Design Conformance Validation Rules	PRD-2.5 Design processing and analysis; DLM-2 Promote data through a quality gate; OVA-1 Quality Management; OVA-2 Metadata Management	Structural, classification, and semantic conformance rules, content-preservation checks, exception handling policy
Design Standardization Registration and Privacy Framework	DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-4 Process Data Management	Data classification, access control design, lineage model, pseudonymization design, audit logging specification

Design Target Data Models and Schemas

Refines: PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-1 Plan and design data assets; DLM-9 Manage reference and master data.

Standardization starts with the canonical structure that all data must conform to. This sub-process specifies the target data model of Standardized Data (State #2): entities, fields, data types, formats, cardinalities, and referential constraints. It sets canonical representations for common concepts, such as dates, geographies, currencies, and identifiers of persons and businesses. Data from surveys, registers, and external sources can then be expressed in the same way. The team also defines the structural conformance contract, which states when a dataset is structurally valid against the target model. Where several sources describe the same population, the design defines a common key structure, such as the key of the statistical business register. The actual linkage of records takes place in VS-03; VS-02 only aligns the keys.

Activity	Key outputs	SAF process	Other reference
Define the target data model of Standardized Data (State #2): entities, fields, types, formats	Target data model specification	PRD-2.2 Design variables; DLM-1 Plan and design data assets	1.3.1 Data standardization
Define canonical representations for common concepts: dates, geography, currency, identifiers	Canonical concept catalog	PRD-2.2 Design variables; DLM-9 Manage reference and master data	SAF-P09 No data without metadata and classification
Define the structural conformance contract and referential constraints	Structural conformance rules	PRD-2.5 Design processing and analysis	
Define the key structure for aligning several sources	Integration key design	PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	2.1.4 Statistical register management

Design Classification Mapping Strategy

Refines: PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data.

Once the target structure is set, the classification steward designs how source codes are mapped to standard classifications. This sub-process selects the authoritative classifications and code lists that the NSI conforms to, such as NACE, ISCO, COICOP, NUTS, and SDMX code lists. It defines the version policy: which

version applies, and how a transition between versions is handled. The mapping is deterministic. A maintained correspondence table translates a value that already carries a source code into the matching standard code, and the original value is kept. The design also specifies what happens to values without a correspondence. They are flagged and passed to VS-03, which carries out any interpretive coding of free text. The reference and master data that anchor the mapping are identified, so that the correspondence tables stay aligned with authoritative sources.

Activity	Key outputs	SAF process	Other reference
Select the authoritative classifications and code lists	Classification standards register	PRD-2.2 Design variables; DLM-9 Manage reference and master data	1.3.1 Data standardization
Define the classification version policy and the handling of transitions	Classification version policy	DLM-9 Manage reference and master data	SAF-P09 No data without metadata and classification
Design deterministic one-to-one correspondence tables from source codes to standard codes	Set of correspondence tables	PRD-2.5 Design processing and analysis	
Define the handling of unmapped values and free text: flag and route to VS-03 Data Editing and Enrichment	Routing rules for unmapped values	PRD-2.5 Design processing and analysis	
Identify the reference and master data sources for mapping	Reference data catalog	DLM-9 Manage reference and master data	CSDA Reference Data

Interpretive coding is not part of VS-02

Coding a written job title to ISCO, or a free-text product name to COICOP, requires judgment. That judgment can change the informational content, so it needs review and an edit record. It is therefore carried out in VS-03. VS-02 only applies deterministic correspondences from a source code to a standard code.

Design Metadata and Semantic Harmonization Rules

Refines: PRD-2.2 Design variables; DLM-1 Plan and design data assets; OVA-2 Metadata Management.

This sub-process defines how descriptive metadata is harmonized, so that data from different sources can be understood in the same way. No value is changed by these rules. The team designs the harmonized definitions of concepts and variables. It defines the metadata that records the unit of measure, the currency, and the reference-period basis of each variable; these are recorded, not converted. Semantic mapping rules link source variables to canonical concepts in the concept system. Where sources use different definitions for the same concept, the design states that the difference is recorded as metadata and flagged for VS-03. Finally, the catalog structure for standardized artifacts is designed so that lineage from State #1 to State #2 can be fully traced. This makes active metadata a foundation for automation downstream, as emphasized by the Metadata Management capability of CSDA.

Activity	Key outputs	SAF process	Other reference
Define harmonized definitions of concepts and variables	Harmonized concept system	PRD-2.2 Design variables; OVA-2 Metadata Management	2.1.2 Metadata management

Activity	Key outputs	SAF process	Other reference
Define the unit, currency, and reference-period metadata to record per variable (no conversion)	Unit and period metadata standards	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Design semantic mapping rules from source variables to canonical concepts	Semantic mapping rules	OVA-2 Metadata Management	CSDA Metadata Management
Design the rules to record differing definitions as metadata and flag them for VS-03 Data Editing and Enrichment	Definition-difference policy	OVA-2 Metadata Management	
Design the catalog structure and the lineage model for standardized artifacts	Catalog registration template	DLM-1 Plan and design data assets; OVA-2 Metadata Management	CSDA Metadata Management

Design Conformance Validation Rules

Refines: PRD-2.5 Design processing and analysis; DLM-2 Promote data through a quality gate; OVA-1 Quality Management; OVA-2 Metadata Management.

Before any data is standardized, it must be clear when the result counts as standardized. This sub-process sets the structural validation rules: field types, formats, value ranges, cardinalities, and referential integrity against the target model. It sets the classification conformance rules: every mapped code belongs to the correct classification version, and there are no orphan codes. It sets the semantic conformance rules: units, definitions, and reference periods are recorded for every variable. The coverage threshold is agreed, which is the maximum acceptable share of unmapped values. Content-preservation checks verify that record counts and key measures are identical before and after standardization, and that no value has changed. Exception handling rules distinguish between rejection, conditional acceptance with a conformance flag, and routing of unmapped values to VS-03. Together, these rules make up the VS-02 part of the quality gate of State #2.

Activity	Key outputs	SAF process	Other reference
Define structural validation rules against the target model	Structural validation rule set	PRD-2.5 Design processing and analysis	
Define classification conformance rules: valid codes, correct version, no orphans	Classification conformance rules	PRD-2.5 Design processing and analysis	
Define semantic conformance rules: units, definitions, reference periods	Semantic conformance rules	PRD-2.5 Design processing and analysis; OVA-2 Metadata Management	
Define the coverage threshold and the content-preservation checks	Content-preservation checks, coverage threshold	DLM-2 Promote data through a quality gate; OVA-1 Quality Management	GAMSO Quality Management
Design exception handling rules: rejection, conditional acceptance, routing to VS-03 Data Editing and Enrichment	Exception handling policy	DLM-2 Promote data through a quality gate	

Design Standardization Registration and Privacy Framework

Refines: DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-4 Process Data Management.

This last design sub-process makes sure that standardized artifacts are registered correctly, can be traced, and comply with privacy rules. Each standardized dataset receives a classification for availability, integrity, and confidentiality, in line with SAF-P09. Role-based access controls for the standardized data repository are designed in line with SAF-P10. The lineage model makes sure that every standardized field can be traced to its raw value and to the mapping rule that was applied. Privacy protection follows SAF-P11. State #2 is the point where direct identifiers leave the working data. The design therefore decides per identifier whether it is removed or pseudonymized, which method is used, and how the keys are managed separately from the data. Audit logging requirements, which define the mapping and coercion events that must be recorded, complete the design.

Activity	Key outputs	SAF process	Other reference
Define the data classification (availability, integrity, confidentiality) for standardized datasets	Data classification policy	DLM-4 Classify and protect data	SAF-P09 No data without metadata and classification
Design role-based access control for the standardized data repository	Access control design	DLM-1 Plan and design data assets; DLM-4 Classify and protect data	SAF-P10 Security by design; 3.3.4 Logical identity and access control
Design the lineage model: standardized field, raw value, mapping rule	Lineage model specification	DLM-1 Plan and design data assets	CSDA Traceability
Design identifier removal and pseudonymization: which identifiers, method, key management, minimization	Pseudonymization design	DLM-4 Classify and protect data	SAF-P11 Privacy by design; 3.3.1 Privacy management
Design audit logging requirements for mapping and coercion	Audit logging specification	OVA-4 Process Data Management	SAF-P10 Security by design

PLACEHOLDER: VS-02 design process decomposition: inputs (State #1 schema, classification standards, framework standards, legal and privacy requirements), the five design sub-processes in order, and the design artifact each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

The implementation process type takes the metadata artifacts from Design and **builds, configures, and tests** the standardization pipeline until it is ready for production. Nothing runs operationally until this process type is complete. Components are reused where possible and built only where needed, in line with SAF-P01. Classification and mapping tooling in particular is a good candidate for sharing between NSIs.

Implementation sub-process	Refines	Main outputs
Build Profiling and Mapping Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Profiling module, configured mapping and format-coercion component, versioned mapping repository

Implementation sub-process	Refines	Main outputs
Configure Classification Mapping Services	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data	Configured mapping services, version-controlled classifications, routing of unmapped values
Build Conformance Validation Pipelines and Metadata Registration	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-2 Promote data through a quality gate; DLM-3 Register and catalog data; DLM-4 Classify and protect data; OVA-2 Metadata Management	Automated conformance pipeline, pseudonymization component, metadata propagation, events
Test and Validate the Pipeline	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems	Test reports, mapping verification, production-ready pipeline, operational procedures

Build Profiling and Mapping Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

This sub-process turns the target data model and the mapping designs into operational tools. Profiling components are built to analyze the structure of incoming raw data. They infer types and detect cardinalities, value distributions, and structural anomalies, so the pipeline can report how far each source is from the target model. The mapping and format-coercion component is built or configured to apply the designed value-preserving operations: renaming, restructuring, type casting, reformatting, and key alignment. It does not convert units, currencies, or reference periods. The component is declarative and records, for each output field, which input field and which rule produced it. All mapping logic is version-controlled, so that any standardized record can be reproduced from its raw version and the recorded rule versions.

Activity	Key outputs	SAF process	Other reference
Build data profiling components: type inference, cardinality, distributions, anomalies	Production-ready profiling module	PRD-3.2 Reuse or build processing and analysis components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Build or configure the mapping and format-coercion component	Configured mapping component	PRD-3.2 Reuse or build processing and analysis components	1.3.1 Data standardization
Configure lineage capture for every mapping and coercion operation	Lineage-aware mapping modules	PRD-3.4 Configure workflows	CSDA Traceability
Put all mapping and coercion logic under version control	Versioned mapping repository	PRD-3.4 Configure workflows	SAF-P10 Security by design

Configure Classification Mapping Services

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data.

In this sub-process the designed correspondence tables become automated mapping services. The correspondence tables are loaded and connected to the authoritative reference data. A value that already carries a

source code is then translated one-to-one into the matching standard code. Version control of classifications is configured, so that the correct version is applied and transitions between versions are managed. Values without a matching correspondence, and free-text values, are routed to the list for VS-03 instead of being coded here. Connections to reference and master data are configured, so that the mapping stays aligned with the authoritative registers.

Activity	Key outputs	SAF process	Other reference
Load the correspondence tables for deterministic mapping from source codes to standard codes	Configured mapping services	PRD-3.2 Reuse or build processing and analysis components; DLM-9 Manage reference and master data	
Configure version control of classifications and the handling of transitions	Version-controlled classifications	DLM-9 Manage reference and master data	SAF-P09 No data without metadata and classification
Configure the routing of unmapped values and free text to VS-03 Data Editing and Enrichment	Routing of unmapped values	PRD-3.4 Configure workflows	
Configure connections to reference and master data	Synchronized reference data	DLM-9 Manage reference and master data	CSDA Reference Data

Build Conformance Validation Pipelines and Metadata Registration

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-2 Promote data through a quality gate; DLM-3 Register and catalog data; DLM-4 Classify and protect data; OVA-2 Metadata Management.

This sub-process implements the conformance rules and the registration of standardized artifacts. The structural, classification, semantic, and content-preservation checks are implemented as automated validations that run on every standardized dataset. The removal or pseudonymization of identifiers is implemented according to the privacy design. Catalog entries are built from the designed metadata model. Automatic propagation is configured, so that mapping lineage, classification versions, concept links, and conformance scores flow into the catalog without manual work. Event notifications are configured, so that VS-03 is alerted when standardized data is registered. Exception workflows route non-conforming data to the right queue.

Activity	Key outputs	SAF process	Other reference
Implement the conformance checks as automated validations	Automated conformance pipeline	PRD-3.2 Reuse or build processing and analysis components; DLM-2 Promote data through a quality gate	
Implement identifier removal and pseudonymization according to the privacy design	Pseudonymization component	PRD-3.2 Reuse or build processing and analysis components; DLM-4 Classify and protect data	SAF-P11 Privacy by design
Configure automatic propagation of metadata and lineage into the catalog	Automated metadata pipelines	PRD-3.4 Configure workflows; DLM-3 Register and catalog data; OVA-2 Metadata Management	CSDA Metadata Management

Activity	Key outputs	SAF process	Other reference
Configure the *Standardized Data Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows	SAF-P07 Implement loosely coupled processes and systems
Configure exception handling workflows	Routing of non-conforming data, review queues	PRD-3.4 Configure workflows; DLM-2 Promote data through a quality gate	

Test and Validate the Pipeline

Refines: PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems.

Before the pipeline goes into production, it must be thoroughly verified. Technical testing covers all individual components: profilers, the mapping component, mapping services, validators, and the catalog. Integration testing checks the end-to-end flow from the *Raw Data Registered* event through mapping and validation to catalog registration. A pilot with real or synthetic data checks that source codes are mapped correctly, that lineage is complete, and that conformance criteria are enforced. The pilot also confirms that no value changes, that unmapped values are routed to VS-03, and that identifier removal works as designed. The correspondence tables are verified against the source code lists. Performance testing confirms that the pipeline handles the expected volumes within the processing window. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	
Test integration: end-to-end flow from raw intake to registration	Integration test report	PRD-3.5 Test production systems	
Run a pilot with real or synthetic data: lineage, no loss of content, routing, identifier removal	Pilot report, validated lineage	PRD-3.6 Test statistical business process	SAF-P11 Privacy by design
Verify the correspondence tables against the source code lists	Mapping verification report	PRD-3.6 Test statistical business process	
Test performance against expected volumes	Performance test report	PRD-3.5 Test production systems	
Finalize the production system and document operational procedures	Operational procedures manual, production-ready pipeline	PRD-3.7 Finalize production systems	

PLACEHOLDER: VS-02 implementation pipeline: the five design artifacts feeding the three build sub-processes, followed by technical testing, pilot and mapping verification, and performance and lineage testing, resulting in a production-ready standardization pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-02 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when VS-01 publishes the *Raw Data Registered* event, or when a rule changes and an existing #1 version must be standardized again. The stages then run in order for each raw version. Execution applies the designs and the solution as they are; a new source structure or a missing correspondence goes back to Design. Every run ends in one of three outcomes: registered, registered with conformance flags, or returned for re-mapping.

Aspect	VS-02
Trigger	*Raw Data Registered*; a new version of a correspondence table, classification, or target model
Cadence	Per registered raw version
Performed by	ROL-14 Data quality analyst, ROL-03 Data steward, ROL-09 Classification steward
Input	Raw data version (#1), target data model, correspondence tables, privacy design
Output	Registered standardized data version (#2), its conformance assessment, and the list of unmapped values for VS-03 Data Editing and Enrichment
Control point	Gate of State #2 in VS-02.4 Validate Structural and Semantic Conformance
Exception path	Re-mapping or correction of a correspondence table; structural change at the source back to Design; failed identifier protection escalated to ROL-19 Compliance officer [**VC13**]

Management: provides quality reports

The management process type runs alongside and after execution. It monitors the performance of the standardization pipeline, assesses the conformance and classification quality of standardized data, makes sure the domain complies with the rules, and feeds improvements back into Design. This closes the governance cycle.

Management sub-process	Refines	Main outputs
Monitor Process Performance	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard, throughput report, paradata records
Assess Standardization Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Mapping completeness report, conformance quality report, trend analysis
Manage Classification and Metadata Lifecycle	DLM-9 Manage reference and master data; OVA-2 Metadata Management; DLM-8 Monitor data quality, lineage and use	Classification version and correspondence register, metadata quality report, lineage audit
Audit and Compliance Reporting	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Audit trail summary, privacy and standards compliance reports to VS-07 Governance and Compliance
Continuous Improvement	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, improvement tracking log

Monitor Process Performance

Refines: OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs.

The operational performance of VS-02 must be tracked systematically across production cycles. The process owner and the quality manager collect and aggregate the key performance indicators (see attribute 9).

They monitor throughput against the processing window. They track paradata: who performed each step, when, how long it took, which engine and classification versions were used, and which exceptions occurred. This information feeds dashboards that give the process owner a current view of the health of the domain. The approach follows the Quality Management capability of GAMSO for monitoring performance across the production cycle.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per production cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Monitor throughput against the processing window	Throughput report	OVA-4 Process Data Management	
Track paradata: operator, timing, engine and classification versions, exceptions	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Assess Standardization Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the standardized data itself is evaluated against the criteria from *Design Conformance Validation Rules*. Mapping completeness covers the share of source codes mapped one-to-one and the share of values routed to VS-03. Classification conformance covers orphan codes and the use of the correct version. Structural conformance covers alignment with the target model. Metadata completeness covers whether units, reference periods, and definitions are recorded. Content preservation confirms that no value changed and that record counts and key measures are identical. Quality indicators are produced per dataset, per source, and per production cycle. Trends are tracked to detect deteriorating sources, growing shares of unmapped values, or problems with classification versions. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Assess mapping completeness and the share of unmapped values	Mapping completeness report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess classification conformance and the rate of orphan codes	Classification conformance report	OVA-1 Quality Management	CSDA Data Quality
Assess structural conformance, metadata completeness, and content preservation	Conformance quality report	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across sources, classifications, and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality

Manage Classification and Metadata Lifecycle

Refines: DLM-9 Manage reference and master data; OVA-2 Metadata Management; DLM-8 Monitor data quality, lineage and use.

The classifications and metadata that support standardization must stay complete, accurate, and current. This sub-process governs the life cycle of classification versions. It adopts new versions, such as a revision of NACE or COICOP, manages the transition period, and maintains the correspondence tables between versions. It verifies that every standardized dataset carries its full mapping lineage, classification versions, and concept links, as a compliance check on SAF-P09. The concept system, code lists, and reference data are maintained, so that the mapping stays anchored to authoritative sources. Lineage from State #1 to State #2 is tracked. Reconciliation across systems makes sure that catalog entries match the artifacts actually stored.

Activity	Key outputs	SAF process	Other reference
Govern the life cycle of classification versions and correspondence tables	Classification version and correspondence register	DLM-9 Manage reference and master data	SAF-P09 No data without metadata and classification
Verify metadata completeness for all standardized datasets	Metadata quality report	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Maintain the concept system, code lists, and reference data	Reference data maintenance log	DLM-9 Manage reference and master data; OVA-2 Metadata Management	CSDA Metadata Management
Track metadata lineage from raw to standardized data	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability
Reconcile metadata across systems	Catalog-artifact reconciliation	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

Audit and Compliance Reporting

Refines: OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use.

This sub-process produces the evidence that VS-02 operates within legal, regulatory, and organizational requirements. Audit trail reports show the full traceability of every standardized value, from the raw value through the mapping rules that were applied. This supports reproducibility and proves that no content was lost. Privacy compliance reporting confirms that direct identifiers were removed or pseudonymized at the transition from State #1 to State #2, and that data minimization was applied. Standards compliance reporting confirms conformance to the adopted classifications and exchange standards, such as SDMX. Security compliance reporting verifies that the designed access controls and audit logging were applied. These reports meet the obligation of SAF capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance*.

Activity	Key outputs	SAF process	Other reference
Produce audit trail reports: full lineage from raw value to standardized value	Audit trail summary	OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	4.1.5 Accountability
Report on privacy compliance: identifier removal or pseudonymization, minimization	Privacy compliance record	OVA-1 Quality Management	SAF-P11 Privacy by design; 3.3.1 Privacy management
Report on standards compliance: classifications, SDMX exchange	Standards compliance report	OVA-1 Quality Management	4.1.5 Accountability

Activity	Key outputs	SAF process	Other reference
Report on security compliance: measures applied, breaches, incidents	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10 Security by design; 3.3.3 Security incident management

Continuous Improvement

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This last management sub-process closes the governance cycle by analyzing performance and quality data and driving improvements. Trends across production cycles are examined: is mapping coverage rising, is the share of unmapped values falling, and are fewer datasets returned for re-mapping? Improvement opportunities are identified. Examples are extending correspondence tables to raise coverage, refining mappings that produce orphan codes, preparing the adoption of a new classification version, and simplifying conformance rules that produce too many false alarms. A recurring volume of unmapped values from one source is reported to VS-03, because the solution is interpretive coding, not a wider VS-02. Recommendations are fed back to the design sub-processes, prioritized by impact and feasibility, and tracked through the following cycles. Where a new method is needed, such as machine-learning support for coding, the recommendation goes to VS-08. This is SAF capability 4.1.4 *Improvement management* in practice.

Activity	Key outputs	SAF process	Other reference
Analyze quality trends across production cycles	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities: extension of correspondence tables, refined mappings, version adoption	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Feed recommendations back to the design sub-processes, to VS-03 Data Editing and Enrichment, or to VS-08 Innovation and Methodological Development	Design feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management

PLACEHOLDER: VS-02 management feedback loop: data and paradata from Execution flowing through the five management sub-processes, with dashboards to the process owner, quality reports to the quality manager, compliance reports to VS-07, coding needs to VS-03, and improvement recommendations back to Design

Metadata and paradata per process type

In VS-02, metadata, and classification in particular, is not a by-product but the central asset being created. The table shows what each process type contributes. Data metadata describes the datasets. Process metadata, also called paradata, describes how the process itself ran: who, when, and with which tools.

Process type	Data metadata produced	Process metadata (paradata) produced
Design	Target data models, classification schemes, code lists, correspondence tables, conformance criteria, harmonized definitions	Log of design decisions, classification version policy, review and approval records
Implementation	Mapping configurations, parameters of the mapping services, parameters of validation rules, catalog templates	Build log, test results, records of configuration changes, deployment approvals
Execution	Mapping lineage per dataset, applied classification versions, recorded units and periods, conformance scores, profiling statistics, list of unmapped values	Operator identity, timing, volumes, mapping coverage, number of unmapped values, engine version
Management	Conformance scores, classification quality assessments, coverage reports, lineage assessments, trend analyses	Conformance dashboards, classification quality trends, improvement decisions, audit findings

PLACEHOLDER: VS-02 business object lifecycle: design artifacts govern implementation assets, which produce the pipeline that turns Raw Data (#1) into Standardized Data (#2), generating mapping lineage, conformance, and audit records that management assesses and feeds back to design

5. Actors and roles

VS-02 is an internal value stream. Its only external actor is the body that maintains the classification and exchange standards the NSI conforms to. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the system configurator in the original VS-02 description is ROL-11 Platform engineer. VS-01, which supplies the raw data, is a neighboring value stream and not an actor; the relation is governed by the handover interface in attribute 8. **[VC14]**

Actor or role	ID	Process type	Responsibility in VS-02 Data Standardization and Integration
International statistical organization	ACT-07	External	Maintains classification and exchange standards, such as NACE, ISCO, COICOP, and SDMX (for example Eurostat and UNSD)
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-02 Data Standardization and Integration
Data owner	ROL-02	Management	Accountable for State #2, for conformance, and for applying the privacy design

Actor or role	ID	Process type	Responsibility in VS-02 Data Standardization and Integration
Data steward	ROL-03	Execution	Registers standardized datasets, maintains catalog entries, applies access controls, resolves gate exceptions
Data custodian	ROL-04	Execution	Operates storage, access control, and retention; manages pseudonymization keys separately from the data
Methodologist	ROL-07	Design	Designs the mapping method and the harmonization of definitions across sources
Metadata and data architect	ROL-08	Design	Designs the target data model, canonical concepts, semantic mappings, and the metadata schema
Classification steward	ROL-09	Design, Execution	Owns the classification strategy, code lists, and correspondence tables; maintains classification versions
Data engineer	ROL-10	Implementation	Builds and configures the profiling, mapping, and coercion components and the pipeline
Platform engineer	ROL-11	Implementation	Configures classification services, connections to reference data, and access controls
Test manager	ROL-12	Implementation	Plans and carries out technical, conformance, and integration testing of the pipeline
Data quality analyst	ROL-14	Execution	Profiles data, assesses conformance, and logs unmapped values and exceptions
Quality manager	ROL-18	Management	Monitors key performance indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures privacy and standards compliance and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-02 relies on, and what role they play. NSIs can use it to map the value stream to their own organization.

Capability group	Capabilities	Role in VS-02 Data Standardization and Integration
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the target data model and the mapping strategy, and build the standardization pipeline
1.3 Processing	1.3.1 Data standardization	Carry out the standardization: format coercion, one-to-one code mapping, semantic alignment **[VC15]**
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage data and metadata through standardization, harmonize semantics, assess conformance, align keys with reference registers
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain mapping components, classification services, and validation pipelines
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.4 Logical identity and access control	Remove or pseudonymize direct identifiers and build security into the pipeline
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward the classification standards, plan improvements, report on compliance

Stages, processes, and functions

The table shows which SAF business processes realize each stage of VS-02 and which business functions they mainly use. It follows the mapping in *Business Processes, Mapping to value stream stages.* The activity tables in attribute 3 can name additional supporting processes. **[VC25]**

Stage	Realized by processes	Main functions
VS-02.1 Profile and Assess Data Structure	PRD-5.3 Review and validate	FUN-3.3 Process statistical data
VS-02.2 Apply Standard Classifications and Formats	PRD-5.2 Classify and code; DLM-9 Manage reference and master data	FUN-3.3 Process statistical data; FUN-7.4 Manage reference and master data
VS-02.3 Align Metadata and Semantics	OVA-2 Metadata Management	FUN-7.2 Manage metadata
VS-02.4 Validate Structural and Semantic Conformance	PRD-5.3 Review and validate; DLM-2 Promote data through a quality gate	FUN-3.3 Process statistical data; FUN-7.3 Manage data lifecycle and storage
VS-02.5 Register Standardized Artifacts	DLM-3 Register and catalog data; DLM-4 Classify and protect data; DLM-5 Store and version data	FUN-7.2 Manage metadata; FUN-7.3 Manage data lifecycle and storage; FUN-7.5 Protect data

VS-02 carries out only part of PRD-5.1, PRD-5.2, and PRD-5.3: key alignment without record linkage, deterministic coding without interpretive coding, and structural checks without checks on the correctness of values. The remaining parts of these processes take place in VS-03. **[VC16]**

PLACEHOLDER: ArchiMate view VS-02 (b) stages realized by processes and functions

Framework mappings

This reference table maps each process type of VS-02 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.2, PRD-2.5, DLM-1, DLM-2, DLM-4, DLM-9, OVA-1, OVA-2, OVA-4	Phase 2 Design (2.2, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Reference and Master Data	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.4 Statistical register management
Implementation	PRD-3.2, PRD-3.4 to PRD-3.7, DLM-2, DLM-3, DLM-4, DLM-9, OVA-2	Phase 3 Build (3.2, 3.4 to 3.7)	Capability Management	Data Integration (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-5.1 to PRD-5.3 (partly), DLM-2 to DLM-5, DLM-9, OVA-1, OVA-2, OVA-4	Phase 5 Process (5.1 key alignment only, 5.2 deterministic coding only, 5.3 structural checks only)	Production (execution)	Data Integration (execution)	1.3.1 Data standardization, 2.1.2 Metadata management, 2.1.4 Statistical register management
Management	OVA-1, OVA-2, OVA-4, DLM-8, DLM-9, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-02 most directly.

Principle	Statement	Application in VS-02 Data Standardization and Integration
SAF-P01	Reuse before open source, open source before buying, buying before building	Mapping, coding, and SDMX tooling is reused across products first. Where new tooling is needed, open-source solutions, preferably developed with other NSIs, come before purchase or own development. **[VC3]**

Principle	Statement	Application in VS-02 Data Standardization and Integration
SAF-P07	Implement loosely coupled processes and systems	Standardization is independent from acquisition and editing. Standardized Data (State #2) is the decoupling interface: downstream consumers use standardized, classified data without knowing the original source structure.
SAF-P08	Data is shared property	Standardized data, once registered, is available to all authorized consumers. A source is standardized once and then reused by every product that needs it. Standardizing the same source per product wastes effort and leads to inconsistent coding.
SAF-P09	No data without metadata and classification	VS-02 Data Standardization and Integration is the main point where this principle is fulfilled. Every standardized dataset carries its mappings, classification versions, harmonized definitions, recorded units and periods, and a classification for availability, integrity, and confidentiality. The principle does not require raw data to arrive classified; it requires that data handed over from State #2 onward carries this metadata.
SAF-P10	Security by design	Access controls, integrity verification, and audit logging are built into the pipeline, so every mapping and coercion operation can be traced and tampering can be detected.
SAF-P11	Privacy by design	Direct identifiers are removed or pseudonymized during standardization, at the latest. Data minimization and purpose limitation are applied by design, and pseudonymization keys are kept separately from the data.

i Principles that apply across the process types

Several principles apply to all four process types. SAF-P09 is the defining principle of VS-02: Design defines the target classifications and mappings, Execution applies and records them, and Management assesses classification quality and closes remaining gaps. SAF-P11 applies to *Design Standardization Registration and Privacy Framework*, to *Build Conformance Validation Pipelines and Metadata Registration*, and to the stages VS-02.4 and VS-02.5. SAF-P07 governs both the intake of State #1 from VS-01 and the handover of State #2 to VS-03.

8. Interfaces, services, and events

Boundary interfaces

VS-02 is loosely coupled (SAF-P07) to its neighbors through two boundary interfaces. Specifying each boundary explicitly makes the loose-coupling contract concrete and testable: the handover artifact, the metadata it must carry, the event that triggers it, and the point where responsibility transfers. Each interface exchanges a registered, immutable artifact plus its metadata, never internal working data. Neither side therefore needs knowledge of the other's internal processing.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-01 Data Acquisition and Ingestion → VS-02 Data Standardization and Integration (intake)	Registered raw data version (#1)	Minimum metadata set of State #1: provenance, source, collection method, receipt timestamp, file hash, agreement reference, quality assessment, classification	*Raw Data Registered*	VS-01 Data Acquisition and Ingestion keeps custody of the raw artifact; VS-02 Data Standardization and Integration takes responsibility for standardization on receipt of the event
VS-02 Data Standardization and Integration → VS-03 Data Editing and Enrichment (handover)	Registered standardized data version (#2)	Minimum metadata set of State #2, including the list of unmapped values, the recorded units, currencies, and reference periods, the definition-difference records, and the pseudonymization record	*Standardized Data Registered*	VS-02 Data Standardization and Integration keeps custody of the standardized artifact; VS-03 Data Editing and Enrichment takes responsibility for editing, interpretive coding, conversion, and reconciliation on receipt of the event

Business services

Service	Offered to	Realized in	Interfaces
Standardization service	VS-01 Data Acquisition and Ingestion (as source), standardization teams	VS-02.1 Profile and Assess Data Structure; VS-02.2 Apply Standard Classifications and Formats	Event subscription (*Raw Data Registered*)
Classification mapping service	Standardization teams, methodologists, other domains that need code mapping	VS-02.2 Apply Standard Classifications and Formats	Mapping interface (source code to standard code), SDMX code list exchange, correspondence table lookup
Semantic harmonization service	Metadata architects, downstream analysts	VS-02.3 Align Metadata and Semantics	Concept system and metadata catalog
Conformance validation service	Process owner, quality manager	VS-02.4 Validate Structural and Semantic Conformance	Conformance reports and dashboards

Service	Offered to	Realized in	Interfaces
Standardized data provisioning service	VS-03 Data Editing and Enrichment, other products reusing the source	VS-02.5 Register Standardized Artifacts	Event notification (*Standardized Data Registered*) and a catalog query interface to find standardized datasets and their lineage
Classification catalog service	All data consumers in the NSI	Management sub-process *Manage Classification and Metadata Lifecycle*	Search and browse interface for classifications, code lists, and concepts, with metadata exchange compliant with SDMX
Design and methodology service	Methodologists, data architects, classification stewards	All Design sub-processes of VS-02 Data Standardization and Integration	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artifacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Raw Data Registered	Start of VS-02.1 Profile and Assess Data Structure for that raw version (published by VS-01 Data Acquisition and Ingestion)
Classification version changed	Review of correspondence tables; possible restandardization of existing #1 versions
Standardization Conformed / *Standardization Non-Conformant*	Outcome of VS-02.4 Validate Structural and Semantic Conformance; non-conformance starts re-mapping or conditional acceptance
Standardized Data Registered	Completion of VS-02.5 Register Standardized Artifacts; triggers VS-03 Data Editing and Enrichment
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-02 event choreography between VS-01, the standardization teams, and process management across Design, Implementation, Execution (profile, map and coerce, validate, conformed or non-conformant, register), and Management (key performance indicators, mapping quality, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-02	Receives from VS-02
VS-07 Governance and Compliance	Privacy policy, classification and standards policy	Audit trails, privacy and standards compliance evidence
VS-08 Innovation and Methodological Development	New mapping methods, support for coding	Recurring mapping problems, improvement needs
VS-09 Infrastructure and Platform Enablement	Mapping and classification services, storage, catalog, event interfaces	Capacity and service requirements

Value stream	Provides to VS-02	Receives from VS-02
VS-10 Customer and Stakeholder Engagement	Requirements of users for classifications and comparability	Information on the classifications and versions applied

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
Mapping coverage	Share of source codes mapped one-to-one to a valid standard code	Per dataset, per classification
Unmapped-value rate	Share of values without a correspondence, routed to VS-03 Data Editing and Enrichment	Per source, per cycle
Conformance pass rate	Share of datasets that pass the gate of State #2 on the first run	Per source, per cycle
Format conformance rate	Share of records that conform to the target structure and format	Per dataset, per cycle
Processing time	Time from the *Raw Data Registered* event to the registration of #2	Per dataset, per cycle
Re-mapping rate	Share of datasets returned for re-mapping or design change	Per source, per cycle

10. Toolkit

Examples from NSI practice, the maturity checklist, and end-to-end scenarios for VS-02 are part of the SAF Toolkit.

Toolkit: SAF-TK-V02 VS-02 Data Standardization and Integration companion **[VC17]**

VS-03 Data Editing and Enrichment

1. Purpose and value created

VS-03 is where the quality of the data is actively improved. Standardized data is rarely clean. It contains errors, inconsistencies, gaps, and values that are not yet coded. VS-03 detects these problems, corrects them, fills the gaps, and derives the variables that analysis needs. Unlike VS-02, it deliberately changes values to do so. What makes this safe is full traceability: every edit, imputation, conversion, and derivation is rule-based, documented with a reason, attributable, and reversible. The value created is **reliability and analytical readiness**. Every estimate, model, and publication downstream rests on the microdata that VS-03 produces.

VS-03 corresponds to the business domain **Editing, derivation, and non-response correction** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

- **Data quality analyst, subject-matter statistician, or operations staff:** the stages in attribute 3, which describe what happens in each production cycle.
- **Methodologist or statistical designer:** the Design processes in attribute 4.
- **System builder or IT staff:** the Implementation processes in attribute 4.
- **Quality or compliance manager:** the Management processes in attribute 4.
- **Manager who wants the big picture:** attributes 1, 2, and 8. Examples and end-to-end scenarios are part of the SAF Toolkit.

Signature discipline: values change, but never silently

VS-03 is the domain that changes values. The guarantee is not that values stay the same, but that no value changes without a record. For every change, the original value, the rule or decision applied, the reason, the actor, and the time are kept. Imputed values are always flagged, so they can be told apart from observed values. Every quality version can be reproduced from its standardized input and the recorded rules. The invariant is: *no undocumented or irreversible change.*

Work inherited from VS-02

VS-02 deliberately leaves all value-changing work to VS-03. VS-03 receives Standardized Data together with the metadata VS-02 attached: the recorded units, currencies, and reference periods, the definition-difference records, and the list of unmapped values. VS-03 then carries out the operations that VS-02 deferred. These are the interpretive coding of free text, the conversion of units and currencies to a common basis, the alignment of reference periods, the reconciliation of values across sources with different definitions, and the derivation of new variables.

PLACEHOLDER: ArchiMate view VS-03 (a) value stream and capabilities

2. Position and state transition

VS-03 is the third primary value stream. It sits between standardization (VS-02) and estimation (VS-04). It receives Standardized Data when VS-02 publishes the *Standardized Data Registered* event, and it hands over Processed Data to VS-04.

VS-03 takes data from State #2 to State #3. Before VS-03, the data is aligned in structure and coding, but its values are unchanged and may contain errors and gaps. After VS-03, it is a registered, immutable quality version of microdata whose content has been checked, corrected, completed, and enriched. It is linked with other sources where applicable, it carries derived variables, and its lineage is fully documented. The quality gate, the minimum metadata set, and the protection of State #3 are described in *State #3: Processed Data* and are not repeated here.

State #3 is the only steady state that is mandatory for all statistical production. Where a source skips State #2, VS-03 takes it directly from State #1. Processed Data is also a product in its own right. Researchers and other statistical domains reuse it directly, and research datasets for microdata services are prepared from it.

From	To	Gate	Event out
#2 Standardized Data (or #1 Raw Data where #2 is skipped)	#3 Processed Data	Gate of State #3 (content quality, residual rule violations, imputation flags, edit lineage, derived-variable metadata, disclosure risk of linkage, reproducibility)	*Processed Data Registered*

PLACEHOLDER: ArchiMate view VS-03 (c) state handover and events: Standardized Data (#2) in the upstream domain, the editing service at the boundary (error detection, editing, imputation, derivation, quality evaluation), Processed Data (#3) inside the editing domain

3. Stages

The stages describe the value that VS-03 adds in every production cycle. Together they form the execution process type of VS-03. In a mature steady state, the editing rules, the imputation methods, and the derivation specifications have been designed and built in advance (see attribute 4). Execution then focuses on applying them, involving analysts where judgment is needed, and recording every change.

Stage	Name	Value added
VS-03.1	Conduct Error Detection and Quality Diagnostics	It is clear where the data is wrong, incomplete, or inconsistent, and which problems matter
VS-03.2	Perform Data Editing and Correction	Errors are corrected and free text is coded, with every change documented and reversible
VS-03.3	Impute Missing or Non-Response Data	Gaps are filled with the best available estimate, clearly flagged as imputed
VS-03.4	Derive Additional Variables and Indicators	The data carries the converted values, links, and derived variables that analysis needs
VS-03.5	Perform Quality Evaluation and Versioning	A traceable Processed Data version with measured quality is registered (State #3)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-03.1 Conduct Error Detection and Quality Diagnostics

Value created: *A clear, prioritized picture of where the data is wrong, incomplete, or inconsistent.*

The stage starts when VS-02 publishes the *Standardized Data Registered* event. The designed detection rules run on the incoming version. They cover consistency between fields and with reference data, value ranges, outliers, duplicates, and completeness. Selective-editing scores are computed, so that effort goes to the errors that materially affect the results. The outcome is a diagnostic report with the location, type, and severity of each suspected error. Each error also gets a disposition: correct automatically, flag for review, or accept. No value is changed in this stage; correction happens in VS-03.2.

Activity	Key outputs	SAF process	Other reference
Run consistency, range, outlier, duplicate, and completeness checks	Error diagnostic report	PRD-5.3 Review and validate	1.3.3 Check content
Compute selective-editing scores and prioritize errors	Prioritized error list	PRD-5.3 Review and validate	
Assign a disposition: automatic correction, review, or accept	Disposition list	PRD-5.3 Review and validate	
Log all detected issues in the edit audit trail	Audit trail entries	OVA-4 Process Data Management	SAF-P10 Security by design

VS-03.2 Perform Data Editing and Correction

Value created: *Errors corrected and inconsistencies resolved, with every change documented and reversible.*

This stage corrects the errors found in VS-03.1. Automatic correction rules are applied to the records that the design marked for automatic treatment. Flagged records go to subject-matter statisticians, who review and correct them. For every change, the original value, the rule or decision, a reason code, the actor, and the time are recorded. The stage also carries out the interpretive coding handed over from VS-02, such as coding free-text occupation to ISCO; cases with low confidence are reviewed. Where linked sources disagree on a value, the designed reconciliation rules are applied. No edit may leave a record that violates the validation rules.

Activity	Key outputs	SAF process	Other reference
Apply automatic corrections according to the designed rules	Corrected records	PRD-5.4 Edit and impute	1.3.4 Adjusting and imputing
Route flagged records to subject-matter review	Reviewed corrections	PRD-5.4 Edit and impute	
Code free-text values interpretively, for example occupation to ISCO	Coded values, confidence and review records	PRD-5.2 Classify and code	
Apply reconciliation rules where linked sources disagree	Reconciled values	PRD-5.1 Integrate data; PRD-5.4 Edit and impute	
Record original value, rule, reason, actor, and time for every change	Edit lineage records	OVA-4 Process Data Management	SAF-P10 Security by design

PLACEHOLDER: VS-03 editing flow: detected errors split by selective-editing score into automatic correction, subject-matter review, and accepted records; free-text values passing interpretive coding with a review loop for low confidence; every change writing original value, rule, reason, actor, and time to the edit audit trail

VS-03.3 Impute Missing or Non-Response Data

Value created: *Complete datasets in which gaps are filled with the best available estimate, clearly flagged as imputed.*

This stage fills missing values and non-response with the designed imputation methods. The method is chosen per variable and pattern of missingness: deterministic rules, donor methods, model-based methods, or external sources such as administrative registers. Acceptance constraints make sure that imputed values still satisfy the edit rules. Every imputed value receives an imputation flag and a method code, so imputed and observed values can always be told apart downstream. The inputs of each imputation are recorded in the lineage. Imputed values never overwrite observed values; they only fill genuine gaps.

Activity	Key outputs	SAF process	Other reference
Select and apply the designed imputation method per variable and missingness pattern	Imputed values	PRD-5.4 Edit and impute	1.3.4 Adjusting and imputing
Enforce acceptance constraints: imputed values satisfy the edit rules	Checked imputations	PRD-5.4 Edit and impute	

Activity	Key outputs	SAF process	Other reference
Flag every imputed value with its method code and record its inputs	Imputation flags and lineage	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Log imputation actions in the edit audit trail	Audit trail entries	OVA-4 Process Data Management	SAF-P10 Security by design

VS-03.4 Derive Additional Variables and Indicators

Value created: *Enriched datasets that carry the converted values, links, and analytical variables that downstream production needs.*

This stage calculates the derived variables and indicators specified in Design. Each derived variable is computed from its documented formula and inputs, receives its metadata and classification, and is linked in the lineage to the inputs it came from. The stage also carries out the conversions that VS-02 deferred. Examples are converting turnover from thousands of euro to euro, or aligning a fiscal-year figure to the calendar year. Each conversion is documented with its rule and rate, so the original value can always be recovered. Where datasets are combined, records are linked on the designed keys, and the linkage is recorded in the lineage.

Activity	Key outputs	SAF process	Other reference
Calculate derived variables and indicators from the designed formulas	Derived variables with metadata	PRD-5.5 Derive new variables and units	1.3.2 Deriving variables and units
Convert units, currencies, and reference periods from the bases recorded by VS-02 Data Standardization and Integration	Converted values with conversion lineage	PRD-5.5 Derive new variables and units	
Link datasets on the designed keys where applicable	Integrated dataset with linkage lineage	PRD-5.1 Integrate data	2.1.4 Statistical register management
Record derivation and conversion formulas and inputs as lineage	Derivation lineage	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

VS-03.5 Perform Quality Evaluation and Versioning

Value created: *A trustworthy, traceable Processed Data version with measured quality improvement.*

This stage evaluates the result and registers the official quality version. Quality is measured against the baseline of State #2: the improvement in completeness and accuracy, the edit and imputation rates, and the derived variables added. The edit lineage is checked for completeness, so that every change is attributable and reversible. The data must then pass the quality gate of State #3. A persistent, immutable quality version is created with a unique identifier and registered in the data catalog with its lineage, imputation flags, derived-variable metadata, and quality assessment. Access permissions are applied. A correction or redelivery later in the cycle always creates a new version; it never overwrites an earlier one. The stage ends by publishing the *Processed Data Registered* event, which tells VS-04 and other consumers that new processed data is available.

Output: Processed Data (State #3), the official handover artifact to the rest of the statistical value chain. Its structure and minimum metadata set are described in *State #3: Processed Data*.

Activity	Key outputs	SAF process	Other reference
Evaluate quality improvement against the baseline of State #2	Quality evaluation report	PRD-5.8 Finalize data files; OVA-1 Quality Management	2.1.3 Statistical quality management
Verify that the edit lineage is complete and reversible	Lineage completeness check	DLM-2 Promote data through a quality gate	SAF-P10 Security by design
Create an immutable, identified quality version of Processed Data	Versioned Processed Data	PRD-5.8 Finalize data files; DLM-5 Store and version data	CSDA Traceability
Register metadata: edit lineage, imputation flags, derived-variable metadata, quality assessment	Catalog registration	DLM-3 Register and catalog data; OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Apply access permissions according to the designed framework	Access-controlled dataset	DLM-4 Classify and protect data	SAF-P10 Security by design, SAF-P11 Privacy by design
Publish the *Processed Data Registered* event to notify VS-04 Estimation, Analysis, and Statistical Modeling	Event notification	DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

PLACEHOLDER: VS-03 quality version structure: the processed data version linked to its edit lineage, imputation record, derived-variable definitions, linkage lineage, and quality assessment

4. The four process types

VS-03 runs through four process types. They do not form a linear sequence but a governance cycle: each process type produces its own output, and that output feeds the others. Design produces metadata, Implementation produces the execution process, Execution produces the statistical data, and Management produces quality reports that feed back into Design. The generic pattern is described in the Overview of this chapter.

The sub-processes below are specific to VS-03. Their names describe the work in the context of editing and enrichment. Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. This keeps every VS-03 sub-process traceable to the reference process model in *Business Processes*.

Design: provides metadata

The design process type produces the **metadata artifacts** that govern all other process types. In VS-03 these are the editing rules, the imputation methods, the derivation and coding specifications, the linkage rules, and the versioning policy. Without them, editing becomes ad hoc and cannot be traced, and management has no baseline to measure quality improvement against. Design runs when a new product or source is introduced, when a method changes, or when Management feeds back an improvement.

Design sub-process	Refines	Main outputs
Design Error-Detection and Validation Rules	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Validation rule set, selective-editing strategy, error taxonomy and disposition policy

Design sub-process	Refines	Main outputs
Design Imputation Methodology	PRD-2.5 Design processing and analysis	Imputation methodology, donor and matching specification, imputation flagging standard
Design Derivation and Coding Specifications	PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	Derived-variable catalog, coding methodology, conversion specification
Design Integration and Linkage Rules	PRD-2.5 Design processing and analysis; DLM-4 Classify and protect data	Linkage rule set, reconciliation policy, disclosure-risk assessment design
Design Quality Evaluation, Versioning, and Audit Framework	DLM-1 Plan and design data assets; DLM-2 Promote data through a quality gate; DLM-4 Classify and protect data; OVA-1 Quality Management; OVA-4 Process Data Management	Quality criteria, edit audit-trail specification, versioning policy, access control and reproducibility design

Design Error-Detection and Validation Rules

Refines: PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

Editing starts with a clear definition of what counts as an error. This sub-process sets the validation and detection rules. They cover logical and consistency checks within a record, between fields, and against reference data, as well as ranges, outliers, duplicates, and completeness. A selective-editing strategy is designed, so that effort concentrates on errors that materially affect the results. Score functions and thresholds decide which records are corrected automatically, which go to review, and which are accepted as they are. The team also designs the error taxonomy and the disposition options. Together, these rules make up the VS-03 part of the quality gate of State #3.

Activity	Key outputs	SAF process	Other reference
Define consistency, range, and outlier detection rules	Validation rule set	PRD-2.5 Design processing and analysis	1.3.3 Check content
Define duplicate and completeness detection rules	Validation rule set	PRD-2.5 Design processing and analysis	
Design selective-editing score functions and review thresholds	Selective-editing strategy	PRD-2.5 Design processing and analysis	
Define the error taxonomy and disposition options: automatic correction, review, accept	Error taxonomy and disposition policy	OVA-1 Quality Management	GAMSO Quality Management

Design Imputation Methodology

Refines: PRD-2.5 Design processing and analysis.

This sub-process designs how missing values and non-response are completed. The team selects an imputation method per variable and type of missingness. Options are deterministic rules, donor methods such as hot-deck or nearest neighbor, model-based methods such as regression or ratio imputation, and the use of external or administrative sources. Donor pools, matching variables, and acceptance constraints are defined; an imputed value must still satisfy the edit rules. The design requires an imputation flag and a method code on every imputed value. Imputed and observed values can then always be told apart downstream, which VS-04 needs to estimate variance correctly.

Activity	Key outputs	SAF process	Other reference
Select imputation methods per variable and missingness type	Imputation methodology document	PRD-2.5 Design processing and analysis	1.3.4 Adjusting and imputing
Define donor pools, matching variables, and acceptance constraints	Donor and matching specification	PRD-2.5 Design processing and analysis	
Specify the use of external and administrative sources for imputation	Rules for imputation from external sources	PRD-2.5 Design processing and analysis	SAF-P08 Data is shared property
Require imputation flags and method codes on every imputed value	Imputation flagging standard	PRD-2.5 Design processing and analysis	SAF-P09 No data without metadata and classification

Design Derivation and Coding Specifications

Refines: PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data.

This sub-process specifies the new variables and indicators to derive, and the interpretive coding handed over from VS-02. Each derived variable is defined by a formula, its input variables, its unit, and the classification it belongs to. For interpretive coding, the design sets the classification, such as ISCO for occupation or COICOP for product names. It also sets the approach: manual, computer-assisted, or assisted by machine learning with confidence thresholds. A review workflow is designed for cases with low confidence. Finally, the team designs how units, currencies, and reference periods are converted, using the bases that VS-02 recorded as metadata.

Activity	Key outputs	SAF process	Other reference
Specify derived variables and indicators: formulas, inputs, units, classification	Derived-variable catalog	PRD-2.2 Design variables	1.3.2 Deriving variables and units
Design the interpretive coding approach and its review workflow	Coding methodology	PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	
Design the conversion rules for units, currencies, and reference periods	Conversion specification	PRD-2.5 Design processing and analysis	
Define metadata and classification for every derived variable	Derived-variable metadata standard	PRD-2.2 Design variables	SAF-P09 No data without metadata and classification

Design Integration and Linkage Rules

Refines: PRD-2.5 Design processing and analysis; DLM-4 Classify and protect data.

Where Processed Data is combined with other datasets, this sub-process designs the linkage. It specifies the linkage keys, such as the key of the statistical business register that VS-02 aligned. It defines deterministic and probabilistic matching rules. It sets the rules for resolving conflicts where linked sources disagree on a value. It also sets the reconciliation policy for the definition differences that VS-02 recorded as metadata. Linkage of pseudonymized microdata can increase the risk of re-identification. The design therefore includes an assessment of disclosure risk per linkage, in line with SAF-P11.

Activity	Key outputs	SAF process	Other reference
Define linkage keys and deterministic and probabilistic matching rules	Linkage rule set	PRD-2.5 Design processing and analysis	2.1.4 Statistical register management
Design conflict resolution and value reconciliation across sources	Reconciliation policy	PRD-2.5 Design processing and analysis	
Design the disclosure-risk assessment for linkage and enrichment	Disclosure-risk assessment design	DLM-4 Classify and protect data	SAF-P11 Privacy by design; 3.3.1 Privacy management
Define lineage requirements for linked and reconciled values	Linkage lineage specification	PRD-2.5 Design processing and analysis	CSDA Traceability

Design Quality Evaluation, Versioning, and Audit Framework

Refines: DLM-1 Plan and design data assets; DLM-2 Promote data through a quality gate; DLM-4 Classify and protect data; OVA-1 Quality Management; OVA-4 Process Data Management.

This last design sub-process makes sure that quality is measured and that every change can be traced. It defines the criteria for quality evaluation: the improvement in completeness and accuracy against the baseline of State #2. It defines the requirements of the edit audit trail: the original value, the rule, the reason, the actor, and the time of every change. It defines the versioning policy: how versions are identified, what triggers a new version, and how immutability and retention are handled. Role-based access controls for processed microdata are designed in line with SAF-P10. Reproducibility is designed in from the start: a quality version must be regenerable from its standardized input and the recorded rules.

Activity	Key outputs	SAF process	Other reference
Define quality evaluation criteria against the baseline of State #2	Quality criteria document	OVA-1 Quality Management; DLM-2 Promote data through a quality gate	GAMSO Quality Management
Define the edit audit-trail requirements: original value, rule, reason, actor, time	Edit audit-trail specification	OVA-4 Process Data Management	SAF-P10 Security by design
Define the versioning policy: identification, triggers, immutability, retention	Versioning policy	DLM-1 Plan and design data assets	CSDA Traceability
Design role-based access control and reproducibility for processed microdata	Access control and reproducibility design	DLM-4 Classify and protect data	SAF-P10 Security by design; 3.3.4 Logical identity and access control

PLACEHOLDER: VS-03 design process decomposition: inputs (State #2 metadata, methodology standards, quality requirements, legal and privacy requirements), the five design sub-processes in order, and the design artifact each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

The implementation process type takes the metadata artifacts from Design and **builds, configures, and tests** the editing and enrichment pipeline until it is ready for production. Nothing runs operationally until this process type is complete. Components are reused where possible and built only where needed, in line with SAF-P01. Editing and imputation tools developed in the NSI community are good candidates for reuse.

Implementation sub-process	Refines	Main outputs
Build Error-Detection and Editing Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Detection module, configured editing engine, edit audit-trail capture
Configure Imputation and Coding Services	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data	Imputation services, coding service and review queue, conversion service
Build Derivation, Integration, and Versioning Pipelines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data	Derivation pipeline, integration component, versioning store, events
Test and Validate the Pipeline	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems	Test reports, edit-impact and reproducibility reports, production-ready pipeline

Build Error-Detection and Editing Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

This sub-process turns the designed validation and editing rules into operational tools. Detection components are built to run the consistency, range, outlier, duplicate, and completeness checks and to compute the selective-editing scores. The editing engine is built or configured to apply automatic corrections according to the designed rules. It routes flagged records to the review workflow for subject-matter statisticians. For every change, it captures the original value, the rule, a reason code, and the actor. All editing logic is version-controlled, so that every edit can be reproduced and attributed.

Activity	Key outputs	SAF process	Other reference
Build detection components: consistency, range, outlier, duplicate, completeness	Production-ready detection module	PRD-3.2 Reuse or build processing and analysis components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Build or configure the editing engine: automatic correction and routing to review	Configured editing engine	PRD-3.2 Reuse or build processing and analysis components	1.3.4 Adjusting and imputing
Configure the capture of edit lineage for every change	Edit audit-trail capture	PRD-3.4 Configure workflows	SAF-P10 Security by design
Put all editing logic under version control	Versioned editing repository	PRD-3.4 Configure workflows	SAF-P10 Security by design

Configure Imputation and Coding Services

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data.

In this sub-process the designed imputation and coding methods become automated services. Imputation services are configured for each method, with donor pools and matching variables loaded and acceptance constraints enforced. Every imputed value is flagged with its method code. The interpretive coding service is configured for free text, with computer-assisted or machine-learning-assisted coding and the designed confidence thresholds. Cases with low confidence go to a manual review queue. The conversion routines for units, currencies, and reference periods are configured to use the bases recorded by VS-02.

Activity	Key outputs	SAF process	Other reference
Configure imputation services per method, with donor pools and constraints	Configured imputation services	PRD-3.2 Reuse or build processing and analysis components	1.3.4 Adjusting and imputing
Enforce imputation flags and method codes	Flagged imputation outputs	PRD-3.4 Configure workflows	SAF-P09 No data without metadata and classification
Configure the interpretive coding service and its review queue	Coding service and review queue	PRD-3.2 Reuse or build processing and analysis components; DLM-9 Manage reference and master data	
Configure the conversion routines for units, currencies, and reference periods	Conversion service	PRD-3.4 Configure workflows	

Build Derivation, Integration, and Versioning Pipelines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data.

This sub-process implements derivation, linkage, and quality versioning. Calculators for derived variables are built from the designed formulas, and each produces its variable together with its metadata and classification. The linkage component is configured with the designed keys and matching rules. It applies the conflict-resolution and reconciliation policy and records linkage lineage. The versioning store is built so that each accepted Processed Data version is immutable, identified, reproducible, and accompanied by its full edit lineage. Event notifications are configured, so that VS-04 and other consumers are alerted when a new quality version is registered.

Activity	Key outputs	SAF process	Other reference
Build calculators for derived variables from the designed formulas	Derivation pipeline	PRD-3.2 Reuse or build processing and analysis components	1.3.2 Deriving variables and units
Configure linkage with conflict resolution and reconciliation	Integration component	PRD-3.2 Reuse or build processing and analysis components	2.1.4 Statistical register management
Build the versioning store: immutable, identified, reproducible	Versioning store	DLM-5 Store and version data	CSDA Traceability
Configure the *Processed Data Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows; DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

Test and Validate the Pipeline

Refines: PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems.

Before the pipeline goes into production, it must be thoroughly verified. Technical testing covers all individual components: detectors, the editing engine, imputation and coding services, derivation calculators, the linkage component, and the versioning store. Integration testing checks the end-to-end flow from the *Standardized Data Registered* event to the registration of a quality version. An edit-impact test confirms that edits and imputations move the quality indicators in the intended direction, without over-editing. A reproducibility test confirms that a quality version can be regenerated identically from its input and the recorded rules. Imputation is evaluated against a holdout sample with known values. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	
Test integration: end-to-end flow from standardized intake to versioned registration	Integration test report	PRD-3.5 Test production systems	
Run the edit-impact test: intended quality change, no over-editing	Edit-impact report	PRD-3.6 Test statistical business process	
Run the reproducibility test: regenerate a version from input and rules	Reproducibility report	PRD-3.6 Test statistical business process	CSDA Traceability
Evaluate imputation against a holdout sample with known values	Imputation evaluation report	PRD-3.6 Test statistical business process	
Finalize the production system and document operational procedures	Operational procedures manual, production-ready pipeline	PRD-3.7 Finalize production systems	

PLACEHOLDER: VS-03 implementation pipeline: the five design artifacts feeding the three build sub-processes, followed by technical testing, edit-impact testing, and reproducibility testing, resulting in a production-ready editing pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-03 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when VS-02 publishes the *Standardized Data Registered* event, or when a redelivery or correction requires a new quality version. The stages then run in order, with a loop between review and correction until no rule violations remain. Execution applies the designs and the solution as they are; a new type of error or a method that no longer works goes back to Design. Every run ends in one of three outcomes: registered, registered with documented residual quality issues, or held for further review.

Aspect	VS-03
Trigger	*Standardized Data Registered*; a redelivery or correction of an earlier version
Cadence	Per production cycle; per redelivery
Performed by	ROL-14 Data quality analyst, ROL-15 Subject-matter statistician, ROL-09 Classification steward, ROL-03 Data steward

Aspect	VS-03
Input	Standardized data version (#2) and its metadata, editing rules, imputation methods, derivation specifications
Output	Registered Processed Data quality version (#3), its edit lineage, and its quality evaluation
Control point	Gate of State #3 in VS-03.5 Perform Quality Evaluation and Versioning
Exception path	Unresolved records held for subject-matter review; a new error pattern or failing method back to Design; disclosure risk of linkage above the threshold escalated to ROL-16 Disclosure control specialist

Management: provides quality reports

The management process type runs alongside and after execution. It monitors the performance of the editing pipeline, assesses the quality improvement achieved, makes sure that every change can be audited, and feeds improvements back into Design. This closes the governance cycle.

Management sub-process	Refines	Main outputs
Monitor Process Performance	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard, throughput report, paradata records
Assess Data Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Quality improvement report, imputation quality report, trend analysis
Manage Methodology and Version Lifecycle	OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use; DLM-10 Archive and preserve data	Methodology version register, metadata quality report, version store audit
Audit and Compliance Reporting	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Edit audit summary, disclosure-risk and methodology compliance reports to VS-07 Governance and Compliance
Continuous Improvement	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, improvement tracking log

Monitor Process Performance

Refines: OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs.

The operational performance of VS-03 must be tracked systematically across production cycles. The process owner and the quality manager collect and aggregate the key performance indicators (see attribute 9). They monitor throughput against the processing window. They track paradata: who edited what, when, with which rule or model version, and how many manual interventions took place. This information feeds dashboards that give the process owner a current view of the health of the domain. The approach follows the Quality Management capability of GAMSO.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per production cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management

Activity	Key outputs	SAF process	Other reference
Monitor throughput against the processing window	Throughput report	OVA-4 Process Data Management	
Track paradata: operator, timing, rule and model versions, manual interventions	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Assess Data Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality improvement itself is evaluated against the criteria from *Design Quality Evaluation, Versioning, and Audit Framework*. Accuracy improvement compares error rates before and after editing. Completeness improvement compares missingness before and after imputation. Imputation quality is assessed against a holdout sample or known values. Consistency checks that no rule violations remain, and coherence checks the derived variables. Quality indicators are produced per dataset, per source, and per cycle. Trends are tracked to detect over-editing, deteriorating sources, or imputation models that need recalibration. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Evaluate accuracy and completeness improvement against the baseline	Quality improvement report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess imputation quality against a holdout sample or known values	Imputation quality report	OVA-1 Quality Management	CSDA Data Quality
Check for residual rule violations and over-editing	Consistency and over-editing report	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across sources, variables, and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality

Manage Methodology and Version Lifecycle

Refines: OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use; DLM-10 Archive and preserve data.

The editing methods and the quality versions they produce must stay current, complete, and well governed. This sub-process governs the life cycle of editing rules, imputation models, and derivation specifications. It versions them, manages transitions, and retires rules that are no longer needed. It verifies that every processed dataset carries its full edit lineage, imputation flags, and derived-variable metadata, as a compliance check on SAF-P09. It governs the versioning store: retention, immutability, the handling of revisions, and the reconciliation of catalog entries with stored versions. Because State #3 is usually subject to national archival requirements, the transfer of versions to long-term preservation is also governed here. **[VC19]**

Activity	Key outputs	SAF process	Other reference
Govern the life cycle and versions of editing rules, imputation models, and derivations	Methodology version register	OVA-2 Metadata Management	SAF-P03 Active life cycle management
Verify metadata completeness: edit lineage, imputation flags, derived-variable metadata	Metadata quality report	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Govern the versioning store: retention, immutability, revisions, reconciliation	Version store audit	DLM-5 Store and version data; DLM-7 Revise data	CSDA Traceability
Track metadata lineage from standardized to processed data	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability
Transfer quality versions to long-term preservation according to retention rules	Archive transfer record	DLM-10 Archive and preserve data	2.1.1 Data lifecycle management

Audit and Compliance Reporting

Refines: OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use.

This sub-process produces the evidence that VS-03 operates within methodological, legal, and organizational requirements. Edit audit reports show that every value change is attributable and reversible, which supports reproducibility and methodological transparency. Disclosure-risk reporting covers linkage and enrichment. It confirms that linkage of pseudonymized microdata stays within the assessed risk of re-identification, in line with SAF-P11. Methodology compliance reporting confirms that approved editing and imputation methods were applied. Security compliance reporting verifies that the designed access controls and the tamper-evident audit trail were applied. These reports meet the obligation of SAF capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance*.

Activity	Key outputs	SAF process	Other reference
Produce edit audit reports: every change attributable and reversible	Edit audit summary	OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	4.1.5 Accountability
Report on disclosure-risk compliance for linkage and enrichment	Disclosure-risk report	OVA-1 Quality Management	SAF-P11 Privacy by design; 3.3.1 Privacy management
Report on methodology compliance: approved editing and imputation methods	Methodology compliance report	OVA-1 Quality Management	4.1.5 Accountability
Report on security compliance: access controls, integrity of the audit trail	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10 Security by design; 3.3.3 Security incident management

Continuous Improvement

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This last management sub-process closes the governance cycle by analyzing performance and quality data and driving improvements. Trends across production cycles are examined: are error rates falling, is over-editing decreasing, is imputation quality improving, and is the share of coded free text rising? Improvement opportunities are identified. Examples are refining edit rules that produce false alarms, recalibrating imputation models, automating manual edits that recur, improving the coding service, and tuning selective-editing thresholds. Recurring errors that originate at the source are reported to VS-01, so they can be raised with the data provider. Recommendations are fed back to the design sub-processes, prioritized by impact and feasibility, and tracked through the following cycles. Where a new method is needed, the recommendation goes to VS-08. This is SAF capability 4.1.4 *Improvement management* in practice.

Activity	Key outputs	SAF process	Other reference
Analyze quality trends across production cycles	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities: rule refinement, model recalibration, threshold tuning	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Feed recommendations back to the design sub-processes, to VS-01 Data Acquisition and Ingestion, or to VS-08 Innovation and Methodological Development	Design feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management

PLACEHOLDER: VS-03 management feedback loop: data and paradata from Execution flowing through the five management sub-processes, with dashboards to the process owner, quality reports to the quality manager, compliance reports to VS-07, source issues to VS-01, and improvement recommendations back to Design

Metadata and paradata per process type

In VS-03, metadata is what keeps value changes transparent: every edit, imputation, and derivation must describe itself. The table shows what each process type contributes. Data metadata describes the datasets. Process metadata, also called paradata, describes how the process itself ran: who, when, and with which tools.

Process type	Data metadata produced	Process metadata (paradata) produced
Design	Editing and validation rule sets, imputation methods, derivation formulas, linkage rules, versioning policy	Log of design decisions, methodology version, review and approval records
Implementation	Engine configurations, rule parameters, parameters of imputation models, derivation definitions, versioning templates	Build log, test results, edit-impact test outcomes, deployment approvals

Process type	Data metadata produced	Process metadata (paradata) produced
Execution	Edit lineage per record (original value, rule, reason), imputation flags and methods, derived-variable formulas and inputs, linkage lineage, quality version identifiers	Operator identity, timing, volumes, edit and imputation rates, rule and model versions, records of manual overrides
Management	Quality improvement scores against the baseline, edit and imputation rate reports, lineage assessments, trend analyses	Editing dashboards, quality trends, improvement decisions, audit findings

PLACEHOLDER: VS-03 business object lifecycle: design artifacts govern implementation assets, which produce the pipeline that turns Standardized Data (#2) into Processed Data (#3), generating edit lineage, imputation flags, and audit records that management assesses and feeds back to design

5. Actors and roles

VS-03 is an internal value stream in which methodological judgment plays a large role. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the editing methodologist and the imputation specialist are ROL-07 Methodologist, and the coding specialist is ROL-09 Classification steward. VS-02, which supplies the standardized data, is a neighboring value stream and not an actor; the relation is governed by the handover interface in attribute 8. [**VC20**]

Actor or role	ID	Process type	Responsibility in VS-03 Data Editing and Enrichment
International statistical organization	ACT-07	External	Issues methodological guidance on editing, imputation, and classification (for example Eurostat)
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-03 Data Editing and Enrichment
Data owner	ROL-02	Management	Accountable for State #3; decides on access, reuse, and retention
Data steward	ROL-03	Execution	Registers quality versions, maintains catalog entries, applies access controls, resolves gate exceptions
Data custodian	ROL-04	Execution	Operates storage, access control, retention, and transfer to long-term preservation
Statistical designer	ROL-06	Design	Specifies derived variables and indicators, their formulas, and their inputs

Actor or role	ID	Process type	Responsibility in VS-03 Data Editing and Enrichment
Methodologist	ROL-07	Design	Designs error-detection rules, the selective-editing strategy, and the imputation methods
Metadata and data architect	ROL-08	Design	Designs the metadata model for edit lineage, imputation flags, and derived variables
Classification steward	ROL-09	Design, Execution	Designs and carries out the interpretive coding of free text; reviews cases with low confidence
Data engineer	ROL-10	Implementation	Builds and configures the editing, imputation, derivation, and linkage components
Platform engineer	ROL-11	Implementation	Configures editing services, connections to reference and donor data, and access controls
Test manager	ROL-12	Implementation	Plans and carries out technical, edit-impact, and reproducibility testing
Data quality analyst	ROL-14	Execution	Runs error detection, evaluates quality improvement, and logs edit lineage and exceptions
Subject-matter statistician	ROL-15	Execution	Reviews flagged records, decides on corrections, and signs off domain-specific edits
Disclosure control specialist	ROL-16	Design, Management	Assesses the disclosure risk of linkage and enrichment
Quality manager	ROL-18	Management	Monitors key performance indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures auditability of edits and compliance with disclosure-risk rules, and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-03 relies on, and what role they play. NSIs can use it to map the value stream to their own organization.

Capability group	Capabilities	Role in VS-03 Data Editing and Enrichment
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the editing and imputation methods and the derivation specifications, and build the editing pipeline
1.3 Processing	1.3.2 Deriving variables and units, 1.3.3 Check content, 1.3.4 Adjusting and imputing	Carry out error detection, editing, imputation, and derivation [**VC21**]
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage quality versions and edit metadata, assess quality improvement, link with statistical registers
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain editing, imputation, derivation, and linkage components
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.4 Logical identity and access control	Manage the disclosure risk of linkage and secure the edit audit trail
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward the editing methods, plan improvements, report on auditability and compliance

Stages, processes, and functions

The table shows which SAF business processes realize each stage of VS-03 and which business functions they mainly use. It follows the mapping in *Business Processes, Mapping to value stream stages*. The activity tables in attribute 3 can name additional supporting processes. [**VC25**]

Stage	Realized by processes	Main functions
VS-03.1 Conduct Error Detection and Quality Diagnostics	PRD-5.3 Review and validate	FUN-3.3 Process statistical data
VS-03.2 Perform Data Editing and Correction	PRD-5.2 Classify and code; PRD-5.4 Edit and impute	FUN-3.3 Process statistical data
VS-03.3 Impute Missing or Non-Response Data	PRD-5.4 Edit and impute	FUN-3.3 Process statistical data; FUN-4.1 Manage statistical and analytical methodology
VS-03.4 Derive Additional Variables and Indicators	PRD-5.1 Integrate data; PRD-5.5 Derive new variables and units	FUN-3.3 Process statistical data
VS-03.5 Perform Quality Evaluation and Versioning	PRD-5.8 Finalize data files; DLM-2 Promote data through a quality gate; DLM-3 Register and catalog data	FUN-3.3 Process statistical data; FUN-5.2 Assure and review quality; FUN-7.3 Manage data lifecycle and storage

PLACEHOLDER: ArchiMate view VS-03 (b) stages realized by processes and functions

Framework mappings

This reference table maps each process type of VS-03 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.2, PRD-2.5, DLM-1, DLM-2, DLM-4, DLM-9, OVA-1, OVA-4	Phase 2 Design (2.2, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Data Quality	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Implementation	PRD-3.2, PRD-3.4 to PRD-3.7, DLM-3, DLM-5, DLM-9	Phase 3 Build (3.2, 3.4 to 3.7)	Capability Management	Data Processing (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-5.1 to PRD-5.5, PRD-5.8, DLM-2 to DLM-5, OVA-1, OVA-2, OVA-4	Phase 5 Process (5.1 Integrate data, 5.2 Classify and code, 5.3 Review and validate, 5.4 Edit and impute, 5.5 Derive new variables and units, 5.8 Finalize data files)	Production (execution)	Data Processing (execution)	1.3.2 Deriving variables and units, 1.3.3 Check content, 1.3.4 Adjusting and imputing
Management	OVA-1, OVA-2, OVA-4, DLM-5, DLM-7, DLM-8, DLM-10, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-03 most directly.

Principle	Statement	Application in VS-03 Data Editing and Enrichment
SAF-P01	Reuse before open source, open source before buying, buying before building	Editing, imputation, and coding tools are reused across products first. Where new tooling is needed, open-source solutions developed with other NSIs, such as selective-editing engines and imputation libraries, come before purchase or own development. **[VC3]**
SAF-P07	Implement loosely coupled processes and systems	Editing is independent from standardization and estimation. Processed Data (State #3) is the decoupling interface: downstream consumers use edited and enriched microdata without knowing how it was corrected or imputed.
SAF-P08	Data is shared property	Processed data, once registered, is available to all authorized consumers. Editing and enrichment are not repeated per product when the same quality version can serve several purposes. Researchers and other domains reuse #3 directly.
SAF-P09	No data without metadata and classification	Every edit, imputation, and derived variable carries metadata: the rule applied, the original value, the imputation method and flag, the derivation formula, and the classification of derived variables. This is what makes the changes transparent and reproducible.
SAF-P10	Security by design	Access controls, integrity verification, and a tamper-evident edit audit trail are built into the pipeline, so every change to a value can be attributed and traced.
SAF-P11	Privacy by design	VS-03 Data Editing and Enrichment works on pseudonymized microdata. Linkage and enrichment with other datasets raise the risk of re-identification, so every linkage is assessed for disclosure risk and data minimization by design.

i Principles that apply across the process types

Several principles apply to all four process types. SAF-P09 is the backbone of the editing discipline: Design defines the rules and the required edit metadata, Execution records the original value, rule, and reason for every change, and Management assesses edit quality and closes gaps. SAF-P10

governs the tamper-evident edit audit trail across all stages. SAF-P11 applies in particular to *Design Integration and Linkage Rules* and to the stages VS-03.2 and VS-03.4, where linkage can raise the risk of re-identification. SAF-P07 governs both the intake of State #2 from VS-02 and the handover of State #3 to VS-04.

8. Interfaces, services, and events

Boundary interfaces

VS-03 is loosely coupled (SAF-P07) to its neighbors through two boundary interfaces. Specifying each boundary explicitly makes the loose-coupling contract concrete and testable: the handover artifact, the metadata it must carry, the event that triggers it, and the point where responsibility transfers. Each interface exchanges a registered, immutable artifact plus its metadata, never internal working data. Neither side therefore needs knowledge of the other's internal processing.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-02 Data Standardization and Integration → VS-03 Data Editing and Enrichment (intake)	Registered standardized data version (#2)	Minimum metadata set of State #2, including the list of unmapped values, the recorded units, currencies, and reference periods, the definition-difference records, and the pseudonymization record	*Standardized Data Registered*	VS-02 Data Standardization and Integration keeps custody of the standardized artifact; VS-03 Data Editing and Enrichment takes responsibility for editing, imputation, derivation, interpretive coding, and conversion on receipt of the event
VS-03 Data Editing and Enrichment → VS-04 Estimation, Analysis, and Statistical Modeling (handover)	Registered Processed Data quality version (#3)	Minimum metadata set of State #3: edit lineage, imputation flags and methods, derived-variable formulas and classification, linkage lineage, quality assessment, quality version identifier, classification	*Processed Data Registered*	VS-03 Data Editing and Enrichment keeps custody of the quality version; VS-04 Estimation, Analysis, and Statistical Modeling takes responsibility for weighting, estimation, and analysis on receipt of the event

Business services

Service	Offered to	Realized in	Interfaces
Error detection service	Editing teams	VS-03.1 Conduct Error Detection and Quality Diagnostics	Diagnostic reports, prioritized error lists
Editing and correction service	Editing teams, subject-matter statisticians	VS-03.2 Perform Data Editing and Correction	Review workbench for analysts, query interface for the edit audit trail

Service	Offered to	Realized in	Interfaces
Imputation service	Editing teams, methodologists	VS-03.3 Impute Missing or Non-Response Data	Imputation services per method
Derivation and integration service	Editing teams, downstream analysts	VS-03.4 Derive Additional Variables and Indicators	Derivation catalog, linkage interface
Processed data provisioning service	VS-04 Estimation, Analysis, and Statistical Modeling, VS-06 Dissemination and Service Delivery (microdata services), other products reusing the data	VS-03.5 Perform Quality Evaluation and Versioning	Event notification (*Processed Data Registered*) and a catalog query interface to find quality versions and their edit lineage
Quality version catalog service	All data consumers in the NSI	Management sub-process *Manage Methodology and Version Lifecycle*	Search and browse interface for quality versions, edit lineage, imputation flags, and derived-variable metadata
Design and methodology service	Methodologists, statistical designers	All Design sub-processes of VS-03 Data Editing and Enrichment	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artifacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Standardized Data Registered	Start of VS-03.1 Conduct Error Detection and Quality Diagnostics for that version (published by VS-02 Data Standardization and Integration)
Errors detected	Outcome of VS-03.1 Conduct Error Detection and Quality Diagnostics; starts automatic correction and review
Edits adjudicated	Completion of subject-matter review in VS-03.2 Perform Data Editing and Correction
Processed Data Registered	Completion of VS-03.5 Perform Quality Evaluation and Versioning; triggers VS-04 Estimation, Analysis, and Statistical Modeling and other consumers
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-03 event choreography between VS-02, the editing teams, subject-matter statisticians, and process management across Design, Implementation, Execution (detect, correct and review, impute, derive and link, evaluate and register), and Management (key performance indicators, quality improvement, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-03	Receives from VS-03
VS-07 Governance and Compliance	Methodology and privacy policy, approval of methods	Edit audit trails, methodology and disclosure-risk compliance evidence
VS-08 Innovation and Methodological Development	New editing, imputation, and coding methods	Methods that no longer work, improvement needs
VS-09 Infrastructure and Platform Enablement	Editing and imputation services, versioning store, catalog, event interfaces	Capacity and service requirements
VS-10 Customer and Stakeholder Engagement	Requirements of researchers and other users for processed microdata	Information on quality and on the methods applied

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
Edit rate	Share of values changed by editing	Per variable, per cycle
Imputation rate	Share of values imputed	Per variable, per cycle
Manual review rate	Share of records routed to subject-matter review	Per source, per cycle
Over-editing indicator	Share of edits with negligible impact on the results	Per cycle
Coding completion rate	Share of free-text values coded successfully	Per classification, per cycle
Processing time	Time from the *Standardized Data Registered* event to the registration of #3	Per dataset, per cycle

10. Toolkit

Examples from NSI practice, the maturity checklist, and end-to-end scenarios for VS-03 are part of the SAF Toolkit.

Toolkit: SAF-TK-V03 VS-03 Data Editing and Enrichment companion [**VC17**]

VS-04 Estimation, Analysis, and Statistical Modeling

1. Purpose and value created

VS-04 is where data becomes statistical information. It takes processed microdata and applies statistical methods, estimation techniques, and analytical models to produce results that describe a population: estimates, aggregates, indices, time series, and model outputs. VS-04 does not change microdata values. It computes new objects from them. Every result carries the method and parameters that produced it, a measure of its accuracy, and its lineage to the inputs. The value created is **analytical insight and statistical credibility**. Any figure that VS-04 produces can be regenerated and defended, and its fitness for purpose and coherence with related statistics have been checked.

VS-04 corresponds to the business domain **Estimation and closed-loop** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

- **Subject-matter statistician or analyst:** the stages in attribute 3, which describe what happens in each production cycle.

- **Methodologist or statistical designer:** the Design processes in attribute 4.
- **System builder or IT staff:** the Implementation processes in attribute 4.
- **Quality or compliance manager:** the Management processes in attribute 4.
- **Manager who wants the big picture:** attributes 1, 2, and 8. Examples and end-to-end scenarios are part of the SAF Toolkit.

i Signature discipline: insight, transparently produced

VS-04 guarantees methodological transparency, reproducibility, and statistical soundness. Every estimate carries the methodology and parameters that produced it. It carries an accuracy measure, such as a sampling error, a confidence interval, or a model diagnostic. It carries its lineage to the processed inputs, so any figure can be regenerated from State #3. Fitness for purpose and coherence with related and earlier statistics are explicit acceptance criteria.

i Metadata inherited from VS-03

VS-04 receives Processed Data together with the metadata VS-03 attached: the edit lineage, the definitions of derived variables, and the imputation flags. The imputation flags matter here. Estimation must reflect the extra uncertainty that imputed values introduce, for example through variance estimation that accounts for imputation. The edit and derivation lineage is carried through, so every estimate stays traceable to its observed and imputed inputs.

PLACEHOLDER: ArchiMate view VS-04 (a) value stream and capabilities

2. Position and state transition

VS-04 is the fourth primary value stream. It sits between editing (VS-03) and product composition (VS-05). It receives Processed Data when VS-03 publishes the *Processed Data Registered* event, and it hands over Statistics to VS-05.

VS-04 takes data from State #3 to State #4, with the status *analytical*. Before VS-04, the data is edited, complete, and enriched microdata. After VS-04, it is a registered, immutable version of statistics. The estimates are calculated, accuracy and quality indicators are produced, the lineage is documented, internal review is complete, and initial disclosure control has been applied. The results can potentially be released, but they are not yet released. The quality gate, the minimum metadata set, and the protection of State #4 are described in *State #4: Statistics* and are not repeated here.

State #4 has two gate moments. VS-04 passes the analytical gate, which confirms that the results are methodologically sound, accurate, coherent, and initially disclosure-controlled. VS-05 then passes the approval gate, which creates a new version with the status *approved*. Some processes start in VS-04 without passing VS-01 to VS-03 of their own. Macro statistics, such as national accounts, often start from Processed Data or Statistics that other domains have registered (see *Applying the steady states pragmatically*).

From	To	Gate	Event out
#3 Processed Data	#4 Statistics (status *analytical*)	Analytical gate of State #4 (methodological soundness, accuracy measures, coherence, model diagnostics, initial disclosure control, reproducibility, metadata and lineage)	*Output Data Registered*

PLACEHOLDER: ArchiMate view VS-04 (c) state handover and events: Processed Data (#3) in the upstream domain, the estimation service at the boundary (weighting, estimation, modeling, validation, initial disclosure control), Statistics (#4, analytical) inside the estimation domain

3. Stages

The stages describe the value that VS-04 adds in every production cycle. Together they form the execution process type of VS-04. In a mature steady state, the estimation strategy, the weighting method, and the models have been designed and built in advance (see attribute 4). Execution then focuses on running the cycle, computing weights, estimates, and models, and recording the methodology, parameters, and accuracy, so that every output can be reproduced and defended.

Stage	Name	Value added
VS-04.1	Confirm Estimation and Analytical Approach	The right, approved method is selected for this cycle's data and objective
VS-04.2	Apply Weighting and Calibration	Bias from sampling, non-response, and coverage is corrected
VS-04.3	Estimate and Aggregate Data	Representative statistics and aggregates are available, each with its accuracy
VS-04.4	Perform Model-Based Analysis	Relationships, trends, and small-domain estimates add insight beyond direct estimates
VS-04.5	Validate and Document Results	Sound, coherent, documented, and initially protected statistics are registered (State #4)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-04.1 Confirm Estimation and Analytical Approach

Value created: *The right, approved method selected for this cycle's data and objective.*

The stage starts when VS-03 publishes the *Processed Data Registered* event, or when a registered input from another domain is available for a downstream process. In a mature steady state, the methodology has been designed already. Execution in each cycle therefore confirms the approved approach for the data at hand, rather than choosing a new one. The team verifies that the calibration totals and reference sources are current. It confirms that the target parameters and domains of estimation still apply. A significant change, such as a new domain or a revision of the method, goes back to Design instead of being solved during the cycle.

Activity	Key outputs	SAF process	Other reference
Confirm or select the approved estimation and analytical approach for the cycle	Confirmed approach for the cycle	PRD-2.5 Design processing and analysis	1.3.5 Estimating population parameters
Verify that calibration totals and reference sources are current	Reference source confirmation	DLM-9 Manage reference and master data	CSDA Reference Data
Confirm the target parameters and domains of estimation	Confirmed estimation scope	PRD-5.6 Calculate weights	

Activity	Key outputs	SAF process	Other reference
Flag changes in methodology for design review	Methodology change exception (if any)	OVA-1 Quality Management	GAMSO Quality Management

VS-04.2 Apply Weighting and Calibration

Value created: *Bias from sampling, non-response, and coverage corrected, so that results represent the population.*

This stage computes the weights. The weighting engine applies the design weights and the non-response adjustment. The calibration engine then calibrates the weights to the confirmed population totals, using the designed auxiliary variables and enforcing the weight bounds. Convergence and the resulting distribution of weights are checked against the designed constraints. The calibration totals, auxiliary variables, and convergence diagnostics are recorded, so that the weighting can be reproduced. For register-based statistics without a sample, this stage may only apply coverage adjustments, or it may be skipped and the reason recorded.

Activity	Key outputs	SAF process	Other reference
Compute design weights and the non-response adjustment	Adjusted weights	PRD-5.6 Calculate weights	1.3.5 Estimating population parameters
Calibrate to the confirmed population totals with the designed auxiliary variables	Calibrated weights	PRD-5.6 Calculate weights	2.1.4 Statistical register management
Check convergence and the weight distribution against the bounds	Weight diagnostics	PRD-5.6 Calculate weights	
Record calibration totals, auxiliary variables, and diagnostics as lineage	Weighting lineage	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

VS-04.3 Estimate and Aggregate Data

Value created: *Representative statistics and aggregates, each carrying its accuracy.*

This stage computes the target estimates and aggregates. The estimation engine computes the target parameters with the calibrated weights. The aggregation pipeline computes the specified aggregates across the designed domains, with the agreed conventions for units and rounding. The variance estimation computes sampling errors and confidence intervals. It reflects both the survey design and the extra uncertainty from imputation, which the imputation flags of VS-03 signal. Every estimate is produced with its methodology, weights, and accuracy measure attached.

Activity	Key outputs	SAF process	Other reference
Compute the target estimates with the calibrated weights	Point estimates	PRD-5.7 Calculate aggregates	1.3.5 Estimating population parameters
Compute aggregates across the designed domains, with units and rounding	Output aggregates	PRD-5.7 Calculate aggregates	
Compute sampling errors and confidence intervals, including imputation uncertainty	Accuracy measures	PRD-6.1 Prepare draft outputs	2.1.3 Statistical quality management

Activity	Key outputs	SAF process	Other reference
Attach methodology, weights, and accuracy to every estimate	Estimate metadata	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

PLACEHOLDER: VS-04 estimation flow: processed microdata with imputation flags passing weighting and calibration against register totals, then estimation and aggregation per domain, with variance estimation producing an accuracy measure attached to every estimate

VS-04.4 Perform Model-Based Analysis

Value created: *Insight beyond direct estimates: relationships, trends, and estimates for small domains.*

This stage runs the model-based analyses specified in Design. Examples are regression models, time-series and forecasting models, and small-area models. The diagnostics of each model are computed and checked against the designed bounds. Seasonal adjustment is applied where it applies. Results are interpreted in context, and their assumptions and limitations are documented. As with direct estimates, every model output carries its specification, parameters, diagnostics, and uncertainty, so it can be reproduced and defended. Not every product needs this stage; where no models are specified, the stage is skipped.

Activity	Key outputs	SAF process	Other reference
Run the designed analytical, forecasting, and small-area models	Model outputs	PRD-6.1 Prepare draft outputs	1.4.1 Statistical (output) analysis
Compute model diagnostics and check them against the designed bounds	Model diagnostics	PRD-6.2 Validate outputs	
Apply seasonal adjustment or other time-series treatment where applicable	Adjusted series	PRD-6.1 Prepare draft outputs	
Interpret results and document assumptions and limitations	Analytical commentary	PRD-6.3 Interpret and explain outputs	

VS-04.5 Validate and Document Results

Value created: *Statistically sound, coherent, documented, and initially protected statistics, ready for handover.*

This stage validates the results and registers the official version. Results are checked for statistical soundness and consistency, and for coherence with earlier periods and related statistics. Significant movements are explained, and revisions are handled according to the revision policy. Initial statistical disclosure control is applied according to the designed policy. Disclosure risk is assessed, and protection methods such as suppression, rounding, or perturbation are applied, so that the output can potentially be released. The data must then pass the analytical gate of State #4. A persistent, immutable version is created with the status *analytical*, a unique identifier, and full lineage, and it is registered in the data catalog. Access permissions are applied. The stage ends by publishing the *Output Data Registered* event, which tells VS-05 that new validated statistics are available.

Output: Statistics (State #4, status *analytical*), the handover artifact to VS-05. Its structure and minimum metadata set are described in *State #4: Statistics*.

Activity	Key outputs	SAF process	Other reference
Validate soundness and coherence, explain movements, handle revisions	Validation report	PRD-6.2 Validate outputs; DLM-7 Revise data	2.1.3 Statistical quality management
Apply initial statistical disclosure control according to the designed policy	Disclosure-controlled outputs	PRD-6.4 Apply disclosure control	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Document the methodology and the quality indicators	Methodology and quality documentation	PRD-6.5 Finalize outputs; OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Create an immutable, identified version with full lineage (status *analytical*)	Versioned Statistics	PRD-6.5 Finalize outputs; DLM-2 Promote data through a quality gate; DLM-5 Store and version data	CSDA Traceability
Register metadata and apply access permissions	Catalog registration, access-controlled dataset	DLM-3 Register and catalog data; DLM-4 Classify and protect data	SAF-P10 Security by design
Publish the *Output Data Registered* event to notify VS-05 Statistical Product Composition	Event notification	DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

PLACEHOLDER: VS-04 statistics version structure: the analytical statistics version linked to its methodology and parameters, weights, accuracy measures, model diagnostics, disclosure control record, and lineage to Processed Data (#3)

4. The four process types

VS-04 runs through four process types. They do not form a linear sequence but a governance cycle: each process type produces its own output, and that output feeds the others. Design produces metadata, Implementation produces the execution process, Execution produces the statistical data, and Management produces quality reports that feed back into Design. The generic pattern is described in the Overview of this chapter.

The sub-processes below are specific to VS-04. Their names describe the work in the context of estimation and analysis. Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. This keeps every VS-04 sub-process traceable to the reference process model in *Business Processes*.

Design: provides metadata

The design process type produces the **metadata artifacts** that govern all other process types. In VS-04 these are the estimation strategy, the weighting and calibration method, the aggregation and model specifications, and the validation and disclosure-control policy. Without them, estimation becomes ad hoc and cannot be defended, and management has no baseline to measure accuracy and coherence against. Design runs when a new product is introduced, when a method is revised, or when Management feeds back an improvement.

Design sub-process	Refines	Main outputs
Design Estimation Strategy and Accuracy Approach	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Estimation strategy, estimation specification, accuracy measurement plan, coherence criteria
Design Weighting and Calibration Methodology	PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	Weighting specification, calibration specification, weight constraint policy, register of calibration-total sources
Design Aggregation and Model Specifications	PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis	Aggregation specification, model specification, output-statistic metadata standard, time-series treatment
Design Validation and Disclosure-Control Policy	PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Validation policy, disclosure-control policy, output documentation standard, reproducibility design
Design Quality-Measurement and Versioning Framework	DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-1 Quality Management	Quality indicator set, versioning and revision policy, lineage specification, access control design

Design Estimation Strategy and Accuracy Approach

Refines: PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

The starting point is the choice of how the statistics will be estimated. This sub-process selects the estimation strategy that fits the source and the analytical objective. Options are design-based estimation for sample surveys, aggregation for register-based statistics, and model-assisted or model-based estimation for small domains. It defines the target parameters, the reference populations, and the domains of estimation. It also defines how accuracy is measured: how sampling errors and confidence intervals are computed, and how the uncertainty introduced by imputation in VS-03 is reflected in variance estimation. Finally, the team defines the coherence criteria with related and earlier statistics.

Activity	Key outputs	SAF process	Other reference
Select the estimation strategy per source and objective: design-based, register-based, or model-based	Estimation strategy document	PRD-2.5 Design processing and analysis	1.3.5 Estimating population parameters
Define target parameters, reference populations, and domains of estimation	Estimation specification	PRD-2.5 Design processing and analysis	
Define how accuracy is measured, including the uncertainty from imputation	Accuracy measurement plan	PRD-2.5 Design processing and analysis	2.1.3 Statistical quality management
Define coherence criteria with related and earlier statistics	Coherence criteria	OVA-1 Quality Management	GAMSO Quality Management

Design Weighting and Calibration Methodology

Refines: PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data.

This sub-process designs how bias from sampling, non-response, and coverage is corrected through weighting. The team specifies the design weights and the model for non-response adjustment. It specifies the calibration

method, such as calibration to known population totals from registers. The auxiliary variables and the calibration totals are defined, together with their authoritative sources and versions. Bounds and acceptance constraints on weights are defined to prevent extreme weights. Recording the source and version of every calibration total makes the weighting reproducible later.

Activity	Key outputs	SAF process	Other reference
Specify design weights and the non-response adjustment	Weighting specification	PRD-2.5 Design processing and analysis	1.3.5 Estimating population parameters
Specify the calibration method, auxiliary variables, and calibration totals	Calibration specification	PRD-2.5 Design processing and analysis	
Define weight bounds and acceptance constraints	Weight constraint policy	PRD-2.5 Design processing and analysis	
Define the authoritative source and version of each calibration total	Register of calibration-total sources	DLM-9 Manage reference and master data	CSDA Reference Data; 2.1.4 Statistical register management

Design Aggregation and Model Specifications

Refines: PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis.

This sub-process specifies what is computed. For aggregation, each output statistic is defined by its formula, its domains and breakdowns, and its conventions for units and rounding. For analysis, the model-based methods are specified with their inputs, assumptions, and the diagnostics that must be produced and checked. Examples are regression, time-series and forecasting models, and small-area models. Where series are seasonally adjusted, the method and its parameters are specified. Each output statistic receives its metadata and classification, in line with SAF-P09.

Activity	Key outputs	SAF process	Other reference
Specify output aggregates: formulas, domains and breakdowns, units, rounding	Aggregation specification	PRD-2.1 Design outputs and dissemination	
Specify analytical and forecasting models: inputs, assumptions, diagnostics	Model specification	PRD-2.5 Design processing and analysis	1.4.1 Statistical (output) analysis
Define metadata and classification for every output statistic	Output-statistic metadata standard	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification
Define seasonal adjustment and other time-series treatment where applicable	Time-series treatment specification	PRD-2.5 Design processing and analysis	

Design Validation and Disclosure-Control Policy

Refines: PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

This sub-process designs how results are validated and protected before handover. The validation criteria cover statistical soundness: plausibility, consistency, and model diagnostics within bounds. They also cover coherence with related and earlier statistics, and the revision policy. The policy for statistical disclosure control is designed. It sets the criteria for disclosure risk of tables and output microdata, and the protection

methods, such as suppression, rounding, and perturbation. Reproducibility is designed in: an output must be regenerable from the processed input and the recorded methodology and parameters. Together, these rules make up the VS-04 part of the analytical gate of State #4.

Activity	Key outputs	SAF process	Other reference
Define validation criteria: soundness, coherence, model diagnostics, revision policy	Validation policy	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	GAMSO Quality Management
Define disclosure-risk criteria and protection methods	Disclosure-control policy	PRD-2.1 Design outputs and dissemination	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Define the requirements for output metadata and methodology documentation	Output documentation standard	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification
Design reproducibility: regenerate outputs from inputs, methodology, and parameters	Reproducibility design	PRD-2.5 Design processing and analysis	SAF-P10 Security by design

Design Quality-Measurement and Versioning Framework

Refines: DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-1 Quality Management.

This last design sub-process makes sure that accuracy is measured and that outputs can be traced. It defines the quality indicators to produce, such as the sampling error, the coefficient of variation, response and coverage rates, and model fit. It defines the versioning and revision policy: how versions are identified, what triggers a revision, and how immutability and retention are handled. It defines the lineage from each statistic back to its processed inputs and the method applied. Role-based access controls for outputs are designed in line with SAF-P10. Unreleased statistics are confidential until release, so access is limited to the roles that need them.

Activity	Key outputs	SAF process	Other reference
Define quality indicators: sampling error, coefficient of variation, response and coverage, model fit	Quality indicator set	OVA-1 Quality Management	GAMSO Quality Management
Define the versioning and revision policy for outputs	Versioning and revision policy	DLM-1 Plan and design data assets	CSDA Traceability
Define the lineage from each statistic to its processed inputs and method	Lineage specification	DLM-1 Plan and design data assets	CSDA Traceability
Design role-based access control for unreleased outputs	Access control design	DLM-4 Classify and protect data	SAF-P10 Security by design; 3.3.4 Logical identity and access control

PLACEHOLDER: VS-04 design process decomposition: inputs (State #3 metadata, methodology standards, register totals, legal and confidentiality requirements), the five design sub-processes in order, and the design artifact each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

The implementation process type takes the metadata artifacts from Design and **builds, configures, and tests** the estimation and analysis pipeline until it is ready for production. Nothing runs operationally until this process type is complete. Components are reused where possible and built only where needed, in line with SAF-P01. Calibration, variance-estimation, and seasonal-adjustment tools shared in the NSI community are good candidates for reuse.

Implementation sub-process	Refines	Main outputs
Build Weighting and Calibration Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data	Weighting engine, calibration engine, connection to calibration totals
Build Estimation and Aggregation Pipelines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Estimation pipeline, variance component, aggregation pipeline
Build Modeling Environment, Validation, and SDC Tooling	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data	Modeling environment, validation tooling, disclosure-control tooling, versioning store, events
Test and Validate the Pipeline	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems	Test reports, reproducibility and benchmark reports, production-ready pipeline

Build Weighting and Calibration Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data.

This sub-process turns the weighting and calibration designs into operational tools. The weighting engine is built or configured to compute design weights and non-response adjustments. The calibration engine is built or configured to calibrate to the designed totals with the specified auxiliary variables, enforcing the weight bounds. The connections to calibration totals are configured to the authoritative register and reference sources, with the version of each total fixed and recorded. All weighting logic is version-controlled in line with SAF-P10. Any set of weights can therefore be reproduced from its inputs and the recorded versions.

Activity	Key outputs	SAF process	Other reference
Build or configure the weighting engine: design weights, non-response adjustment	Configured weighting engine	PRD-3.2 Reuse or build processing and analysis components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes

Activity	Key outputs	SAF process	Other reference
Build or configure the calibration engine with auxiliary variables and bounds	Configured calibration engine	PRD-3.2 Reuse or build processing and analysis components	1.3.5 Estimating population parameters
Configure connections to versioned register sources for calibration totals	Integration of calibration totals	DLM-9 Manage reference and master data	CSDA Reference Data
Put all weighting and calibration logic under version control	Versioned methodology repository	PRD-3.4 Configure workflows	SAF-P10 Security by design

Build Estimation and Aggregation Pipelines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

In this sub-process the designed estimation and aggregation specifications become automated services. The estimation engine computes the target parameters with the designed accuracy measures. The variance component computes sampling errors and confidence intervals. It accounts for the survey design and for the uncertainty from imputation, using the imputation flags of VS-03. The aggregation pipeline computes the specified aggregates across the designed domains, with the agreed units and rounding. Every output is produced with its methodology, parameters, weights, and accuracy measure attached.

Activity	Key outputs	SAF process	Other reference
Build the estimation engine for the designed target parameters	Estimation pipeline	PRD-3.2 Reuse or build processing and analysis components	1.3.5 Estimating population parameters
Build the variance component: design and imputation uncertainty	Variance and accuracy component	PRD-3.2 Reuse or build processing and analysis components	2.1.3 Statistical quality management
Build the aggregation pipeline: domains, units, rounding	Aggregation pipeline	PRD-3.2 Reuse or build processing and analysis components	
Attach methodology, weights, and accuracy measure to every output	Binding of output metadata	PRD-3.4 Configure workflows	SAF-P09 No data without metadata and classification

Build Modeling Environment, Validation, and SDC Tooling

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data.

This sub-process implements model-based analysis, validation, and disclosure control. The modeling environment is built or configured to run the specified models with their diagnostics, and seasonal adjustment where it applies. Validation tooling implements the designed checks on soundness and coherence, including comparison with earlier periods and related statistics. The tooling for statistical disclosure control implements the designed risk checks and protection methods. The versioning store is built so that each accepted output version is immutable, identified, reproducible, and accompanied by its lineage. Event notifications are configured, so that VS-05 is alerted when a new version of Statistics is registered.

Activity	Key outputs	SAF process	Other reference
Build or configure the modeling environment: models, diagnostics, seasonal adjustment	Modeling environment	PRD-3.2 Reuse or build processing and analysis components	1.4.1 Statistical (output) analysis

Activity	Key outputs	SAF process	Other reference
Implement validation tooling: soundness, coherence, comparison with earlier periods	Validation tooling	PRD-3.2 Reuse or build processing and analysis components	
Implement disclosure-control tooling: risk checks, suppression, rounding, perturbation	Disclosure-control tooling	PRD-3.2 Reuse or build processing and analysis components	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Build the versioning store: immutable, identified, reproducible	Versioning store	DLM-5 Store and version data	CSDA Traceability
Configure the *Output Data Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows; DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

Test and Validate the Pipeline

Refines: PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems.

Before the pipeline goes into production, it must be thoroughly verified. Technical testing covers all individual components: weighting, calibration, estimation, variance, aggregation, modeling, validation, disclosure control, and the versioning store. Integration testing checks the end-to-end flow from the *Processed Data Registered* event to the registration of Statistics. A reproducibility test confirms that an output can be regenerated identically from its input and the recorded methodology and parameters. A benchmark test compares results with a known benchmark or with an earlier series. Accuracy measures and model diagnostics are checked against expectations. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	
Test integration: end-to-end flow from processed intake to versioned registration	Integration test report	PRD-3.5 Test production systems	
Run the reproducibility test: regenerate an output from inputs and methodology	Reproducibility report	PRD-3.6 Test statistical business process	CSDA Traceability
Run the benchmark test against known or earlier results	Benchmark test report	PRD-3.6 Test statistical business process	
Check accuracy measures and model diagnostics against expectations	Accuracy verification report	PRD-3.6 Test statistical business process	2.1.3 Statistical quality management
Finalize the production system and document operational procedures	Operational procedures manual, production-ready pipeline	PRD-3.7 Finalize production systems	

PLACEHOLDER: VS-04 implementation pipeline: the five design artifacts feeding the three build sub-processes, followed by technical testing, reproducibility testing, and benchmark testing, resulting in a production-ready estimation pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-04 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when VS-03 publishes the *Processed Data Registered* event, when a registered input from another domain is available, or when a revision is due. The stages run in order; VS-04.4 is skipped where no models are specified. Execution applies the designs and the solution as they are; a new domain or a method that no longer fits goes back to Design. Every run ends in one of three outcomes: registered as *analytical*, registered with documented quality restrictions, or returned for re-estimation.

Aspect	VS-04
Trigger	*Processed Data Registered*; a registered input from another domain; a scheduled revision
Cadence	Per production cycle, in line with the release calendar
Performed by	ROL-15 Subject-matter statistician, ROL-07 Methodologist, ROL-03 Data steward
Input	Processed Data quality version (#3) or other registered input, calibration totals, approved methodology
Output	Registered version of Statistics (#4, *analytical*), its methodology and accuracy documentation
Control point	Analytical gate of State #4 in VS-04.5 Validate and Document Results
Exception path	Implausible results back to VS-04.2 Apply Weighting and Calibration or VS-04.3 Estimate and Aggregate Data; a data problem back to VS-03 Data Editing and Enrichment; a method problem back to Design; disclosure risk above the threshold escalated to ROL-16 Disclosure control specialist

Management: provides quality reports

The management process type runs alongside and after execution. It monitors the performance of the estimation pipeline, assesses the accuracy and coherence of the outputs, makes sure that every result can be reproduced and is disclosure-safe, and feeds improvements back into Design. This closes the governance cycle.

Management sub-process	Refines	Main outputs
Monitor Process Performance	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard, throughput report, paradata records
Assess Output Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Accuracy report, coherence report, revision analysis, trend analysis
Manage Methodology and Output-Version Lifecycle	OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use	Methodology version register, metadata quality report, version store audit

Management sub-process	Refines	Main outputs
Audit and Compliance Reporting	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Reproducibility audit, disclosure-control and methodology compliance reports to VS-07 Governance and Compliance
Continuous Improvement	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, improvement tracking log

Monitor Process Performance

Refines: OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs.

The operational performance of VS-04 must be tracked systematically across production cycles. The process owner and the quality manager collect and aggregate the key performance indicators (see attribute 9). They monitor throughput against the release calendar, because VS-04 is often on the critical path to publication. They track paradata: which methods, parameter versions, and software versions were run, when, by whom, and how many manual interventions took place. This information feeds dashboards that give the process owner a current view of the health of the domain. The approach follows the Quality Management capability of GAMSO.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per production cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Monitor throughput against the release calendar	Throughput report	OVA-4 Process Data Management	
Track paradata: methods, parameter and software versions, interventions	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Assess Output Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the outputs is evaluated against the criteria from *Design Validation and Disclosure-Control Policy* and *Design Quality-Measurement and Versioning Framework*. Accuracy covers sampling error, the coefficient of variation, and model fit. Coherence covers consistency with related and earlier statistics. Timeliness compares production with the release calendar. Reliability is assessed through revision analysis, which compares preliminary and final figures. Quality indicators are produced per statistic, per output, and per cycle. Trends are tracked to detect falling accuracy, growing revisions, or models that need recalibration. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Evaluate accuracy: sampling error, coefficient of variation, model fit	Accuracy report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess coherence with related and earlier statistics	Coherence report	OVA-1 Quality Management	CSDA Data Quality

Activity	Key outputs	SAF process	Other reference
Carry out revision analysis: preliminary against final figures	Revision analysis	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across statistics and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality

Manage Methodology and Output-Version Lifecycle

Refines: OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use.

The estimation methods and the output versions they produce must stay current, complete, and well governed. This sub-process governs the life cycle of estimation methods, weighting and calibration specifications, and models. It versions them, manages transitions, and retires methods that are no longer used. A change of method often has consequences for earlier periods, so the need to revise back series is assessed with every change. The sub-process verifies that every output carries its full methodology, weights, accuracy measures, and lineage, as a compliance check on SAF-P09. It also governs the versioning store: retention, immutability, revisions, and the reconciliation of catalog entries with stored versions.

Activity	Key outputs	SAF process	Other reference
Govern the life cycle and versions of estimation methods, weighting, and models	Methodology version register	OVA-2 Metadata Management	SAF-P03 Active life cycle management
Assess the impact of method changes on back series	Back-series impact assessment	DLM-7 Revise data	
Verify metadata completeness: methodology, weights, accuracy, lineage	Metadata quality report	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Govern the versioning store: retention, immutability, revisions, reconciliation	Version store audit	DLM-5 Store and version data; DLM-7 Revise data	CSDA Traceability
Track metadata lineage from processed data to statistics	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability

Audit and Compliance Reporting

Refines: OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use.

This sub-process produces the evidence that VS-04 operates within methodological, legal, and organizational requirements. Reproducibility audit reports show that every figure can be regenerated from the processed input and the recorded methodology and parameters. This supports methodological transparency and accountability. Disclosure-control reporting confirms that disclosure control was applied and that outputs meet the risk criteria, in line with SAF-P11. Methodology compliance reporting confirms that approved estimation and analytical methods were applied. Security compliance reporting verifies that the designed access controls and the tamper-evident record of methods were applied. These reports meet the obligation of SAF capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance*.

Activity	Key outputs	SAF process	Other reference
Produce reproducibility audit reports: every figure can be regenerated	Reproducibility audit summary	OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	4.1.5 Accountability
Report on the disclosure control applied and its outcomes	Disclosure-control report	OVA-1 Quality Management	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Report on methodology compliance: approved methods applied	Methodology compliance report	OVA-1 Quality Management	4.1.5 Accountability
Report on security compliance: access controls, integrity of the methodology record	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10 Security by design; 3.3.3 Security incident management

Continuous Improvement

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This last management sub-process closes the governance cycle by analyzing performance and quality data and driving improvements. Trends across production cycles are examined: are sampling errors falling, are revisions getting smaller, is coherence improving, and do the models still hold? Improvement opportunities are identified. Examples are refining the calibration to reduce variance, recalibrating or replacing models, improving the non-response adjustment, adopting better variance estimators, and shortening the time to publication. Recommendations are fed back to the design sub-processes, prioritized by impact and feasibility, and tracked through the following cycles. Problems that originate in the data are reported to VS-03. Where a new method is needed, the recommendation goes to VS-08. This is SAF capability 4.1.4 *Improvement management* in practice.

Activity	Key outputs	SAF process	Other reference
Analyze quality trends across production cycles	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities: calibration, models, variance estimators, timeliness	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Feed recommendations back to the design sub-processes, to VS-03 Data Editing and Enrichment, or to VS-08 Innovation and Methodological Development	Design feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management

PLACEHOLDER: VS-04 management feedback loop: outputs and paradata from Execution flowing through the five management sub-processes, with dashboards to the process owner, accuracy and coherence reports to the quality manager, compliance reports to VS-07, data issues to VS-03, and improvement recommendations back to Design

Metadata and paradata per process type

In VS-04, metadata is what makes a published figure defensible: every estimate must carry the method, parameters, weights, and accuracy that produced it. The table shows what each process type contributes. Data metadata describes the outputs. Process metadata, also called paradata, describes how the process itself ran: who, when, and with which tools.

Process type	Data metadata produced	Process metadata (paradata) produced
Design	Estimation and weighting methodology, calibration totals and their sources, aggregation specifications, model specifications, validation and disclosure-control policy	Log of design decisions, methodology version, review and approval records
Implementation	Engine configurations, weighting and estimation parameters, model parameters, variance settings, disclosure-control parameters	Build log, test results, reproducibility test outcomes, deployment approvals
Execution	Methodology per estimate, weights, accuracy measures, model diagnostics, reference population, record of disclosure control applied	Operator identity, timing, software versions, parameter values, run identifiers
Management	Accuracy and coherence assessments, revision analyses, lineage assessments, trend analyses	Accuracy dashboards, coherence reports, improvement decisions, audit findings

PLACEHOLDER: VS-04 business object lifecycle: design artifacts govern implementation assets, which produce the pipeline that turns Processed Data (#3) into Statistics (#4, analytical), generating methodology, accuracy, and audit records that management assesses and feeds back to design

5. Actors and roles

VS-04 is an internal value stream driven by methodology and subject-matter expertise. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the estimation methodologist is ROL-07 Methodologist, the model designer is ROL-06 Statistical designer, and the survey statistician and the subject-matter reviewer are ROL-15 Subject-matter statistician. VS-03, which supplies the processed data, is a neighboring value stream and not an actor; the relation is governed by the handover interface in attribute 8. **[VC20]**

Actor or role	ID	Process type	Responsibility in VS-04 Estimation, Analysis, and Statistical Modeling
International statistical organization	ACT-07	External	Issues methodological guidance and regulations on estimation, seasonal adjustment, and accuracy reporting (for example Eurostat)
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-04 Estimation, Analysis, and Statistical Modeling
Data owner	ROL-02	Management	Accountable for analytical versions of State #4; decides on internal access and reuse
Data steward	ROL-03	Execution	Registers output versions, maintains catalog entries, applies access controls, resolves gate exceptions
Data custodian	ROL-04	Execution	Operates storage, access control, and retention of unreleased statistics
Statistical designer	ROL-06	Design	Specifies output aggregates and analytical and forecasting models, with their inputs and diagnostics
Methodologist	ROL-07	Design, Execution	Designs the estimation strategy, weighting and calibration, and accuracy measurement; confirms the approach per cycle
Metadata and data architect	ROL-08	Design	Designs the metadata model for methodology, accuracy, and lineage of outputs
Data engineer	ROL-10	Implementation	Builds and configures the weighting, estimation, aggregation, and modeling components
Platform engineer	ROL-11	Implementation	Configures estimation services, connections to calibration-total sources, and access controls
Test manager	ROL-12	Implementation	Plans and carries out technical, reproducibility, and benchmark testing

Actor or role	ID	Process type	Responsibility in VS-04 Estimation, Analysis, and Statistical Modeling
Subject-matter statistician	ROL-15	Execution	Computes estimates, runs models, reviews results for plausibility and coherence, and signs off
Disclosure control specialist	ROL-16	Design, Execution	Designs the disclosure-control policy and assesses the disclosure risk of outputs
Quality manager	ROL-18	Management	Monitors accuracy and coherence indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures reproducibility and compliance with disclosure-risk rules, and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-04 relies on, and what role they play. NSIs can use it to map the value stream to their own organization.

Capability group	Capabilities	Role in VS-04 Estimation, Analysis, and Statistical Modeling
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the estimation and weighting methods and the models, and build the estimation pipeline
1.3 Processing	1.3.5 Estimating population parameters, 1.3.7 Statistical disclosure	Compute weights, estimates, and aggregates, and apply initial disclosure control **[VC21]**
1.4 Analysis	1.4.1 Statistical (output) analysis	Run model-based analysis, validate, and interpret results **[VC21]**
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage output versions and methodology metadata, assess accuracy and coherence, use registers for calibration totals
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain weighting, estimation, aggregation, and modeling components
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.4 Logical identity and access control	Assess the disclosure risk of outputs and protect unreleased statistics

Capability group	Capabilities	Role in VS-04 Estimation, Analysis, and Statistical Modeling
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward the estimation methods, plan improvements, report on reproducibility and compliance

Stages, processes, and functions

The table shows which SAF business processes realize each stage of VS-04 and which business functions they mainly use. It follows the mapping in *Business Processes, Mapping to value stream stages*. The activity tables in attribute 3 can name additional supporting processes. **[VC25]**

Stage	Realized by processes	Main functions
VS-04.1 Confirm Estimation and Analytical Approach	PRD-2.4 Design frame and sample; PRD-2.5 Design processing and analysis	FUN-3.1 Design and build statistical products; FUN-4.1 Manage statistical and analytical methodology
VS-04.2 Apply Weighting and Calibration	PRD-5.6 Calculate weights	FUN-3.4 Estimate and analyze
VS-04.3 Estimate and Aggregate Data	PRD-5.7 Calculate aggregates	FUN-3.4 Estimate and analyze
VS-04.4 Perform Model-Based Analysis	PRD-6.1 Prepare draft outputs; PRD-6.3 Interpret and explain outputs	FUN-3.4 Estimate and analyze
VS-04.5 Validate and Document Results	PRD-6.2 Validate outputs; PRD-6.4 Apply disclosure control; OVA-2 Metadata Management	FUN-3.4 Estimate and analyze; FUN-3.5 Protect statistical confidentiality; FUN-7.2 Manage metadata

PLACEHOLDER: ArchiMate view VS-04 (b) stages realized by processes and functions

Framework mappings

This reference table maps each process type of VS-04 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.1, PRD-2.5, DLM-1, DLM-4, DLM-9, OVA-1	Phase 2 Design (2.1, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Data Quality	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Implementation	PRD-3.2, PRD-3.4 to PRD-3.7, DLM-3, DLM-5, DLM-9	Phase 3 Build (3.2, 3.4 to 3.7)	Capability Management	Data Analysis (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-5.6, PRD-5.7, PRD-6.1 to PRD-6.5, DLM-2 to DLM-5, DLM-7, DLM-9, OVA-1, OVA-2	Phase 5 Process (5.6 Calculate weights, 5.7 Calculate aggregates); Phase 6 Analyze (6.1 to 6.5, with 6.4 as initial disclosure control)	Production (execution)	Data Analysis (execution)	1.3.5 Estimating population parameters, 1.3.7 Statistical disclosure, 1.4.1 Statistical (output) analysis
Management	OVA-1, OVA-2, OVA-4, DLM-5, DLM-7, DLM-8, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-04 most directly.

Principle	Statement	Application in VS-04 Estimation, Analysis, and Statistical Modeling
SAF-P01	Reuse before open source, open source before buying, buying before building	Estimation, calibration, variance, and seasonal-adjustment tools are reused across products first. Where new tooling is needed, open-source solutions developed with other NSIs come before purchase or own development. **[VC3]**
SAF-P07	Implement loosely coupled processes and systems	Estimation is independent from editing and composition. Statistics (State #4) is the decoupling interface: downstream consumers use validated statistics without knowing the internals of estimation.

Principle	Statement	Application in VS-04 Estimation, Analysis, and Statistical Modeling
SAF-P08	Data is shared property	Statistics and output microdata, once validated, are available to all authorized internal consumers. Estimation is not repeated per product when the same validated output can serve several purposes, for example as input for macro statistics.
SAF-P09	No data without metadata and classification	Every estimate carries metadata: the methodology and parameters, the weights, the accuracy measures, the reference population, and the classification of derived statistics. This is what makes results transparent and reproducible.
SAF-P10	Security by design	Access controls, integrity verification, and a tamper-evident record of methods and parameters are built into the pipeline, so every figure can be attributed and reproduced. Unreleased statistics are protected against early disclosure.
SAF-P11	Privacy by design	Initial statistical disclosure control is applied to outputs, and the disclosure risk of estimates and output microdata is assessed by design, so the results can potentially be released.

i Principles that apply across the process types

Several principles apply to all four process types. SAF-P09 is the backbone of methodological transparency: Design defines the methodology and the required output metadata, Execution records method, parameters, weights, and accuracy for every estimate, and Management assesses output quality and coherence. SAF-P10 governs the tamper-evident record of methods and the reproducibility of outputs across all stages. SAF-P11 applies in particular to *Design Validation and Disclosure-Control Policy* and to VS-04.5, where disclosure risk is assessed before handover. SAF-P07 governs both the intake of State #3 from VS-03 and the handover of State #4 to VS-05.

8. Interfaces, services, and events

Boundary interfaces

VS-04 is loosely coupled (SAF-P07) to its neighbors through two boundary interfaces. Specifying each boundary explicitly makes the loose-coupling contract concrete and testable: the handover artifact, the metadata it must carry, the event that triggers it, and the point where responsibility transfers. Each interface exchanges a registered, immutable artifact plus its metadata, never internal working data. Neither side therefore needs knowledge of the other's internal processing.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-03 Data Editing and Enrichment → VS-04 Estimation, Analysis, and Statistical Modeling (intake)	Registered Processed Data quality version (#3)	Minimum metadata set of State #3, including edit lineage, derived-variable definitions, imputation flags and methods, quality assessment, and quality version identifier	*Processed Data Registered*	VS-03 Data Editing and Enrichment keeps custody of the quality version; VS-04 Estimation, Analysis, and Statistical Modeling takes responsibility for weighting, estimation, modeling, and validation on receipt of the event
VS-04 Estimation, Analysis, and Statistical Modeling → VS-05 Statistical Product Composition (handover)	Registered version of Statistics (#4, *analytical*)	Methodology and parameters, weights, accuracy measures, model diagnostics, record of disclosure control applied, quality indicators, version identifier, lineage to the processed source, classification	*Output Data Registered*	VS-04 Estimation, Analysis, and Statistical Modeling keeps custody of the analytical version; VS-05 Statistical Product Composition takes responsibility for composition and approval on receipt of the event

Business services

Service	Offered to	Realized in	Interfaces
Weighting and calibration service	Estimation teams	VS-04.2 Apply Weighting and Calibration	Weighting and calibration engines, connection to calibration totals
Estimation and aggregation service	Estimation teams, analysts	VS-04.3 Estimate and Aggregate Data	Estimation interface that returns estimates with accuracy measures; query interface for methodology and parameters
Modeling and analysis service	Analysts, methodologists	VS-04.4 Perform Model-Based Analysis	Modeling environment, diagnostics reports
Validation and disclosure-control service	Quality managers, disclosure control specialists	VS-04.5 Validate and Document Results	Validation and disclosure-control reports
Statistics provisioning service	VS-05 Statistical Product Composition, other internal products using the statistics as input	VS-04.5 Validate and Document Results	Event notification (*Output Data Registered*) and a catalog query interface to find output versions, their accuracy, and methodology

Service	Offered to	Realized in	Interfaces
Output version catalog service	All data consumers in the NSI	Management sub-process *Manage Methodology and Output-Version Lifecycle*	Search and browse interface for output versions, accuracy measures, methodology documentation, and lineage
Design and methodology service	Methodologists, statistical designers	All Design sub-processes of VS-04 Estimation, Analysis, and Statistical Modeling	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artifacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Processed Data Registered	Start of VS-04.1 Confirm Estimation and Analytical Approach for that version (published by VS-03 Data Editing and Enrichment)
Calibration totals updated	Review of calibration in VS-04.1 Confirm Estimation and Analytical Approach before the next run
Results validated / *Results rejected*	Outcome of validation in VS-04.5 Validate and Document Results; rejection starts re-estimation or a return to VS-03 Data Editing and Enrichment
Output Data Registered	Completion of VS-04.5 Validate and Document Results; triggers VS-05 Statistical Product Composition
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-04 event choreography between VS-03, the estimation teams, subject-matter statisticians, and process management across Design, Implementation, Execution (confirm approach, weight and calibrate, estimate, model, validate and register), and Management (key performance indicators, accuracy and coherence, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-04	Receives from VS-04
VS-07 Governance and Compliance	Methodology and confidentiality policy, approval of methods	Reproducibility audits, methodology and disclosure-control compliance evidence
VS-08 Innovation and Methodological Development	New estimation, modeling, and seasonal-adjustment methods	Methods that no longer fit, improvement needs
VS-09 Infrastructure and Platform Enablement	Estimation and modeling environments, versioning store, catalog, event interfaces	Capacity requirements, in particular around release dates
VS-10 Customer and Stakeholder Engagement	Requirements of users for accuracy, breakdowns, and timeliness	Information on accuracy and on methods applied

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
On-time output rate	Share of outputs produced within the window of the release calendar	Per output, per cycle
Average sampling error or coefficient of variation	Typical accuracy of key estimates	Per statistic, per cycle
Coherence check pass rate	Share of outputs that pass the coherence checks on the first run	Per cycle
Revision magnitude	Size of revisions against previously published figures	Per statistic, per cycle
Reproducibility rate	Share of outputs that can be regenerated identically from inputs and methodology	Per cycle
Processing time	Time from the *Processed Data Registered* event to the registration of #4	Per output, per cycle

10. Toolkit

Examples from NSI practice, the maturity checklist, and end-to-end scenarios for VS-04 are part of the SAF Toolkit.

Toolkit: SAF-TK-V04 VS-04 Estimation, Analysis, and Statistical Modeling companion **[VC17]**

VS-05 Statistical Product Composition

1. Purpose and value created

VS-05 turns separate analytical results into finished statistical products. It takes validated statistics from VS-04 and brings them together into coherent products: integrated tables, composite indicators, indices, time series, and analytical publications with their narrative and visualizations. VS-05 does not re-estimate. It composes validated outputs as they are received, and it carries their accuracy and methodology into the product. Every product is checked for internal consistency and coherence with related statistics, explained in plain language, and approved against dissemination and confidentiality standards. The value created is **coherence and interpretability**. Users receive a product that is consistent, understandable, and signed off by the NSI.

VS-05 corresponds to the business domain **Statistically composed and composite products** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

- **Subject-matter statistician, author, or release manager:** the stages in attribute 3, which describe what happens in each production cycle.
- **Statistical designer or methodologist:** the Design processes in attribute 4.
- **System builder or IT staff:** the Implementation processes in attribute 4.
- **Quality or compliance manager:** the Management processes in attribute 4.
- **Manager who wants the big picture:** attributes 1, 2, and 8. Examples and end-to-end scenarios are part of the SAF Toolkit.

> **i** Signature discipline: coherence and interpretability
>
> VS-05 composes; it does not re-estimate. Composite products are internally consistent, methodologically sound, coherent with related and earlier statistics, and accompanied by the metadata and interpretation that users need. Composition never silently changes a component estimate. Any

adjustment for coherence is documented and traceable. Every product carries its lineage to the component outputs and the composition method, and it is approved before it is registered.

i Metadata inherited from VS-04

VS-05 receives Statistics together with the metadata VS-04 attached: the methodology, the weights, the accuracy measures, and the record of the disclosure control already applied. VS-05 carries this accuracy and quality metadata into the composite product. It does not discard or recompute it. It uses the metadata to weight or caveat composed figures, and to run coherence checks across the combined outputs.

PLACEHOLDER: ArchiMate view VS-05 (a) value stream and capabilities

2. Position and state transition

VS-05 is the fifth primary value stream. It sits between estimation (VS-04) and dissemination (VS-06). It receives Statistics when VS-04 publishes the *Output Data Registered* event, and it hands over approved statistical products to VS-06.

VS-05 works within State #4. Before VS-05, the data is a set of separate versions of Statistics with the status *analytical*. After VS-05, it is a registered, immutable product version with the status *approved*. That version refers to the analytical versions it was composed from, so approval never changes an analytical version. The product is coherent across sources and time, explained, checked against dissemination and confidentiality standards, and signed off. The two gate moments, the minimum metadata set, and the protection of State #4 are described in *State #4: Statistics* and are not repeated here. **[VC22]**

A product can draw on analytical versions from several domains, surveys, and periods. Composition then often reveals incoherence that no single domain could see. In that case the discrepancy is explained or the component goes back to VS-04; it is never silently adjusted in VS-05.

From	To	Gate	Event out
#4 Statistics (status *analytical*)	#4 Statistics (status *approved*)	Approval gate of State #4 (component lineage, propagated accuracy, internal consistency, coherence, interpretive content, confidentiality including combination risk, approval record)	*Statistical Product Registered*

PLACEHOLDER: ArchiMate view VS-05 (c) state handover and events: analytical Statistics (#4) from several domains, the composition service (selection and integration, composite construction, coherence checks, interpretive content, approval), approved Statistics (#4) inside the composition domain

3. Stages

The stages describe the value that VS-05 adds in every production cycle. Together they form the execution process type of VS-05. In a mature steady state, the product specifications, the composite indicators, and the coherence checks have been designed and built in advance (see attribute 4). Execution then focuses on composing the products, checking coherence, writing the content, and obtaining approval.

Stage	Name	Value added
VS-05.1	Select and Integrate Source Datasets	The right validated outputs are brought together as the basis for a coherent product
VS-05.2	Construct Composite Indicators or Indices	Indices, ratios, and time series summarize the data and add meaning
VS-05.3	Perform Consistency and Coherence Checks	The product is internally consistent and fits the wider statistical picture
VS-05.4	Generate Analytical and Interpretive Outputs	Users can understand the product through narrative, key messages, and visualizations
VS-05.5	Approve and Register Statistical Products	An approved product is registered and ready for dissemination (State #4, *approved*)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-05.1 Select and Integrate Source Datasets

Value created: *The right validated outputs brought together as the basis for a coherent product.*

The stage starts when VS-04 publishes the *Output Data Registered* event for a component that a product needs, or when all components for a scheduled release are available. The designed selection rules pick the correct analytical versions of State #4 by domain, period, and accuracy. The components are integrated into the product structure. Their accuracy and methodology metadata is carried forward into the product. The exact component versions used are recorded, so the product can be reproduced later. Components from several domains, surveys, or periods are combined according to the product specification.

Activity	Key outputs	SAF process	Other reference
Select the correct component versions according to the product specification	Selected components	PRD-6.1 Prepare draft outputs; DLM-6 Provide data for reuse	SAF-P08 Data is shared property
Integrate the components into the product structure	Integrated product base	PRD-6.1 Prepare draft outputs	1.3.6 Integrate statistics
Carry component accuracy and methodology metadata into the product	Propagated metadata	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Record component lineage: which versions were used	Component lineage	OVA-2 Metadata Management	CSDA Traceability

VS-05.2 Construct Composite Indicators or Indices

Value created: *Derived measures, such as indices, ratios, and time series, that summarize the data and add meaning.*

This stage computes the designed composite indicators, indices, and time-series aggregates from the integrated components. Indicators and indices are calculated with the designed formulas and weights. Time series are chained, linked, and benchmarked according to the specification. Each composite receives an uncertainty measure derived from the accuracy of its components. No component estimate is re-estimated;

composites are built from the validated outputs as received. The composition method and its inputs are recorded in the lineage. Products without composites skip this stage, and the reason is recorded.

Activity	Key outputs	SAF process	Other reference
Calculate composite indicators and indices with the designed formulas	Composite indicators	PRD-6.1 Prepare draft outputs	1.3.6 Integrate statistics
Construct time series: chain, link, benchmark	Composite time series	PRD-6.1 Prepare draft outputs	
Derive the uncertainty of composites from component accuracy	Composite accuracy measures	PRD-6.1 Prepare draft outputs	2.1.3 Statistical quality management
Record the composition method and inputs as lineage	Composite lineage	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

VS-05.3 Perform Consistency and Coherence Checks

Value created: *Confidence that the product is internally consistent and fits the wider statistical picture.*

This stage runs the designed checks. Internal consistency checks confirm that totals reconcile with their components and that breakdowns add up to totals. Coherence checks compare the product with related statistics, such as the national accounts, and with earlier releases, within the designed tolerances. Each discrepancy receives a disposition according to policy: accept, flag, or return for review. Significant discrepancies are explained. No silent adjustment is made; any reconciliation is documented and traceable. A discrepancy that points to a component error goes back to VS-04.

Activity	Key outputs	SAF process	Other reference
Run internal consistency checks: totals and breakdowns reconcile	Consistency results	PRD-6.2 Validate outputs	
Run coherence checks against related and earlier statistics	Coherence results	PRD-6.2 Validate outputs	1.3.6 Integrate statistics
Assign a disposition to discrepancies (accept, flag, return) and explain them	Discrepancy dispositions	PRD-6.2 Validate outputs	GAMSO Quality Management
Log all check results in the product audit trail	Audit trail entries	OVA-4 Process Data Management	SAF-P10 Security by design

PLACEHOLDER: VS-05 coherence check flow: integrated components and composites passing internal consistency and coherence checks against related and earlier statistics; discrepancies accepted, flagged with an explanation, or returned to VS-04; all results written to the product audit trail

VS-05.4 Generate Analytical and Interpretive Outputs

Value created: *Products that users can understand: narrative, key messages, and visualizations grounded in the data.*

This stage produces the content that people read. The analytical narrative and commentary are written according to the templates. Key messages are stated together with the evidence that supports them. Tables and visualizations are produced. The content goes through the editorial and quality review workflow, which checks that it is accurate, written in plain language, accessible, and consistent with the data and its caveats.

The accuracy carried from VS-04 is reflected in the wording: a change within the margin of error is not presented as a real change.

Activity	Key outputs	SAF process	Other reference
Write the analytical narrative and commentary according to the templates	Product narrative	PRD-6.3 Interpret and explain outputs	1.4.1 Statistical (output) analysis
State key messages with their supporting evidence	Key messages	PRD-6.3 Interpret and explain outputs	
Produce tables and visualizations	Tables and visualizations	PRD-7.2 Produce dissemination products	
Run the editorial and quality review of the content	Reviewed content	OVA-1 Quality Management	GAMSO Quality Management

VS-05.5 Approve and Register Statistical Products

Value created: *A validated, approved, registered product, ready for dissemination.*

This stage validates, approves, and registers the product. The product is checked against the dissemination and confidentiality standards. This includes the disclosure risk that arises from combining outputs, because figures that are safe on their own can reveal confidential data when they are combined. The product is then submitted to the designated approval authority. On approval, the data passes the approval gate of State #4, and a persistent, immutable product version with the status *approved* is created. It carries a unique identifier and full lineage: components, method, coherence results, and the approval record. It is registered in the data catalog, and access permissions are applied. The stage ends by publishing the *Statistical Product Registered* event, which tells VS-06 that an approved product is available.

Output: Approved statistical product (State #4, status *approved*), the handover artifact to the dissemination domain. Its structure and minimum metadata set are described in *State #4: Statistics*.

Activity	Key outputs	SAF process	Other reference
Check the product against dissemination and confidentiality standards, including combination risk	Validation result	PRD-6.4 Apply disclosure control; PRD-6.5 Finalize outputs	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Submit the product to the approval authority and obtain sign-off	Approval record	PRD-6.5 Finalize outputs; DLM-2 Promote data through a quality gate	4.1.5 Accountability
Create an immutable, identified product version with full lineage (status *approved*)	Versioned product	DLM-5 Store and version data	CSDA Traceability
Register metadata and apply access permissions	Catalog registration, access-controlled product	DLM-3 Register and catalog data; DLM-4 Classify and protect data	SAF-P10 Security by design
Publish the *Statistical Product Registered* event to notify VS-06 Dissemination and Service Delivery	Event notification	DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

PLACEHOLDER: VS-05 product version structure: the approved product version linked to its component versions, composition method, propagated accuracy, coherence results, interpretive content, and approval record

4. The four process types

VS-05 runs through four process types. They do not form a linear sequence but a governance cycle: each process type produces its own output, and that output feeds the others. Design produces metadata, Implementation produces the execution process, Execution produces the statistical data, and Management produces quality reports that feed back into Design. The generic pattern is described in the Overview of this chapter.

The sub-processes below are specific to VS-05. Their names describe the work in the context of product composition. Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. This keeps every VS-05 sub-process traceable to the reference process model in *Business Processes*.

Design: provides metadata

The design process type produces the **metadata artifacts** that govern all other process types. In VS-05 these are the product specifications, the composite-indicator definitions, the coherence checks, the content templates, and the approval and confidentiality policy. Without them, composition becomes ad hoc and incoherent, and management has no baseline to measure coherence and interpretability against. Design runs when a product is introduced or changed, when an index is rebased, or when Management feeds back an improvement.

Design sub-process	Refines	Main outputs
Design the Statistical Product Portfolio	PRD-1.3 Establish output objectives; PRD-2.1 Design outputs and dissemination	Product specifications, component selection rules, metadata propagation rules
Design Composite Indicators and Indices	PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis	Composite-indicator specification, time-series specification, uncertainty representation
Design Consistency and Coherence Checks	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Consistency and coherence check sets, tolerance policy, revision policy
Design Analytical and Interpretive Content	PRD-2.1 Design outputs and dissemination	Content templates, visualization specification, key-message guidelines, editorial review policy
Design Approval, Confidentiality, and Versioning Framework	PRD-2.1 Design outputs and dissemination; DLM-1 Plan and design data assets; DLM-4 Classify and protect data	Approval policy, confidentiality standard, versioning and lineage policy, access control design

Design the Statistical Product Portfolio

Refines: PRD-1.3 Establish output objectives; PRD-2.1 Design outputs and dissemination.

The starting point is the definition of what products will be produced. This sub-process specifies each statistical product: its purpose, target users, content, structure, breakdowns, and release frequency. It defines the analytical versions of State #4 that each product draws on, and the rules for selecting them by domain, period, and accuracy. It also defines how the accuracy and quality metadata of the components is carried into the product. Each product receives its classification and versioning identity. Reuse of existing validated outputs comes before new analysis, in line with SAF-P08.

Activity	Key outputs	SAF process	Other reference
Specify each product: purpose, users, content, structure, breakdowns, frequency	Product specification	PRD-1.3 Establish output objectives; PRD-2.1 Design outputs and dissemination	1.1.2 Design statistics product
Define the component outputs each product draws on and the selection rules	Component selection specification	PRD-2.1 Design outputs and dissemination	SAF-P08 Data is shared property
Define how component accuracy and quality metadata is carried into the product	Metadata propagation rules	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification
Define the product classification and versioning identity	Product metadata standard	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification

Design Composite Indicators and Indices

Refines: PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis.

This sub-process specifies the derived measures that products present: composite indicators, indices, ratios, and time-series aggregates. Each is defined by its formula, its component inputs, its weighting or aggregation method, and its base period. For time series, the rules for chaining, linking, and benchmarking are specified. Where a composite combines outputs of different accuracy, the design specifies how its uncertainty is represented. Each composite receives its metadata and classification. International manuals, such as those for price and production indices, are followed where they apply.

Activity	Key outputs	SAF process	Other reference
Specify composite indicators and indices: formula, inputs, weighting, base period	Composite-indicator specification	PRD-2.5 Design processing and analysis	1.3.6 Integrate statistics
Specify the construction of time series: chaining, linking, benchmarking	Time-series specification	PRD-2.5 Design processing and analysis	
Define how component uncertainty is represented in composites	Uncertainty representation rules	PRD-2.5 Design processing and analysis	SAF-P09 No data without metadata and classification
Define metadata and classification for each composite	Composite metadata standard	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification

Design Consistency and Coherence Checks

Refines: PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

This sub-process designs the checks that make sure integrated products are sound. It specifies internal consistency checks: totals reconcile with their components, and breakdowns add up to totals. It specifies coherence checks against related statistics, such as the national accounts or earlier products, and across time. It sets the tolerance thresholds and the disposition rules: accept, flag, or return for review. It also sets the revision policy for products. Together, these rules make up the coherence part of the approval gate of State #4.

Activity	Key outputs	SAF process	Other reference
Define internal consistency checks: totals and breakdowns reconcile	Consistency check set	PRD-2.5 Design processing and analysis	
Define coherence checks against related statistics and across time	Coherence check set	OVA-1 Quality Management	GAMSO Quality Management
Define tolerance thresholds and disposition rules	Coherence tolerance policy	PRD-2.5 Design processing and analysis	
Define the revision policy for products	Revision policy	OVA-1 Quality Management	GAMSO Quality Management

Design Analytical and Interpretive Content

Refines: PRD-2.1 Design outputs and dissemination.

This sub-process designs the content that makes products understandable. It specifies the structure of the analytical narrative and commentary. It specifies the tables and visualizations. It sets the guidelines for key messages and the evidence each message must rest on. It designs the editorial and quality review workflow for interpretive content. Standards for accessibility and plain language are designed in from the start, so that the product can be used by the public as well as by experts.

Activity	Key outputs	SAF process	Other reference
Specify the structure of the analytical narrative and commentary	Content template	PRD-2.1 Design outputs and dissemination	1.4.1 Statistical (output) analysis
Specify tables and visualizations	Visualization specification	PRD-2.1 Design outputs and dissemination	
Define key messages and the evidence they require	Key-message guidelines	PRD-2.1 Design outputs and dissemination	
Define the editorial and quality review workflow, including accessibility and plain language	Editorial review policy	PRD-2.1 Design outputs and dissemination	GAMSO Quality Management

Design Approval, Confidentiality, and Versioning Framework

Refines: PRD-2.1 Design outputs and dissemination; DLM-1 Plan and design data assets; DLM-4 Classify and protect data.

This last design sub-process makes sure that products are approved, protected, and traceable. It defines the approval workflow and the authority that signs off each type of product. It defines the confidentiality standards that a product must meet before registration, including the risk that arises from combining outputs. It defines the versioning policy for products: identification, triggers for revision, immutability, and retention. It defines the lineage from each product to its component versions. Role-based access controls are designed in line with SAF-P10, because approved products remain confidential until their release.

Activity	Key outputs	SAF process	Other reference
Define the approval workflow and authority	Approval policy	PRD-2.1 Design outputs and dissemination	4.1.5 Accountability
Define confidentiality standards for products, including combination risk	Confidentiality standard	DLM-4 Classify and protect data	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure

Activity	Key outputs	SAF process	Other reference
Define the versioning and lineage requirements for products	Versioning and lineage policy	DLM-1 Plan and design data assets	CSDA Traceability
Design role-based access control for products before release	Access control design	DLM-4 Classify and protect data	SAF-P10 Security by design; 3.3.4 Logical identity and access control

PLACEHOLDER: VS-05 design process decomposition: inputs (user needs, analytical versions of State #4, standards and index manuals, confidentiality rules), the five design sub-processes in order, and the design artifact each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

The implementation process type takes the metadata artifacts from Design and **builds, configures, and tests** the composition pipeline until it is ready for production. Nothing runs operationally until this process type is complete. Components are reused where possible and built only where needed, in line with SAF-P01. Index calculation and visualization tools shared in the NSI community are good candidates for reuse.

Implementation sub-process	Refines	Main outputs
Build Integration and Composition Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Integration engine, composition engine, version-fixed connection to the catalog
Build Indicator-Construction and Time-Series Engines	PRD-3.2 Reuse or build processing and analysis components	Indicator engine, time-series engine, composite accuracy component
Build Coherence-Check, Content, and Approval Tooling	PRD-3.2 Reuse or build processing and analysis components; PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data	Coherence tooling, content tooling, approval and versioning store, events
Test and Validate the Pipeline	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems	Test reports, coherence and reproducibility reports, production-ready pipeline

Build Integration and Composition Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

This sub-process turns the product designs into operational tools. The integration engine is built or configured to select and combine the designed component outputs into the product structure, together with their accuracy and quality metadata. The composition engine is built to assemble products from their components according to the specifications. Connections to the catalog of State #4 are configured with fixed versions, so that the exact component versions used are recorded. All composition logic is version-controlled in line with SAF-P10. Any product can therefore be reproduced from its component versions and the recorded method.

Activity	Key outputs	SAF process	Other reference
Build or configure the integration engine: select and combine component outputs	Configured integration engine	PRD-3.2 Reuse or build processing and analysis components	1.3.6 Integrate statistics
Build or configure the composition engine: assemble products from components	Configured composition engine	PRD-3.2 Reuse or build processing and analysis components	
Configure version-fixed connections to the catalog of State #4	Integration with the output catalog	PRD-3.4 Configure workflows	CSDA Metadata Management
Put all composition logic under version control	Versioned composition repository	PRD-3.4 Configure workflows	SAF-P10 Security by design

Build Indicator-Construction and Time-Series Engines

Refines: PRD-3.2 Reuse or build processing and analysis components.

In this sub-process the designed composite indicators and time series become automated services. Calculators for indicators and indices are built from the designed formulas and weights. Time-series engines are configured for chaining, linking, and benchmarking. The representation of uncertainty is implemented, so that every composite carries an appropriate accuracy measure. Every composite output is produced with its method and inputs attached. Tests on historical data confirm that the engines reproduce existing published series before they replace older tools.

Activity	Key outputs	SAF process	Other reference
Build calculators for indicators and indices from the designed formulas	Indicator construction engine	PRD-3.2 Reuse or build processing and analysis components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Configure chaining, linking, and benchmarking of time series	Time-series engine	PRD-3.2 Reuse or build processing and analysis components	
Implement the representation of uncertainty for composites	Composite accuracy component	PRD-3.2 Reuse or build processing and analysis components	SAF-P09 No data without metadata and classification
Attach method and inputs to every composite	Binding of composite metadata	PRD-3.2 Reuse or build processing and analysis components	SAF-P09 No data without metadata and classification

Build Coherence-Check, Content, and Approval Tooling

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data.

This sub-process implements the coherence checks, the tooling for content, and the approval and versioning store. Coherence tooling implements the designed consistency checks and the checks across sources and time, with their tolerances. Content tooling supports the narrative, tables, and visualizations according to the templates, with the editorial review workflow. The approval and versioning store is built so that each approved product version is immutable, identified, reproducible, and accompanied by its lineage and approval record. Event notifications are configured, so that VS-06 is alerted when a product is approved and registered.

Activity	Key outputs	SAF process	Other reference
Implement consistency and coherence checks with their tolerances	Coherence tooling	PRD-3.2 Reuse or build processing and analysis components	
Build content and visualization tooling with the editorial workflow	Content tooling	PRD-3.3 Reuse or build dissemination components	
Build the approval and versioning store: immutable, identified, reproducible	Approval and versioning store	DLM-5 Store and version data	CSDA Traceability
Configure the *Statistical Product Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows; DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

Test and Validate the Pipeline

Refines: PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems.

Before the pipeline goes into production, it must be thoroughly verified. Technical testing covers all components: integration, composition, indicator and time-series engines, coherence checks, content tooling, and the approval store. Integration testing checks the end-to-end flow from the *Output Data Registered* event to the registration of an approved product. A coherence test confirms that composites reconcile with their components and fit earlier products. A reproducibility test confirms that a product can be regenerated identically from its component versions and the composition method. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	
Test integration: end-to-end flow from output intake to registration	Integration test report	PRD-3.5 Test production systems	
Run the coherence test: composites reconcile with components and earlier products	Coherence test report	PRD-3.6 Test statistical business process	
Run the reproducibility test: regenerate a product from components and method	Reproducibility report	PRD-3.6 Test statistical business process	CSDA Traceability
Finalize the production system and document operational procedures	Operational procedures manual, production-ready pipeline	PRD-3.7 Finalize production systems	

PLACEHOLDER: VS-05 implementation pipeline: the five design artifacts feeding the three build sub-processes, followed by technical testing, coherence testing, and reproducibility testing, resulting in a production-ready composition pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-05 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when all components for a scheduled release have been registered by VS-04, or when a revision of a published product is due. The stages run in order, with a loop back to VS-05.1 or VS-04 when a coherence check fails. Execution applies the designs and the solution as they are; a new product or a change in an index base goes back to Design. Every run ends in one of three outcomes: approved and registered, returned to VS-04 for a component, or held by the approval authority.

Aspect	VS-05
Trigger	All components of a scheduled release registered (*Output Data Registered*); a scheduled revision
Cadence	Per release, according to the release calendar
Performed by	ROL-15 Subject-matter statistician, ROL-10 Data engineer, ROL-17 Release manager, ROL-03 Data steward
Input	Analytical versions of Statistics (#4), product specifications, content templates
Output	Registered approved product version (#4, *approved*), its lineage, and its approval record
Control point	Approval gate of State #4 in VS-05.5 Approve and Register Statistical Products
Exception path	Coherence failure back to VS-04 Estimation, Analysis, and Statistical Modeling; combination risk above the threshold to ROL-16 Disclosure control specialist; refusal by the approval authority back to VS-05.3 Perform Consistency and Coherence Checks or VS-05.4 Generate Analytical and Interpretive Outputs

Management: provides quality reports

The management process type runs alongside and after execution. It monitors the performance of the composition pipeline, assesses the coherence and interpretability of products, makes sure that every product is approved and can be reproduced, and feeds improvements back into Design. This closes the governance cycle.

Management sub-process	Refines	Main outputs
Monitor Process Performance	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard, throughput report, paradata records
Assess Product Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Coherence report, interpretability review, completeness report, trend analysis
Manage Product and Version Lifecycle	OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use	Product version register, metadata quality report, version store audit
Audit and Compliance Reporting	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Reproducibility audit, approval and confidentiality reports to VS-07 Governance and Compliance
Continuous Improvement	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, improvement tracking log

Monitor Process Performance

Refines: OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs.

The operational performance of VS-05 must be tracked systematically across production cycles. The process owner and the quality manager collect and aggregate the key performance indicators (see attribute 9). They monitor throughput against the release calendar, because a late product delays publication. They track paradata: which component versions and methods were used, when, by whom, and how many approval rounds were needed. This information feeds dashboards that give the process owner a current view of the health of the domain. The approach follows the Quality Management capability of GAMSO.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per production cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Monitor throughput against the release calendar	Throughput report	OVA-4 Process Data Management	
Track paradata: component versions, methods, approval rounds	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Assess Product Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the products is evaluated against the criteria from Design. Coherence covers internal consistency and consistency with related and earlier statistics. Interpretability covers the clarity and correctness of the narrative and visualizations. Completeness checks that all designed content is present. Accuracy propagation checks that the accuracy of components was carried correctly into the product. Quality indicators are produced per product and per cycle. Trends are tracked to detect recurring coherence problems or weaknesses in interpretation. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management.*

Activity	Key outputs	SAF process	Other reference
Evaluate coherence: internal, and with related and earlier statistics	Coherence report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess the interpretability of narrative and visualizations	Interpretability review	OVA-1 Quality Management	CSDA Data Quality
Check completeness and correct propagation of accuracy	Product completeness report	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across products and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality

Manage Product and Version Lifecycle

Refines: OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use.

The product portfolio and the versions it produces must stay current, complete, and well governed. This sub-process governs the life cycle of product specifications, composite-indicator definitions, and content

templates. It versions them, manages transitions such as the rebasing of an index, and retires products that are no longer needed. It verifies that every product carries its full component lineage, composition method, coherence results, and approval record, as a compliance check on SAF-P09. It also governs the versioning store: retention, immutability, revisions, and the reconciliation of catalog entries with stored versions. Lineage from analytical versions through composition to approved products is tracked.

Activity	Key outputs	SAF process	Other reference
Govern the life cycle and versions of products, indicators, and templates	Product version register	OVA-2 Metadata Management	SAF-P03 Active life cycle management
Verify metadata completeness: component lineage, method, coherence, approval	Metadata quality report	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Govern the versioning store: retention, immutability, revisions, reconciliation	Version store audit	DLM-5 Store and version data; DLM-7 Revise data	CSDA Traceability
Track metadata lineage from analytical versions to approved products	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability

Audit and Compliance Reporting

Refines: OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use.

This sub-process produces the evidence that VS-05 operates within methodological, legal, and organizational requirements. Reproducibility audit reports show that every product can be regenerated from its component versions and the composition method. Approval and confidentiality reporting confirms that every registered product was approved by the designated authority and meets the confidentiality standard, including the risk of combining outputs. Standards compliance reporting confirms conformance to product and exchange standards, such as SDMX. Security compliance reporting verifies the access controls and the integrity of the approval record. These reports meet the obligation of SAF capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance*.

Activity	Key outputs	SAF process	Other reference
Produce reproducibility audit reports: every product can be regenerated	Reproducibility audit summary	OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	4.1.5 Accountability
Report on approval and confidentiality compliance of products	Approval and confidentiality report	OVA-1 Quality Management	SAF-P11 Privacy by design
Report on standards compliance: products, SDMX exchange	Standards compliance report	OVA-1 Quality Management	4.1.5 Accountability
Report on security compliance: access controls, integrity of the approval record	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10 Security by design; 3.3.3 Security incident management

Continuous Improvement

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This last management sub-process closes the governance cycle by analyzing performance and quality data and driving improvements. Trends across production cycles are examined: are coherence problems decreasing, are fewer approval rounds needed, is the reuse of components rising, and is interpretability improving? Improvement opportunities are identified. Examples are refining indicator methods, improving coherence checks that produce false alarms, streamlining the approval workflow, improving visualizations, and reusing more validated outputs. Feedback from users through VS-10 is part of the input. Recommendations are fed back to the design sub-processes, prioritized by impact and feasibility, and tracked through the following cycles. This is SAF capability 4.1.4 *Improvement management* in practice.

Activity	Key outputs	SAF process	Other reference
Analyze quality trends across production cycles, including user feedback	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities: indicator methods, coherence checks, approval, visualization, reuse	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Feed recommendations back to the design sub-processes or to VS-04 Estimation, Analysis, and Statistical Modeling	Design feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management

PLACEHOLDER: VS-05 management feedback loop: products and paradata from Execution flowing through the five management sub-processes, with dashboards to the process owner, coherence and interpretability reports to the quality manager, compliance reports to VS-07, component issues to VS-04, and improvement recommendations back to Design

Metadata and paradata per process type

In VS-05, metadata is what makes a composite product interpretable: each product must carry its components, methodology, coherence results, and approval. The table shows what each process type contributes. Data metadata describes the products. Process metadata, also called paradata, describes how the process itself ran: who, when, and with which tools.

Process type	Data metadata produced	Process metadata (paradata) produced
Design	Product specifications, composite-indicator definitions, coherence criteria, content templates, approval and confidentiality policy	Log of design decisions, methodology version, review and approval records
Implementation	Integration configurations, parameters of indicator construction and coherence checks, product templates	Build log, test results, coherence test outcomes, deployment approvals

Process type	Data metadata produced	Process metadata (paradata) produced
Execution	Component lineage per product, composition and indicator method, coherence results, interpretive content, approval record	Operator identity, timing, software versions, trail of the approval workflow
Management	Coherence and interpretability assessments, product quality scores, lineage assessments, trend analyses	Coherence dashboards, product quality trends, improvement decisions, audit findings

PLACEHOLDER: VS-05 business object lifecycle: design artifacts govern implementation assets, which produce the pipeline that turns analytical Statistics (#4) into approved products (#4), generating component lineage, coherence, and approval records that management assesses and feeds back to design

5. Actors and roles

VS-05 brings subject-matter expertise, communication, and formal approval together. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the product manager and the coherence designer are ROL-06 Statistical designer, the indicator methodologist is ROL-07 Methodologist, the integration specialist is ROL-10 Data engineer, the subject-matter author is ROL-15 Subject-matter statistician, and the approval authority is ROL-17 Release manager. VS-04, which supplies the statistics, is a neighboring value stream and not an actor; the relation is governed by the handover interface in attribute 8. **[VC20]**

Actor or role	ID	Process type	Responsibility in VS-05 Statistical Product Composition
International statistical organization	ACT-07	External	Maintains product, indicator, and exchange standards and manuals, such as SDMX and index manuals
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-05 Statistical Product Composition
Data owner	ROL-02	Management	Accountable for approved versions of State #4; decides on internal access before release
Data steward	ROL-03	Execution	Registers approved products, maintains catalog entries, applies access controls, resolves gate exceptions
Data custodian	ROL-04	Execution	Operates storage, access control, and retention of approved products before release

Actor or role	ID	Process type	Responsibility in VS-05 Statistical Product Composition
Statistical designer	ROL-06	Design	Designs the product portfolio, product specifications, coherence checks, and content templates
Methodologist	ROL-07	Design	Specifies composite indicators, indices, and the construction of time series
Metadata and data architect	ROL-08	Design	Designs the product metadata model and the propagation of component metadata
Data engineer	ROL-10	Implementation, Execution	Builds the composition components; selects and integrates components into the product structure
Platform engineer	ROL-11	Implementation	Configures composition services, connections to the output catalog, and access controls
Test manager	ROL-12	Implementation	Plans and carries out technical, coherence, and reproducibility testing
Subject-matter statistician	ROL-15	Execution	Runs coherence checks, explains discrepancies, and writes the analytical and interpretive content
Disclosure control specialist	ROL-16	Design, Execution	Assesses the confidentiality of products, including the risk of combining outputs
Release manager	ROL-17	Execution	Approves products against dissemination and confidentiality standards and signs them off for registration
Quality manager	ROL-18	Management	Monitors coherence and interpretability indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures confidentiality and standards compliance of products, and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-05 relies on, and what role they play. NSIs can use it to map the value stream to their own organization.

Capability group	Capabilities	Role in VS-05 Statistical Product Composition
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the products and composite indicators, and build the composition pipeline
1.3 Processing	1.3.6 Integrate statistics, 1.3.7 Statistical disclosure	Integrate outputs into products and check the confidentiality of combined outputs [**VC21**]
1.4 Analysis	1.4.1 Statistical (output) analysis	Check coherence and produce interpretive content [**VC21**]
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage product versions and metadata, assess coherence, integrate across registers and time
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain integration, indicator construction, and content tooling
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.4 Logical identity and access control	Check products against confidentiality standards and protect them before release
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward product approval, plan improvements, report on compliance

Stages, processes, and functions

The table shows which SAF business processes realize each stage of VS-05 and which business functions they mainly use. It follows the mapping in *Business Processes, Mapping to value stream stages*. The activity tables in attribute 3 can name additional supporting processes. [**VC25**]

Stage	Realized by processes	Main functions
VS-05.1 Select and Integrate Source Datasets	PRD-5.1 Integrate data; PRD-6.1 Prepare draft outputs	FUN-3.4 Estimate and analyze
VS-05.2 Construct Composite Indicators or Indices	PRD-5.5 Derive new variables and units; PRD-5.7 Calculate aggregates	FUN-3.3 Process statistical data; FUN-3.4 Estimate and analyze
VS-05.3 Perform Consistency and Coherence Checks	PRD-6.2 Validate outputs	FUN-3.4 Estimate and analyze; FUN-5.2 Assure and review quality
VS-05.4 Generate Analytical and Interpretive Outputs	PRD-6.3 Interpret and explain outputs; PRD-7.2 Produce dissemination products	FUN-3.4 Estimate and analyze; FUN-3.6 Disseminate statistics
VS-05.5 Approve and Register Statistical Products	PRD-6.4 Apply disclosure control; PRD-6.5 Finalize outputs; DLM-3 Register and catalog data	FUN-3.4 Estimate and analyze; FUN-3.5 Protect statistical confidentiality; FUN-7.2 Manage metadata

PLACEHOLDER: ArchiMate view VS-05 (b) stages realized by processes and functions

Framework mappings

This reference table maps each process type of VS-05 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-1.3, PRD-2.1, PRD-2.5, DLM-1, DLM-4, OVA-1	Phase 1 Specify Needs (1.3); Phase 2 Design (2.1, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Data Quality	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Implementation	PRD-3.2 to PRD-3.7, DLM-3, DLM-5	Phase 3 Build (3.2 to 3.7)	Capability Management	Data Analysis (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-6.1 to PRD-6.5, PRD-7.2 (content), DLM-2, DLM-3, DLM-5, DLM-6, OVA-1, OVA-2, OVA-4	Phase 6 Analyze (6.1 to 6.5); 7.2 Produce dissemination products (content only)	Production (execution)	Data Analysis (execution)	1.3.6 Integrate statistics, 1.3.7 Statistical disclosure, 1.4.1 Statistical (output) analysis
Management	OVA-1, OVA-2, OVA-4, DLM-5, DLM-7, DLM-8, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-05 most directly.

Principle	Statement	Application in VS-05 Statistical Product Composition
SAF-P01	Reuse before open source, open source before buying, buying before building	Composition, index, and visualization tools are reused across products first. Where new tooling is needed, open-source solutions developed with other NSIs come before purchase or own development. **[VC3]**
SAF-P07	Implement loosely coupled processes and systems	Composition is independent from estimation and dissemination. The approved product is the decoupling interface: dissemination uses it without knowing how it was composed.
SAF-P08	Data is shared property	Approved products and their components, once registered, are available to all authorized consumers. Composition reuses existing validated outputs instead of repeating analysis.
SAF-P09	No data without metadata and classification	Every product carries metadata: its component outputs and their accuracy, the composition and indicator method, the coherence results, the classification, and the approval record. This is what makes products interpretable and trustworthy.
SAF-P10	Security by design	Access controls, integrity verification, and a tamper-evident record of composition and approval are built into the pipeline, so every product can be attributed and reproduced. Products are protected against early disclosure.
SAF-P11	Privacy by design	Products are checked against confidentiality standards before registration. The disclosure risk of combining outputs is assessed by design, because combined figures can reveal what single figures do not.

ℹ Principles that apply across the process types

Several principles apply to all four process types. SAF-P09 underpins interpretability: Design defines the product and the required metadata, Execution records component lineage, composition method, and coherence results, and Management assesses product quality. SAF-P10 governs the tamper-evident record of composition and approval across all stages. SAF-P11 applies in particular to *Design Approval, Confidentiality, and Versioning Framework* and to VS-05.5, where confidentiality and combination risk are confirmed before registration. SAF-P07 governs both the intake of analytical versions from VS-04 and the handover of approved products to VS-06.

8. Interfaces, services, and events

Boundary interfaces

VS-05 is loosely coupled (SAF-P07) to its neighbors through two boundary interfaces. Specifying each boundary explicitly makes the loose-coupling contract concrete and testable: the handover artifact, the metadata it must carry, the event that triggers it, and the point where responsibility transfers. Each interface exchanges a registered, immutable artifact plus its metadata, never internal working data. Neither side therefore needs knowledge of the other's internal processing.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-04 Estimation, Analysis, and Statistical Modeling → VS-05 Statistical Product Composition (intake)	Registered version of Statistics (#4, *analytical*)	Methodology and parameters, weights, accuracy measures, model diagnostics, record of disclosure control applied, version identifier, lineage to the processed source, classification	*Output Data Registered*	VS-04 Estimation, Analysis, and Statistical Modeling keeps custody of the analytical version; VS-05 Statistical Product Composition takes responsibility for composition on receipt of the event
VS-05 Statistical Product Composition → VS-06 Dissemination and Service Delivery (handover)	Registered approved product version (#4, *approved*)	Component lineage, composition and indicator method, propagated accuracy, coherence results, interpretive content, approval record, confidentiality status, product version identifier, classification	*Statistical Product Registered*	VS-05 Statistical Product Composition keeps custody of the product version; VS-06 Dissemination and Service Delivery takes responsibility for final disclosure control, release, and user support on receipt of the event

Business services

Service	Offered to	Realized in	Interfaces
Integration and composition service	Composition teams	VS-05.1 Select and Integrate Source Datasets	Integration and composition engines, version-fixed catalog connection
Indicator and index service	Composition teams, methodologists	VS-05.2 Construct Composite Indicators or Indices	Interface for composite construction, query interface for methodology and parameters
Coherence validation service	Quality managers, subject-matter statisticians	VS-05.3 Perform Consistency and Coherence Checks	Coherence reports and discrepancy lists
Analytical content service	Authors, editors, VS-06 Dissemination and Service Delivery	VS-05.4 Generate Analytical and Interpretive Outputs	Content templates, editorial workflow

Service	Offered to	Realized in	Interfaces
Product provisioning service	VS-06 Dissemination and Service Delivery	VS-05.5 Approve and Register Statistical Products	Event notification (*Statistical Product Registered*) and a catalog query interface to find approved products, their lineage, and their accuracy
Product catalog service	All data consumers in the NSI	Management sub-process *Manage Product and Version Lifecycle*	Search and browse interface for products, component lineage, coherence results, and approval records
Design and methodology service	Statistical designers, methodologists	All Design sub-processes of VS-05 Statistical Product Composition	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artifacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Output Data Registered	Availability of a component; start of VS-05.1 Select and Integrate Source Datasets when all components of a release are available (published by VS-04 Estimation, Analysis, and Statistical Modeling)
Coherence check failed	Return of a component to VS-04 Estimation, Analysis, and Statistical Modeling, or explanation of the discrepancy
Product approved / *Product rejected*	Outcome of approval in VS-05.5 Approve and Register Statistical Products
Statistical Product Registered	Completion of VS-05.5 Approve and Register Statistical Products; triggers VS-06 Dissemination and Service Delivery
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-05 event choreography between VS-04, the composition teams, the release manager, and process management across Design, Implementation, Execution (select and integrate, construct composites, check coherence, write content, approve and register), and Management (key performance indicators, coherence, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-05	Receives from VS-05
VS-07 Governance and Compliance	Approval and confidentiality policy, standards policy	Approval records, confidentiality and standards compliance evidence
VS-08 Innovation and Methodological Development	New indicators, index methods, and forms of visualization	Product gaps and improvement needs
VS-09 Infrastructure and Platform Enablement	Composition and content tooling, versioning store, catalog, event interfaces	Capacity requirements around release dates

Value stream	Provides to VS-05	Receives from VS-05
VS-10 Customer and Stakeholder Engagement	User needs for products, breakdowns, and interpretation	Planned products and their content for consultation

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
On-time product rate	Share of products composed within the window of the release calendar	Per product, per cycle
Coherence check pass rate	Share of products that pass the coherence checks on the first run	Per product, per cycle
Number of approval rounds	Number of approval cycles before sign-off	Per product, per cycle
Component reuse rate	Share of products that reuse existing validated outputs	Per cycle
Reproducibility rate	Share of products that can be regenerated from components and method	Per cycle
Processing time	Time from the availability of all components to product registration	Per product, per cycle

10. Toolkit

Examples from NSI practice, the maturity checklist, and end-to-end scenarios for VS-05 are part of the SAF Toolkit.

Toolkit: SAF-TK-V05 VS-05 Statistical Product Composition companion [**VC17**]

VS-06 Dissemination and Service Delivery

1. Purpose and value created

VS-06 is the exit point of the statistical value chain. It takes approved statistical products and makes them available to users through trusted, secure channels: publications, open data, APIs, dashboards, deliveries to international organizations, and access to research datasets in a secure environment. VS-06 does not change the statistics. It protects them with final disclosure control, formats them for each channel, publishes them at the announced moment, and supports users in their correct use. The value created is **impact, accessibility, and trust**. Users receive accurate, documented statistics at a moment that is the same for everyone, and no respondent can be identified.

VS-06 corresponds to the business domain **Dissemination** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

- **Release manager, dissemination staff, or user support:** the stages in attribute 3, which describe what happens in each release cycle.
- **Dissemination and service manager or disclosure control specialist:** the Design processes in attribute 4.
- **System builder or IT staff:** the Implementation processes in attribute 4.
- **Quality or compliance manager:** the Management processes in attribute 4.
- **Manager who wants the big picture:** attributes 1, 2, and 8. Examples and end-to-end scenarios are part of the SAF Toolkit.

Signature discipline: accessibility, trust, and responsible use

VS-06 converts statistical output into value for society. The guarantee is accurate, timely, and understandable delivery through trusted channels. It is gated by final statistical disclosure control: nothing is released without confidentiality clearance. Equal access is an explicit obligation, which means managed release timing and strict embargo rules. Accessibility is an explicit obligation too: formats, languages, and machine-readability. Documentation and user support make responsible use of the statistics possible.

Metadata inherited from VS-05

VS-06 receives approved products together with the metadata VS-05 attached: the component lineage, the composition method, the propagated accuracy, the coherence results, the approval record, and the confidentiality status. VS-06 applies the final disclosure control for release, which comes on top of the initial control of VS-04 and the check of VS-05. It formats the product for each channel and publishes it. It protects, formats, and delivers the statistics; it does not alter them.

Reliance on the supporting value streams

Dissemination depends strongly on the supporting value streams. VS-07 makes sure that all dissemination complies with data protection and confidentiality law. VS-09 provides the portals, APIs, and secure research environments. VS-10 makes sure that the dissemination services meet the real needs of users. The usage data and feedback collected in VS-06.5 flow back to VS-10 and across the chain.

PLACEHOLDER: ArchiMate view VS-06 (a) value stream and capabilities

2. Position and state transition

VS-06 is the sixth and last primary value stream. It receives approved products when VS-05 publishes the *Statistical Product Registered* event. It has no downstream primary value stream. Its handover is to external users, and its feedback path runs to VS-10.

VS-06 takes data from State #4, with the status *approved*, to State #5. Before VS-06, the product is approved but still confidential and not yet public. After VS-06, it is released data: final disclosure control has been applied, the product is documented for correct interpretation, and it is published or delivered through trusted channels. Once released, the data is used outside the NSI as boundary position #6, *Consumed Data*, beyond the NSI's direct control. The quality gate, the minimum metadata set, and the protection of State #5 are described in *State #5: Released Data* and are not repeated here.

If nothing changes between the approved #4 version and the release, the same artifact can be marked as released instead of being registered again. The release steps still take place and are recorded: release authorization, the final disclosure check, and the release itself (see *Applying the steady states pragmatically*). VS-06 also covers controlled access to microdata for research, using the processes for microdata services, and the exchange of data in data spaces. **[VC24]**

From	To	Gate	Event out
#4 Statistics (status *approved*)	#5 Released Data	Gate of State #5 (release authorization, final disclosure control, documentation and metadata, channel conformance, equal access)	*Statistics Released*

From	To	Gate	Event out
#5 Released Data	#6 Consumed Data (boundary position)	No gate; use outside the NSI	Usage and feedback return to VS-06.5 Monitor Usage and Feedback and VS-10 Customer and Stakeholder Engagement

PLACEHOLDER: ArchiMate view VS-06 (c) state handover and events: approved Statistics (#4) inside the NSI, the release service at the boundary (final disclosure control, formatting, managed release), Released Data (#5) in public and controlled channels, and Consumed Data (#6) outside the NSI

3. Stages

The stages describe the value that VS-06 adds in every release cycle. Together they form the execution process type of VS-06. In a mature steady state, the dissemination strategy, the disclosure rules, and the channels have been designed and built in advance (see attribute 4). Execution then focuses on running the release, serving users, and monitoring use, with final disclosure control as the gate that must be passed before anything is published.

Stage	Name	Value added
VS-06.1	Plan Dissemination and Access Strategy	The right channels, formats, and timing are confirmed for this release **[VC23]**
VS-06.2	Prepare Data for Dissemination	Products are disclosure-safe and formatted for each channel
VS-06.3	Publish and Provide Access Services	Statistics are released through trusted channels, equally and on time (State #5)
VS-06.4	Support and Engage Users	Users can find, understand, and correctly use the statistics
VS-06.5	Monitor Usage and Feedback	It is known how the statistics are used and valued, which feeds improvement

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-06.1 Plan Dissemination and Access Strategy

Value created: *The right channels, formats, and timing confirmed for this release.*

The stage starts when VS-05 publishes the *Statistical Product Registered* event. In a mature steady state, the dissemination strategy has been designed already. Execution in each cycle therefore confirms the plan for the product rather than creating a new one. The team confirms the channels, formats, and slot in the release calendar. It verifies the arrangements for embargo and equal access, including any pre-release access that the law allows. It confirms the target user groups and the communication plan, together with VS-10. A significant change, such as a new channel or a change of policy, goes back to Design instead of being solved during the cycle.

Activity	Key outputs	SAF process	Other reference
Confirm channels, formats, and the release calendar slot for the product	Confirmed release plan	PRD-7.1 Update output systems	1.5.1 Publication management
Verify the arrangements for embargo and equal access	Confirmed access arrangements	PRD-7.3 Release dissemination products	4.1.5 Accountability
Confirm target user groups and the communication plan	Confirmed audience and communication	PRD-7.4 Promote dissemination products	1.5.2 Promotion to use
Flag changes to the plan for design review	Plan change exception (if any)	OVA-1 Quality Management	GAMSO Quality Management

VS-06.2 Prepare Data for Dissemination

Value created: *Products made disclosure-safe and formatted for each channel.*

This stage prepares the product for release. Final disclosure control is applied. For tables and dashboards this means suppression, rounding, or perturbation on top of the protection already applied upstream. For released microdata, it means de-identification to the designed standard. The release is checked against the disclosure-risk criteria, and nothing proceeds if the criteria are not met. The product is then formatted for each channel: publication, dataset or API, dashboard, or transmission to an international organization. The documentation, metadata, and methodology notes are assembled according to the documentation standard. For research access, the research dataset for an approved project is prepared from State #3 or #4.

Activity	Key outputs	SAF process	Other reference
Apply final disclosure control to tables and dashboards	Disclosure-controlled outputs	PRD-6.4 Apply disclosure control	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
De-identify released microdata and prepare research datasets for approved projects	De-identified microdata, research datasets	PRD-6.4 Apply disclosure control; MDS-3 Prepare research datasets	SAF-P11 Privacy by design
Check the release against the disclosure-risk criteria (gate)	Disclosure clearance	DLM-2 Promote data through a quality gate	SAF-P11 Privacy by design
Format the product for each channel and assemble documentation and metadata	Channel-ready products and documentation	PRD-7.2 Produce dissemination products	SAF-P09 No data without metadata and classification

> **!** Release gate
>
> No product is published until it passes the disclosure-risk criteria. A release that fails the gate is returned for additional protection. Confidentiality is not negotiable at the boundary of State #5.

VS-06.3 Publish and Provide Access Services

Value created: *Statistics released to users through trusted, secure channels, equally and on time.*

This stage carries out the release. The product and its metadata are published through the confirmed channels: the portal and publications, open data and APIs, and dashboards. Deliveries to international organizations are transmitted under the agreed programs, and data offerings in data spaces are published and exchanged under usage control. Accredited researchers receive mediated access to research datasets

in a secure processing environment, and their results are checked before they leave it. Release timing is managed according to the calendar, with embargo and equal access enforced: no privileged early access outside the policy. The release is promoted according to the communication plan, together with VS-10. A persistent, identified released version is recorded with lineage to the approved product, and the *Statistics Released* event is published.

Output: Released Data (State #5), published statistics, de-identified microdata, and access services. Its structure and minimum metadata set are described in *State #5: Released Data*.

Activity	Key outputs	SAF process	Other reference
Publish products and metadata through the confirmed channels: portal, open data, API, dashboards	Published release	PRD-7.3 Release dissemination products	1.5.1 Publication management; 1.5.3 Facilitate data access
Transmit deliveries to international organizations and exchange offerings in data spaces	Transmitted deliveries, data space exchanges	PRD-7.3 Release dissemination products; DSP-2 Publish data offerings; DSP-4 Exchange data under usage control	4.1.6 Collaboration
Provide mediated access in a secure processing environment and check outputs before release	Research access, checked research outputs	MDS-4 Provide a secure processing environment; MDS-7 Check and release output	SAF-P10 Security by design; 2.2.4 Data services
Manage release timing, embargo, and equal access	Managed release	PRD-7.3 Release dissemination products	4.1.5 Accountability
Promote the release according to the communication plan (with VS-10 Customer and Stakeholder Engagement)	Promotion carried out	PRD-7.4 Promote dissemination products	1.5.2 Promotion to use
Record the released version with lineage and publish the *Statistics Released* event	Registered release, event notification	DLM-3 Register and catalog data; DLM-5 Store and version data	SAF-P07 Implement loosely coupled processes and systems

PLACEHOLDER: VS-06 channel architecture: the disclosure-cleared product fanning out to publications and portal, open data and API, dashboards, transmissions to international organizations, data space offerings, and the secure processing environment for accredited researchers, all at one managed release moment

VS-06.4 Support and Engage Users

Value created: *Users helped to find, understand, and correctly use the statistics.*

This stage supports users after release. Documentation, metadata, and user guides are made available with the release. The service desk answers questions from users. Training and guidance help users interpret the figures correctly. Researchers working in the secure environment are supported in using the data and the environment correctly. Engagement activities, together with VS-10, collect questions and needs. Errata and clarifications are issued according to the correction policy. The aim is responsible use: helping users avoid misreading estimates and their uncertainty.

Activity	Key outputs	SAF process	Other reference
Provide documentation, metadata, and user guides with the release	Published documentation	PRD-7.5 Provide user support	SAF-P09 No data without metadata and classification; 1.5.4 Statistical information management
Operate the service desk and answer questions from users	Answers to users	PRD-7.5 Provide user support	3.4.1 Services management
Provide training and guidance for correct interpretation, including support for researchers	User guidance	PRD-7.5 Provide user support; MDS-6 Support researchers	
Issue errata and clarifications according to the correction policy	Corrections (if any)	PRD-7.5 Provide user support; DLM-7 Revise data	4.1.5 Accountability

VS-06.5 Monitor Usage and Feedback

Value created: *Insight into how statistics are used and valued, feeding continuous improvement.*

This last stage closes the loop with users. Use is tracked across channels: downloads, API calls, dashboard views, and publication reads. The use of research datasets and the compliance of researchers with the conditions of access are monitored, and research projects are closed in a controlled way when they end. User satisfaction and feedback are collected. The insights are analyzed to understand reach, value, and unmet needs. They are routed back to Design, to VS-10, and across the value chain. An example is demand for a new breakdown, which can only be met in earlier value streams.

Activity	Key outputs	SAF process	Other reference
Track use across channels: downloads, API calls, views	Usage statistics	PRD-7.4 Promote dissemination products; DLM-8 Monitor data quality, lineage and use	2.2.5 User research
Monitor the use of research data and close research projects	Use and compliance records, closed projects	MDS-8 Monitor use and compliance; MDS-9 Close projects; DSP-5 Monitor compliance and report	SAF-P11 Privacy by design
Collect user satisfaction and feedback	Feedback log	PRD-8.1 Gather evaluation inputs	2.2.5 User research
Analyze reach, value, and unmet needs, and route insights to Design, VS-10 Customer and Stakeholder Engagement, and the chain	Usage insight report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management

4. The four process types

VS-06 runs through four process types. They do not form a linear sequence but a governance cycle: each process type produces its own output, and that output feeds the others. Design produces metadata, Implementation produces the execution process, Execution produces the released statistics and services, and Management produces quality reports that feed back into Design. The generic pattern is described in the Overview of this chapter.

The sub-processes below are specific to VS-06. Their names describe the work in the context of dissemination. Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. This keeps every VS-06 sub-process traceable to the reference process model in *Business Processes*.

Design: provides metadata

The design process type produces the **metadata artifacts** that govern all other process types. In VS-06 these are the dissemination and access strategy, the release and disclosure policy, the channel and format specifications, the documentation standard, and the model for user support. Without them, release becomes ad hoc, with a risk of confidentiality breaches or unequal access, and management has no baseline to measure accessibility and impact against. Design runs when a channel or service is introduced, when a policy changes, or when Management feeds back an improvement.

Design sub-process	Refines	Main outputs
Design the Dissemination and Access Strategy	PRD-2.1 Design outputs and dissemination; DSP-1 Join a data space	User segmentation, channel and format strategy, release calendar and access policy, product-channel map
Design Disclosure Control and De-identification for Release	PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis; MDS-1 Authorize institutions and researchers	Release disclosure rules, de-identification standard, disclosure-risk criteria, protection per channel
Design Channels, Formats, and Documentation	PRD-2.1 Design outputs and dissemination; PRD-2.6 Design production systems and workflows	Format and accessibility standard, API and dataset specification, service design, documentation standard
Design User Support, Promotion, and Feedback Model	PRD-2.1 Design outputs and dissemination	User support model, communication plan, usage and feedback model, correction policy
Design Release Governance and Audit Framework	PRD-2.1 Design outputs and dissemination; DLM-1 Plan and design data assets; DLM-4 Classify and protect data	Release approval policy, equal-access controls, versioning and lineage policy, access control design

Design the Dissemination and Access Strategy

Refines: PRD-2.1 Design outputs and dissemination; DSP-1 Join a data space.

The starting point is the decision on who the products are for and how they will be reached. This sub-process defines the target user groups and their needs, together with VS-10. It defines the access channels and release formats: open data, APIs, research access, publications, dashboards, transmissions to international organizations, and participation in data spaces. It sets the release calendar and the policy for equal access and embargo. Each product is mapped to its channels and formats. The strategy keeps open access as the default and restricted access as the exception that needs a reason.

Activity	Key outputs	SAF process	Other reference
Define target user groups and their needs (with VS-10 Customer and Stakeholder Engagement)	User segmentation	PRD-2.1 Design outputs and dissemination	2.2.5 User research
Define access channels and release formats, including data spaces	Channel and format strategy	PRD-2.1 Design outputs and dissemination; DSP-1 Join a data space	1.5.3 Facilitate data access

Activity	Key outputs	SAF process	Other reference
Define the release calendar and the policy for equal access and embargo	Release calendar and access policy	PRD-2.1 Design outputs and dissemination	4.1.5 Accountability
Map each product to its channels and formats	Product-channel map	PRD-2.1 Design outputs and dissemination	SAF-P08 Data is shared property

Design Disclosure Control and De-identification for Release

Refines: PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis; MDS-1 Authorize institutions and researchers.

This sub-process designs the final confidentiality protection before release. It specifies the disclosure-control rules for tables and dashboards, on top of the protection already applied upstream. It specifies the standard for de-identifying released microdata. It sets the disclosure-risk criteria that every release must meet. Protection is designed per channel. For example, more detailed microdata is only available in a secure research environment, and only to institutions and researchers authorized for it. The rules for checking research outputs before they leave the environment are part of this design.

Activity	Key outputs	SAF process	Other reference
Define the final disclosure-control rules for tables and dashboards	Release disclosure rules	PRD-2.5 Design processing and analysis	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Define the de-identification standard for released microdata	De-identification standard	PRD-2.5 Design processing and analysis	SAF-P11 Privacy by design
Define the disclosure-risk criteria that every release must meet	Disclosure-risk criteria	PRD-2.1 Design outputs and dissemination	SAF-P11 Privacy by design
Design protection per channel, authorization of researchers, and output checking	Protection per channel, authorization criteria	MDS-1 Authorize institutions and researchers	SAF-P10 Security by design

Design Channels, Formats, and Documentation

Refines: PRD-2.1 Design outputs and dissemination; PRD-2.6 Design production systems and workflows.

This sub-process designs how products are delivered. It specifies the publication formats and the accessibility standards: machine-readable formats, languages, and web accessibility. It specifies the structures of APIs and datasets, for example according to SDMX. It specifies the dashboards and visualizations, and the model for access to the secure research environment. It sets the documentation standard: the methodology notes, metadata, and user guides that must come with every release. The design builds on the platforms that VS-09 provides.

Activity	Key outputs	SAF process	Other reference
Specify publication formats and accessibility standards	Format and accessibility standard	PRD-2.1 Design outputs and dissemination	1.5.1 Publication management
Specify the structures of APIs and datasets, for example SDMX	API and dataset specification	PRD-2.6 Design production systems and workflows	CSDA Dissemination

Activity	Key outputs	SAF process	Other reference
Specify dashboards, visualizations, and the model for research access	Service design	PRD-2.6 Design production systems and workflows	3.4.4 Self-service
Define the documentation standard that accompanies every release	Documentation standard	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification

Design User Support, Promotion, and Feedback Model

Refines: PRD-2.1 Design outputs and dissemination.

This sub-process designs how users are supported and heard. It specifies the model for user support: documentation, service desk, and training. It specifies the approach to promoting releases and communicating about them, together with VS-10. It designs the model for monitoring use and collecting feedback: what is measured, how feedback is collected, and how it flows back into improvement. It sets the policy for corrections, errata, and re-release. Together these make sure that users are not left alone with the figures once they are published.

Activity	Key outputs	SAF process	Other reference
Define the user support model: documentation, service desk, training	User support model	PRD-2.1 Design outputs and dissemination	3.4.1 Services management
Define the promotion and communication approach (with VS-10 Customer and Stakeholder Engagement)	Communication plan	PRD-2.1 Design outputs and dissemination	1.5.2 Promotion to use; 3.1.4 Communication management
Define the model for monitoring use and collecting feedback	Usage and feedback model	PRD-2.1 Design outputs and dissemination	2.2.5 User research
Define the policy for corrections, errata, and re-release	Correction policy	PRD-2.1 Design outputs and dissemination	4.1.5 Accountability

Design Release Governance and Audit Framework

Refines: PRD-2.1 Design outputs and dissemination; DLM-1 Plan and design data assets; DLM-4 Classify and protect data.

This last design sub-process makes sure that releases are governed, equal, and auditable. It defines the workflow for release authorization and sign-off. It defines the controls for equal access, so that no one gets privileged early access outside the policy. It defines the versioning policy for releases: identification, revisions and corrections, and retention. It defines the lineage from each release back to its approved product. Role-based access controls for the access services are designed in line with SAF-P10.

Activity	Key outputs	SAF process	Other reference
Define the workflow for release authorization and sign-off	Release approval policy	PRD-2.1 Design outputs and dissemination	4.1.5 Accountability
Define equal-access controls and embargo governance	Equal-access controls	PRD-2.1 Design outputs and dissemination	4.1.5 Accountability

Activity	Key outputs	SAF process	Other reference
Define the versioning and lineage requirements for releases	Versioning and lineage policy	DLM-1 Plan and design data assets	CSDA Traceability
Design role-based access control for the access services	Access control design	DLM-4 Classify and protect data	SAF-P10 Security by design; 3.3.4 Logical identity and access control

PLACEHOLDER: VS-06 design process decomposition: inputs (user needs from VS-10, approved products, confidentiality law, platforms of VS-09), the five design sub-processes in order, and the design artifact each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

The implementation process type takes the metadata artifacts from Design and **builds, configures, and tests** the dissemination and access pipeline until it is ready for production, on the platforms that VS-09 provides. Nothing is published operationally until this process type is complete. Components are reused where possible and built only where needed, in line with SAF-P01. Dissemination platforms, SDMX tools, and disclosure-control software shared in the NSI community are good candidates for reuse.

Implementation sub-process	Refines	Main outputs
Build Publication and Access Channels	PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows; MDS-4 Provide a secure processing environment	Publication and API services, dashboards, research access, SDMX exchange
Configure Disclosure-Control and Release Tooling	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Disclosure-control tooling, de-identification component, release gate, release management tooling
Build Documentation, Support, and Usage-Monitoring Services	PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data	Documentation services, user support services, usage and feedback tooling, release store, events
Test and Validate the Pipeline	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems	Test reports, accessibility and disclosure reports, release rehearsal, production-ready pipeline

Build Publication and Access Channels

Refines: PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows; MDS-4 Provide a secure processing environment.

This sub-process turns the channel and format designs into operational services. The publication channels, such as the website and the publication system, are built or configured on the platforms of VS-09. The open-data and API services and the dashboards are configured according to the designed formats and accessibility standards. Access to the secure research environment is configured, with the designed controls. Exchange of datasets and metadata according to SDMX is configured, as are the connectors for transmission to international organizations and for data spaces. Access controls are applied according to the designed model.

Activity	Key outputs	SAF process	Other reference
Build or configure publication channels and the open-data and API services	Configured publication and API services	PRD-3.3 Reuse or build dissemination components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Build or configure dashboards and visualization services	Configured dashboards	PRD-3.3 Reuse or build dissemination components	3.4.4 Self-service
Configure access to the secure research environment	Research access service	MDS-4 Provide a secure processing environment	SAF-P10 Security by design
Configure SDMX exchange and the connectors for transmission and data spaces	Standard exchange interfaces	PRD-3.4 Configure workflows	CSDA Dissemination

Configure Disclosure-Control and Release Tooling

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

In this sub-process the designed disclosure rules and release policy are implemented. The disclosure-control tooling is configured to apply the designed suppression, rounding, and perturbation to tables and dashboards. De-identification of released microdata is configured to the designed standard. The disclosure-risk checks are enforced as a gate before publication, so a release cannot proceed without clearance. The release tooling is configured for release at a managed moment, for embargo handling, and for enforcing equal access. Every release action is logged, so that equal access can be demonstrated afterwards.

Activity	Key outputs	SAF process	Other reference
Configure the final disclosure-control tooling: suppression, rounding, perturbation	Configured disclosure-control tooling	PRD-3.2 Reuse or build processing and analysis components	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Configure de-identification of microdata to the designed standard	De-identification component	PRD-3.2 Reuse or build processing and analysis components	SAF-P11 Privacy by design
Enforce the disclosure-risk checks before publication	Release disclosure gate	PRD-3.4 Configure workflows	SAF-P11 Privacy by design
Configure release at a managed moment, embargo handling, and equal-access enforcement	Release management tooling	PRD-3.4 Configure workflows	4.1.5 Accountability

Build Documentation, Support, and Usage-Monitoring Services

Refines: PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data.

This sub-process implements the services for users and the feedback loop. Documentation and metadata services are built, so that every release comes with methodology notes, reference metadata, and user guides. The service desk and training are set up according to the support model. Monitoring of use and collection of feedback are implemented, so that usage statistics, satisfaction, and feedback are captured and routed into the improvement loop and to VS-10. The release store is built so that each release is identified, immutable, and traceable to its approved product. Event notifications are configured to signal each release.

Activity	Key outputs	SAF process	Other reference
Build documentation and metadata services for every release	Documentation services	PRD-3.3 Reuse or build dissemination components	SAF-P09 No data without metadata and classification
Set up the service desk and training	User support services	PRD-3.4 Configure workflows	3.4.1 Services management
Implement monitoring of use and collection of feedback	Usage and feedback tooling	PRD-3.3 Reuse or build dissemination components	2.2.5 User research
Build the release store and configure the *Statistics Released* event	Release store, event publication	DLM-3 Register and catalog data; DLM-5 Store and version data	CSDA Traceability

Test and Validate the Pipeline

Refines: PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems.

Before the pipeline goes into production, it must be thoroughly verified. Technical testing covers all components: publication channels, APIs, dashboards, research access, disclosure-control tooling, release tooling, and the services for documentation and support. Integration testing checks the end-to-end flow from the *Statistical Product Registered* event through preparation, disclosure control, and release to access and support. An accessibility test confirms that formats meet the accessibility standard. A security and disclosure test confirms the access controls and that no output breaches the disclosure-risk criteria. A release rehearsal confirms that managed timing and embargo work as designed. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	
Test integration: end-to-end flow from intake to access and support	Integration test report	PRD-3.5 Test production systems	
Run the accessibility test: formats, languages, machine-readability	Accessibility test report	PRD-3.6 Test statistical business process	
Run the security and disclosure test: access controls, disclosure-risk criteria	Security and disclosure test report	PRD-3.5 Test production systems	SAF-P10 Security by design, SAF-P11 Privacy by design
Run a release rehearsal: managed timing, embargo, equal access	Release rehearsal report	PRD-3.6 Test statistical business process	
Finalize the production system and document operational procedures	Operational procedures manual, production-ready pipeline	PRD-3.7 Finalize production systems	

PLACEHOLDER: VS-06 implementation pipeline: the five design artifacts feeding the three build sub-processes on VS-09 platforms, followed by technical testing, accessibility testing, security and disclosure testing, and release rehearsal, resulting in a production-ready dissemination pipeline and operational procedures

Execution: provides the released statistics and services

Execution runs the stages of VS-06 in every release cycle. The stages and their activities are described in attribute 3. A cycle starts when VS-05 publishes the *Statistical Product Registered* event, when a delivery to an international organization is due, or when an approved research project needs its dataset. VS-06.1 to VS-06.3 run per release; VS-06.4 and VS-06.5 run continuously after it. Execution applies the designs and the solution as they are; a new channel or a policy change goes back to Design. Every release ends in one of three outcomes: released on time, released with a documented delay, or returned for additional protection.

Aspect	VS-06
Trigger	*Statistical Product Registered*; a due transmission to an international organization; an approved research project
Cadence	Per release, according to the release calendar; continuous for support and monitoring
Performed by	ROL-17 Release manager, ROL-22 Dissemination and service manager, ROL-16 Disclosure control specialist, ROL-23 Relationship manager
Input	Approved product version (#4, *approved*), release calendar, channel specifications
Output	Released Data (#5), access services, documentation, usage and feedback data
Control point	Gate of State #5 in VS-06.2 Prepare Data for Dissemination, before VS-06.3 Publish and Provide Access Services
Exception path	Failed disclosure gate back for additional protection; an error found after release handled through the correction policy and DLM-7 Revise data; a breach of embargo escalated to ROL-19 Compliance officer

Management: provides quality reports

The management process type runs alongside and after execution. It monitors the performance of the dissemination pipeline, assesses the accessibility and impact of releases, makes sure that every release is compliant and equal, and feeds improvements back into Design and to VS-10. This closes the governance cycle.

Management sub-process	Refines	Main outputs
Monitor Process Performance	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard, punctuality report, paradata records
Assess Dissemination Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Accessibility report, punctuality report, reach and clarity report, trend analysis
Manage Release and Channel Lifecycle	OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-10 Archive and preserve data	Channel version register, metadata quality report, release store audit
Audit and Compliance Reporting	OVA-1 Quality Management; OVA-4 Process Data Management; MDS-8 Monitor use and compliance; DSP-5 Monitor compliance and report	Disclosure, equal-access, accessibility, and security compliance reports to VS-07 Governance and Compliance
Continuous Improvement	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, improvement tracking log

Monitor Process Performance

Refines: OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs.

The operational performance of VS-06 must be tracked systematically across release cycles. The process owner and the quality manager collect and aggregate the key performance indicators (see attribute 9). They monitor adherence to the release calendar, because punctuality is a visible commitment to users. They track paradata: which channels were used, when each release went live, how embargoes were handled, and how many questions from users followed. This information feeds dashboards that give the process owner a current view of the health of the domain. The approach follows the Quality Management capability of GAMSO.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per release cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Monitor adherence to the release calendar	Release calendar adherence report	OVA-4 Process Data Management	
Track paradata: channels, timing, embargo handling, questions from users	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Assess Dissemination Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of dissemination itself is evaluated against the criteria from Design. Accessibility covers formats, languages, machine-readability, and compliance with web accessibility rules. Timeliness and punctuality are measured against the release calendar. Reach and relevance compare use with expectations and with the coverage of user groups. Clarity is judged from questions and feedback: do users interpret the figures correctly? Quality indicators are produced per release and per cycle, and trends are tracked. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Assess the accessibility compliance of releases	Accessibility report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess timeliness and punctuality against the release calendar	Punctuality report	OVA-1 Quality Management	CSDA Data Quality
Assess reach, relevance, and clarity from use and feedback	Reach and clarity report	DLM-8 Monitor data quality, lineage and use	2.2.5 User research
Identify quality trends across releases and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management

Manage Release and Channel Lifecycle

Refines: OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-10 Archive and preserve data.

The channels, formats, and releases must stay current, complete, and well governed. This sub-process governs the life cycle of channels, formats, and documentation standards. It versions them, manages transitions such as an upgrade of an API version, and retires channels that are no longer needed. It verifies

that every release carries its full documentation, disclosure record, and lineage, as a compliance check on SAF-P09. It governs the release store: retention, immutability, the handling of corrections, and the reconciliation of catalog entries with what is actually published. Released statistics are part of the national record, so their transfer to long-term preservation is also governed here.

Activity	Key outputs	SAF process	Other reference
Govern the life cycle and versions of channels, formats, and documentation	Channel version register	OVA-2 Metadata Management	SAF-P03 Active life cycle management
Verify metadata completeness: documentation, disclosure record, lineage	Metadata quality report	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Govern the release store: retention, immutability, corrections	Release store audit	DLM-5 Store and version data; DLM-7 Revise data	CSDA Traceability
Transfer releases to long-term preservation according to retention rules	Archive transfer record	DLM-10 Archive and preserve data	3.1.8 Information management and archive management

Audit and Compliance Reporting

Refines: OVA-1 Quality Management; OVA-4 Process Data Management; MDS-8 Monitor use and compliance; DSP-5 Monitor compliance and report.

This sub-process produces the evidence that VS-06 operates within legal, regulatory, and organizational requirements. Disclosure compliance reports confirm that every release passed the disclosure-risk gate and that released microdata meets the de-identification standard. Equal-access compliance reports confirm that release timing and the embargo policy were respected. Accessibility compliance reports confirm that formats met the accessibility standard. Research access reports confirm that researchers and data space partners complied with their conditions of use. Security compliance reports verify the access controls and the integrity of the secure research environment. These reports, produced together with VS-07, meet the obligation of SAF capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance*.

Activity	Key outputs	SAF process	Other reference
Produce disclosure compliance reports: every release passed the gate	Disclosure compliance summary	OVA-1 Quality Management	SAF-P11 Privacy by design
Report on equal-access compliance: timing, embargo	Equal-access compliance report	OVA-4 Process Data Management	4.1.5 Accountability
Report on accessibility compliance	Accessibility compliance report	OVA-1 Quality Management	4.1.5 Accountability
Report on compliance of researchers and data space partners with conditions of use	Research and data space compliance report	MDS-8 Monitor use and compliance; DSP-5 Monitor compliance and report	SAF-P11 Privacy by design
Report on security compliance: access controls, integrity of the research environment	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10 Security by design; 3.3.3 Security incident management

Continuous Improvement

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This last management sub-process closes the governance cycle by analyzing performance, usage, and feedback and driving improvements. Trends across release cycles are examined: is punctuality holding, is reach growing, is satisfaction improving, and are questions from users decreasing? Improvement opportunities are identified. Examples are adding or improving channels and formats, rewriting documentation that users find unclear, streamlining the release process, improving accessibility, and signaling demand for new statistics or breakdowns. Recommendations are fed back to the design sub-processes, to VS-10, and, where a need starts earlier in the chain, to the value stream concerned. They are prioritized by impact and feasibility and tracked through the following cycles. This is SAF capability 4.1.4 *Improvement management* in practice.

Activity	Key outputs	SAF process	Other reference
Analyze trends in use, satisfaction, and quality across cycles	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities: channels, documentation, accessibility, release process	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Feed recommendations back to the design sub-processes, to VS-10 Customer and Stakeholder Engagement, and to the chain	Design and engagement feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management

PLACEHOLDER: VS-06 management feedback loop: releases, usage, and feedback flowing through the five management sub-processes, with dashboards to the process owner, accessibility and punctuality reports to the quality manager, compliance reports to VS-07, user needs to VS-10 and the chain, and improvement recommendations back to Design

Metadata and paradata per process type

In VS-06, metadata is what makes responsible use possible: every release must carry the documentation, methodology, and reference metadata that users need. The table shows what each process type contributes. Data metadata describes the released products. Process metadata, also called paradata, describes how the process itself ran: who, when, and with which tools.

Process type	Data metadata produced	Process metadata (paradata) produced
Design	Dissemination and access strategy, channel and format specifications, release and disclosure policy, documentation standards	Log of design decisions, policy version, review and approval records

Process type	Data metadata produced	Process metadata (paradata) produced
Implementation	Channel configurations, API and dataset templates, disclosure-control parameters, documentation templates	Build log, test results, accessibility test outcomes, deployment approvals
Execution	Documentation per release, record of disclosure control applied, channel and format published, release timestamp, metadata of access services	Operator identity, adherence to the release calendar, embargo handling, publication run identifiers
Management	Usage statistics, accessibility assessments, user satisfaction measures, release quality reports	Usage dashboards, feedback logs, improvement decisions, audit findings

PLACEHOLDER: VS-06 business object lifecycle: design artifacts govern implementation assets, which produce the pipeline that turns approved Statistics (#4) into Released Data (#5), generating disclosure records, release records, and usage data that management assesses and feeds back to design and VS-10

5. Actors and roles

VS-06 is the value stream where the NSI meets its users. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the dissemination manager, the channel designer, the open-data and API product owner, and the user support lead are ROL-22 Dissemination and service manager, and the communications officer is ROL-23 Relationship manager. VS-05, which supplies the approved products, is a neighboring value stream and not an actor; the relation is governed by the handover interface in attribute 8. The relation with accredited researchers is organized in the collaboration COL-04 *Research access partnership*. **[VC20]**

Actor or role	ID	Process type	Responsibility in VS-06 Dissemination and Service Delivery
Data user	ACT-05	External	Uses released data: policy makers, researchers, media, businesses, and the public
International statistical organization	ACT-07	External	Receives transmissions under agreed programs, for example Eurostat or the OECD
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-06 Dissemination and Service Delivery
Data owner	ROL-02	Management	Accountable for State #5; decides on access to restricted and research data
Data steward	ROL-03	Execution	Registers releases, maintains catalog entries, applies access controls

Actor or role	ID	Process type	Responsibility in VS-06 Dissemination and Service Delivery
Data custodian	ROL-04	Execution	Operates the release store, the secure research environment, retention, and archiving
Metadata and data architect	ROL-08	Design	Designs the structures of APIs and datasets and the reference metadata
Data engineer	ROL-10	Implementation	Builds and configures publication channels, APIs, and access services
Platform engineer	ROL-11	Implementation	Configures release tooling, access controls, and disclosure-control tooling on the platforms of VS-09 Infrastructure and Platform Enablement
Test manager	ROL-12	Implementation	Plans and carries out technical, accessibility, security, and release testing
Disclosure control specialist	ROL-16	Design, Execution	Designs and applies final disclosure control and de-identification; checks research outputs
Release manager	ROL-17	Execution	Manages the release: timing, embargo, equal access, and publication
Quality manager	ROL-18	Management	Monitors usage and accessibility indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures compliance with data protection, confidentiality, and equal access, together with VS-07 Governance and Compliance
Dissemination and service manager	ROL-22	Design, Execution	Designs the dissemination strategy and channels; operates open-data, API, and dashboard services and user support
Relationship manager	ROL-23	Execution	Promotes releases, maintains relationships with key users, and collects feedback, together with VS-10 Customer and Stakeholder Engagement

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-06 relies on, and what role they play. NSIs can use it to map the value stream to their own organization.

Capability group	Capabilities	Role in VS-06 Dissemination and Service Delivery
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design dissemination products and services, and build the release and access pipeline
1.3 Processing	1.3.7 Statistical disclosure	Apply final disclosure control and de-identification **[VC21]**
1.5 Dissemination	1.5.1 Publication management, 1.5.2 Promotion to use, 1.5.3 Facilitate data access, 1.5.4 Statistical information management	Publish, promote, and give access to statistics, and manage their documentation **[VC21]**
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management	Manage released data and its metadata, and the quality of release and access
2.2 Research	2.2.4 Data services, 2.2.5 User research	Provide research access to microdata and understand user needs
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain publication channels, APIs, and dashboards, with the platforms of VS-09 Infrastructure and Platform Enablement
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.4 Logical identity and access control	Secure access to released data and research environments
3.4 Customer and chain interaction	3.4.1 Services management, 3.4.2 Contact management, 3.4.4 Self-service	Operate user services, the service desk, and self-service channels
4.1 Strategy and leadership	4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability, 4.1.6 Collaboration	Govern release, manage the policy for equal access, plan improvements, report on compliance, cooperate in international transmission

Stages, processes, and functions

The table shows which SAF business processes realize each stage of VS-06 and which business functions they mainly use. It follows the mapping in *Business Processes, Mapping to value stream stages*. The activity tables in attribute 3 can name additional supporting processes. **[VC25]**

Stage	Realized by processes	Main functions
VS-06.1 Plan Dissemination and Access Strategy	PRD-2.1 Design outputs and dissemination	FUN-3.1 Design and build statistical products; FUN-3.6 Disseminate statistics
VS-06.2 Prepare Data for Dissemination	PRD-6.4 Apply disclosure control; PRD-7.1 Update output systems	FUN-3.5 Protect statistical confidentiality; FUN-3.6 Disseminate statistics

Stage	Realized by processes	Main functions
VS-06.3 Publish and Provide Access Services	PRD-7.2 Produce dissemination products; PRD-7.3 Release dissemination products; MDS-1 Authorize institutions and researchers; MDS-2 Assess access applications; MDS-3 Prepare research datasets; MDS-4 Provide a secure processing environment; MDS-5 Facilitate privacy-preserving analysis; MDS-7 Check and release output; DSP-2 Publish data offerings	FUN-3.6 Disseminate statistics; FUN-8.1 Manage research access; FUN-8.2 Prepare research data; FUN-8.3 Operate secure research environments; FUN-8.4 Control research output; FUN-8.5 Enable privacy-preserving analysis; FUN-9.1 Manage data space participation
VS-06.4 Support and Engage Users	PRD-7.4 Promote dissemination products; PRD-7.5 Provide user support; MDS-6 Support researchers	FUN-3.6 Disseminate statistics; FUN-8.3 Operate secure research environments; FUN-10.9 Manage users and services
VS-06.5 Monitor Usage and Feedback	PRD-8.1 Gather evaluation inputs; MDS-8 Monitor use and compliance; MDS-9 Close projects	FUN-8.1 Manage research access; FUN-8.3 Operate secure research environments; FUN-10.9 Manage users and services

PLACEHOLDER: ArchiMate view VS-06 (b) stages realized by processes and functions

Framework mappings

This reference table maps each process type of VS-06 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.1, PRD-2.5, PRD-2.6, DLM-1, DLM-4, MDS-1, DSP-1	Phase 2 Design (2.1, 2.5, 2.6)	Production (design activities), Capability Management	Data Governance, Metadata Management, Dissemination	1.1.2 Design statistics product, 1.5.1 Publication management, 2.1.2 Metadata management
Implementation	PRD-3.2 to PRD-3.7, DLM-3, DLM-5, MDS-4	Phase 3 Build (3.2 to 3.7)	Capability Management	Dissemination (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Execution	PRD-6.4 (final), PRD-7.1 to PRD-7.5, PRD-8.1, PRD-8.2, DLM-2, DLM-3, DLM-5, DLM-7, DLM-8, MDS-3, MDS-4, MDS-6 to MDS-9, DSP-2, DSP-4, DSP-5	6.4 Apply disclosure control; Phase 7 Disseminate (7.1 to 7.5); 8.1 Gather evaluation inputs	Production (execution)	Dissemination (execution)	1.3.7 Statistical disclosure, 1.5.1 to 1.5.4, 2.2.4 Data services
Management	OVA-1, OVA-2, OVA-4, DLM-5, DLM-7, DLM-8, DLM-10, MDS-8, DSP-5, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-06 most directly.

Principle	Statement	Application in VS-06 Dissemination and Service Delivery
SAF-P01	Reuse before open source, open source before buying, buying before building	Dissemination platforms, APIs, SDMX tools, and disclosure-control software are reused across products first. Where new tooling is needed, open-source solutions developed with other NSIs come before purchase or own development. **[VC3]**
SAF-P07	Implement loosely coupled processes and systems	Dissemination uses approved products through the interface of State #4 without knowing how they were composed. It exposes released data through stable, versioned access services.
SAF-P08	Data is shared property	Released data is made broadly and equally available to all users through open and standard channels. Open access is the default; restricted access is the exception that needs a reason.

Principle	Statement	Application in VS-06 Dissemination and Service Delivery
SAF-P09	No data without metadata and classification	Every release carries the metadata, methodology notes, and documentation that users need to interpret it correctly. Released datasets come with structural and reference metadata, for example according to SDMX.
SAF-P10	Security by design	Access controls, secure channels, and integrity protection are built into publication and access services, including the secure research environment for sensitive microdata.
SAF-P11	Privacy by design	Final disclosure control and de-identification are applied before release. The disclosure risk of every released product and microdata file is assessed and cleared by design.

i Principles that apply across the process types

Several principles apply to all four process types. SAF-P11 is the release gate: disclosure control is designed in *Design Disclosure Control and De-identification for Release*, implemented in *Configure Disclosure-Control and Release Tooling*, and applied in VS-06.2 before anything is published. SAF-P08 and SAF-P09 drive open, well-documented, and equal release. SAF-P10 governs the access services and the secure research environment in VS-06.3 and VS-06.4. SAF-P07 governs the intake of approved products from VS-05 and the access interface for external users.

8. Interfaces, services, and events

Boundary interfaces

VS-06 is loosely coupled (SAF-P07) to its upstream neighbor and to external users. Specifying each boundary explicitly makes the loose-coupling contract concrete and testable: the handover artifact, the metadata it must carry, the event that triggers it, and the point where responsibility transfers. As the last primary value stream, VS-06 has no downstream primary value stream. Its handover is to external users, and its feedback path runs to VS-10.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-05 Statistical Product Composition → VS-06 Dissemination and Service Delivery (intake)	Registered approved product version (#4, *approved*)	Component lineage, composition method, propagated accuracy, coherence results, approval record, confidentiality status, product version identifier, classification	*Statistical Product Registered*	VS-05 Statistical Product Composition keeps custody of the product version; VS-06 Dissemination and Service Delivery takes responsibility for preparation, release, and service delivery on receipt of the event
VS-06 Dissemination and Service Delivery → users and society (release)	Released Data (#5): published statistics, de-identified microdata, access services	Record of final disclosure control, documentation and methodology notes, reference and structural metadata, channel and format, release timestamp, release version identifier, lineage to the approved product	*Statistics Released*	The NSI publishes and serves; external users use the data as #6 Consumed Data, outside the NSI's control. Usage and feedback return to VS-06.5 Monitor Usage and Feedback and VS-10 Customer and Stakeholder Engagement

Business services

Service	Offered to	Realized in	Interfaces
Dissemination preparation service	Release teams	VS-06.2 Prepare Data for Dissemination	Disclosure-control tooling, channel formatting
Publication and open data service	Data users, the public, businesses	VS-06.3 Publish and Provide Access Services	Web publication and bulk download of open data with machine-readable metadata; release timing for equal access enforced
API and dashboard service	Researchers, developers, analysts	VS-06.3 Publish and Provide Access Services	API compliant with SDMX for released series and datasets with their metadata; interactive dashboards fed from released datasets
Secure research access service	Accredited researchers	VS-06.3 Publish and Provide Access Services	Mediated access to de-identified microdata in a controlled environment, with output checking
Transmission and data space service	International statistical organizations, data space partners	VS-06.3 Publish and Provide Access Services	Agreed transmission channels, data space connectors

Service	Offered to	Realized in	Interfaces
User support service	All data users	VS-06.4 Support and Engage Users	Documentation, service desk, training
Usage and feedback service	Process owners, VS-10 Customer and Stakeholder Engagement, the chain	VS-06.5 Monitor Usage and Feedback	Usage dashboards, feedback channels
Design and strategy service	Dissemination and service managers	All Design sub-processes of VS-06 Dissemination and Service Delivery	Design documents, policy advice

Business events

Event	Triggers or marks
Design artifacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Statistical Product Registered	Start of VS-06.1 Plan Dissemination and Access Strategy for that product (published by VS-05 Statistical Product Composition)
Disclosure cleared / Disclosure gate failed	Outcome of VS-06.2 Prepare Data for Dissemination; failure returns the product for additional protection
Statistics Released	Completion of VS-06.3 Publish and Provide Access Services at the announced moment; start of use as #6
Erratum issued	Correction of a release under the correction policy
Feedback received	Input for VS-06.5 Monitor Usage and Feedback and VS-10 Customer and Stakeholder Engagement
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-06 event choreography between VS-05, the release teams, external users, VS-10, and process management across Design, Implementation, Execution (plan, prepare and clear, release at the announced moment, support, monitor), and Management (key performance indicators, accessibility, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-06	Receives from VS-06
VS-07 Governance and Compliance	Data protection and confidentiality policy, release policy, audit	Disclosure, equal-access, and accessibility compliance evidence
VS-08 Innovation and Methodological Development	New disclosure-control methods, new forms of dissemination	Unmet user needs and improvement needs
VS-09 Infrastructure and Platform Enablement	Portals, APIs, secure research environments, release store	Availability and capacity requirements, in particular at release moments
VS-10 Customer and Stakeholder Engagement	User needs, user groups, communication with users	Usage data, feedback, and user questions

9. Key performance indicators

The key performance indicators are collected per release cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
On-time release rate	Share of releases published exactly at the announced moment	Per release, per cycle
Disclosure gate pass rate	Share of releases that pass the disclosure-risk gate on the first run	Per release, per cycle
Channel availability	Availability of APIs and access services	Per channel
Use and reach	Downloads, API calls, and dashboard views per release	Per release, per channel
User satisfaction	Satisfaction measure from feedback	Per cycle
Support response time	Time to resolve questions from users	Per cycle

10. Toolkit

Examples from NSI practice, the maturity checklist, and end-to-end scenarios for VS-06 are part of the SAF Toolkit.

Toolkit: SAF-TK-V06 VS-06 Dissemination and Service Delivery companion [**VC17**]

Supporting value streams

VS-07 Governance and Compliance

1. Purpose and value created

VS-07 makes sure that every process, dataset, and product of the NSI operates under clear rules, standards, and accountability. It defines the governance frameworks and policies, maps the applicable law to concrete procedures, assigns the data governance roles, monitors quality and compliance, and audits and certifies the results. VS-07 does not move data along the chain. It oversees the value streams that do, and it is the place where the data principles SAF-P09, SAF-P10, and SAF-P11 are enforced across the whole organization. Its frame of reference includes the European Statistics Code of Practice, the GDPR, the national statistics law, and international quality standards. The value created is **integrity and accountability**. Users, respondents, and regulators can trust that the NSI works ethically, transparently, and within the law.

VS-07 is a supporting value stream. It aligns with the GAMSO areas *Strategy and Leadership* and *Corporate Support* rather than with the production phases of GSBPM. This section describes its business layer for all four process types. The application and technology controls that implement governance are provided with VS-09.

- **Compliance officer, quality manager, or auditor:** the stages in attribute 3, which describe the continuous governance loop.
- **Chief data officer or policy maker:** the Design processes in attribute 4.
- **Process owners of other value streams:** attributes 2 and 8, which describe what VS-07 provides and what it expects in return.
- **Manager who wants the big picture:** attributes 1, 2, and 9. Examples and scenarios are part of the SAF Toolkit.

> **i** How a supporting value stream differs from a primary one
>
> VS-07 has no steady-state data transition. It does not take data from one state to the next, and it does not appear in the chain from #1 to #6. It runs continuously, not per production cycle, so no *Data Registered* event triggers it. Its output is an enabling capability rather than a data artifact: governance frameworks, compliance mechanisms, assurance, and audit results. It foregrounds the principles SAF-P02 to SAF-P06, and it enforces the data principles of the primary value streams.

PLACEHOLDER: ArchiMate view VS-07 (a) value stream and capabilities

2. Position and interaction with the chain

VS-07 is not a stage in the production line. It runs horizontally across all six primary value streams and the other supporting streams. It interacts with them in two directions. Downward, it issues governance frameworks, policies, roles, and controls. Upward, it receives compliance evidence, quality indicators, and audit findings. The *Audit and Compliance Reporting* sub-process of every primary value stream feeds directly into VS-07.

Counterpart	VS-07 provides	VS-07 receives
VS-01 to VS-06 (primary value streams)	Frameworks, policies, data governance roles, controls, quality criteria	Compliance evidence, quality indicators, audit findings
VS-08 Innovation and Methodological Development	Compliance criteria that a new method must meet before production	Requests to assess and approve new methods
VS-09 Infrastructure and Platform Enablement	Security and continuity policy to enforce	Security posture, incident evidence
VS-10 Customer and Stakeholder Engagement	Commitments on transparency and accountability	Signals on stakeholder trust, complaints
External parties: regulators, peer reviewers, the public	Certifications, assurance statements, transparency reports	Regulatory requirements and changes, peer review findings

PLACEHOLDER: ArchiMate view VS-07 (c) interaction with the chain: VS-07 across the six primary value streams, with frameworks, policies, and controls flowing down, and compliance evidence, quality indicators, and audit findings flowing up; certification and assurance to external parties

3. Stages

The stages of VS-07 form a continuous governance loop: plan, build, operate, and improve. The loop never stops. Rules are set, embedded in the work of the value streams, checked, and revised as the law, the organization, and the evidence change.

Stage	Name	Value added
VS-07.1	Define Governance Frameworks and Policies	The organization works under one clear rulebook
VS-07.2	Establish Legal and Regulatory Compliance Mechanisms	Every process stays within the law through concrete procedures
VS-07.3	Implement Data and Metadata Governance	Governance is built into how the value streams actually work
VS-07.4	Monitor Quality and Compliance Indicators	The status of quality and compliance is visible at all times
VS-07.5	Audit, Certify, and Report Compliance	There is independent assurance that the statistics can be trusted
VS-07.6	Update and Improve Governance Mechanisms	Governance keeps pace with change and lessons learned

The strategy model labels two stages as VS-07.5. They are distinct activities; this section treats the second as VS-07.6. [**VC27**]

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. Several governance activities have no reference process in

the SAF; they are carried out through NSI-specific management processes, which the column shows as *NSI-specific*. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-07.1 Define Governance Frameworks and Policies

Value created: *One clear rulebook under which the whole organization works.*

The loop begins with the frameworks and policies that set the rules. These are the charters and principles for data, metadata, ethics, confidentiality, and statistical quality. They translate the European Statistics Code of Practice and the NSI's mission into rules that the value streams can apply. Each policy has an owner, a scope, a version, and a review date. The data policy defines which data assets the NSI holds, who is accountable for them, and how they are classified. Policies are written so that they can be checked: a rule that cannot be monitored is rewritten until it can.

Activity	Key outputs	SAF process	Other reference
Define charters and principles for data, metadata, ethics, confidentiality, and quality	Governance framework	NSI-specific	4.1.1 Strategy and governance; European Statistics Code of Practice
Define the data policy: data assets, accountability, classification	Data policy	DLM-1 Plan and design data assets	SAF-P09 No data without metadata and classification
Define the quality framework and quality criteria for products and processes	Quality framework	OVA-1 Quality Management	2.1.3 Statistical quality management
Assign an owner, scope, version, and review date to every policy	Policy register	NSI-specific	SAF-P03 Active life cycle management

VS-07.2 Establish Legal and Regulatory Compliance Mechanisms

Value created: *Every process stays within the law through concrete, assigned procedures.*

In parallel, VS-07 identifies the regulations that apply to the NSI. Examples are the GDPR, the national statistics law, archival law, and the conditions in data sharing agreements and data spaces. For each regulation, it defines the procedures that keep the processes within the law. It maintains the register of processing activities and the requirements for data protection impact assessments. It defines how security classification and protection follow from the law. The result is a compliance register that links each legal obligation to the procedures, the value streams, and the roles that fulfill it.

Activity	Key outputs	SAF process	Other reference
Identify the applicable regulations and agreements	Regulatory inventory	NSI-specific	3.1.7 Legal support
Define procedures that keep each process within the law	Compliance procedures	NSI-specific	SAF-P11 Privacy by design
Maintain the register of processing activities and the requirements for impact assessments	Register of processing activities	DLM-4 Classify and protect data	3.3.1 Privacy management
Link each obligation to procedures, value streams, and roles	Compliance register	NSI-specific	4.1.5 Accountability

VS-07.3 Implement Data and Metadata Governance

Value created: *Governance built into how the value streams work, rather than living only on paper.*

The frameworks become operational when roles and processes are in place. VS-07 assigns the data governance roles per data domain and per steady state: data owner, data steward, and data custodian. It defines the processes for creating and maintaining metadata, for tracking lineage, and for granting access to data for reuse. It sets the obligations that every registration in a steady state must meet, in line with SAF-P09. It coordinates the data governance board, where data owners and stewards resolve issues across domains. From this point on, every value stream can show who is accountable for each dataset.

Activity	Key outputs	SAF process	Other reference
Assign data owner, data steward, and data custodian per domain and steady state	Role assignments	OVA-3 Data Management; DLM-1 Plan and design data assets	2.1.1 Data lifecycle management
Define processes for metadata creation, maintenance, and lineage	Metadata and lineage rules	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Define the rules for access to registered data for reuse	Reuse and access rules	DLM-6 Provide data for reuse	SAF-P08 Data is shared property
Operate the data governance board	Board decisions	NSI-specific	4.1.6 Collaboration

VS-07.4 Monitor Quality and Compliance Indicators

Value created: *A current, visible picture of quality and compliance across the organization.*

With controls in place, VS-07 monitors whether they work. It collects the quality indicators of processes and outputs, such as accuracy, timeliness, and comparability, from the management sub-processes of the value streams. It tracks compliance status on scorecards and dashboards. It monitors data quality, lineage, and use across registered data. It also receives the compliance reports on research access and data spaces. Deviations are raised with the process owner concerned and followed until they are resolved.

Activity	Key outputs	SAF process	Other reference
Collect quality indicators of processes and outputs from all value streams	Quality scorecard	OVA-1 Quality Management	2.1.3 Statistical quality management
Monitor data quality, lineage, and use across registered data	Data governance dashboard	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality
Monitor compliance of research access and data space exchanges	Research and data space compliance status	MDS-8 Monitor use and compliance; DSP-5 Monitor compliance and report	SAF-P11 Privacy by design
Raise deviations with the process owners and track them to resolution	Deviation log	OVA-1 Quality Management	4.1.5 Accountability

VS-07.5 Audit, Certify, and Report Compliance

Value created: *Independent assurance that the statistics and the processes behind them can be trusted.*

Periodically, VS-07 checks independently whether the frameworks are followed. It plans and conducts internal audits and supports external audits and peer reviews, such as those under the European Statistics

Code of Practice. It carries out quality reviews of products and processes. It documents adherence and issues assurance statements and transparency reports. Audit findings are assigned to an owner and a deadline. The audit trails that the primary value streams produce, such as edit lineage and disclosure records, are the evidence used here.

Activity	Key outputs	SAF process	Other reference
Plan and conduct internal audits; support external audits and peer reviews	Audit reports	OVA-1 Quality Management	4.1.5 Accountability
Carry out quality reviews of products and processes	Quality review reports	PRD-8.2 Conduct evaluation	GAMSO Quality Management
Issue certifications, assurance statements, and transparency reports	Assurance statements, transparency reports	NSI-specific	4.1.5 Accountability
Assign each finding to an owner with a deadline	Findings register	OVA-1 Quality Management	

VS-07.6 Update and Improve Governance Mechanisms

Value created: *Governance that evolves with the organization and the law rather than growing rigid.*

The loop closes with the revision of the governance mechanisms. Feedback, regulatory changes, audit findings, and lessons learned are incorporated into the policies and controls. Action plans are agreed with the process owners and tracked. Policies and controls that are no longer needed are formally retired. Governance of retention is also applied here: data that must be kept long term is transferred to archives, and data at the end of its retention period is disposed of under control. Each revised policy is issued as a new version, and the value streams are notified.

Activity	Key outputs	SAF process	Other reference
Incorporate feedback, regulatory changes, and findings into policies and controls	Revised policies and controls	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Agree and track action plans with process owners	Action plans	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Govern archiving and disposal according to retention rules	Archive and disposal decisions	DLM-10 Archive and preserve data; DLM-11 Dispose of data	3.1.8 Information management and archive management
Retire obsolete policies and notify the value streams of new versions	*Policy update issued*	NSI-specific	SAF-P03 Active life cycle management

4. The four process types

VS-07 is itself a capability that must be designed, built, operated, and improved. The four process types therefore apply to VS-07 as they do to the primary value streams, but they govern the governance capability rather than a data pipeline. Design produces the governance model and its metadata, Implementation builds the roles, procedures, and tooling, Execution runs the stages of attribute 3, and Management evaluates whether governance itself works. [**VC26**]

Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. Where no reference process exists, the sub-process is marked as NSI-specific.

Design: provides the governance model

Design sub-process	Refines	Main outputs
Design the Governance Operating Model	DLM-1 Plan and design data assets; NSI-specific	Governance operating model, role model, decision rights
Design the Quality and Compliance Indicator Set	OVA-1 Quality Management	Indicator set, scorecard design, audit program design

Design the Governance Operating Model

Refines: DLM-1 Plan and design data assets; NSI-specific management processes.

This sub-process designs how governance is organized. It defines the governance bodies, such as the data governance board and the quality board, and their decision rights. It defines the role model for data governance and how the roles relate to the reference roles of the SAF. It defines how policies are proposed, approved, issued, and retired. It also defines the escalation paths from the value streams to the governance bodies. A good operating model keeps governance light enough to be followed and strong enough to be enforced, in line with SAF-P04.

Activity	Key outputs	SAF process	Other reference
Define governance bodies and their decision rights	Governance operating model	NSI-specific	4.1.1 Strategy and governance
Define the role model for data governance	Role model	DLM-1 Plan and design data assets	2.1.1 Data lifecycle management
Define the life cycle of policies: proposal, approval, issue, retirement	Policy life cycle	NSI-specific	SAF-P03 Active life cycle management

Design the Quality and Compliance Indicator Set

Refines: OVA-1 Quality Management.

This sub-process designs what VS-07 measures. It selects the quality indicators for processes and outputs, based on the quality framework and the European quality dimensions. It defines the compliance indicators for each policy and legal obligation. It designs the scorecards and dashboards, and which indicators the value streams must supply. It designs the audit program: which areas are audited, how often, and by whom. The indicator set is kept small enough to be supplied without excessive burden on the value streams.

Activity	Key outputs	SAF process	Other reference
Select quality indicators for processes and outputs	Quality indicator set	OVA-1 Quality Management	2.1.3 Statistical quality management
Define compliance indicators per policy and obligation	Compliance indicator set	OVA-1 Quality Management	4.1.5 Accountability
Design the scorecards, dashboards, and audit program	Scorecard design, audit program	OVA-1 Quality Management	GAMSO Quality Management

Implementation: provides the governance mechanisms

Implementation sub-process	Refines	Main outputs
Establish Governance Roles and Procedures	OVA-3 Data Management; NSI-specific	Appointed role holders, operating procedures, trained staff
Build Governance Tooling	PRD-3.4 Configure workflows; OVA-2 Metadata Management	Policy register, compliance register, dashboards, with VS-09 Infrastructure and Platform Enablement

Establish Governance Roles and Procedures

Refines: OVA-3 Data Management; NSI-specific management processes.

This sub-process puts the operating model into practice. Data owners, stewards, and custodians are appointed for each domain and steady state. The governance bodies are set up with their members and meeting cycles. Operating procedures are written for issuing policies, handling deviations, and escalating issues. Role holders are trained, so that they know their responsibilities and the evidence they must supply. Without named people in these roles, governance remains a document rather than a practice.

Activity	Key outputs	SAF process	Other reference
Appoint data owners, stewards, and custodians	Appointed role holders	OVA-3 Data Management	2.1.1 Data lifecycle management
Set up the governance bodies and their meeting cycles	Governance bodies in operation	NSI-specific	4.1.6 Collaboration
Write operating procedures and train role holders	Procedures, trained staff	NSI-specific	3.1.1 Personnel management (HR)

Build Governance Tooling

Refines: PRD-3.4 Configure workflows; OVA-2 Metadata Management.

This sub-process builds the tools that governance needs, together with VS-09. It implements the policy register and the compliance register. It configures the dashboards that collect indicators from the value streams. It makes sure that the data catalog records owners, classifications, and lineage, so that governance can be checked automatically where possible. It configures the workflow for audit findings and action plans. Automated checks reduce the burden on the value streams and make compliance visible in real time.

Activity	Key outputs	SAF process	Other reference
Implement the policy register and the compliance register	Registers	NSI-specific	SAF-P03 Active life cycle management
Configure dashboards that collect indicators from the value streams	Quality and compliance dashboards	PRD-3.4 Configure workflows	2.1.3 Statistical quality management
Make the catalog record owners, classifications, and lineage	Governance-ready catalog	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

Execution: runs the governance loop

Execution runs the stages of attribute 3 continuously. VS-07.1 to VS-07.3 run when policies are created or revised; VS-07.4 runs continuously; VS-07.5 runs according to the audit program; VS-07.6 runs after audits, incidents, and regulatory changes.

Aspect	VS-07
Trigger	*Regulatory change*; *Audit due*; *Incident reported*; review date of a policy
Cadence	Continuous monitoring; periodic audits; annual policy review
Performed by	ROL-05 Chief data officer, ROL-19 Compliance officer, ROL-18 Quality manager, ROL-21 Auditor
Input	Laws and regulations, Code of Practice, compliance evidence and quality indicators from all value streams
Output	Policies, compliance register, role assignments, dashboards, audit reports, assurance statements
Control point	Approval of policies by the governance bodies; audit conclusions
Exception path	Serious non-compliance escalated to NSI management (ACT-01); confidentiality or security incident handled with ROL-20 Security officer

Management: provides governance effectiveness reports

Management sub-process	Refines	Main outputs
Evaluate Governance Effectiveness	PRD-8.1 Gather evaluation inputs; PRD-8.2 Conduct evaluation	Effectiveness report, burden assessment
Improve the Governance Capability	PRD-8.3 Agree an action plan	Improvement plan for the governance capability

Evaluate Governance Effectiveness

Refines: PRD-8.1 Gather evaluation inputs; PRD-8.2 Conduct evaluation.

This sub-process asks whether governance itself works. It measures the key performance indicators of VS-07 (see attribute 9), such as policy coverage and the closure of audit findings. It assesses the burden that governance places on the value streams, in line with SAF-P04. It checks whether incidents were prevented or only detected afterwards. It collects the views of process owners on the clarity and usefulness of policies. The result is an honest report on where governance adds value and where it only adds work.

Activity	Key outputs	SAF process	Other reference
Measure the key performance indicators of VS-07 Governance and Compliance	Effectiveness report	PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Assess the burden of governance on the value streams	Burden assessment	PRD-8.2 Conduct evaluation	SAF-P04 Maximize the benefits for the organization
Collect the views of process owners on policies	Feedback on policies	PRD-8.1 Gather evaluation inputs	

Improve the Governance Capability

Refines: PRD-8.3 Agree an action plan.

This sub-process turns the evaluation into improvements of the governance capability itself. It simplifies policies that are not followed because they are unclear or too heavy. It automates checks that are now done by hand. It strengthens areas where incidents or findings recur. It prioritizes the improvements by impact

and feasibility and tracks them through the following cycles. This is SAF capability 4.1.4 *Improvement management* applied to governance.

Activity	Key outputs	SAF process	Other reference
Identify improvements: simplification, automation, strengthening	Improvement plan	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Prioritized improvement list	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	4.1.4 Improvement management

5. Actors and roles

VS-07 defines the data governance roles for the whole organization, and it relies on specialists in law, quality, and audit. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the chief governance officer is ROL-05 Chief data officer, and legal counsel is ROL-19 Compliance officer. The data governance roles are organized in the collaboration COL-06 *Data governance board*.

Actor or role	ID	Responsibility in VS-07 Governance and Compliance
National Statistical Institute	ACT-01	Its management is accountable for the governance framework and decides on escalated issues
Supervisory and legal authority	ACT-08	Sets the legal requirements and supervises compliance, for example the data protection authority
International statistical organization	ACT-07	Sets common standards and carries out peer reviews, for example under the European Statistics Code of Practice
Process owners	ROL-01	Accountable, in their own value streams, for working within the controls of VS-07 Governance and Compliance and supplying evidence
Data owner, data steward, data custodian	ROL-02, ROL-03, ROL-04	The data governance roles that VS-07 Governance and Compliance defines and assigns across domains
Chief data officer	ROL-05	Accountable for the governance framework and the data policy of the organization
Quality manager	ROL-18	Defines quality indicators and monitors quality and compliance status
Compliance officer	ROL-19	Ensures compliance with the Code of Practice, the GDPR, and the statistics law; interprets regulation

Actor or role	ID	Responsibility in VS-07 Governance and Compliance
Security officer	ROL-20	Defines and monitors information security and continuity, together with VS-09 Infrastructure and Platform Enablement
Auditor	ROL-21	Conducts internal audits and supports external audits and certifications

6. Capabilities, processes, and functions

Capability foundations

Capability group	Capabilities	Role in VS-07 Governance and Compliance
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management	Govern data, metadata, and quality across the organization
3.1 Business operations	3.1.7 Legal support, 3.1.8 Information management and archive management	Interpret regulation and govern retention and archiving
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.3 Security incident management	Set the privacy and security policy and handle incidents
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability, 4.1.6 Collaboration	Govern the organization, issue policies, improve governance, account to society

Stages, processes, and functions

The table follows the mapping in *Business Processes, Mapping to value stream stages.*

Stage	Realized by processes	Main functions
VS-07.1 Define Governance Frameworks and Policies	DLM-1 Plan and design data assets	FUN-1.2 Govern and steer the organization; FUN-5.1 Manage the quality framework; FUN-7.1 Govern data
VS-07.2 Establish Legal and Regulatory Compliance Mechanisms	DLM-4 Classify and protect data	FUN-10.5 Manage legal affairs; FUN-10.8 Manage information security and privacy
VS-07.3 Implement Data and Metadata Governance	OVA-2 Metadata Management; OVA-3 Data Management; DLM-1 Plan and design data assets; DLM-6 Provide data for reuse	FUN-7.1 Govern data; FUN-7.2 Manage metadata
VS-07.4 Monitor Quality and Compliance Indicators	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use; MDS-8 Monitor use and compliance; DSP-5 Monitor compliance and report	FUN-5.2 Assure and review quality; FUN-7.1 Govern data

Stage	Realized by processes	Main functions
VS-07.5 Audit, Certify, and Report Compliance	OVA-1 Quality Management; PRD-8.2 Conduct evaluation	FUN-1.2 Govern and steer the organization; FUN-5.2 Assure and review quality
VS-07.6 Update and Improve Governance Mechanisms	PRD-8.3 Agree an action plan; DLM-10 Archive and preserve data; DLM-11 Dispose of data	FUN-2.1 Plan capability improvements; FUN-7.1 Govern data

PLACEHOLDER: ArchiMate view VS-07 (b) stages realized by processes and functions

Framework mappings

Where the primary value streams map to the production phases of GSBPM, VS-07 maps to the over-arching activities of GAMSO.

GAMSO area	Relevance to VS-07 Governance and Compliance	SAF capabilities
Strategy and Leadership	Set the governance strategy, charters, and accountability; manage organizational and ethical frameworks	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.5 Accountability
Corporate Support	Manage the legal framework, quality management, information security governance, and archives	2.1.3 Statistical quality management, 3.1.7 Legal support, 3.1.8 Information management and archive management, 3.3.2 Security planning
Capability Management	Govern the life cycle of standards, methods, and data assets	2.1.1 Data lifecycle management, 4.1.4 Improvement management

7. Principles

VS-07 enforces the data principles across the whole chain and foregrounds several of the broader architecture principles.

Principle	Statement	How VS-07 Governance and Compliance applies or enforces it
SAF-P03	Active life cycle management	Governs the life cycle of policies, standards, classifications, and data assets, including archiving and disposal
SAF-P04	Maximize the benefits for the organization	Aligns controls with value and accountability for the organization as a whole, and avoids governance for its own sake
SAF-P07	Implement loosely coupled processes and systems	Defines the steady-state interface and event conventions under which the value streams are governed
SAF-P09	No data without metadata and classification	Governs the metadata and classification obligations and audits them across every value stream

Principle	Statement	How VS-07 Governance and Compliance applies or enforces it
SAF-P10	Security by design	Sets and assures the framework for information security and access control
SAF-P11	Privacy by design	Sets the policy for privacy, confidentiality, and disclosure control, and verifies compliance

8. Interfaces, services, and events

Business services

Service	Offered to	Realized in	Interfaces
Policy and framework service	All value streams	VS-07.1 Define Governance Frameworks and Policies	Policy register, published policies
Compliance advice service	Process owners, VS-08 Innovation and Methodological Development	VS-07.2 Establish Legal and Regulatory Compliance Mechanisms	Compliance register, advice on new methods and sources
Data governance service	Data owners, stewards, and custodians	VS-07.3 Implement Data and Metadata Governance	Role assignments, data governance board
Quality and compliance monitoring service	NSI management, process owners	VS-07.4 Monitor Quality and Compliance Indicators	Scorecards and dashboards
Audit and assurance service	NSI management, regulators, the public	VS-07.5 Audit, Certify, and Report Compliance	Audit reports, assurance statements, transparency reports

Business events

Event	Triggers or marks
Regulatory change	Start of VS-07.2 Establish Legal and Regulatory Compliance Mechanisms and VS-07.6 Update and Improve Governance Mechanisms
Policy update issued	Notification to the value streams of a new policy version
Compliance review	Periodic review of the compliance evidence of a value stream
Audit due / Audit completed	Start and end of an audit in VS-07.5 Audit, Certify, and Report Compliance
Method compliance assessment requested	Request from VS-08 Innovation and Methodological Development before a new method goes into production
Incident reported	Start of incident handling and of a review of the controls concerned
Certification granted	Outcome of an external audit or peer review

PLACEHOLDER: VS-07 event choreography between regulators, VS-07, the value streams, and auditors: regulatory change, policy update issued, compliance evidence, audit, findings, and certification

9. Key performance indicators

VS-07 monitors itself with the same discipline it imposes on the chain.

KPI	Description	Measurement
Policy coverage	Share of value streams and data domains working under current, approved policies	Per policy domain, annually
Audit findings closed on time	Share of audit findings resolved within the agreed deadline	Per audit cycle
Compliance incidents	Number of confidentiality, security, or legal incidents	Per quarter (target: zero)
Certification status	Certifications and peer review assessments held and current, for example under the Code of Practice	Annually
Metadata governance compliance	Share of registered datasets that meet the obligations of SAF-P09 No data without metadata and classification	Per cycle, from the compliance evidence of the value streams
Regulatory response time	Time from detecting a regulatory change to issuing an updated policy	Per change

10. Toolkit

Examples from NSI practice and a maturity checklist for VS-07 are part of the SAF Toolkit.

Toolkit: SAF-TK-V07 VS-07 Governance and Compliance companion [**VC17**]

VS-08 Innovation and Methodological Development

1. Purpose and value created

VS-08 makes sure that the NSI does not stand still. It discovers, tests, and introduces new ways of working: new data sources, methods, tools, and technologies. It connects research, methodology, IT, and business users. It keeps innovation controlled, so that only methods that are sound, documented, and compliant reach production. VS-08 does not produce statistics. It improves the methods and tools that the production value streams use, and it builds the skills needed to apply them. The value created is **continuous improvement**: the NSI stays ready for new data, such as sensor data, administrative data, and data spaces, and for new techniques, such as machine learning and small-area estimation.

VS-08 is a supporting value stream. It aligns with the GAMSO area *Capability Management* rather than with the production phases of GSBPM. This section describes its business layer for all four process types.

- **Methodologist, researcher, or data scientist:** the stages in attribute 3, which describe the innovation pipeline.
- **Head of methodology or innovation:** the Design processes in attribute 4.
- **Process owners of the production value streams:** attributes 2 and 8, which describe how needs enter VS-08 and how validated methods return.
- **Manager who wants the big picture:** attributes 1, 2, and 9. Examples and scenarios are part of the SAF Toolkit.

> **i** How a supporting value stream differs from a primary one
>
> VS-08 has no steady-state data transition. It does not move data from one state to the next; it improves the methods and tools that the production value streams use. It runs continuously as a pipeline of research and development, not per production cycle. Its output is an enabling capability rather than a data artifact: validated methods, documented tools, reference implementations, and

trained staff. Where it uses production data for research, it does so under specific authorization and within the rules of VS-07.

PLACEHOLDER: ArchiMate view VS-08 (a) value stream and capabilities

2. Position and interaction with the chain

VS-08 is not a stage in the production line. It runs alongside the chain as a pipeline that feeds improvements into every primary value stream and draws its agenda from their needs. The improvement recommendations of the primary value streams, produced in their *Continuous Improvement* sub-processes, together with the user needs from VS-10, form the innovation backlog of VS-08. After research and evaluation, the transfer gate in VS-08.5 moves a validated method or tool into the production value stream concerned. That transfer requires the approval of VS-07 and runs on the platforms of VS-09.

Counterpart	VS-08 provides	VS-08 receives
VS-01 to VS-06 (primary value streams)	Validated methods and tools transferred to production	Improvement recommendations, problems with current methods, validation of fitness for purpose
VS-07 Governance and Compliance	Methods submitted for approval before production	Compliance criteria and approval decisions
VS-09 Infrastructure and Platform Enablement	Requirements for research and production platforms	Research environments (sandboxes) and production runtime
VS-10 Customer and Stakeholder Engagement	New possibilities for products and services	User needs that require new methods or sources
External parties: universities, other NSIs, international bodies	Shared methods and open-source contributions	Research results, shared methods, and tools

Occasional access to data in the steady states is part of this interaction. Researchers in VS-08 may explore Standardized Data (#2) or Processed Data (#3) to test new methods, under specific authorization (see *State #2: Standardized Data* and *State #3: Processed Data*).

PLACEHOLDER: ArchiMate view VS-08 (c) interaction with the chain: improvement needs from VS-01 to VS-06 and user needs from VS-10 entering the innovation backlog; the pipeline of research, evaluation, and development; the transfer gate to production with approval of VS-07 and platforms of VS-09

3. Stages

The stages of VS-08 form a continuous innovation pipeline: identify, research, evaluate, develop, integrate, and share. Ideas enter at the start, and many of them are stopped along the way. That is intended: a healthy pipeline stops attractive but unsound ideas before they reach production.

Stage	Name	Value added
VS-08.1	Identify Innovation Needs and Opportunities	It is clear which innovations are worth pursuing
VS-08.2	Conduct Research and Proof-of-Concepts	A new method or technology has been tested in a limited scope
VS-08.3	Evaluate Methodological Soundness and Feasibility	Only sound and feasible innovations go forward
VS-08.4	Develop and Document New Methodologies or Tools	A validated method is reproducible and ready to hand over

Stage	Name	Value added
VS-08.5	Integrate Innovation into Production Environments	The method or tool works as part of a production value stream
VS-08.6	Promote Knowledge Sharing and Capability Building	The organization can adopt and sustain the new way of working

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. Research has no reference process in the SAF; it is carried out through innovation projects, which the column shows as *NSI-specific*. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-08.1 Identify Innovation Needs and Opportunities

Value created: *A prioritized backlog of innovations that are worth pursuing.*

The pipeline begins by collecting needs and opportunities. The improvement recommendations of the production value streams and the user needs from VS-10 enter the backlog. The team scans technological, methodological, and policy developments, such as artificial intelligence, new data sources, and cloud services. It checks whether a source or method that already exists elsewhere, in the NSI or at another NSI, can meet the need. Each candidate is assessed for value to the organization, feasibility, and risk. The result is a prioritized innovation backlog, aligned with the strategy of the NSI.

Activity	Key outputs	SAF process	Other reference
Collect improvement needs from the value streams and user needs from VS-10 Customer and Stakeholder Engagement	Raw backlog	NSI-specific	4.1.4 Improvement management
Scan technological, methodological, and policy developments	Horizon scan	NSI-specific	2.2.2 Product innovation
Check whether existing sources or methods can meet the need	Availability check	PRD-1.5 Check data availability and suitability	2.2.1 Data source exploration; SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Assess and prioritize candidates by value, feasibility, and risk	Prioritized innovation backlog	NSI-specific	SAF-P04 Maximize the benefits for the organization

VS-08.2 Conduct Research and Proof-of-Concepts

Value created: *A new method or technology tested in a limited, controlled scope before any commitment.*

Candidates from the backlog enter research. The team defines a research question, a hypothesis, and criteria for success before it starts. It runs controlled experiments and pilots in a research environment provided by VS-09. Where production data is needed, access is requested under specific authorization, and the data is used only for the stated purpose. Results are recorded, including negative results, so that the organization learns from what did not work. A proof-of-concept ends with a clear recommendation: stop, adjust, or proceed to evaluation.

Activity	Key outputs	SAF process	Other reference
Define the research question, hypothesis, and success criteria	Research plan	NSI-specific	2.2.3 Methodology
Run experiments and pilots in a research environment	Experiment and pilot results	NSI-specific	2.2.1 Data source exploration
Request authorized access to production data where needed	Access authorization	DLM-6 Provide data for reuse	SAF-P11 Privacy by design
Record results, including negative results, and recommend a next step	Proof-of-concept report	NSI-specific	

VS-08.3 Evaluate Methodological Soundness and Feasibility

Value created: *Confidence that only sound, feasible, and compliant innovations go forward.*

Promising results are evaluated before any further investment. The evaluation assesses accuracy, bias, robustness, and scalability. It checks alignment with the quality framework and with ethical rules, for example on transparency and explainability of machine-learning methods. It compares the new method with the current method on real or realistic data. Subject-matter experts from the production value stream concerned confirm fitness for purpose. VS-07 assesses the method against its compliance criteria. This gate stops ideas that are attractive but not sound, with a documented rationale.

Activity	Key outputs	SAF process	Other reference
Assess accuracy, bias, robustness, and scalability	Soundness assessment	PRD-3.6 Test statistical business process	2.2.3 Methodology
Compare the new method with the current method on realistic data	Comparison report	PRD-3.6 Test statistical business process	GAMSO Quality Management
Confirm fitness for purpose with the production value stream	Fitness-for-purpose confirmation	PRD-3.6 Test statistical business process	SAF-P05 IT and technology is everyone's business
Obtain the compliance assessment of VS-07 Governance and Compliance	Compliance assessment	NSI-specific	SAF-P11 Privacy by design

VS-08.4 Develop and Document New Methodologies or Tools

Value created: *A validated method that is reproducible and ready to hand over, rather than living in a researcher's notebook.*

What passes the gate is developed into a standard method, algorithm, or reference implementation. The method is documented: its purpose, assumptions, parameters, limitations, and the evidence from the evaluation. The reference implementation is written so that it can be maintained, tested, and run on the production platforms of VS-09. Where possible, it is developed or shared as open source with other NSIs. The method is added to the knowledge base, so that others can find and apply it.

Activity	Key outputs	SAF process	Other reference
Develop the standard method, algorithm, or reference implementation	Reference implementation	PRD-3.2 Reuse or build processing and analysis components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Document purpose, assumptions, parameters, limitations, and evidence	Method documentation	OVA-5 Knowledge Management	SAF-P09 No data without metadata and classification
Prepare the implementation to run on production platforms	Production-ready component	PRD-3.2 Reuse or build processing and analysis components	SAF-P07 Implement loosely coupled processes and systems
Add the method to the knowledge base	Knowledge base entry	OVA-5 Knowledge Management	2.2.3 Methodology

VS-08.5 Integrate Innovation into Production Environments

Value created: *A validated method or tool working as part of a production value stream.*

This is the transfer gate, where innovation becomes part of everyday production. The method or tool is handed over to the production value stream concerned. That value stream takes it up through its own Design and Implementation process types. Approval from VS-07 is confirmed before the transfer. The component is deployed on the platforms of VS-09 and tested in the production context. Ownership is transferred formally, so that the production value stream maintains the method from then on. The transfer is recorded, together with the method version and the date of entry into production.

Activity	Key outputs	SAF process	Other reference
Hand over the method to the Design and Implementation of the production value stream	Handover package	PRD-3.7 Finalize production systems	4.1.3 Change management
Confirm the approval of VS-07 Governance and Compliance	Approval record	NSI-specific	4.1.5 Accountability
Deploy and test on the platforms of VS-09 Infrastructure and Platform Enablement	Production deployment	PRD-3.7 Finalize production systems	3.2.3 Technical management
Transfer ownership formally and record the transfer	Transfer record, *Method transferred to production*	PRD-3.7 Finalize production systems	SAF-P03 Active life cycle management

VS-08.6 Promote Knowledge Sharing and Capability Building

Value created: *An organization that can adopt and sustain the new way of working.*

The pipeline ends by spreading what was learned. Results, best practices, and guidelines are shared inside the NSI and with the international community. Staff in the production value streams are trained in the new method. Communities of practice keep knowledge alive after the project ends. Superseded methods are retired formally. The experience gained surfaces the next round of needs, which returns to VS-08.1.

Activity	Key outputs	SAF process	Other reference
Share results, best practices, and guidelines inside and outside the NSI	Published guidelines	OVA-5 Knowledge Management	4.1.6 Collaboration
Train staff in the production value streams	Trained staff	OVA-5 Knowledge Management	3.1.1 Personnel management (HR)
Support communities of practice	Active communities of practice	OVA-5 Knowledge Management	SAF-P05 IT and technology is everyone's business
Retire superseded methods	Retirement record	OVA-5 Knowledge Management	SAF-P03 Active life cycle management

4. The four process types

VS-08 is itself a capability that must be designed, built, operated, and improved. The four process types therefore apply to VS-08 as they do to the primary value streams, but they govern the innovation capability rather than a data pipeline. Design produces the innovation strategy and the evaluation criteria, Implementation builds the research environment and the pipeline procedures, Execution runs the stages of attribute 3, and Management evaluates whether innovation delivers. **[VC26]**

Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. Where no reference process exists, the sub-process is marked as NSI-specific.

Design: provides the innovation framework

Design sub-process	Refines	Main outputs
Design the Innovation Strategy and Portfolio	NSI-specific	Innovation strategy, portfolio themes, budget rules
Design the Evaluation and Transfer Criteria	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Evaluation criteria, gate criteria, transfer protocol

Design the Innovation Strategy and Portfolio

Refines: NSI-specific strategic planning processes.

This sub-process sets the direction of innovation. It derives innovation themes from the strategy of the NSI, such as the use of new data sources or the automation of coding. It decides how the innovation portfolio is balanced between quick improvements and long-term research. It sets the rules for funding and prioritizing projects. It defines how the NSI cooperates with universities, other NSIs, and international bodies. The strategy keeps innovation focused on value for the organization, in line with SAF-P04, rather than on novelty.

Activity	Key outputs	SAF process	Other reference
Derive innovation themes from the strategy of the NSI	Innovation strategy	NSI-specific	4.1.1 Strategy and governance
Define the balance of the portfolio and the rules for funding and priority	Portfolio rules	NSI-specific	SAF-P04 Maximize the benefits for the organization
Define cooperation with universities, other NSIs, and international bodies	Cooperation framework	NSI-specific	4.1.6 Collaboration

Design the Evaluation and Transfer Criteria

Refines: PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

This sub-process designs the gates of the pipeline. It sets the criteria for soundness: accuracy, bias, robustness, scalability, and explainability. It sets the criteria that a method must meet before transfer, together with VS-07. It designs the transfer protocol: what must be handed over, to whom, and with which documentation. It defines when a method is retired. Clear criteria make gate decisions predictable and defensible.

Activity	Key outputs	SAF process	Other reference
Define criteria for methodological soundness	Evaluation criteria	PRD-2.5 Design processing and analysis	2.2.3 Methodology
Define the criteria for transfer, together with VS-07 Governance and Compliance	Gate criteria	OVA-1 Quality Management	4.1.5 Accountability
Design the transfer protocol and the rules for retirement	Transfer protocol	NSI-specific	SAF-P03 Active life cycle management

Implementation: provides the innovation environment

Implementation sub-process	Refines	Main outputs
Set Up Research Environments and Procedures	PRD-3.4 Configure workflows; DLM-6 Provide data for reuse	Research environment, access procedure for research data, pipeline procedures
Build the Knowledge Base	OVA-5 Knowledge Management	Knowledge base, method catalog, training material

Set Up Research Environments and Procedures

Refines: PRD-3.4 Configure workflows; DLM-6 Provide data for reuse.

This sub-process builds the environment in which research takes place, together with VS-09. It provides research environments, or sandboxes, with the tools that researchers need. It defines the procedure for authorized access to production data for research, with VS-07. It sets up the procedures for the pipeline: how ideas are submitted, how gate decisions are taken, and how results are recorded. It makes sure that experiments can be reproduced. A good environment lets researchers move fast without putting production or confidentiality at risk.

Activity	Key outputs	SAF process	Other reference
Provide research environments with VS-09 Infrastructure and Platform Enablement	Research environment	PRD-3.4 Configure workflows	3.2.5 IT infrastructure
Define the procedure for authorized access to production data	Access procedure for research data	DLM-6 Provide data for reuse	SAF-P11 Privacy by design
Set up the pipeline procedures: submission, gate decisions, recording	Pipeline procedures	NSI-specific	4.1.4 Improvement management

Build the Knowledge Base

Refines: OVA-5 Knowledge Management.

This sub-process builds the place where methods and knowledge are kept. It sets up the method catalog with the documentation of validated methods and their versions. It links each method to the production value streams that use it. It develops training material and learning paths for new methods. It provides the channels for communities of practice. The knowledge base turns individual expertise into an asset of the organization.

Activity	Key outputs	SAF process	Other reference
Set up the method catalog with documentation and versions	Method catalog	OVA-5 Knowledge Management	2.2.3 Methodology
Link methods to the value streams that use them	Method usage map	OVA-5 Knowledge Management	SAF-P09 No data without metadata and classification
Develop training material and support for communities of practice	Training material, community channels	OVA-5 Knowledge Management	3.1.1 Personnel management (HR)

Execution: runs the innovation pipeline

Execution runs the stages of attribute 3 continuously, as a pipeline of projects. VS-08.1 runs as a periodic planning cycle; VS-08.2 to VS-08.5 run per project; VS-08.6 runs continuously.

Aspect	VS-08
Trigger	*Innovation need raised* by a value stream or VS-10 Customer and Stakeholder Engagement; a planning cycle; a new opportunity from the horizon scan
Cadence	Per project; periodic review of the backlog
Performed by	ROL-07 Methodologist, ROL-15 Subject-matter statistician (researcher, data scientist), ROL-10 Data engineer
Input	Improvement needs, user needs, research results from outside, authorized research data
Output	Validated methods and tools, documentation, transfer records, trained staff
Control point	Gate in VS-08.3 Evaluate Methodological Soundness and Feasibility; transfer gate in VS-08.5 Integrate Innovation into Production Environments
Exception path	An unsound method stopped with a documented rationale; a compliance objection returned to research or stopped

Management: provides innovation performance reports

Management sub-process	Refines	Main outputs
Monitor the Innovation Pipeline	PRD-8.1 Gather evaluation inputs	Pipeline dashboard, lead times, gate statistics
Evaluate and Improve the Innovation Capability	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Evaluation of transfers, improvement plan

Monitor the Innovation Pipeline

Refines: PRD-8.1 Gather evaluation inputs.

This sub-process tracks how the pipeline performs. It measures the key performance indicators of VS-08 (see attribute 9), such as the time from idea to production and the share of proofs-of-concept stopped at the gate. It tracks the status of each item in the backlog. It records the use of research data and the compliance of research projects. It reports the progress of the portfolio to management. Visibility of the pipeline helps the value streams see when their needs will be addressed.

Activity	Key outputs	SAF process	Other reference
Measure the key performance indicators of VS-08 Innovation and Methodological Development	Pipeline dashboard	PRD-8.1 Gather evaluation inputs	GAMSO Capability Management
Track the status of backlog items	Backlog status report	PRD-8.1 Gather evaluation inputs	
Record the use of research data and the compliance of research projects	Research compliance record	DLM-8 Monitor data quality, lineage and use	SAF-P11 Privacy by design

Evaluate and Improve the Innovation Capability

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This sub-process asks whether innovation delivers lasting value. It evaluates transferred methods after a period in production: are they still in use, and did they deliver the expected benefits? It holds retrospectives on stopped and completed projects. It adjusts the strategy, the evaluation criteria, and the procedures based on the findings. It prioritizes improvements of the innovation capability and tracks them. This closes the cycle of VS-08 itself, in line with SAF capability 4.1.4 *Improvement management*.

Activity	Key outputs	SAF process	Other reference
Evaluate transferred methods after a period in production	Transfer evaluation	PRD-8.2 Conduct evaluation	GAMSO Capability Management
Hold retrospectives on stopped and completed projects	Lessons learned	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Adjust strategy, criteria, and procedures, and track the improvements	Improvement plan	PRD-8.3 Agree an action plan	4.1.4 Improvement management

5. Actors and roles

VS-08 connects researchers, methodologists, IT staff, and the production value streams. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the head of methodology or innovation is ROL-07 Methodologist, the researcher or data scientist is ROL-15 Subject-matter statistician, and the tool or platform engineer is ROL-10 Data engineer. The trainer or knowledge manager is not part of the reference set. **[VC28]**

Actor or role	ID	Responsibility in VS-08 Innovation and Methodological Development
International statistical organization	ACT-07	Shares methods, standards, and research results, for example through UNECE projects

Actor or role	ID	Responsibility in VS-08 Innovation and Methodological Development
Supplier and service provider	ACT-09	Supplies research partnerships, tools, or data for innovation, for example universities
Process owners of production value streams	ROL-01	Raise needs, confirm fitness for purpose, and take over transferred methods
Methodologist	ROL-07	Owns the innovation agenda and the methodological standards; evaluates soundness and formalizes validated methods
Data engineer	ROL-10	Builds reference implementations and prepares them for production, together with VS-09 Infrastructure and Platform Enablement
Subject-matter statistician	ROL-15	Carries out research and proofs-of-concept; brings the real problem from production and validates fitness for purpose
Compliance officer	ROL-19	Assesses methods against the compliance criteria of VS-07 Governance and Compliance before transfer

6. Capabilities, processes, and functions

Capability foundations

Capability group	Capabilities	Role in VS-08 Innovation and Methodological Development
2.2 Research	2.2.1 Data source exploration, 2.2.2 Product innovation, 2.2.3 Methodology	Explore new sources, develop new products and methods
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Bring validated methods into the design and build of products
3.2 Computerization and automation	3.2.1 Application development, 3.2.3 Technical management	Build reference implementations and deploy them
4.1 Strategy and leadership	4.1.3 Change management, 4.1.4 Improvement management, 4.1.6 Collaboration	Manage the change that innovation brings, improve continuously, cooperate with partners

Stages, processes, and functions

The table follows the mapping in *Business Processes, Mapping to value stream stages.*

Stage	Realized by processes	Main functions
VS-08.1 Identify Innovation Needs and Opportunities	PRD-1.5 Check data availability and suitability	FUN-4.2 Research and innovate
VS-08.2 Conduct Research and Proof-of-Concepts	No reference process (NSI-specific)	FUN-4.2 Research and innovate

Stage	Realized by processes	Main functions
VS-08.3 Evaluate Methodological Soundness and Feasibility	PRD-3.6 Test statistical business process	FUN-4.1 Manage statistical and analytical methodology; FUN-5.2 Assure and review quality
VS-08.4 Develop and Document New Methodologies or Tools	PRD-3.2 Reuse or build processing and analysis components; OVA-5 Knowledge Management	FUN-4.1 Manage statistical and analytical methodology; FUN-10.4 Manage information and records
VS-08.5 Integrate Innovation into Production Environments	PRD-3.7 Finalize production systems	FUN-2.4 Transfer capability improvements to operations; FUN-3.1 Design and build statistical products
VS-08.6 Promote Knowledge Sharing and Capability Building	OVA-5 Knowledge Management	FUN-10.1 Manage human resources; FUN-10.4 Manage information and records

PLACEHOLDER: ArchiMate view VS-08 (b) stages realized by processes and functions

Framework mappings

Where the primary value streams map to the production phases of GSBPM, VS-08 maps to GAMSO *Capability Management*.

GAMSO area	Relevance to VS-08 Innovation and Methodological Development	SAF capabilities
Capability Management	Manage research and innovation, develop methodology, manage change, build staff capability	2.2.1 Data source exploration, 2.2.2 Product innovation, 2.2.3 Methodology, 4.1.3 Change management
Strategy and Leadership	Set the innovation agenda in line with the strategy of the organization	4.1.1 Strategy and governance
Corporate Support	Train staff and manage knowledge	3.1.1 Personnel management (HR), 3.1.8 Information management and archive management

7. Principles

Principle	Statement	How VS-08 Innovation and Methodological Development applies it
SAF-P01	Reuse before open source, open source before buying, buying before building	Innovates on and contributes to shared, open methods and tools rather than closed products from vendors [**VC3**]
SAF-P03	Active life cycle management	Manages methods and tools across their life cycle, from proof-of-concept to production to retirement
SAF-P04	Maximize the benefits for the organization	Prioritizes innovations by value for the whole organization and by scalability, not by novelty

Principle	Statement	How VS-08 Innovation and Methodological Development applies it
SAF-P05	IT and technology is everyone's business	Connects research, methodology, IT, and business users, and builds broad capability to adopt new methods
SAF-P07	Implement loosely coupled processes and systems	Transfers methods that fit the steady-state interfaces, so that the production value streams stay decoupled

8. Interfaces, services, and events

Business services

Service	Offered to	Realized in	Interfaces
Innovation intake service	All value streams, VS-10 Customer and Stakeholder Engagement	VS-08.1 Identify Innovation Needs and Opportunities	Innovation backlog, submission procedure
Research and experimentation service	Production value streams	VS-08.2 Conduct Research and Proof-of-Concepts	Research environments, proof-of-concept reports
Methodology advice and evaluation service	Production value streams, VS-07 Governance and Compliance	VS-08.3 Evaluate Methodological Soundness and Feasibility	Soundness assessments, comparison reports
Method transfer service	Production value streams	VS-08.5 Integrate Innovation into Production Environments	Transfer protocol, handover packages
Knowledge and training service	All staff	VS-08.6 Promote Knowledge Sharing and Capability Building	Method catalog, training, communities of practice

Business events

Event	Triggers or marks
Innovation need raised	Entry of an item into the backlog in VS-08.1 Identify Innovation Needs and Opportunities
Proof-of-concept started / *Proof-of-concept completed*	Start and end of VS-08.2 Conduct Research and Proof-of-Concepts
Method compliance assessment requested	Request to VS-07 Governance and Compliance during VS-08.3 Evaluate Methodological Soundness and Feasibility
Gate passed / *Gate stopped*	Outcome of VS-08.3 Evaluate Methodological Soundness and Feasibility
Platform request	Request to VS-09 Infrastructure and Platform Enablement for a research or production environment
Method transferred to production	Completion of VS-08.5 Integrate Innovation into Production Environments
Method retired	Formal retirement of a superseded method in VS-08.6 Promote Knowledge Sharing and Capability Building

PLACEHOLDER: VS-08 event choreography between the production value streams, VS-08, VS-07, and VS-09: need raised, proof-of-concept, compliance assessment, gate decision, platform request, transfer to production, and retirement

9. Key performance indicators

KPI	Description	Measurement
Idea-to-production lead time	Median time from a backlog item in VS-08.1 Identify Innovation Needs and Opportunities to a completed transfer in VS-08.5 Integrate Innovation into Production Environments	Per transferred innovation
Transfer rate and durability	Methods and tools transferred per year, and the share still in use after 12 months	Annually
Gate effectiveness	Share of proofs-of-concept stopped at the gate of VS-08.3 Evaluate Methodological Soundness and Feasibility with a documented rationale; a healthy gate stops some	Per evaluation
Backlog responsiveness	Share of prioritized backlog items that reach a proof-of-concept within the agreed time	Per planning period
Capability building	Staff trained in new methods, and the rate of adoption in the target value streams	Per year, per transfer

10. Toolkit

Examples from NSI practice and a maturity checklist for VS-08 are part of the SAF Toolkit.

Toolkit: SAF-TK-V08 VS-08 Innovation and Methodological Development companion [**VC17**]

VS-09 Infrastructure and Platform Enablement

1. Purpose and value created

VS-09 provides the technical and digital foundation for every part of the NSI. Where the other value streams focus on data, methods, or governance, VS-09 makes sure that all of these activities can actually run: safely, efficiently, and at scale. It delivers the IT environments, data platforms, tools, and integration services that statistical production needs, from collection to dissemination. It keeps systems secure, resilient, and compliant, while staying flexible enough to adopt new technologies. VS-09 describes the business view of this enablement: the capability of providing platforms and services. The detailed application and technology architecture is the subject of the application and technology layers of the SAF. The value created is **reliable enablement**: infrastructure becomes an active enabler of the statistical mission rather than a background function.

VS-09 is a supporting value stream. It aligns with the GAMSO areas *Capability Management* and *Corporate Support* rather than with the production phases of GSBPM. This section describes its business layer for all four process types.

- **Platform engineer or operations staff:** the stages in attribute 3, which describe the enablement life cycle.
- **Enterprise or platform architect:** the Design processes in attribute 4.

- **Process owners of other value streams:** attributes 2 and 8, which describe what VS-09 provides and what it needs from them.
- **Manager who wants the big picture:** attributes 1, 2, and 9. Examples and scenarios are part of the SAF Toolkit.

> **i** How a supporting value stream differs from a primary one
>
> VS-09 has no steady-state data transition. It does not move data along the chain; it provides the environments in which the chain runs. It runs continuously as an always-on service, not per production cycle. Its output is an enabling capability rather than a data artifact: platforms, environments, integration services, and assured availability. The steady states and their quality gates are business agreements; VS-09 provides the storage, catalog, and event interfaces that implement them.

PLACEHOLDER: ArchiMate view VS-09 (a) value stream and capabilities

2. Position and interaction with the chain

VS-09 is not a stage in the production line. It is the foundation underneath it: every primary and supporting value stream runs on the platforms that VS-09 provides. VS-09 receives requirements that follow from business capabilities, data volumes, and analytical complexity. It receives specific requests from VS-08 for new technology and from VS-06 for dissemination platforms. It provides platforms, integration services, and operational assurance. Its security and continuity are governed by VS-07.

Counterpart	VS-09 provides	VS-09 receives
VS-01 to VS-05 (primary value streams)	Environments for collection, processing, and analysis; steady-state repositories; catalog; event interfaces	Requirements, capacity demand, incidents
VS-06 Dissemination and Service Delivery	Portals, APIs, dashboards, secure research environments, data space connectors	Requirements for channels and scaling, especially at release moments
VS-07 Governance and Compliance	Evidence on security posture and continuity	Security and continuity policy
VS-08 Innovation and Methodological Development	Research environments and production runtime for new technology	Requests to adopt new technology
VS-10 Customer and Stakeholder Engagement	Platforms for user services and communication	Requirements from users on availability and self-service
External parties: suppliers and service providers	Requirements and service agreements	Infrastructure, cloud services, software, and support

PLACEHOLDER: ArchiMate view VS-09 (c) interaction with the chain: VS-09 as the foundation underneath all value streams, providing environments, steady-state repositories, catalog, event interfaces, and dissemination platforms, and receiving requirements, capacity demand, and incidents; security and continuity governed by VS-07

3. Stages

The stages of VS-09 form a continuous enablement life cycle: define, design, provision, integrate, assure, and evolve. Requirements lead to architecture, architecture to platforms, platforms to services that are monitored and kept safe, and experience to modernization.

Stage	Name	Value added
VS-09.1	Define Digital and Technical Requirements	It is clear which technical capabilities the business needs
VS-09.2	Design Platform and Infrastructure Architecture	There is a coherent blueprint for storage, processing, integration, and access
VS-09.3	Provision and Configure Platforms	Controlled environments are available to the value streams
VS-09.4	Enable Integration and Orchestration Services	Processes and value streams are connected through stable interfaces
VS-09.5	Ensure Performance, Security, and Continuity	The chain runs reliably, safely, and can recover from disruption
VS-09.6	Evolve and Modernize the Technical Ecosystem	The foundation stays current and ready for new data and methods

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. Some activities have no reference process in the SAF; they are carried out through NSI-specific IT management processes, which the column shows as *NSI-specific*. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-09.1 Define Digital and Technical Requirements

Value created: *A clear view of the technical capabilities that the business needs.*

The life cycle begins with the requirements. VS-09 translates business capabilities, data volumes, and analytical complexity into technical requirements. It collects the requirements of the value streams for storage, processing, integration, access, and availability. It includes the requirements that follow from the steady states: immutable versions, persistent identifiers, field-level lineage, and the handover events. It adds the requirements of VS-07 for security, privacy, and continuity. Each requirement is linked to the business need it serves, so that platform choices can be explained in business terms.

Activity	Key outputs	SAF process	Other reference
Translate business capabilities and data volumes into technical requirements	Technical requirements	PRD-2.6 Design production systems and workflows	4.1.7 Architecture management
Collect the requirements of the value streams for storage, processing, integration, access, and availability	Requirements inventory	PRD-2.6 Design production systems and workflows	SAF-P05 IT and technology is everyone's business
Include the requirements of the steady states and of VS-07 Governance and Compliance	Requirements for steady states, security, and continuity	PRD-2.6 Design production systems and workflows	SAF-P07 Implement loosely coupled processes and systems; SAF-P10 Security by design
Link each requirement to the business need it serves	Traceability of requirements	NSI-specific	SAF-P04 Maximize the benefits for the organization

VS-09.2 Design Platform and Infrastructure Architecture

Value created: *A coherent blueprint for storage, processing, integration, and secure access.*

The requirements feed the design of the platform and infrastructure architecture. The blueprint covers data storage, processing, integration, and secure access, for example a data fabric, a data marketplace, and compute platforms. It decides which deployment model fits each component: on-premises, private cloud, public cloud, or a hybrid. It defines how the steady states are implemented on the platforms, so that each state has its repository, catalog entry, and access control. Shared platforms are preferred over solutions per product, and reuse comes before own development. The architecture is reviewed by the architecture board before it is realized.

Activity	Key outputs	SAF process	Other reference
Design the blueprint for storage, processing, integration, and access	Platform architecture	PRD-2.6 Design production systems and workflows	4.1.7 Architecture management
Decide the deployment model per component: on-premises, private, public, or hybrid cloud	Deployment decisions	PRD-2.6 Design production systems and workflows	3.2.5 IT infrastructure
Define how the steady states are implemented on the platforms	Implementation pattern for steady states	PRD-2.6 Design production systems and workflows	SAF-P07 Implement loosely coupled processes and systems **[VC29]**
Review the architecture with the architecture board	Architecture decision record	NSI-specific	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes

> **i** On the Medallion architecture
>
> A platform team may use Bronze, Silver, and Gold to organize storage, staging, and table optimization inside the platform. That is an engineering pattern, not a governance model. It describes how data has been refined, not what the data means, what methodological state it is in, or what another statistical process may rely on. The steady states remain the governance model; Medallion, if used at all, sits underneath it as a storage convention. **[VC29]**

VS-09.3 Provision and Configure Platforms

Value created: *Controlled environments that the value streams can use.*

The architecture is realized by provisioning and configuring the platforms. Environments are deployed for development, testing, research, and production, with the necessary governance, monitoring, and security controls. The secure processing environment for researchers is provisioned with its specific controls. Self-service is offered where possible, so that teams can obtain standard environments without waiting. Each environment is documented and registered, with an owner and a service level.

Activity	Key outputs	SAF process	Other reference
Deploy environments for development, testing, research, and production	Deployed environments	PRD-3.4 Configure workflows	3.2.5 IT infrastructure
Configure governance, monitoring, and security controls per environment	Controlled environments	PRD-3.4 Configure workflows	SAF-P10 Security by design
Provision the secure processing environment for researchers	Secure processing environment	MDS-4 Provide a secure processing environment	SAF-P11 Privacy by design

Activity	Key outputs	SAF process	Other reference
Offer standard environments as self-service and register them with an owner and service level	Self-service catalog, environment register	NSI-specific	SAF-P05 IT and technology is everyone's business; 3.4.4 Self-service

VS-09.4 Enable Integration and Orchestration Services

Value created: *Processes and value streams connected through stable, versioned interfaces.*

The platforms are connected through integration and orchestration services. These are the APIs, workflow engines, and automation that link the business services and processes. They include the steady-state event interfaces that the primary value streams rely on, such as *Raw Data Registered* and *Processed Data Registered*. They include the connectors for data spaces and for transmission to international organizations. Interfaces are versioned and documented, so that a change on one side does not break the other, in line with SAF-P07.

Activity	Key outputs	SAF process	Other reference
Provide APIs, workflow engines, and automation	Integration and orchestration services	PRD-3.4 Configure workflows	3.2.1 Application development
Provide the steady-state event interfaces of the primary value streams	Event interfaces	PRD-3.4 Configure workflows	SAF-P07 Implement loosely coupled processes and systems
Provide connectors for data spaces and international transmission	Data space connectors	DSP-4 Exchange data under usage control	4.1.6 Collaboration
Version and document every interface	Interface catalog	NSI-specific	SAF-P03 Active life cycle management

VS-09.5 Ensure Performance, Security, and Continuity

Value created: *A chain that runs reliably and safely and can recover from disruption.*

Day to day, VS-09 keeps the platforms running and safe. It monitors performance and capacity, with special attention to known peaks such as release moments. It implements the technical data protection that the classification of each dataset requires, including the storage of immutable versions. It manages incidents and restores service within the agreed times. It guarantees availability and recovery from disaster through tested continuity plans. Evidence on security posture and continuity is reported to VS-07.

Activity	Key outputs	SAF process	Other reference
Monitor performance and capacity, including known peaks	Performance and capacity reports	PRD-3.5 Test production systems	SAF-P02 Active support
Implement technical data protection and the storage of immutable versions	Protected storage	DLM-5 Store and version data	SAF-P10 Security by design; 3.3.6 IT security services
Manage incidents and restore service	Incident records	NSI-specific	3.3.3 Security incident management
Guarantee availability and disaster recovery through tested continuity plans	Continuity plans, test results	NSI-specific	SAF-P06 Business continuity

VS-09.6 Evolve and Modernize the Technical Ecosystem

Value created: *A foundation that stays current and ready for new data sources, methods, and ways of working.*

The life cycle closes with the modernization of the ecosystem. New technologies are introduced, often after they have been tested in VS-08. Components that are obsolete or no longer supported are retired according to a roadmap. Dependence on single suppliers is reviewed, in line with SAF-P01. The roadmap is aligned with the strategy of the NSI and with the architecture. Changes are planned so that the production value streams can continue without disruption.

Activity	Key outputs	SAF process	Other reference
Introduce new technologies transferred from VS-08 Innovation and Methodological Development	Adopted technologies	No reference process (NSI-specific)	4.1.3 Change management
Retire obsolete and unsupported components according to the roadmap	Retirement records	No reference process (NSI-specific)	SAF-P03 Active life cycle management
Review dependence on single suppliers	Supplier dependence review	No reference process (NSI-specific)	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Maintain the modernization roadmap, aligned with strategy and architecture	Modernization roadmap	No reference process (NSI-specific)	4.1.7 Architecture management

4. The four process types

VS-09 is itself a capability that must be designed, built, operated, and improved. The four process types therefore apply to VS-09 as they do to the primary value streams, but they govern the platform capability rather than a data pipeline. Design produces the operating model and the service catalog, Implementation builds the service management and the supplier arrangements, Execution runs the stages of attribute 3, and Management evaluates whether the platforms meet the needs of the value streams. **[VC26]**

Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. Where no reference process exists, the sub-process is marked as NSI-specific.

Design: provides the platform operating model

Design sub-process	Refines	Main outputs
Design the Platform Operating Model and Service Catalog	PRD-2.6 Design production systems and workflows; NSI-specific	Operating model, service catalog, service levels
Design the Security and Continuity Architecture	PRD-2.6 Design production systems and workflows; DLM-4 Classify and protect data	Security architecture, continuity requirements

Design the Platform Operating Model and Service Catalog

Refines: PRD-2.6 Design production systems and workflows; NSI-specific IT management processes.

This sub-process designs how VS-09 delivers its services. It defines the service catalog: which platforms and services are offered, to whom, and with which service levels. It defines the operating model: which services are run by the NSI, which are shared with other government bodies, and which are sourced from suppliers. It defines how teams request services and how requests are prioritized. It sets the service levels for availability, response, and recovery. The catalog makes the offer of VS-09 visible and comparable for all value streams.

Activity	Key outputs	SAF process	Other reference
Define the service catalog with services, users, and service levels	Service catalog	NSI-specific	3.4.1 Services management
Define the operating model: own, shared, and sourced services	Operating model	NSI-specific	3.1.2 Purchasing and contract management
Define the procedure for requests and their prioritization	Request procedure	PRD-2.6 Design production systems and workflows	SAF-P04 Maximize the benefits for the organization

Design the Security and Continuity Architecture

Refines: PRD-2.6 Design production systems and workflows; DLM-4 Classify and protect data.

This sub-process designs how platforms are protected and kept available. It translates the security and privacy policy of VS-07 into a security architecture: identity and access control, encryption, logging, and separation of environments. It links the protection level of each platform to the classification of the data it holds. It defines the continuity requirements per service, based on the impact of an outage on the chain and on release commitments. It designs the recovery approach and the tests that prove it works.

Activity	Key outputs	SAF process	Other reference
Translate the security and privacy policy into a security architecture	Security architecture	PRD-2.6 Design production systems and workflows	SAF-P10 Security by design; 3.3.2 Security planning
Link platform protection to the classification of the data held	Protection levels per platform	DLM-4 Classify and protect data	3.3.4 Logical identity and access control
Define continuity requirements and the recovery approach per service	Continuity requirements	NSI-specific	SAF-P06 Business continuity

Implementation: provides the service management

Implementation sub-process	Refines	Main outputs
Establish Service Management and Operations	NSI-specific	Service desk, incident and change management, monitoring
Arrange Supplier and Sourcing Agreements	NSI-specific	Contracts, service agreements with suppliers

Establish Service Management and Operations

Refines: NSI-specific IT service management processes.

This sub-process puts the operating model into practice. The service desk is set up for requests and incidents. Incident, problem, and change management are established, with clear roles and escalation paths. Monitoring of availability, performance, capacity, and security is configured across all platforms.

Operational procedures and runbooks are written and tested. Operations staff are trained, so that services are supported actively and not only built once, in line with SAF-P02.

Activity	Key outputs	SAF process	Other reference
Set up the service desk for requests and incidents	Service desk	NSI-specific	3.4.1 Services management
Establish incident, problem, and change management	Service management processes	NSI-specific	SAF-P02 Active support
Configure monitoring and write operational procedures	Monitoring, runbooks	NSI-specific	3.2.3 Technical management

Arrange Supplier and Sourcing Agreements

Refines: NSI-specific procurement processes.

This sub-process arranges the services that VS-09 sources from outside. It selects suppliers according to the operating model and the procurement rules. It negotiates service agreements that match the service levels the NSI promises to its value streams. It includes requirements for security, data location, exit, and the use of open standards. It makes sure that the NSI keeps control over critical components and data, in line with SAF-P01. Agreements are registered and reviewed before they expire.

Activity	Key outputs	SAF process	Other reference
Select suppliers according to the operating model and procurement rules	Supplier selection	NSI-specific	3.1.2 Purchasing and contract management
Negotiate service agreements that match the promised service levels	Service agreements	NSI-specific	3.4.3 Service agreement
Include requirements for security, data location, exit, and open standards	Contract requirements	NSI-specific	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes

Execution: runs the platform services

Execution runs the stages of attribute 3. VS-09.1 and VS-09.2 run when needs change or new value streams are set up; VS-09.3 and VS-09.4 run per request and per release of the platforms; VS-09.5 runs continuously; VS-09.6 runs as a roadmap.

Aspect	VS-09
Trigger	*Platform or service request*; *Incident*; *Capacity change*; *Technology onboarding* from VS-08 Innovation and Methodological Development
Cadence	Continuous operation; per request; roadmap cycle
Performed by	ROL-11 Platform engineer, ROL-10 Data engineer, ROL-20 Security officer
Input	Requirements of the value streams, security and continuity policy, supplier services

Aspect	VS-09
Output	Platforms, environments, integration services, event interfaces, availability and continuity evidence
Control point	Architecture review before realization; acceptance of an environment before production use
Exception path	A major incident escalated according to the continuity plan; a security incident handled with ROL-20 Security officer and VS-07 Governance and Compliance

Management: provides service performance reports

Management sub-process	Refines	Main outputs
Monitor Service Performance	PRD-8.1 Gather evaluation inputs	Service level reports, capacity and security reports
Evaluate and Improve the Platform Capability	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Satisfaction of value streams, improvement plan

Monitor Service Performance

Refines: PRD-8.1 Gather evaluation inputs.

This sub-process tracks how the platforms perform against the agreed service levels. It measures the key performance indicators of VS-09 (see attribute 9), such as availability, time to restore service, and the reliability of event interfaces. It reports capacity headroom ahead of known peaks. It reports the security posture, together with VS-07. It reviews the performance of suppliers against their agreements. The results are shared with the process owners of the value streams that depend on the platforms.

Activity	Key outputs	SAF process	Other reference
Measure the key performance indicators of VS-09 Infrastructure and Platform Enablement	Service level reports	PRD-8.1 Gather evaluation inputs	SAF-P02 Active support
Report capacity headroom and security posture	Capacity and security reports	PRD-8.1 Gather evaluation inputs	SAF-P06 Business continuity; SAF-P10 Security by design
Review supplier performance against agreements	Supplier performance review	NSI-specific	3.4.3 Service agreement

Evaluate and Improve the Platform Capability

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This sub-process asks whether the platforms meet the needs of the value streams. It collects the satisfaction of the process owners with the services. It analyzes recurring incidents and their root causes. It assesses the cost and value of the services. It feeds the findings into the roadmap of VS-09.6 and into the design of the service catalog. Improvements are prioritized by impact and feasibility and tracked, in line with SAF capability 4.1.4 *Improvement management*.

Activity	Key outputs	SAF process	Other reference
Collect the satisfaction of process owners with the services	Satisfaction survey	PRD-8.2 Conduct evaluation	3.4.1 Services management

Activity	Key outputs	SAF process	Other reference
Analyze recurring incidents and their root causes	Problem analysis	PRD-8.2 Conduct evaluation	
Feed findings into the roadmap and the service catalog, and track improvements	Improvement plan	PRD-8.3 Agree an action plan	4.1.4 Improvement management

5. Actors and roles

VS-09 brings architecture, engineering, operations, and security together. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the platform or cloud architect, the DevOps engineer, and service desk and operations staff are ROL-11 Platform engineer, the integration engineer is ROL-10 Data engineer, and the security and continuity officer is ROL-20 Security officer. The chief technology or infrastructure officer is part of the management of the NSI (ACT-01) and a member of the collaboration COL-07 *Architecture board*.

Actor or role	ID	Responsibility in VS-09 Infrastructure and Platform Enablement
National Statistical Institute	ACT-01	Its management, including the chief technology officer, is accountable for the technical foundation and its strategy
Supplier and service provider	ACT-09	Supplies infrastructure, cloud services, software, and support under service agreements
Process owners of all value streams	ROL-01	Define requirements and run their processes on the platforms
Data custodian	ROL-04	Operates storage, access control, retention, and key management on the platforms
Metadata and data architect	ROL-08	Aligns the catalog and repositories with the metadata model and the steady states
Data engineer	ROL-10	Builds APIs, workflow engines, orchestration, and event interfaces
Platform engineer	ROL-11	Designs the platform architecture; provisions, configures, and operates environments; runs incident management
Security officer	ROL-20	Implements security controls, monitoring, and disaster recovery, together with VS-07 Governance and Compliance

6. Capabilities, processes, and functions

Capability foundations

Capability group	Capabilities	Role in VS-09 Infrastructure and Platform Enablement
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management, 3.2.3 Technical management, 3.2.4 Digital workplace management, 3.2.5 IT infrastructure	Build, configure, and operate applications, platforms, and infrastructure
3.3 Security and privacy	3.3.2 Security planning, 3.3.3 Security incident management, 3.3.4 Logical identity and access control, 3.3.5 Physical security and access control, 3.3.6 IT security services	Protect platforms and data, and handle incidents
3.1 Business operations	3.1.2 Purchasing and contract management	Source infrastructure and services from suppliers
3.4 Customer and chain interaction	3.4.1 Services management, 3.4.3 Service agreement, 3.4.4 Self-service	Offer platform services with service levels and self-service
4.1 Strategy and leadership	4.1.3 Change management, 4.1.7 Architecture management	Manage technological change and the architecture of the platforms

Stages, processes, and functions

The table follows the mapping in *Business Processes, Mapping to value stream stages.*

Stage	Realized by processes	Main functions
VS-09.1 Define Digital and Technical Requirements	PRD-2.6 Design production systems and workflows	FUN-2.5 Manage enterprise architecture; FUN-3.1 Design and build statistical products
VS-09.2 Design Platform and Infrastructure Architecture	PRD-2.6 Design production systems and workflows	FUN-2.5 Manage enterprise architecture; FUN-10.3 Manage information technology
VS-09.3 Provision and Configure Platforms	PRD-3.4 Configure workflows; MDS-4 Provide a secure processing environment	FUN-8.3 Operate secure research environments; FUN-10.3 Manage information technology
VS-09.4 Enable Integration and Orchestration Services	PRD-3.4 Configure workflows; DSP-4 Exchange data under usage control	FUN-9.2 Operate data space connectivity; FUN-10.3 Manage information technology
VS-09.5 Ensure Performance, Security, and Continuity	PRD-3.5 Test production systems; DLM-5 Store and version data	FUN-10.3 Manage information technology; FUN-10.8 Manage information security and privacy
VS-09.6 Evolve and Modernize the Technical Ecosystem	No reference process (NSI-specific)	FUN-2.2 Develop capability improvements; FUN-10.3 Manage information technology

PLACEHOLDER: ArchiMate view VS-09 (b) stages realized by processes and functions

Framework mappings

Where the primary value streams map to the production phases of GSBPM, VS-09 maps to the capability and support activities of GAMSO.

GAMSO area	Relevance to VS-09 Infrastructure and Platform Enablement	SAF capabilities
Capability Management	Manage the IT capability; introduce and modernize technology	4.1.3 Change management, 4.1.7 Architecture management
Corporate Support	Manage IT services and operations, security, and business continuity	3.2.1 to 3.2.5, 3.3.2 to 3.3.6, 3.4.1 Services management

7. Principles

Principle	Statement	How VS-09 Infrastructure and Platform Enablement applies it
SAF-P01	Reuse before open source, open source before buying, buying before building	Favors open, shared platform components and open standards, to keep collective control and avoid dependence on single suppliers **[VC3]**
SAF-P02	Active support	Runs platforms as actively supported, monitored services, not one-off builds
SAF-P05	IT and technology is everyone's business	Provides self-service, well-documented platforms, so technology serves all teams
SAF-P06	Business continuity	Guarantees availability, resilience, and recovery from disaster for the whole chain
SAF-P07	Implement loosely coupled processes and systems	Provides the versioned interfaces and events that keep the value streams decoupled
SAF-P10	Security by design	Builds security and access control into every platform and environment from the start

8. Interfaces, services, and events

Business services

Service	Offered to	Realized in	Interfaces
Platform and environment service	All value streams	VS-09.3 Provision and Configure Platforms	Service catalog, self-service portal, service desk
Steady-state storage and catalog service	VS-01 to VS-06 (primary value streams)	VS-09.3 Provision and Configure Platforms; VS-09.5 Ensure Performance, Security, and Continuity	Repositories per steady state, data catalog
Integration and event service	All value streams	VS-09.4 Enable Integration and Orchestration Services	APIs, workflow engines, event interfaces, data space connectors
Dissemination platform service	VS-06 Dissemination and Service Delivery	VS-09.3 Provision and Configure Platforms; VS-09.4 Enable Integration and Orchestration Services	Portals, APIs, dashboards, secure research environment

Service	Offered to	Realized in	Interfaces
Research environment service	VS-08 Innovation and Methodological Development	VS-09.3 Provision and Configure Platforms	Sandboxes with research tooling
Operations and continuity service	All value streams, VS-07 Governance and Compliance	VS-09.5 Ensure Performance, Security, and Continuity	Monitoring, incident management, continuity reports

Business events

Event	Triggers or marks
Platform or service request	Start of VS-09.3 Provision and Configure Platforms or VS-09.4 Enable Integration and Orchestration Services for a request
Technology onboarding	Adoption of a technology transferred from VS-08 Innovation and Methodological Development in VS-09.6 Evolve and Modernize the Technical Ecosystem
Capacity change	Adjustment of capacity in VS-09.5 Ensure Performance, Security, and Continuity, for example ahead of a release
Incident / Service restored	Start and end of incident handling in VS-09.5 Ensure Performance, Security, and Continuity
Security or continuity review	Periodic review with VS-07 Governance and Compliance
Component retired	Retirement according to the roadmap in VS-09.6 Evolve and Modernize the Technical Ecosystem

PLACEHOLDER: VS-09 event choreography between the value streams, VS-09, suppliers, and VS-07: service request, provisioning, incident and restoration, capacity change, security review, technology onboarding, and retirement

9. Key performance indicators

KPI	Description	Measurement
Platform availability	Measured availability against the agreed service level, per platform	Monthly
Incident response	Mean time to restore service after an incident, per severity class	Per incident, trended quarterly
Event interface reliability	Delivery rate of the steady-state handover events that the chain depends on	Per interface, monthly
Capacity headroom	Share of platforms operating within their planned capacity	Monthly, ahead of known peaks
Security posture	Open critical vulnerabilities and time to patch, assessed with VS-07 Governance and Compliance	Continuous, reported quarterly
Modernization progress	Components retired or replaced according to the roadmap	Annually

10. Toolkit

Examples from NSI practice and a maturity checklist for VS-09 are part of the SAF Toolkit.

Toolkit: SAF-TK-V09 VS-09 Infrastructure and Platform Enablement companion **[VC17]**

VS-10 Customer and Stakeholder Engagement

1. Purpose and value created

VS-10 keeps the NSI connected, relevant, and responsive to the people it serves. It builds and maintains mutual understanding and trust between the NSI and its users, data providers, partners, and policy stakeholders. Official statistics exist to create public value, so the NSI must understand what stakeholders need, communicate openly, and keep its products, services, and policies in line with real demand. VS-10 is not a one-way process. It is a continuous dialogue that turns what stakeholders say into better products, better services, and stronger relationships. The value created is **relevance and public value**.

VS-10 is a supporting value stream. It aligns with the GAMSO areas *Strategy and Leadership* and *Corporate Support* rather than with the production phases of GSBPM. This section describes its business layer for all four process types.

- **Relationship manager, user researcher, or service desk staff:** the stages in attribute 3, which describe the engagement cycle.
- **Head of communication or dissemination and service manager:** the Design processes in attribute 4.
- **Process owners of other value streams:** attributes 2 and 8, which describe how needs flow to them and how feedback returns.
- **Manager who wants the big picture:** attributes 1, 2, and 9. Examples and scenarios are part of the SAF Toolkit.

> **i** How a supporting value stream differs from a primary one
>
> VS-10 has no steady-state data transition. It does not move data along the chain; it moves understanding: needs come in, and communication and services go out. It runs continuously as an ongoing dialogue, not per production cycle. Its output is an enabling capability rather than a data artifact: understood needs, a managed service portfolio, relationships, and insight into impact. It uses the usage data of State #5 as a secondary consumer, to measure reach and impact.

> **i** Indicative roles, not a prescribed structure
>
> The roles in attribute 5 are indicative, not a mandated organizational structure. The SAF deliberately leaves organization design to each NSI. An NSI assigns these responsibilities to fit its own size, culture, and operating model.

PLACEHOLDER: ArchiMate view VS-10 (a) value stream and capabilities

2. Position and interaction with the chain

VS-10 is not a stage in the production line. It sits at the edges of the chain. On one side it faces the data providers who feed VS-01; on the other side it faces the users who consume what VS-06 releases. It channels what it learns inward, to shape the whole chain. VS-10 is closely coupled with VS-01, for the relationships with providers, and with VS-06, from which it receives usage and satisfaction data. It translates what it hears into requirements that flow into the design of the primary value streams and into the innovation backlog of VS-08.

Counterpart	VS-10 provides	VS-10 receives
Users and society	Services, support, communication, the service portfolio	Needs, feedback, satisfaction
Data providers and respondents, through VS-01 Data Acquisition and Ingestion	Relationship management, transparency on the use and protection of their data, burden reduction	Willingness to provide data, needs and concerns of providers
VS-06 Dissemination and Service Delivery	Service requirements, user groups, expectations of users	Usage and satisfaction data
VS-01 to VS-05 (design of the primary value streams)	Requirements for new or changed products, breakdowns, and timeliness	Information on what can be delivered, and when
VS-08 Innovation and Methodological Development	User-driven needs for new methods, sources, and services	New and improved products and services
VS-07 Governance and Compliance	Signals on stakeholder trust, complaints	Commitments on transparency and accountability
Partners: other producers of official statistics, international organizations, data space partners	Coordination and joint communication	Joint requirements and agreements

PLACEHOLDER: ArchiMate view VS-10 (c) interaction with the chain: VS-10 at both edges of the chain, facing data providers at VS-01 and users at VS-06, and channeling needs inward to the design of the primary value streams and to VS-08

3. Stages

The stages of VS-10 form a continuous engagement cycle: understand, offer, deliver, and measure. What is learned in the last stage improves the understanding in the first.

Stage	Name	Value added
VS-10.1	Identify and Segment Stakeholders	It is clear who the stakeholders are and what they expect
VS-10.2	Collect and Analyze User Needs	Stakeholder needs are known and translated into requirements
VS-10.3	Define and Manage Service Portfolios	The offer of products and services matches needs and strategy
VS-10.4	Deliver Services and Manage Relationships	Stakeholders are served well and relationships are maintained
VS-10.5	Engage and Communicate with Stakeholders	Stakeholders are aware of the statistics and understand them correctly
VS-10.6	Monitor Impact and Satisfaction	The use, impact, and appreciation of the statistics are known

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-10.1 Identify and Segment Stakeholders

Value created: *A clear view of who the stakeholders are, what role they play, and what they expect.*

The cycle begins with a map of the stakeholder landscape. It covers users, data providers, partners, and policy makers, as well as the media and the public. Stakeholders are grouped into segments with similar needs and expectations, for example policy makers, researchers, businesses, and the general public. Key stakeholders are identified, such as the main ministries, large data providers, and international partners. The roles of other producers of official statistics and of data space partners are included. The map is kept current, because the landscape changes with new data sources, new users, and new partnerships.

Activity	Key outputs	SAF process	Other reference
Map users, data providers, partners, policy makers, media, and the public	Stakeholder map	PRD-1.2 Consult and confirm needs	3.4.2 Contact management
Group stakeholders into segments with similar needs	Stakeholder segments	PRD-1.2 Consult and confirm needs	2.2.5 User research
Identify key stakeholders and partners, including data space partners	Key stakeholder list	DSP-6 Participate in data space governance	4.1.6 Collaboration
Keep the map current	Updated stakeholder map	NSI-specific	SAF-P03 Active life cycle management

VS-10.2 Collect and Analyze User Needs

Value created: *Stakeholder needs known and translated into concrete data or service requirements.*

Next, the needs of stakeholders are collected and analyzed. Feedback is gathered through consultations, user councils, surveys, and the usage data from VS-06. Needs are analyzed for their value, their urgency, and the number of stakeholders they concern. They are translated into concrete requirements: a new statistic, a finer breakdown, a faster release, or a new channel. Each requirement is routed to the value stream that can meet it, or to VS-08 when it needs a new method or source. The needs register records what was asked, what was decided, and why.

Activity	Key outputs	SAF process	Other reference
Gather feedback through consultations, user councils, surveys, and usage data	Collected needs	PRD-1.1 Identify needs; PRD-1.2 Consult and confirm needs	2.2.5 User research
Analyze needs for value, urgency, and reach	Needs analysis	PRD-1.1 Identify needs	1.1.1 Needs inventory
Translate needs into concrete data or service requirements	Requirements	PRD-1.2 Consult and confirm needs	SAF-P04 Maximize the benefits for the organization
Route requirements to the value streams or to VS-08 Innovation and Methodological Development, and record decisions	Needs register, *Need routed to design or innovation*	NSI-specific	4.1.5 Accountability

VS-10.3 Define and Manage Service Portfolios

Value created: *A portfolio of products and services that matches stakeholder needs and the strategy of the NSI.*

The needs shape the service portfolio. The portfolio is the catalog of business services and data products that the NSI offers, from publications and open data to tailored analyses and research access. It is the bridge between what users want and what the chain produces. New services are added when needs and

strategy justify them, and services that are no longer used are retired. Each service has an owner, a target group, and a service standard. The portfolio is aligned with the dissemination strategy of VS-06 and the capacity of the production value streams.

Activity	Key outputs	SAF process	Other reference
Define the catalog of business services and data products	Service and product portfolio	PRD-2.1 Design outputs and dissemination	3.4.1 Services management
Add and retire services based on needs and strategy	Portfolio decisions	PRD-2.1 Design outputs and dissemination	SAF-P03 Active life cycle management
Assign an owner, target group, and service standard to each service	Service descriptions	NSI-specific	3.4.3 Service agreement
Align the portfolio with the dissemination strategy and production capacity	Aligned portfolio	PRD-2.1 Design outputs and dissemination	SAF-P04 Maximize the benefits for the organization

VS-10.4 Deliver Services and Manage Relationships

Value created: *Stakeholders served well and relationships with key partners maintained.*

Services are put into the hands of users. VS-10 operates customer support and account management, together with the user support of VS-06. Relationships with key users, partners, and policy stakeholders are maintained through account managers. Relationships with data providers are maintained together with VS-01, with attention to burden and to transparency about the use of their data. Tailored services, such as custom tables, are delivered under the service standard. Service performance and satisfaction are tracked per service.

Activity	Key outputs	SAF process	Other reference
Operate customer support and account management	Support and account records	PRD-7.5 Provide user support	3.4.1 Services management
Maintain relationships with key users, partners, and policy stakeholders	Relationship plans	NSI-specific	3.4.2 Contact management
Maintain relationships with data providers, together with VS-01 Data Acquisition and Ingestion	Provider relationship records	OVA-6 Data Supplier Management	SAF-P11 Privacy by design
Deliver tailored services and support researchers	Tailored services	PRD-7.5 Provide user support; MDS-6 Support researchers	2.2.4 Data services

VS-10.5 Engage and Communicate with Stakeholders

Value created: *Stakeholders who know the statistics exist and understand them correctly.*

Alongside delivery, VS-10 engages and communicates with stakeholders. It runs outreach, workshops, and events that build awareness of statistical outputs. It works with the media, so that figures are reported correctly, including their uncertainty. It promotes statistical literacy among the public and in education. It communicates openly about how the NSI works, how it protects data, and how it handles errors. This builds the public confidence on which the willingness to respond and the use of statistics depend.

Activity	Key outputs	SAF process	Other reference
Run outreach, workshops, and events	Outreach activities	PRD-7.4 Promote dissemination products	1.5.2 Promotion to use
Work with the media on correct reporting	Media relations	PRD-7.4 Promote dissemination products	3.1.4 Communication management
Promote statistical literacy	Literacy programs	NSI-specific	1.5.2 Promotion to use
Communicate openly on methods, data protection, and corrections	Transparency communication	NSI-specific	SAF-P11 Privacy by design; 4.1.5 Accountability

VS-10.6 Monitor Impact and Satisfaction

Value created: *Knowledge of how statistics and services are used, valued, and what impact they have.*

The cycle closes by measuring impact and satisfaction. Satisfaction surveys are run across user segments and services. The usage data from VS-06 is analyzed to understand reach. Documented uses of official statistics in policy, research, and the media are collected as evidence of impact. Complaints and trust signals are passed to VS-07. The insights are fed back into the needs analysis, the service portfolio, and the innovation backlog, so that the next cycle of needs is better understood.

Activity	Key outputs	SAF process	Other reference
Run satisfaction surveys across segments and services	Satisfaction results	PRD-8.1 Gather evaluation inputs	2.2.5 User research
Analyze usage data from VS-06 Dissemination and Service Delivery	Reach analysis	PRD-8.1 Gather evaluation inputs	
Collect documented uses of statistics in policy, research, and the media	Impact evidence	PRD-8.2 Conduct evaluation	4.1.5 Accountability
Feed insights back into needs, portfolio, and innovation; pass complaints to VS-07 Governance and Compliance	Impact and satisfaction insight	PRD-8.2 Conduct evaluation	4.1.4 Improvement management

4. The four process types

VS-10 is itself a capability that must be designed, built, operated, and improved. The four process types therefore apply to VS-10 as they do to the primary value streams, but they govern the engagement capability rather than a data pipeline. Design produces the engagement strategy and the portfolio model, Implementation builds the channels for support and feedback, Execution runs the stages of attribute 3, and Management evaluates whether engagement creates relevance. [**VC26**]

Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. Where no reference process exists, the sub-process is marked as NSI-specific.

Design: provides the engagement framework

Design sub-process	Refines	Main outputs
Design the Engagement Strategy	NSI-specific	Engagement strategy, segment approach, communication principles

Design sub-process	Refines	Main outputs
Design the Service Portfolio Model	PRD-2.1 Design outputs and dissemination	Portfolio model, service standards, criteria for adding and retiring services

Design the Engagement Strategy

Refines: NSI-specific strategic planning processes.

This sub-process sets the direction of engagement. It defines the goals of engagement in line with the strategy of the NSI, for example more reach among young people or better use of statistics in policy. It defines the approach per stakeholder segment: which channels, how often, and with which message. It sets the principles for communication, such as openness about uncertainty and about errors. It defines how user councils and consultations are organized. A clear strategy keeps engagement focused rather than spread thin.

Activity	Key outputs	SAF process	Other reference
Define the goals of engagement in line with the strategy	Engagement strategy	NSI-specific	4.1.1 Strategy and governance
Define the approach per stakeholder segment	Segment approach	NSI-specific	3.4.2 Contact management
Set communication principles and organize user councils and consultations	Communication principles, consultation model	NSI-specific	3.1.4 Communication management

Design the Service Portfolio Model

Refines: PRD-2.1 Design outputs and dissemination.

This sub-process designs how the service portfolio is managed. It defines the types of services that the NSI offers and the rules that apply to each type, such as free or paid and standard or tailored. It sets the service standards for response and delivery. It defines the criteria for adding a service and for retiring one. It defines how the portfolio is reviewed and by whom. The model makes portfolio decisions consistent and explainable to stakeholders.

Activity	Key outputs	SAF process	Other reference
Define the types of services and the rules that apply to each	Portfolio model	PRD-2.1 Design outputs and dissemination	3.4.1 Services management
Set service standards for response and delivery	Service standards	PRD-2.1 Design outputs and dissemination	3.4.3 Service agreement
Define the criteria for adding and retiring services, and the review procedure	Portfolio criteria	PRD-2.1 Design outputs and dissemination	SAF-P03 Active life cycle management

Implementation: provides the engagement channels

Implementation sub-process	Refines	Main outputs
Build Support and Feedback Channels	PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows	Service desk, feedback channels, needs register, relationship records

Implementation sub-process	Refines	Main outputs
Establish Relationship Management	OVA-6 Data Supplier Management; NSI-specific	Account plans, relationship managers, provider engagement procedures

Build Support and Feedback Channels

Refines: PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows.

This sub-process builds the channels through which stakeholders reach the NSI, together with VS-09. It sets up the service desk and the channels for questions, by phone, e-mail, chat, and web forms. It builds the channels for feedback, such as surveys and comment functions on the website. It sets up the needs register and the system for relationship records. It connects these channels to the usage data of VS-06. Every channel records what it receives, so that needs and feedback are not lost.

Activity	Key outputs	SAF process	Other reference
Set up the service desk and the channels for questions	Service desk and channels	PRD-3.3 Reuse or build dissemination components	3.4.4 Self-service
Build the channels for feedback	Feedback channels	PRD-3.3 Reuse or build dissemination components	2.2.5 User research
Set up the needs register and the relationship records, connected to usage data	Needs register, relationship records	PRD-3.4 Configure workflows	SAF-P09 No data without metadata and classification

Establish Relationship Management

Refines: OVA-6 Data Supplier Management; NSI-specific relationship processes.

This sub-process puts relationship management into practice. Account managers are assigned to key users, partners, and policy stakeholders. Account plans are written for the most important relationships. The procedures for engaging data providers are agreed with VS-01, including how burden is discussed and how providers are informed about the use of their data. Agreements with partners, such as other producers of official statistics, are registered. Staff are trained in engagement and communication.

Activity	Key outputs	SAF process	Other reference
Assign account managers and write account plans	Account plans	NSI-specific	3.4.2 Contact management
Agree procedures for engaging data providers with VS-01 Data Acquisition and Ingestion	Provider engagement procedures	OVA-6 Data Supplier Management	SAF-P11 Privacy by design
Register partner agreements and train staff	Partner register, trained staff	NSI-specific	4.1.6 Collaboration

Execution: runs the engagement cycle

Execution runs the stages of attribute 3 continuously. VS-10.1 and VS-10.3 run as periodic reviews; VS-10.2, VS-10.4, and VS-10.5 run continuously; VS-10.6 runs per survey wave and per reporting period.

Aspect	VS-10
Trigger	*User need captured*; *Service request*; *Feedback received* from VS-06 Dissemination and Service Delivery; a planned consultation or campaign
Cadence	Continuous dialogue; periodic reviews of stakeholders and portfolio; survey waves
Performed by	ROL-23 Relationship manager, ROL-22 Dissemination and service manager
Input	Feedback, consultations, usage data, complaints, strategy of the NSI
Output	Needs register, service portfolio, relationships, communication, impact and satisfaction insight
Control point	Portfolio review; decisions on needs recorded in the needs register
Exception path	A complaint about confidentiality or conduct passed to VS-07 Governance and Compliance; a need that the chain cannot meet explained to the stakeholder

Management: provides engagement performance reports

Management sub-process	Refines	Main outputs
Monitor Engagement Performance	PRD-8.1 Gather evaluation inputs	Engagement dashboard, service performance reports
Evaluate and Improve the Engagement Capability	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Evaluation of engagement, improvement plan

Monitor Engagement Performance

Refines: PRD-8.1 Gather evaluation inputs.

This sub-process tracks how engagement performs. It measures the key performance indicators of VS-10 (see attribute 9), such as user satisfaction and the conversion of needs into requirements. It tracks service performance against the service standards. It tracks reach and participation in consultations and outreach. It reports the results to management and to the value streams concerned. Visible results show stakeholders that their input leads to action.

Activity	Key outputs	SAF process	Other reference
Measure the key performance indicators of VS-10 Customer and Stakeholder Engagement	Engagement dashboard	PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Track service performance against the service standards	Service performance report	PRD-8.1 Gather evaluation inputs	3.4.3 Service agreement
Track reach and participation	Reach report	PRD-8.1 Gather evaluation inputs	1.5.2 Promotion to use

Evaluate and Improve the Engagement Capability

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This sub-process asks whether engagement creates relevance and public value. It evaluates whether captured needs led to changes in products and services, and how long that took. It assesses which segments are reached well and which are not. It reviews the effectiveness of channels and campaigns. It adjusts the strategy, the portfolio model, and the channels based on the findings. Improvements are prioritized and tracked, in line with SAF capability 4.1.4 *Improvement management*.

Activity	Key outputs	SAF process	Other reference
Evaluate whether needs led to changes, and how quickly	Needs follow-up evaluation	PRD-8.2 Conduct evaluation	4.1.5 Accountability
Assess the reach per segment and the effectiveness of channels	Segment and channel evaluation	PRD-8.2 Conduct evaluation	2.2.5 User research
Adjust strategy, portfolio model, and channels, and track improvements	Improvement plan	PRD-8.3 Agree an action plan	4.1.4 Improvement management

5. Actors and roles

VS-10 faces outward more than any other supporting value stream. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the head of communication, the account manager, the user researcher, the communications officer, and the impact analyst are ROL-23 Relationship manager, and the service portfolio manager and service desk staff are ROL-22 Dissemination and service manager. Relationships with data providers are organized in the collaboration COL-01 *Data provision partnership*, and relationships with other producers in COL-02 *National statistical system*.

Actor or role	ID	Responsibility in VS-10 Customer and Stakeholder Engagement
Data user	ACT-05	Expresses needs, uses services, gives feedback
Data provider	ACT-03	Provides data; is engaged on burden, transparency, and the use of their data
Respondent	ACT-04	Provides data through surveys; is informed about the purpose and protection of their data
Other producer of official statistics	ACT-06	Coordinates needs and communication within the national statistical system
International statistical organization	ACT-07	Expresses international requirements and cooperates on communication
Process owners of the production value streams	ROL-01	Receive requirements and report what can be delivered
Dissemination and service manager	ROL-22	Manages the service portfolio and day-to-day user support
Relationship manager	ROL-23	Owns the stakeholder strategy and the relationships with key users, providers, and partners; runs outreach, user research, and impact measurement

6. Capabilities, processes, and functions

Capability foundations

Capability group	Capabilities	Role in VS-10 Customer and Stakeholder Engagement
1.1 Statistical design and build	1.1.1 Needs inventory	Capture and analyze needs as input for the design of products
1.5 Dissemination	1.5.2 Promotion to use	Promote awareness and correct use of statistics
2.2 Research	2.2.5 User research	Understand users, their needs, and their satisfaction
3.1 Business operations	3.1.4 Communication management	Manage external communication and media relations
3.4 Customer and chain interaction	3.4.1 Services management, 3.4.2 Contact management, 3.4.3 Service agreement, 3.4.4 Self-service	Manage the service portfolio, relationships, service standards, and self-service
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.5 Accountability, 4.1.6 Collaboration	Align engagement with strategy, account to society, cooperate with partners

Stages, processes, and functions

The table follows the mapping in *Business Processes, Mapping to value stream stages*.

Stage	Realized by processes	Main functions
VS-10.1 Identify and Segment Stakeholders	PRD-1.2 Consult and confirm needs; DSP-6 Participate in data space governance	FUN-1.3 Manage strategic collaboration; FUN-10.9 Manage users and services
VS-10.2 Collect and Analyze User Needs	PRD-1.1 Identify needs; PRD-1.2 Consult and confirm needs	FUN-3.1 Design and build statistical products; FUN-10.9 Manage users and services
VS-10.3 Define and Manage Service Portfolios	PRD-2.1 Design outputs and dissemination	FUN-10.9 Manage users and services
VS-10.4 Deliver Services and Manage Relationships	PRD-7.5 Provide user support; MDS-6 Support researchers; OVA-6 Data Supplier Management	FUN-6.1 Manage data supplier relationships; FUN-10.9 Manage users and services
VS-10.5 Engage and Communicate with Stakeholders	PRD-7.4 Promote dissemination products	FUN-1.4 Build public confidence; FUN-10.6 Manage communication
VS-10.6 Monitor Impact and Satisfaction	PRD-8.1 Gather evaluation inputs; PRD-8.2 Conduct evaluation	FUN-5.2 Assure and review quality; FUN-10.9 Manage users and services

PLACEHOLDER: ArchiMate view VS-10 (b) stages realized by processes and functions

Framework mappings

Where the primary value streams map to the production phases of GSBPM, VS-10 maps to the leadership and support activities of GAMSO.

GAMSO area	Relevance to VS-10 Customer and Stakeholder Engagement	SAF capabilities
Strategy and Leadership	Manage relationships with stakeholders and customers; align the portfolio with the strategy	4.1.1 Strategy and governance, 4.1.6 Collaboration

GAMSO area	Relevance to VS-10 Customer and Stakeholder Engagement	SAF capabilities
Corporate Support	Manage communication, user support, and service management	3.1.4 Communication management, 3.4.1 Services management, 3.4.2 Contact management
Production (Specify Needs)	Provide the needs that start the design of statistical products	1.1.1 Needs inventory

7. Principles

Principle	Statement	How VS-10 Customer and Stakeholder Engagement applies it
SAF-P02	Active support	Runs user support and service management as actively managed services
SAF-P04	Maximize the benefits for the organization	Keeps the whole chain aligned with real stakeholder value and public benefit
SAF-P08	Data is shared property	Promotes broad and equal access to statistics and services, and their reuse
SAF-P11	Privacy by design	Engages data providers and users openly about how their data is used and protected

8. Interfaces, services, and events

Business services

Service	Offered to	Realized in	Interfaces
Needs intake service	Users, partners, policy stakeholders	VS-10.2 Collect and Analyze User Needs	Consultations, user councils, feedback channels, needs register
Service portfolio service	Users; VS-06 Dissemination and Service Delivery	VS-10.3 Define and Manage Service Portfolios	Service catalog
Relationship management service	Key users, partners, data providers	VS-10.4 Deliver Services and Manage Relationships	Account managers, relationship records
Customer support service	All users	VS-10.4 Deliver Services and Manage Relationships	Service desk, self-service, together with VS-06 Dissemination and Service Delivery
Communication and outreach service	The public, the media, education	VS-10.5 Engage and Communicate with Stakeholders	Events, media relations, literacy programs
Impact and satisfaction service	NSI management, value streams	VS-10.6 Monitor Impact and Satisfaction	Satisfaction surveys, impact reports

Business events

Event	Triggers or marks
User need captured	Entry of a need into the needs register in VS-10.2 Collect and Analyze User Needs
Need routed to design or innovation	Routing of a requirement to a primary value stream or to VS-08 Innovation and Methodological Development
Consultation held	Completion of a consultation or user council meeting
Service request	Start of service delivery in VS-10.4 Deliver Services and Manage Relationships
Feedback received	Receipt of usage and satisfaction data from VS-06 Dissemination and Service Delivery
Provider engagement	Contact with data providers, together with VS-01 Data Acquisition and Ingestion
Complaint received	Handling of a complaint and, where relevant, referral to VS-07 Governance and Compliance

PLACEHOLDER: VS-10 event choreography between users, data providers, VS-10, VS-06, the primary value streams, and VS-08: need captured, consultation, need routed, service request, feedback received, and complaint

9. Key performance indicators

KPI	Description	Measurement
User satisfaction	Satisfaction score across user segments and services	Per survey wave, per segment
Needs conversion	Share of captured needs converted into requirements routed to design or innovation	Per planning period
Service performance	Response and resolution times of support against the service standard	Monthly
Reach and engagement	Participation in consultations and outreach; reach of channels across segments	Per campaign, annually
Impact evidence	Documented uses of official statistics in policy, research, and the media	Annually

10. Toolkit

Examples from NSI practice and a maturity checklist for VS-10 are part of the SAF Toolkit.

Toolkit: SAF-TK-V10 VS-10 Customer and Stakeholder Engagement companion **[VC17]**